# Project Report

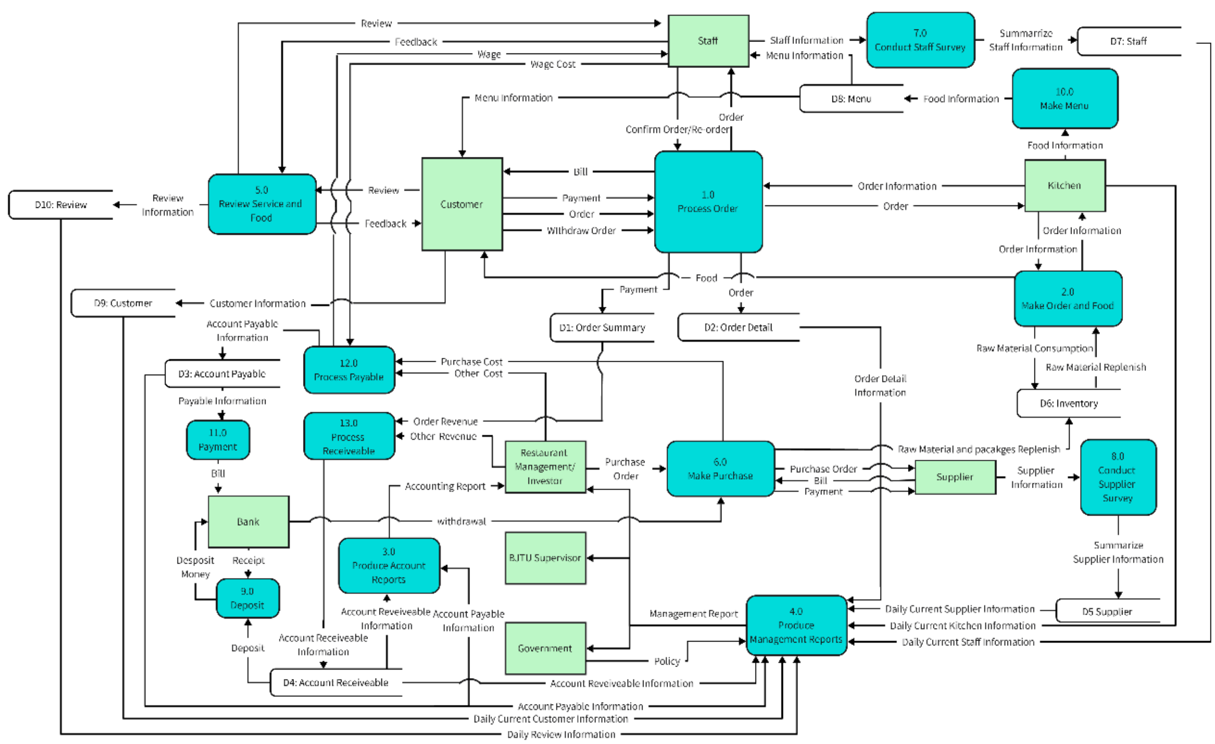
Margaret Western Restaurant
Super Management System

Version: 2.0

# Executive Overview

The main functions of the Margaret Western Restaurant information system include user information, dish information, order information, inventory information, staff information and supply, etc. These functions can help managers and relevant investors better control the inventory situation, ensure the freshness and sufficiency of food, and optimize staff management and menu design to obtain more profits

The relevant data of the restaurant information system is mainly input by employees in different positions, and various outputs and reports are read by managers and investors. The detailed data flow is shown in the following figure DFD:




- When entertaining clients, input information about the clients and the orders
- When making inventory purchases, new inventories are entered through the information of relevant suppliers and the corresponding purchase Item information
- When conducting various management and financial information statistics, inventory information (such as the Waste amount of raw materials) will be recorded in "Good Waste", customer hospitality situations (such as customer reviews) will be recorded in "customer review", and the discount situation of elderly customers will be counted
- When conducting property settlements with banks, recognition and summary are carried out through the input of accounts receivable and accounts payable

Various reports can be output from the above data stream. The eight main reports are the sales report, inventory report, customer analysis report, employee performance report, sales forecast, dish performance, Food waste report and financial summary report. For managers, understanding customers' preferences for dishes, inventory reserves, employees' performance, etc., is very important for calculating income and expenditure

# PART 1: Initial Project Proposal

# I. Background of Margaret Western Restaurant

## 1.1 Restaurant Introduction

MARGARET WESTERN RESTAURANT is a fast-food dining spot that serves convenient and delicious western food. It is located on the second floor of the /  Students' Canteen, which mainly serves the teachers, students and visitors in the campus. Similar to mainstream American fast-food brands such as KFC, Burger King and Pizza Hut, Margaa Restaurant mainly sells hamburgers, fries, pizza, fried chicken, drinks, and so on. It also offers a variety of discount packages for students to choose.

## 1.2 Objective of Margaret Western Restaurant

MARGARET WESTERN RESTAURANT is committed to providing high-quality, convenient Western fast food to meet the diverse dining needs of students. The restaurant features unique Western dishes and a first-class dining environment, aiming to build a popular dining brand on campus and become the preferred choice for teachers and students.

However, as the restaurant's populay continues to grow, its reliance on traditional manual management has led to several problems:

- **Manual Process Limitations:** The restaurant currently relies on manual processes for order management, inventory tracking, and sales statistics. This approach is not only time-consuming but also prone to errors, significantly reducing operational efficiency.

- **Delayed Decision-Making**: The management can only obtain the sales information of the month through the monthly sales report. Meanwhile, the data recorded on paper is difficult to conduct data analysis. As a result, they don't have real-time access to sales data, inventory status, and customer preferences. This information lag hampers timely and effective decision-making.

- **Poor Inventory Control:** The absence of accurate inventory forecasting often results in either expired ingredients or shortages. In some cases, students have experienced health issues, such as diarrhea, after consuming stale food, which has damaged the restaurant's reputation.

- **Limited Digital Services:** The restaurant currently lacks digital solutions, such as online ordering platforms. Many students prefer to place advance orders but are forced to visit the restaurant in person, which contradicts modern consumer expectations for convenience. The long waiting queues also made many people give up dining here.

- **Inefficient people management:** Current staff management relies on manual paper scheduling tables, resulting in low efficiency of shift adjustment and poor communication. Frequent problems such as scheduling conflicts and chaotic attendance records lead to high labor costs and low employee satisfaction.

To address these issues, MARGARET WESTERN RESTAURANT plans to implement a unified data management system with the following objectives: optimize operational efficiency, reduce costs, enhance the user experience, and support data-driven management decisions.

# II. Software Application—Margaret Super Management System

## 2.1 Application Description

The Margaret Restaurant Super Management System is a comprehensive software solution. It aims to collect and analyze all operational information generated by restaurants. It will integrate several key business functions, including order management, inventory tracking, customer relationship management, operational reporting and personnel management.

**Order Management**

The super system enables customers to place orders online through an electronic menu, eliminating the need for manual communication. When an order is submitted, the system automatically records all relevant details, including the order number, items purchased, and any special requests. This information is instantly transmitted to the kitchen, streamlining communication between waitstaff and chefs and reducing the risk of errors or delays.

**Inventory Tracking**

The super system stores the required quantities of raw materials for all menu items and tracks inventory changes in real time. For example, when a hamburger is sold, the system automatically deducts the corresponding ingredients from the inventory. It also sends alerts for low stock levels or approaching expiration dates. Additionally, the system uses data analysis technology to predict raw material demand, helping to prevent overstocking or shortages.

**Customer Relationship Management**

The system collects and stores detailed information about each member, including their name, contact details, and order history. It also supports customer feedback collection, enabling the restaurant to improve service quality. This data is used to offer personalized services, such as loyalty points, special promotions, and birthday discounts, enhancing customer satisfaction and retention.

**Operational Reports**

The system generates comprehensive sales reports on a daily, weekly, and monthly basis. These reports provide valuable insights, such as the most popular dishes, peak dining hours, and revenue breakdowns by menu category. Meanwhile, this system also supports AI-driven automatic decision-making functions.

**Personnel management system**

The system provides intelligent scheduling function, which automatically generates the optimal scheduling table based on the historical passenger flow prediction, and employees can view the scheduling in real time through the mobile terminal and feedback the adjustment needs. Automatically record daily attendance (clocking time, leave records, etc.)

## 2.2 Managerial Information

| Report | Content | Description |
|---|---|---|
| Daily sales | Total Sales | Real-time statistics of the day's gross sales, broken down by food category (e.g. burgers, pizzas, drinks, etc.) |
| | Order volume | Record the total number of orders completed on that day and their categories (in, out, online) |
| | Average customer unit price | Through the total sales and order volume automatic calculation, analysis of customer consumption level |
| | Distribution during peak hours | Order volume and sales peak are counted by hour, and passenger flow rules during lunch (11:00-14:00) and dinner (17:00-19:00) are visualized |
| Stock status | Current stock | Real-time display of the remaining stock of all ingredients (such as beef patties, cheese, French fries, etc.), sorted by shelf life |
| | Shortage warning | Automatic warning is triggered based on the preset safety stock threshold (for example, "beef stock is below 20kg, need to replenish") |
| | Expired ingredients list | Automatically screen the expired ingredients (such as vegetables, sauces) within 3 days, indicating prioy use or promotional treatment |
| Customer preference | Hot dishes ranking | The weekly/monthly TOP10 dishes list is generated according to sales volume and sales volume (such as "spicy chicken burger" has the highest sales volume) |
| | Member repurchase rate | Statistics the proportion of repeated consumption of member users within 30 days to identify high loyalty customer groups |
| | Consumption time preference | Analyze the order time distribution of different customer groups (such as students' preference for lunch, faculty members' preference for evening takeout) |
| Cost-benefit | Food consumption rate | Calculate the proportion of expired or wasted ingredients to the total purchase cost and optimize the purchase plan against historical data |
| | Margin trends | Track net profit on a weekly/monthly basis and analyze profitability drivers in conjunction with promotions and cost fluctuations |
| Personnel situation | Employee attendance statistics | Record daily attendance data (late arrival, early departure, absence), and generate monthly attendance reports |
| | Turnover analysis | Track employee onboarding/turnover rates, job transfer records, and assess team stability and training needs |
| | Scheduling situation | Count employees' working hours and schedule future work |

# III. Purpose and Advantages of Margaret Super Management System

## 3.1 Purpose of Margaret Super Management System

MARGARET WESTERN RESTAURANT aims to implement this super management system to achieve four key goals: optimize operational efficiency, reduce costs, enhance the user experience, and support data-driven management decisions.

## 3.2 Advantages of Margaret Super Management System

**1. Enhanced Accuracy**

Manual ordering and inventory tracking are prone to human error. Automated systems reduce the possibility of order processing and inventory counting errors. By accurately transmitting customer requirements to the kitchen, the system ensures precise order fulfillment. Additionally, automated inventory management minimizes waste and mitigates food safety risks caused by improper stock handling.

**2. Improved Efficiency**

The system automatically transmits orders to the kitchen and updates inventory in real time, enabling faster service and reducing customer wait times. Employees can allocate less time to repetitive tasks and focus more on delivering exceptional customer service, thereby enhancing overall operational productivity.

**3. Data-driven Decision Making**

The detailed sales reports generated by the system provide valuable insights to management. Managers can make informed decisions based on accurate and up-to-date data, rather than relying on guesswork or time-consuming manual data collection. Furthermore, the system's built-in data analysis capabilities provide predictive analytics, enabling proactive planning and strategic decision-making.

| | Manual management | Super management system |
|---|---|---|
| **Order management** | Ordering takes 3 minutes and is error-prone | Automatic ordering takes 1 minute with zero error |
| **Inventory management** | Inventory is counted every week, and the error rate is about 10% | System real-time monitoring, error rate <1% |
| **Decision making** | Decisions depend on experience, with a lag of 3-7 days | Data is updated in real time and decisions can be adjusted on the day |
| **User experience** | No member system, single feedback channel | Member offers, personalized recommendations, and instant feedback collection |

# PART 2: Feasibility Study / Planning Phase

# I. Problem Definition &Project Scope

### 1.1 Inefficient order processing and human error

Margaret Restaurant has long relied on employees to manually record customers' orders and calculate the bill. At peak times, manual transcription is prone to missing orders or errors in the number of dishes. For example, when a student orders a hamburger set (burger and drink), the waiter may misremember the type of drink. At the same time, manual calculation of the total order price frequently occurs price errors. There have been customer complaints due to overcharging, further aggravating the service confusion. The reliance on verbal delivery of order details between the kitchen and the front desk leads to miscommunication in the preparation of dishes (e.g. spiciness requirements are not accurately communicated). These directly affect the customer dining experience.

### 1.2 Delayed decision support and inventory risk

Management lacks real-time sales insights to respond quickly to market changes. Management can only obtain sales data through monthly summary reports. They can't monitor inventory status and customer preferences in real time and adjust strategies. One month, for example, the restaurant failed to recognize a decline in pizza sales, resulting in a large backlog of ingredients that eventually expired. At the same time, the management of food ingredients relies on the chef's experience to judge the freshness, which is highly subjective. The restaurant once caused diarrhea among students due to the use of seasonal beef.

### 1.3 Digital service deficit

Students generally prefer online ordering and delivery services, but the restaurant only supports the in-store ordering mode, which cannot meet the convenience needs. Every meal time, a large number of students line up to buy food. In addition, most western restaurants have a membership system, but the restaurant has not established a membership system, which is difficult to implement personalized marketing (such as birthday offers). In addition, the service lacks an automated refund system, and if customers are not satisfied with their meals, they need to negotiate with employees privately, which is an opaque and inefficient process. For example, a student asked for a refund due to food quality problems, but because there was no standardized process, it took 20 minutes to complete the processing, resulting in a significant decrease in customer satisfaction.

### 1.4 Inefficient manpower scheduling

The restaurant relies on paper scheduling for staff management, and the adjustment of scheduling needs to be manually modified and notified to the staff, which is inefficient and error-prone. For example, one time, because the schedule was not updated in time, only one waiter was on duty during peak hours, and the waiting time for customers was extended to 20 minutes. Attendance records are in the form of attendance tables, there are signs or omissions, attendance statistics error rate is high. In addition, employees do not have real-time access to schedule changes and are often late or absent from work.

### 1.5 Compliance risk and canteen collaboration barriers

The restaurant's hand-recorded cleaning logs and ingredient inspection reports are difficult to cope with government spot checks. The government warned the restaurant for incomplete data, so its food safety rating was only B. In addition, inefficient operation directly affects the overall efficiency of the  canteen. The public areas of the canteen were very crowded due to long queues at lunchtime due to delays in order processing. As a result, many students think that the dining atmosphere of the whole  canteen is poor, and choose to eat in other places, which damages the customer flow of the whole canteen.

Based on the above problems, we developed the following functions for the Margaret Super Management System.

**1. An electronic ordering system.** The waiter selects dishes through the tablet terminal and automatically generates orders. The total price is calculated by the system in real time, with no manual errors. Order data is synchronized to the kitchen display in real time, and special requirements (such as "no Onions") are clearly marked to reduce miscommunication.

**2. Integrated refund management module.** It allows customers to submit refund applications online, and the system automatically reviews and records the processing progress.

**3. Real-time inventory monitoring module.** It will track production date and shelf life of ingredients, and remind the staff 3 days before the deadline. It will also automatically generate a list of near-expired ingredients for prioy use to reduce waste.

**4. AI suggestions system.** This system will dynamically adjust the purchase volume based on historical sales and holiday models.

**5. Online ordering system.** It will support takeout orders and in-store order in advance.

**6. Customer management system.** This system can record consumption data and support credits exchange.

**7. Employee scheduling system.** It can predict manpower demand based on historical passenger flow data, automatically generate optimal schedule and support dynamic adjustment. Employees can check the scheduling information in real time through the system, confirm the shift or submit the shift transfer application, and the system will update it automatically.

**Expected result**

After the implementation of the project, Margaret Western Restaurant will achieve multi-dimensional breakthroughs. Based on our company's previous system development practices, the improvement of operational efficiency could increase the daily order volume by 20%-40%. That is because efficient order processing allows customers to reduce waiting time and increase their purchase intention. At the same time, due to the development of online order system, the proportion of online orders could jump to 40%. The high efficiency management will reduce annual risk losses by 20%-35%. Customer satisfaction score is expected to increase by 20%. Margaret Western Restaurant will become the preferred brand for campus dining.

# II. Context Level

Base on the previous existing defects, we design the new IT systems. Recognizing the existing operational inefficiencies, human errors in order processing, delays in decision support, inventory risks, digital service deficits, manpower scheduling issues, and compliance risks, this system aims to address these challenges by streamlining operations and enhancing data management.

The diagram illustrates the core interactions between the " Margaret Super Management System" and key external entities. It depicts the exchange of customer information (Customer Information, Payment, Invoice, Food Order, Receipt) between the system and the CUSTOMER, facilitating efficient order placement and payment processing. The system also interacts with the BANK for deposit and withdrawal transactions (Deposit, Withdrawal Capital). SUPPLIERS provide raw materials and invoices (Raw Material, Invoice), while the system generates purchase information and payment details (Purchase Information, Payment, Supplier Information). STAFF provides staff information (Staff Information) and receives payment receipts (Payment Receipt). The KITCHEN receives raw material updates, food orders and communicates raw material consumption and food delivery (Raw Material Update, Raw Material Consumption, Food Order, Food Delivery). Regulatory bodies such as the GOVERNMENT provide material and food quality information (Material and Food Quality Information). The RESTAURANT MANAGEMENT/INVESTOR receives sales reports (Sales Reports). Finally, the SUPERVISOR gains insight from management reports (Management Reports).

This context level DFD lays the foundation for a comprehensive information system that aims to mitigate the challenges outlined in the problem definition, improve operational efficiency, and enhance customer satisfaction.

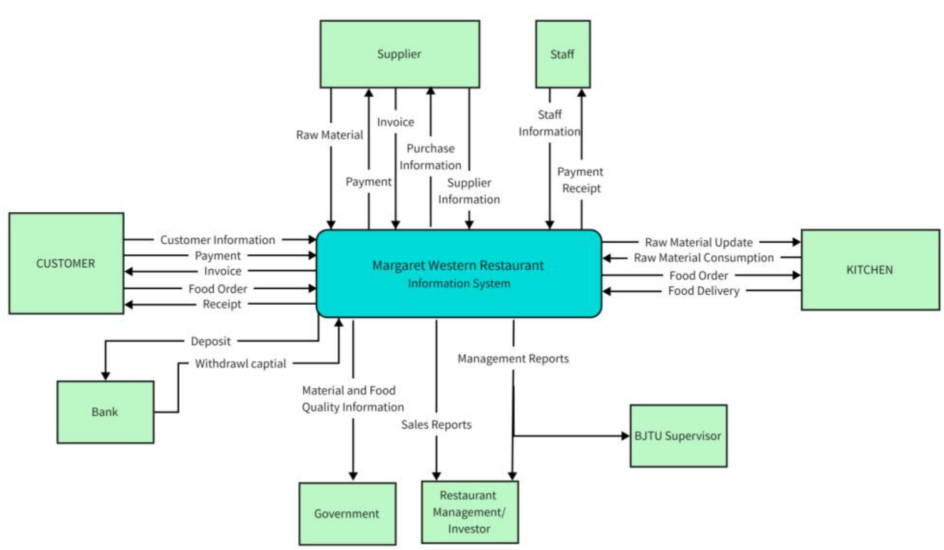



*Figure 1. DFD diagram*

# III. Project Feasibility

### 3.1 Technical Feasibility

We analyzed the feasibility of this technology from the following aspects:

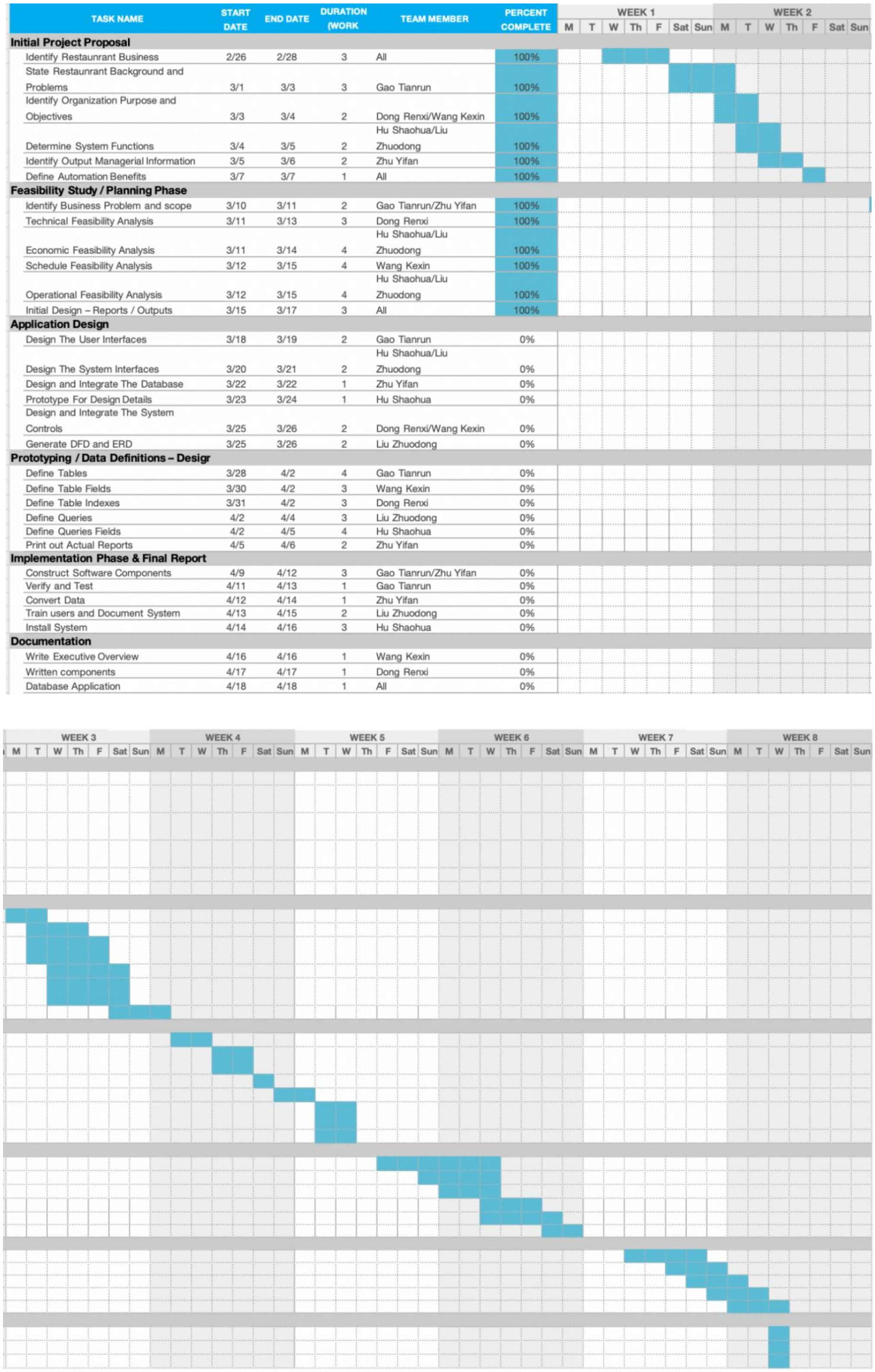
**Necessary Equipment and Infrastructures:**
To function effectively, the system requires a combination of hardware, software, and network resources. Principal hardware components include a good Wi-Fi network supported by routers for real-time transmission of information; one desktop computer with sufficient processing capacity to handle high-volume order and inventory data; tablets for the waitstaff to manage orders; and dedicated order processing screens for the kitchen. Peripheral equipment such as mice and keyboards are also required for normal operations. On the software side, the system will use Microsoft Access or MySQL for database management, Python for development, and Pandas/NumPy libraries for advanced analysis and reporting. For administrative tasks, the system will utilize Microsoft Office.

**Technique Availability:**
Our team has the necessary skills to deliver a solid solution. We have hands-on experience in Python development, API integration, and database design. Moreover, we have successfully implemented similar systems in the foodservice sector. The chosen technologies—Python, Access, and MySQL—are mature and widely used in the hospitality industry. These technologies benefit from large developer communities, which lowers compatibility risks. To ensure scalability, we will utilize cloud platforms such as AWS or Alibaba Cloud. Additionally, MySQL is an open-source tool, which reduces licensing costs. Locally sourced hardware like tablets and routers, along with commercial software like Microsoft Access, will facilitate seamless integration into existing workflows.

**Realization and Maintainability:**
The system focuses on secuy and adaptability. Sensitive data will be encrypted in transit using SSL/TLS and AES-256 encryption at rest, with regular key rotation to minimize potential breaches. Firewalls and intrusion detection systems will monitor network traffic, and daily automated backups to external cloud storage will ensure quick recovery in case of emergencies. The cloud-based structure is designed for scalability, making expansion easy—new servers can be added for new branches, and the existing setup can handle ten times the current order volume. Service agreements with vendors like AWS and Microsoft ensure 24/7 tech support, and the modular design allows for targeted updates to systems like inventory management without disrupting overall operations.

In summary, the Margaret Super Management System is technologically feasible based on careful consideration of resources required for development, procurement, deployment, and operation.

### 3.2 Schedule Feasibility



| TASK NAME | START DATE | END DATE | DURATION (WORK | TEAM MEMBER | PERCENT COMPLETE |
|---|---|---|---|---|---|
| **Initial Project Proposal** | | | | | |
| Identify Restaunrant Business | 2/26 | 2/28 | 3 | All | 100% |
| State Restaunrant Background and Problems | 3/1 | 3/3 | 3 | Gao Tianrun | 100% |
| Identify Organization Purpose and Objectives | 3/3 | 3/4 | 2 | Dong Renxi/Wang Kexin | 100% |
| Determine System Functions | 3/4 | 3/5 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| Identify Output Managerial Information | 3/5 | 3/6 | 2 | Zhu Yifan | 100% |
| Define Automation Benefits | 3/7 | 3/7 | 1 | All | 100% |
| **Feasibility Study / Planning Phase** | | | | | |
| Identify Business Problem and scope | 3/10 | 3/11 | 2 | Gao Tianrun/Zhu Yifan | 100% |
| Technical Feasibility Analysis | 3/11 | 3/13 | 3 | Dong Renxi | 100% |
| Economic Feasibility Analysis | 3/11 | 3/14 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Schedule Feasibility Analysis | 3/12 | 3/15 | 4 | Wang Kexin | 100% |
| Operational Feasibility Analysis | 3/12 | 3/15 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Initial Design – Reports / Outputs | 3/15 | 3/17 | 3 | All | 100% |
| **Application Design** | | | | | |
| Design The User Interfaces | 3/18 | 3/19 | 2 | Gao Tianrun | 0% |
| Design The System Interfaces | 3/20 | 3/21 | 2 | Hu Shaohua/Liu Zhuodong | 0% |
| Design and Integrate The Database | 3/22 | 3/22 | 1 | Zhu Yifan | 0% |
| Prototype For Design Details | 3/23 | 3/24 | 1 | Hu Shaohua | 0% |
| Design and Integrate The System Controls | 3/25 | 3/26 | 2 | Dong Renxi/Wang Kexin | 0% |
| Generate DFD and ERD | 3/25 | 3/26 | 2 | Liu Zhuodong | 0% |
| **Prototyping / Data Definitions – Desigr** | | | | | |
| Define Tables | 3/28 | 4/2 | 4 | Gao Tianrun | 0% |
| Define Table Fields | 3/30 | 4/2 | 3 | Wang Kexin | 0% |
| Define Table Indexes | 3/31 | 4/2 | 3 | Dong Renxi | 0% |
| Define Queries | 4/2 | 4/4 | 3 | Liu Zhuodong | 0% |
| Define Queries Fields | 4/2 | 4/5 | 4 | Hu Shaohua | 0% |
| Print out Actual Reports | 4/5 | 4/6 | 2 | Zhu Yifan | 0% |
| **Implementation Phase & Final Report** | | | | | |
| Construct Software Components | 4/9 | 4/12 | 3 | Gao Tianrun/Zhu Yifan | 0% |
| Verify and Test | 4/11 | 4/13 | 1 | Gao Tianrun | 0% |
| Convert Data | 4/12 | 4/14 | 1 | Zhu Yifan | 0% |
| Train users and Document System | 4/13 | 4/15 | 2 | Liu Zhuodong | 0% |
| Install System | 4/14 | 4/16 | 3 | Hu Shaohua | 0% |
| **Documentation** | | | | | |
| Write Executive Overview | 4/16 | 4/16 | 1 | Wang Kexin | 0% |
| Written components | 4/17 | 4/17 | 1 | Dong Renxi | 0% |
| Database Application | 4/18 | 4/18 | 1 | All | 0% |

*Figure 2. Gantt chart*

Our team carefully planned the project schedule according to the workload of the task. We also took into account each team member's schedule and the customer's requirements for delivery times. With the project start date of February 26, 2025, and the project delivery date of April 19, 2025, we are committed to ensuring the rationality and timeliness of the project schedule. Besides, our team has rich financial and human resources. We are a team of six people consisting of Gao Tianrun as the project manager and Zhu Yifan, Dong Renxi, Wang Kexin, Liu Zhuodong, and Hu Shaohua. Each person allocates tasks according to their skills and schedule.

**A. The project task is divided into several phases:**

1. **Initial Project Proposal:** The project manager leads the team to identify the restaurant business, set a clear division of labor, and define important elements like background, issues, organizational goals, system functions, managing information output, and the benefits of automation.

2. **Feasibility Study and Planning:** This stage is divided into multiple parts: determining the business problem and scope, performing economic, schedule, operational feasibility analysis, and completing the initial design. Gao Tianrun leads the overall process, while each team member focuses on different aspects, ensuring the project's feasibility.

3. **Application Design:** This crucial stage involves the design of user interfaces, system interfaces, database integration, prototype design details, system control integration, and the generation of DFD and ERD. The team carefully plans and schedules these tasks to ensure the system is effectively designed.

4. **Prototyping and Data Definitions:** The detailed design stage, where tables, table fields, indexes, queries, and reports are defined. These tasks are scheduled and assigned to team members for efficient completion, ensuring the design phase progresses smoothly.

5. **Implementation and Final Report:** In this phase, the team builds software components, tests the system, converts data, trains users, and installs the system. The implementation phase ensures that the system is ready for use, and the final report solidifies the project's completion.

6. **Documentation:** The final stage involves wing the executive overview, completing the wten components, and finalizing the database application. This is the final step in ensuring the project's success and completing all documentation required.

**B. Resource Allocating Stipulation**

Our team ensures proper allocation of resources, including financial and human resources, to maximize efficiency. We ensure that all tasks are properly divided and deadlines are met through effective communication and flexibility in task assignments. If any team member is unable to meet deadlines, other members are ready to step in and help complete the task on time.

**C. Related Risks and Management**

We track project progress carefully and address any issues promptly. Risks associated with delays are managed through collaborative work, careful time management, and quick adaptation to unforeseen challenges. We ensure that any changes in customer requirements or market conditions are promptly incorporated into the schedule to avoid any impact on the project's completion.

### 3.3 Economic Feasibility

#### 3.3.1 Initial Development Cost

The MIS economic feasibility analysis of Margrethe Restaurant mainly carries out systematic budget and evaluation from three aspects: manpower cost, training and implementation, hardware and software.

### A. Personnel

The total personnel cost of the project is estimated at approximately ¥83,500, based on the workload and hourly rates for various roles. The project manager oversees the overall planning and coordination of the project. The system architect focuses on ensuring the system's structure and performance. The full-stack developer and database engineer handle the core tasks related to front-end development and database management, respectively. Business analysts, UI/UX designers, and quality assurance specialists play essential supporting roles, with business analysts researching requirements, UI/UX designers working on interface design, and quality assurance specialists ensuring testing and quality control throughout the project.

*Table 1. Personnel cost*

| Personnel | Cost (CNY) |
|---|---|
| 1 Business Analyst (5 hours/ea ¥180.00/hr) | ￥900 |
| 1 Project Manager (10 hours/ea ¥220.00/hr) | ￥2200 |
| 1 System Architect (5 hours/ea ¥300.00/hr) | ￥1500 |
| 1 Full Stack Developers (5 hours/ea ¥180.00/hr) | ￥900 |
| 1 UI/UX Designer (5 hours/ea ¥180.00/hr) | ￥900 |
| 1 Database Engineer (5 hours/ea ¥200.00/hr) | ￥1,500 |
| 1 QA Specialist (3 hours/ea ¥150.00/hr) | ￥450 |
| Total | ￥8,350 |

### B. Training & Implementation

The total cost of training and implementation is about 4,500 CNY, which includes staff training and ongoing support after the system goes live. The training focuses on key staff at the restaurant, including front desk, kitchen and management staff. The goal is to ensure that employees in different departments can use MIS systems smoothly to improve the quality of customer service and collaboration among colleagues. In addition, a portion of the budget is allocated for consultation and implementation.

*Table 2. Training & Implementation cost*

| Training & Implementation | Cost (CNY) |
|---|---|

| | |
|---|---|
| Staff Training (front-of-house, kitchen, management) (5 employees × ¥80/ea) | ￥400 |
| Additional Consultancy/Implementation (5 hours × ¥100.00/hr) | ￥500 |
| Total | ￥900 |

**C. Hardware**

The hardware investment of 6699 CNY is mainly used to purchase management computer equipment, network infrastructure and a barcode scanner. This part of the expenditure can ensure the smooth process and reduce the restaurant's error rate at the hardware level, and provide the necessary equipment support for inventory management, incoming order processing and data aggregation and mining functions. The addition of barcode scanners can more accurately track incoming and outgoing goods, further improving the accuracy and efficiency of inventory management. The tablet is used to place and manage orders, as well as collaborate to view and manage inventory.

*Table 3. Hardware & Software fee*

| Hardware & Software | Cost (CNY) |
|---|---|
| Management computer | ￥4,199 |
| Network equipment (wireless router, switch and necessary cabling) | ￥1,500 |
| Barcode scanner (for inventory management) | ￥200 |
| Tablet (for Ordering) | ￥800 |
| Total | ￥6,699 |

**D. Development Cost**

Summarizing all the above development costs, the total initial development cost of this MIS is ¥19,549. The final figure for the total development cost is ￥28,101.

*Table 4. Total development cost*

| Category | Cost (CNY) |
|---|---|
| Personnel Costs | ￥8,350 |
| Training & Implementation | ￥4,500 |
| Hardware & Software | ￥6,699 |
| **Total Initial Development Cost** | ￥19,549 |
| Contingency fee (15%) | ￥2,932 |

| | |
|---|---|
| Subtotal | ￥22481 |
| Overhead (25%) | ￥5,620 |
| **Total development cost** | ￥28,101 |

### 3.3.2 Monthly Operating Costs

#### A. Personnel

Next, considering the impact of MIS system on continuous operation, we first analyze the additional operating costs.
The company provides optional personnel services, including a part-time system administrator (3 hours/time, ¥100.00/ hour) and a part-time data analyst (5 hours/time, ¥100.00/ hour) for a total fee of ¥800. The system administrator is responsible for daily system maintenance, such as server update, data backup and emergency problem handling; Data analysts perform data analysis and data mining, provide report analysis, and provide recommendations for restaurant sustainable operations and user discovery.

Table 5. Monthly Personnel Cost

| **Personnel** | **Cost (CNY)** |
|---|---|
| 1 System Administrator (**Selectable**, part-time, 3 hours/ea ¥100.00/hr) | ￥300 |
| 1 Data Analyst (**Selectable**, part-time, 5 hours/ea ¥100.00/hr) | ￥500 |
| Total | ￥800 |

#### B. Expenses

In terms of continuous expenditures on software and hardware, the subscription fee for the POS system software is ¥300, the hardware maintenance insurance is ¥100, and the licensing fee for professional office software (Microsoft 365, including Access, OneDrive, etc.) is ¥200, totaling ¥600. Hardware maintenance insurance provides protection in the event of equipment failure, reducing the risk and cost of temporary repair or replacement; The professional office software suite further enhances data management and team collaboration, facilitates collaboration between analysts and communication between merchants and analysts, and also helps to unify data within the restaurant.

*Table 6. Monthly Expenses*

| Expenses | Cost (CNY) |
|---|---|
| POS System Software (Subscription) | ￥300 |
| Hardware maintenance insurance | ￥100 |
| Office Software (Microsoft 365 Apps, including Access, OneDrive) | ￥200 |
| Total | ￥600 |

**C. Total Monthly Operating Costs**

Considering all services have been selected, we have obtained the additional monthly operating costs brought by the MIS system:

*Table 7. Total Monthly Operating Costs*

| Category | Description | Cost (CNY) |
|---|---|---|
| Personnel | System Administrator (part-time, 3 hrs @ ¥100/hr) | ￥300 |
| | Data Analyst (part-time, 5 hrs @ ¥100/hr) | ￥500 |
| Total Personnel | | ￥800 |
| Expenses | POS System Software (Subscription) | ￥300 |
| | Hardware maintenance | ￥100 |
| | Office Software (Microsoft 365 Apps, including Access, OneDrive) | ￥200 |
| Total Expenses | | ￥600 |
| Total Monthly Operating Costs | | ￥1,400 |

### 3.3.3 Benefit

**A. Tangible Benefits**

By implementing the MIS system, we anticipate a significant boost in both revenue and turnover for the restaurant. Based on our experience with past project deployments, we expect a 5% to 15% increase in sales due to the system's impact. Additionally, improvements in cost and error reduction are projected to contribute a 0.5% to 1% increase in revenue. The systems enhancements in operational efficiency and throughput are expected to generate another 1% to 2% rise in revenue. Moreover, the more effective use of staff time will not only improve the employee experience but also indirectly lead to an additional 0.5% to 1% increase in revenue.

*Table 8. Tangible Benefits*

| Benefit Category | Expected Revenue Increase (%) |
|---|---|
| Increased Sales | 5% - 15% |
| Cost/Error Reductions | 0.5% - 1% |
| Increased Efficiency/Throughput | 1% - 2% |
| More Effective Use of Staff Time | 0.5% - 1% |
| Total Tangible Benefits | 7% - 19% |

**B. Intangible Benefits**

In addition to tangible benefits, the deployment of an MIS system can also bring the following intangible benefits:

*Table 9. Intangible Benefits*

| Benefit Category | Description | Timeframe |
|---|---|---|
| Increased Flexibility of Operation | Quick adaptation to menu changes, promotions, and seasonal demand | Continuous |
| Higher Quality Products/Services | Improved food quality and service driven by better data and process insights | Medium to Long-Term |
| Better Customer Relations | Enhanced loyalty through improved communication, online reservations, and loyalty programs | Long-Term |
| Improved Staff Morale | Higher employee satisfaction and lower turnover from streamlined processes | Short to Long-Term |

**3.3.4 Cost recovery analysis**

After the deployment of the MIS system, this restaurant will increase its monthly operating costs by 1,400 yuan. Due to the campus restaurant's pricing restrictions, the restaurant's monthly turnover is 50,000 yuan. According to previous estimates, in the most conservative scenario, the deployment of the MIS system will bring an additional 7% tangible benefits, and in the most ideal scenario, it will bring an additional 19% tangible benefits. Therefore, based on the calculation, it will take 3.5 to 8.5 months to recover the cost and generate net income after the deployment of the MIS system.

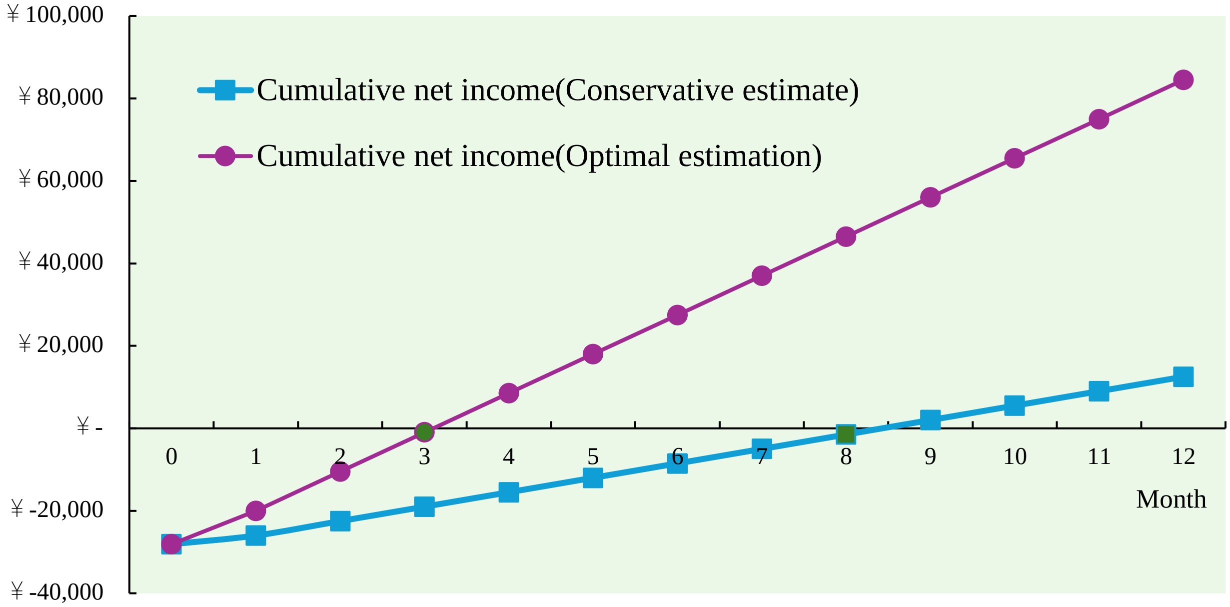



*Figure 2. Cost recovery analysis*

### 3.4 Operational Feasibility

#### 3.4.1 Daily Operations

**A. Front-end Operational Feasibility**

The front-end operation constitutes the core touchpoint of the restaurant system, directly affecting customer experience and operational efficiency. Currently, Margaret Western Restaurant employs traditional handwten order-taking methods, which not only exhibits low efficiency but is also prone to errors. According to monitoring of existing operational processes, manual order-taking requires an average of 90 seconds to complete a single order, with peak period customer queuing times reaching 15-20 minutes, resulting in low customer satisfaction. The new system will fundamentally transform this process through touchscreen ordering terminals, projecting a reduction in single order processing time to 30 seconds or less, enhancing efficiency by approximately 67%.

The system will support integrated diversified payment methods, including campus cards, WeChat Pay, Alipay, and others, covering 99% of student payment preferences. This flexibility not only accelerates transaction speeds but also reduces queue delays caused by payment issues. Data indicates that in similar restaurant environments, payment diversification can shorten checkout times by an average of 40%, thereby further improving front-end operational efficiency.

Regarding peak-period load capacity, the system architecture has been specifically designed to accommodate concurrent demands during lunch peak hours (11:30-13:00), capable of simultaneously processing 8-10 concurrent orders without diminishing response speed. Validated through load testing, the system can maintain 95% of response times within 1 second under simulated peak conditions of 200 persons/hour, dramatically reducing student waiting times from an average of 15 minutes to under 5 minutes. The restructuring of front-end operational processes is expected to significantly enhance customer experience while alleviating staff pressure.

#### 3.4.2 Human Resource Management Feasibility

**Kitchen Operations Feasibility**

Kitchen operational efficiency directly determines food supply speed and quality. Currently, the communication between Margaret Western Restaurant's kitchen and front-end relies on paper order transmission, resulting in information delays and frequent errors. Orders require an average of 2-3 minutes to transit from the front-end to the kitchen, even longer during peak periods, causing subsequent production delays. The new system will implement electronic order transmission, reducing delivery time from placement to kitchen display to less than 3 seconds, achieving near-instantaneous communication.

The raw material warning system is a powerful tool to enhance kitchen operations. This system will automatically monitor existing inventory levels and trigger an alert when the stock of key ingredients falls to 20% of the threshold. This will significantly reduce the frequency of emergency procurement, from the current 3-4 times a week to 1-2 times a month, greatly decreasing the likelihood of unexpected situations. In addition, precise inventory management can effectively prevent the misuse of expired ingredients, fundamentally improving the freshness and safety of the food.3.4.2 Human Resource Management Feasibility

**Staff Scheduling and Management**

The efficiency of staff scheduling plays a crucial role in managing labor costs and maintaining service quality. At present, the restaurant relies on paper-based scheduling, requiring managers to invest significant time in manual adjustments, which makes it challenging to adapt swiftly to workforce changes. To address this, a newly designed system will feature an intelligent scheduling module that predicts peak-hour staffing needs based on historical customer flow data, automatically generating optimized work schedules.

This system will process at least three months of operational data to identify traffic trends and anticipate staffing demands for the next one to two weeks. When there is a sudden surge in customers, the system will issue an alert, suggesting that additional support staff be reassigned to front-end service. Conversely, Tuesday and Thursday afternoons experience reduced customer flow, allowing for appropriate reductions in staffing. This data-driven intelligent scheduling is expected to reduce personnel redundancy by approximately 18%, potentially saving monthly labor costs of approximately 8,000-10,000 yuan, while ensuring uncompromised service quality during peak periods.

Attendance management will also undergo digital transformation. Transitioning from traditional paper sign-ins to an electronic timekeeping system, employees can quickly complete check-in/check-out procedures through fingerprint or facial recognition. This will reduce attendance error rates from 10% to below 0.5%, while simultaneously increasing salary calculation accuracy to 99.9%. Automated attendance will also generate detailed reports, assisting management in identifying attendance issues and work efficiency patterns for targeted improvements.

Employee training constitutes a cical phase in system implementation. Front-end staff require 8 hours of system operation training, including ordering interface operation, payment processing, order modification, and other functionalities, with an anticipated 2-week adaptation period. Kitchen personnel require 4 hours of basic system training, primarily focusing on order reception and processing procedures, with an adaptation period of approximately 1 week. Management personnel require more comprehensive 12-hour training, encompassing system administration, data analysis, and report interpretation, among other advanced functions. Training will combine theoretical instruction with practical operation, ensuring employees can proficiently utilize the new system.

**Workflow Restructuring**

System implementation will bring comprehensive workflow restructuring, significantly improving operational efficiency. The restaurant's current order processing involves 11 manual steps from customer ordering to food service, including handwten recording, order transmission, manual amount calculation, etc., prone to errors and time-consuming. The new system will simplify this process to 5 systematic steps, from electronic ordering, automatic pricing, order transmission to service notification, with full digital processing throughout.

The position structure will also adjust accordingly. Preliminary assessment indicates that after system implementation, one dedicated cashier position can be eliminated as front-end service staff can directly complete cashier work through terminals. Simultaneously, one data analysis position will be created, responsible for system data mining and decision support. This position adjustment will not result in layoffs but rather improve human resource utilization efficiency through internal transfers. Work focus redistribution represents another significant change. Front-end staff will be liberated from tedious manual recording tasks, allowing more energy to be invested in customer service and experience enhancement.

### 3.4.3 Customer Experience Feasibility

**Online Service Capability**

Online service has become standard in modern catering, especially for campus restaurants primarily serving students. Margaret Western Restaurant currently offers only physical ordering, failing to meet students' requirements for convenience. According to campus questionnaire surveys, 85% of students express preference for online ordering services to save queuing time.

The new system will develop a dedicated WeChat mini-program allowing students to browse menus, place orders, and make payments anytime, anywhere. Based on data analysis from similar campus restaurants, approximately 30% of orders are expected to convert to online completion within three months after system launch, potentially reaching 50% after six months. Peak period on-site queuing is expected to reduce by approximately 100 people, substantially improving dining experience. The mini-program will also provide personalized recommendation functionality, intelligently suggesting potentially interesting dishes based on user order history, projected to increase average order value by 5-8%.

The reservation and pickup system constitute a core feature of online service. Students can pre-order meals from 30 minutes to 24 hours in advance, with the system automatically arranging production schedules based on reservation times. For collection, students need only present electronic order numbers or scan codes for quick meal retrieval, reducing average waiting times from 8 minutes to 3 minutes. The reservation system will also help the restaurant better distribute customer flow peaks, estimated to reduce peak period customer flow by an average of 15-20%, alleviating congestion issues.

### 3.4.4 Operational Data Visualization and Decision Support

**Management Reports and Analysis**

Data-driven decision-making represents a core competitive advantage in modern catering management. Margaret Western Restaurant currently can only understand operational status through monthly manual statistical reports, with severely lagged data making timely decision support difficult. The new system will fundamentally change this situation, providing real-time sales monitoring and multi-dimensional data analysis capabilities.

The system will deploy intuitive management dashboards displaying key business indicators such as real-time sales, customer flow, average order value, best-selling dishes, etc. Data delay will reduce from the current 30 days to minute-level, allowing management to grasp restaurant operational status at any time. Dashboards support multi-dimensional data filtering and drill-down, such as analyzing sales by time period, dish category, or payment method, identifying potential problems and opportunities.

Best-selling dish analysis represents an important system function. By analyzing historical sales data, the system can predict daily demand for various dishes, guiding the kitchen to formulate more precise production plans. This will reduce ingredient preparation waste by approximately 30% while avoiding customer dissatisfaction caused by popular dishes selling out. The system will also identify underperforming dishes, providing basis for menu optimization.

Cost control analysis functionality will help the restaurant better manage cost structures. The system automatically tracks ingredient costs, labor costs, and profit margins for each dish, improving accuracy from ±10% to ±2%. By identifying high-cost, low-profit dishes, management can timely adjust pricing or recipes. The system will also monitor ingredient utilization efficiency, identifying waste segments, projected to uncover 25% of cost optimization opportunities, saving annual costs of 4,000-5,000 yuan.

**Integration with  Management Systems**

As part of 's dining facilities, integration with school management systems is crucial for resource optimization. Current restaurant systems remain completely isolated from school ERP systems, resulting in uncoordinated resource allocation affecting overall operational efficiency. The new system will achieve data sharing and collaborative work with school systems through API interfaces.

The restaurant system will also partially integrate with the school's student information system, achieving more personalized service. For example, with student consent, the system can obtain basic student information and dietary preferences, providing more precise service and marketing. This deep integration not only enhances restaurant service quality but also provides the school with anonymized student lifestyle data, supporting campus life improvement plans.

### 3.4.5. Emergency Response Mechanisms

**System Failure Response**

Any information system faces failure risks; comprehensive emergency mechanisms are key to ensuring business continuity. The new system will adopt multi-level failure response strategies to minimize impacts. First, the system will implement local caching mechanisms; even during network interruptions, cical data will be stored in local devices, ensuring basic operations can continue for over 3 hours.

To address more serious system failures, the restaurant will preset emergency manual operation procedures, including paper order forms, temporary cashier processes, and simplified inventory recording methods. Staff will receive specialized training, enabling transition from system operations to manual mode within 10 minutes, ensuring uninterrupted service. Once the system recovers, manually recorded data will be batch imported, maintaining data integy.

Data secuy represents another key consideration. The system will implement strict backup strategies, including hourly incremental backups and daily complete backups, with all backup data encrypted and stored both locally and in the cloud. In data loss situations, the Recovery Time Objective (RTO) is set within 2 hours, with Recovery Point Objective (RPO) for the most recent hour's data, ensuring minimized business impact. Historical data indicates that systems adopting such dual backup strategies achieve data recovery success rates of 99.95%.

The system will also establish automatic monitoring and alert mechanisms, real-time monitoring system performance and abnormal behavior. When potential issues are detected, the system automatically sends alerts to IT support teams, achieving early problem discovery and handling. This mechanism is expected to reduce average fault repair time from 6 hours to within 2 hours, significantly improving system reliability.

**Peak Period/Special Situation Response**

Peak periods and special events present severe tests for the system. The new system adopts elastic architecture design, capable of dynamically allocating server resources to respond to load changes. Under normal loads, the system utilizes only 50% of computing resources, reserving ample capacity for peak values. Load testing indicates the system can easily handle 200% of baseline load without affecting response speeds, sufficient for special peak periods such as semester beginnings and finals.

Queue management mechanisms represent another system highlight. When concurrent requests exceed thresholds, the system automatically initiates intelligent queuing algohms, ensuring cical functions (such as payment processing) operate with prioy while reasonably delaying other requests. This mechanism prevents system overload crashes, maintaining normal operation of core functions even in extreme situations. Regarding user experience, the system displays real-time status and estimated completion times to waiting customers, reducing anxiety caused by uncertainty.

For unexpected events (such as ingredient shortages, equipment failures), the system designs rapid response mechanisms. For example, when certain ingredients suddenly become scarce, the system automatically updates menu displays, marking relevant dishes as "temporarily unavailable" while pushing notifications to management, avoiding embarrassing situations where customers discover unavailability only after ordering. This mechanism is expected to reduce over 80% of similar complaints, enhancing service transparency and reliability.

# Initial Design – Reports / Outputs

Depending on the application and function, we provide the following reports:

*Table 10. Report Summary*

| Report Name | Frequency | Type of User | Type of Information |
|---|---|---|---|
| Raw material availability report | Daily | Manager | Operational |
| Margaret Restaurant Personnel Daily Report | Daily | Manager | Operational |
| Customer flow intelligent forecast Report | Daily | Manager & Clerk | Tactical |
| Daily Sales Report | Daily | Manager | Operational |
| Weekly Sales Report | Weekly | Manager | Tactical |
| Monthly Sales Report | Monthly | Manager | Strategic |
| Monthly customer report | Monthly | Manager | Strategic |
| Weekly Popular Dishes Analysis Report | Weekly | Manager & Chef | Tactical |
| Weekly Order Pattern Analysis Report | Weekly | Manager & Staff | Tactical |
| Hourly Order Pattern Analysis Report | Daily | Manager & Staff | Operational |

### 1. Raw material availability report

*Table 11. Raw material availability report*

| Order ID | Supplier | Supplied Ingredient | Ingredient Specification | Quantity | Unit | Unit Price (CNY) | Notes |
|---|---|---|---|---|---|---|---|
| **2001** | Beijing Farm Fresh Co. | Potato | Fresh Yellow Potato | 250 | kg | 5 | No sprouting, clean soil |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **2002** | Golden Grain Group | Flour | High-gluten Wheat Flour | 125 | kg | 10 | Food-grade packaging |
| **2003** | Green Valley Produce | Tomato | Italian Variety | 50 | kg | 30 | Ripened but not overripe |
| **2004** | Happy Hen Poultry | Egg | Grade AA Fresh Eggs | 10 | case | 100 | 180 eggs/case, same-day lay |

**2. Personnel management system**

The Margaret Restaurant Personnel Daily Report presents the duty roster of the staff and the corresponding evaluation indicators.

*Table 12. Raw material availability report*

| Employee ID | Employee Name | Position | Shift | Date | Start Time | End Time | Working Hours | Contact Info | Attendance Status | Shift Sales (Production) | Feedback Rating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 001 | Zhang Ming | Chef | Early | 2025/3/5 | 7:00 | 15:00 | 8 | 154XXXXXX | Present | ￥5,000 | 4.8 |
| 002 | Li Hua | Chef | Late | 2025/3/5 | 13:00 | 21:00 | 8 | 163XXXXXX | Present | ￥4,800 | 4.6 |
| 003 | Wang Qiang | Waiter | Early | 2025/3/5 | 7:00 | 15:00 | 8 | 187XXXXXX | Present | ￥2,000 | 4.9 |
| 004 | Liu Min | Waiter | Early | 2025/3/5 | 7:00 | 15:00 | 8 | 138XXXXXX | Present | ￥1,500 | 4.7 |
| 005 | Chen Fang | Waiter | Late | 2025/3/5 | 13:00 | 21:00 | 8 | 139XXXXXX | Present | ￥1,200 | 4.8 |
| 006 | Zhao Lei | Waiter | Late | 2025/3/5 | 13:00 | 21:00 | 8 | 137XXXXXX | Present | ￥700 | 4.7 |

The Customer Flow Intelligent Forecast Report provides daily customer flow predictions and recent macro trends, including changes in customer flow at the same time each day.

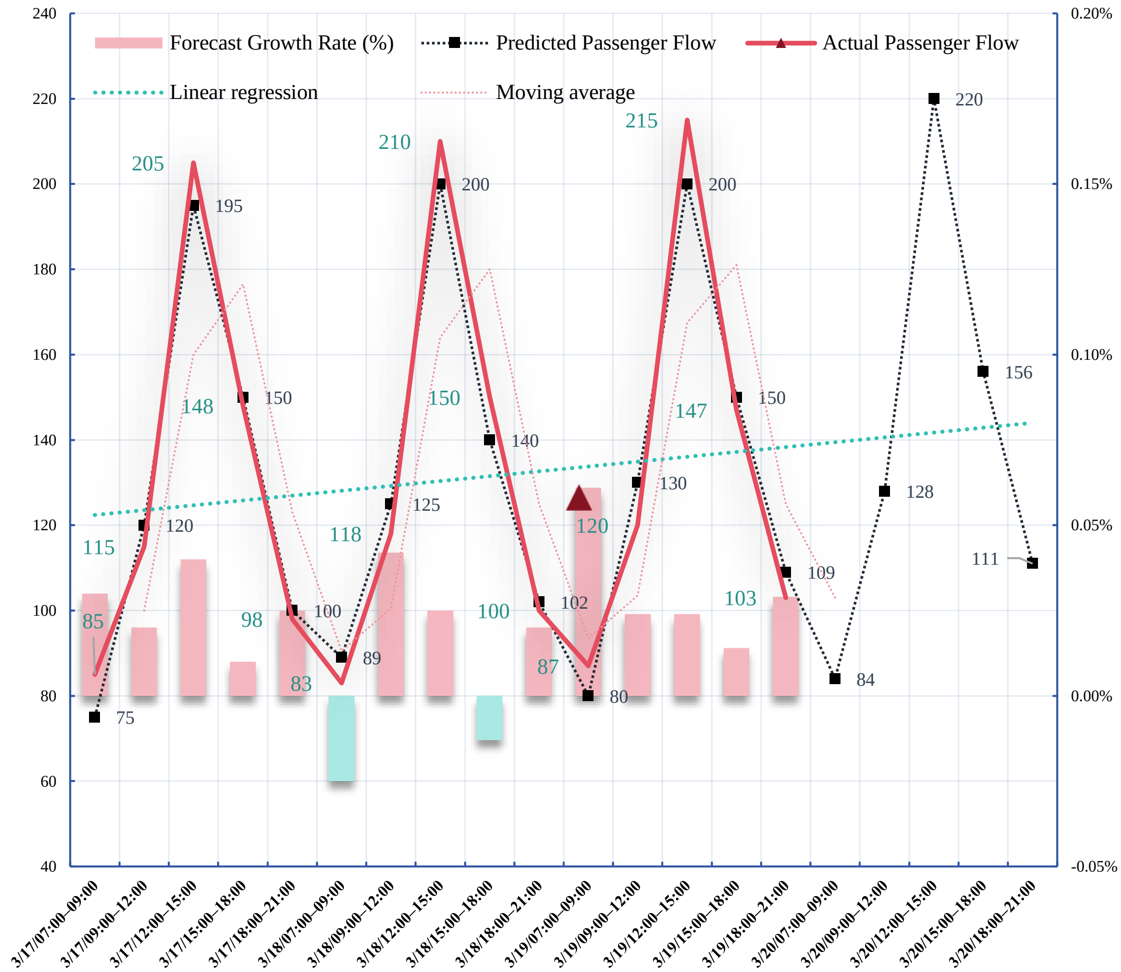



Figure 3. Customer Flow Intelligent Forecast

Table 13. Customer Flow Intelligent Forecast Report

| Date | Time Period | Predicted Passenger Flow | Actual Passenger Flow | Day-on-day Growth Rate (%) |
|---|---|---|---|---|
| 2025/3/18 | 3/18/07:00–09:00 | 89 | 83 | 2.50% |
| 2025/3/18 | 3/18/09:00–12:00 | 125 | 118 | 4.20% |
| 2025/3/18 | 3/18/12:00–15:00 | 200 | 210 | 2.50% |
| 2025/3/18 | 3/18/15:00–18:00 | 140 | 150 | -1.30% |
| 2025/3/18 | 3/18/18:00–21:00 | 102 | 100 | 2.00% |
| 2025/3/19 | 3/19/07:00–09:00 | 80 | 87 | 6.10% |
| 2025/3/19 | 3/19/09:00–12:00 | 130 | 120 | 2.40% |
| 2025/3/19 | 3/19/12:00–15:00 | 200 | 215 | 2.40% |
| 2025/3/19 | 3/19/15:00–18:00 | 150 | 147 | 1.40% |
| 2025/3/19 | 3/19/18:00–21:00 | 109 | 103 | 2.90% |
| 2025/3/20 (upcoming) | 3/20/07:00–09:00 | 84 | - | - |
| 2025/3/20 (upcoming) | 3/19/09:00–12:00 | 128 | - | - |
| 2025/3/20 (upcoming) | 3/19/12:00–15:00 | 220 | - | - |
| 2025/3/20 (upcoming) | 3/19/15:00–18:00 | 156 | - | - |
| 2025/3/20 (upcoming) | 3/19/18:00–21:00 | 111 | - | - |

### 3. Order management

The Order Management system efficiently tracks and manages customer orders by automatically recording details such as order ID, customer information, items purchased, and special requests.

*Table 14. Order report*

| Order ID | Customer Name | Contact Info | Order Time | Items Purchased | Special Requests | Order Status | Kitchen Receipt Time |
|---|---|---|---|---|---|---|---|
| ORD - 001 | Zhang San | Zhang San@example.com, 123 - 456 - 7890 | 2025/3/17 12:30 | Burger, Fries, Cola | No pickles, extra ketchup | Submitted | 2025/3/17 12:30 |
| ORD - 002 | Li Si | Li Si@example.net, 987 - 654 - 3210 | 2025/3/17 13:15 | Pizza (Pepperoni), Salad | Gluten - free crust, light on cheese | Submitted | 2025/3/17 13:15 |
| ORD - 003 | Wang Wu | Wang Wu@example.org, 555 - 123 - 4567 | 2025/3/17 14:00 | Spaghetti Bolognese, Garlic Bread | No onions, extra meat sauce | Submitted | 2025/3/17 14:00 |

**4. Sales Report**

Daily, weekly, and monthly sales reports, along with peak order patterns and popular dish analysis, provide insights into sales trends, customer preferences, and operational efficiency, enabling the restaurant to optimize menus, staffing, and overall performance.

*Table 15. Daily Sales Report*

| Date | Sales Revenue (¥) | Order Count |
|---|---|---|
| 2025/03/01 | ¥ 1,500 | 80 |
| 2025/03/02 | ¥ 1,700 | 85 |
| 2025/03/03 | ¥ 1,600 | 88 |
| 2025/03/04 | ¥ 1,650 | 90 |
| 2025/03/05 | ¥ 1,550 | 75 |

Daily sales reports display daily revenue and order counts, aiding restaurants in assessing performance and adjusting strategies.

*Table 16. Weekly Sales Report*

| Week Start | Sales Revenue (¥) | Order Count | Most Popular Dish |
|---|---|---|---|
| 2025/03/01 | ¥10,500 | 450 | Angus Beef Burger Set |
| 2025/03/08 | ¥11,200 | 475 | Classic Fried Chicken |
| 2025/03/15 | ¥10,800 | 460 | Classic Tomato Pasta Set |
| 2025/03/22 | ¥11,300 | 490 | Snowflake Chicken Fillet |

Weekly sales reports analyze trends and popular dishes, optimizing menus and staff schedules.

*Table 17. Monthly Sales Report*

| Month | Sales Revenue (¥) | Order Count | Most Popular Dish | Monthly Sales Growth |
|---|---|---|---|---|
| 2025 March | ¥45,000 | 2000 | Angus Beef Burger Set | +8% |
| 2025 February | ¥41,500 | 1900 | Classic Fried Chicken | +6% |

Monthly sales reports provide long-term trend data. The weekly order pattern identifies peak and off-peak days for resource allocation.

*Table 18. Weekly Popular Dishes*

| Dish Name | Order Count | Sales Revenue (¥) | Percentage of Total Sales (%) |
|---|---|---|---|
| Angus Beef Burger Set | 110 | ¥3,080 | 22.9 |
| Classic Fried Chicken | 90 | ¥1,800 | 18.8 |
| Classic Tomato Pasta Set | 85 | ¥2,295 | 17.7 |
| Snowflake Chicken Fillet | 80 | ¥800 | 16.7 |
| Mushroom Risotto | 55 | ¥660 | 11.5 |
| Others | 60 | ¥900 | 12.5 |

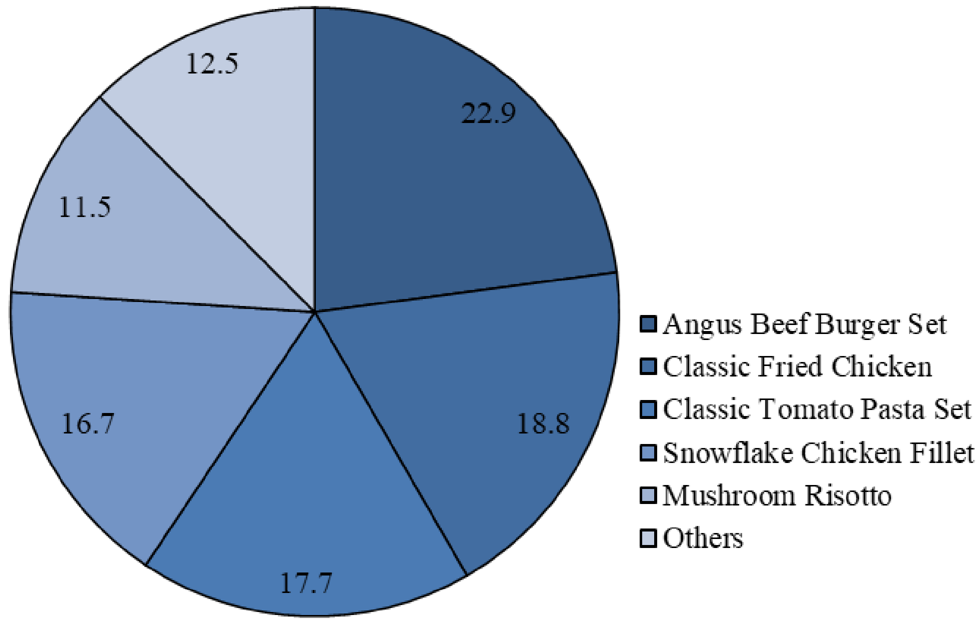



*Figure 4. Pie Chart of Weekly Sales Distribution by Dish*

The weekly popular dishes list details best-selling items, including order counts, revenue, and sales percentage, guiding menu and promotional decisions.

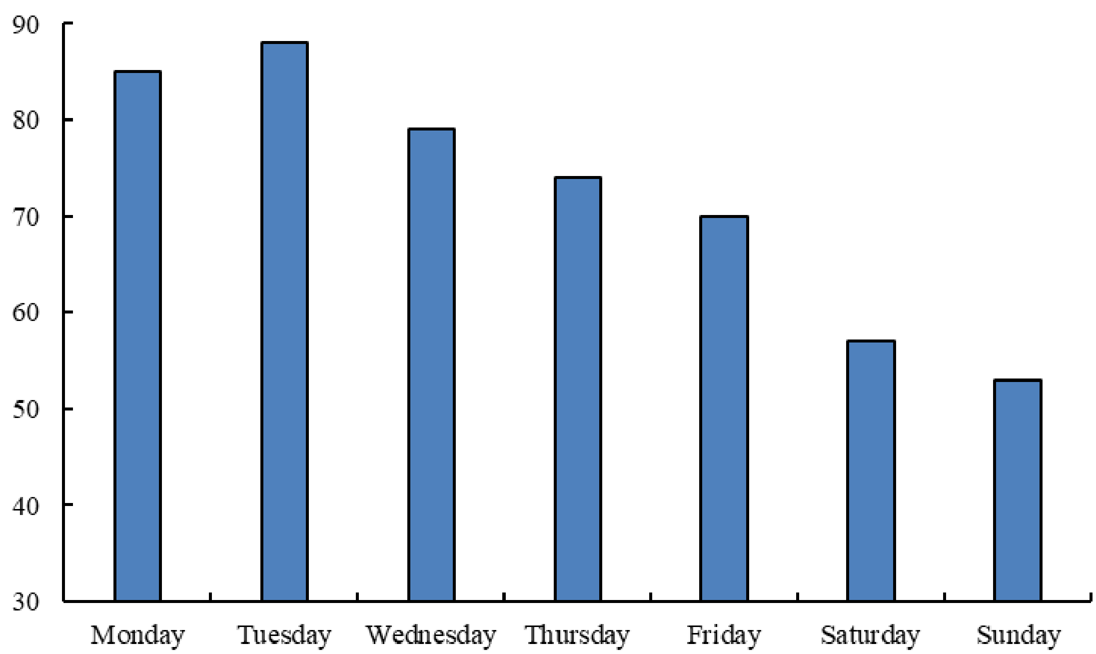



*Figure 5. Weekly Order Pattern with Peak Days*

This figure illustrates the distribution of orders across different days of the week, highlighting peak and off-peak days. It helps restaurants identify which days experience the highest and lowest order volumes, enabling better resource allocation and staffing decisions.

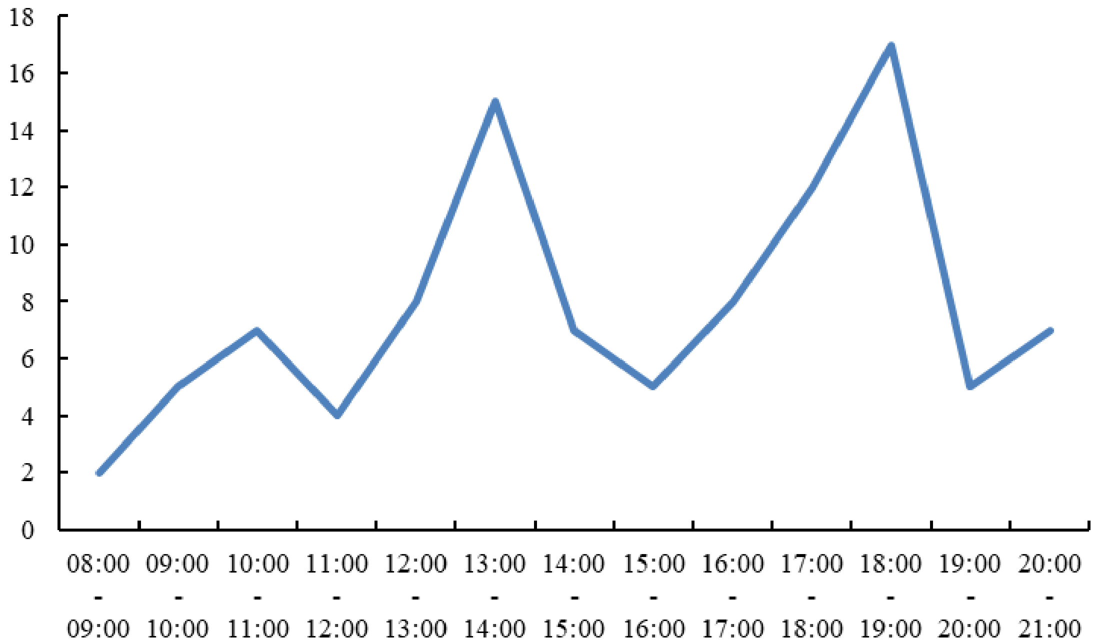



*Figure 6. Hourly Order Pattern with Peak Times*

This figure shows the number of orders received each hour, pinpointing the busiest times of the day. By understanding these peak hours, restaurants can optimize staff schedules and improve service efficiency during high-demand periods.

**5. Monthly customer report**

The monthly customer report is an important management tool for restaurants, providing in-depth insights into each customer based on information such as visit frequency, dining preferences, and ratings. Supported by data, it enables quick identification of customer value, spending trends, and potential hidden issues within the restaurant.

*Table 19. Monthly customer report*

| Customer ID | Customer Name | Total Spending | Visit Count | Dine-in Count | Most Ordered Dish | Peak Time | Customer Value |
|---|---|---|---|---|---|---|---|
| C10045 | James Wilson | 245.8 | 8 | 3 | Grilled Salmon | Dinner | A |
| C10078 | Emma Johnson | 380.5 | 5 | 4 | Beef Wellington | Lunch | A+ |
| C10103 | David Chen | 125.3 | 4 | 1 | Spicy Chicken Wings | Dinner | B |
| C10156 | Sarah Miller | 64.75 | 2 | 2 | Caesar Salad | Lunch | C |
| C10187 | Michael Brown | 210.4 | 7 | 5 | Seafood Pasta | Dinner | A |
| C10234 | Lisa Garcia | 455.6 | 6 | 6 | Wagyu Steak | Dinner | A+ |
| C10267 | Robert Taylor | 98.2 | 3 | 1 | Margherita Pizza | Lunch | B |
| C10302 | Jennifer Lee | 86.4 | 3 | 3 | Thai Green Curry | Dinner | C |
| C10345 | Thomas Wright | 178.5 | 6 | 2 | Club Sandwich | Lunch | B+ |
| C10389 | Olivia Rodriguez | 395.7 | 4 | 3 | Lobster Thermidor | Dinner | A |

## 6. Operation safety inspection report

Business safety inspection report is an important tool to control the safety of food and restaurant. The purpose of the report is to monitor the safety and hygiene of food from the source to the table.

By combing through key data, such as qualification audits of food suppliers, food storage temperature records, and compliance with hygienic practices during processing, it helps to quickly identify food safety risk points. This report can handle the inspection of the government and the cafeteria management.

*Table 20. Operation safety inspection report*

| Inspection Items | Key Inspection Entries |
|---|---|
| Perishable Foods | Meat and seafood have no spoilage odor, dairy products are within the shelf life, and fruits and vegetables are free from decay |
| Refrigerator Temperature | Refrigerator temperature is 2 - 8℃, freezer temperature is below -18℃ |
| Environmental Hygiene | The restaurant floor is clean without garbage, the kitchen operation area is tidy without oil stains, and the trash cans are emptied in a timely manner |
| Food Storage | Separate raw and cooked foods, store foods off the ground and away from the wall, and there are no expired foods |
| Tableware Disinfection | Tableware is cleaned thoroughly, the disinfection method complies with regulations, and it is stored in a dedicated cleaning cabinet |
| Personal Hygiene of Staff | Employees hold valid health certificates, meet personal hygiene standards, and there are training records |
| Equipment and Facilities | Cooking, refrigeration and freezing equipment are in normal operation, and fire-fighting equipment is in good condition |
| Food Safety System | The management system is sound, and there are food traceability and complaint handling mechanisms |
| Purchase Food Qualification Certificates | All types of purchased foods are equipped with corresponding qualification certificates, |

# PART 3: Application Design

# I. Database Design (Tables)

### 1. Supplier

This table is used to record information about each supplier. In this table, Its attributes include SupplierID, SupplierName, SupplierProperty, SupplierCity, SupplierProvince, ContactFirstName, ContactLastName and ContactEmail. The primary key is SupplierID for each supplier. It can help us construct supplier reports and improve management reports. In addition, it can also provide information about suppliers, which is very helpful for us to contact them in time.

**2. Inventory**

This table is used to record updates to each item of inventory. In this table, its properties include InventoryID, InventoryName, InventorySupplierID, InventoryType, InventoryQuantity, InventoryMargin, InventoryAccountID, OrderDate, DueDate, and Description. The primary key is InventoryID. Inventory change as they are procured or consumed and serve as a source of evidence for the availability of a certain dish. At the same time, the Inventory table helps to construct the inventory report, which can provide the quantity and time of raw materials and packages update.

**3. Recipe (consumption of individual ingredients used in each dish)**

This table is used to record the consumption of certain ingredients required for each dish, and a dish often requires multiple recipe lists because it requires multiple ingredients (such as two slices of bread, one piece of beef, etc.). At the same time it is a bridge table. In this table, its properties include RecipeID, MenuItemID, MaterialID, and QuantityUsed. The primary key is RecipeID. This table helps staff and the kitchen to know what ingredients are used in a dish, and allows users to know the ingredients for a dish.

**4. MenuItem**

This table is used to record basic information about each menu item (ingredient). In this table, its properties include MenuItemID, MenuItemName, Price, Description, State, and Category. The primary key is MenuItemID for each menu item. This table helps construct a menu item list report that lets users know what dishes they can choose. In addition, it can provide the status of menu items.

**5. Customer Review**

This table is used to record customer comments and is connected to the customer table. In this table, its properties include ReviewID, CustomerID, ReviewDate, and ReviewContext. The primary key is ReviewID. It provides specific content about each customer service review.

**6. Customer**

This table is used to record basic information about each customer. In this table, its properties include CustomerID, CustomerLastName, CustomerFirstName,CustomerGender, CustomerPhone, CustomerEmail, CustomerType, CustomerBirthday, and CustomerCard. The primary key is CustomerID for each customer. This table is used to store the customer's personal information, It can help us to maintain customer relationships through customer information and also allows us to have a clearer understanding of customer distribution.

**7. Order**

This table records the order of each customer, which is generated from orders of users. In this table, its properties include OrderID, OrderDate, CustomerID, OrderPayment, MenuItemID, StaffID, AccountID, Note. The primary key is OrderID. This table helps us maintain order information, construct order reports, and improve management reports.

**8. Staff**

This table is used to record basic information about each staff member. In this table, its properties include StaffID, StaffLastName, StaffFirstName, StaffPhone, StaffEmail, StaffPosition, and StaffAddress. The primary key is StaffID for each staff member. This helps in managing staff information and reporting the composition of staff members.

**9. Staff Working**

This table is used to record Staff working situation information. In this table, its properties include StaffID, StaffWage(Yuan/hour), WorkDate, StaffWorkHours, PayableID and Note. The primary key is StaffID for each staff member. This helps in recording working situation of staff and distribute precise wages.

**10. Item**

This table is used to record the whole items the restaurant need, from raw materials to packages. In this table, its properties include ItemID, ItemName, ItemType, ItemUnitPrice, MinmumQuantity, SupplierID, State, and Note. The primary key is ItemID. The purpose of this table is to record information about each item, allowing for accurate purchase management and supplier analysis.

**11. Account Receivable**

This table is used to record account receiveable associated with the business. In this table, its properties include ReceiveableID, ReceiveableName, ReceiveableType, ReceiveableAmount and Note. The primary key is ReceiveableID. This table helps with financial reporting and tracking in the aspect of revenues.

**12. Account Payable**

This table is used to track financial account payable associated with the business. In this table, its properties include PayableID, Invoice, PayableName, PayableType, PayableDueDate, PayableDatePaid, PayableAmount and Note. The primary key is AccountID. This table helps with financial reporting and tracking in the aspect of costs.

*Table 1. Summary of System's Tables*

| Name of Tables | Primary Key | Source of Data | Number of Records |
|---|---|---|---|
| Customer | CustomerID | Customer Survey | 200 |

| | | | |
|---|---|---|---|
| Staff | StaffID | Staff Survey | 6 |
| Staff Working | StaffWorkingID | Staff Survey | 120 |
| Order | OrderID | Customer Order | 603 |
| MenuItem | MenuItemID | Kitchen | 15 |
| Recipe | RecipeID | Kitchen | 4 |
| Supplier | SupplierID | Supplier Survey | 8 |
| Item | ItemID | Supplier Survey | 19 |
| Inventory | MaterialID | Purchase | 32 |
| Account Receivable | ReceivableID | Order and so on | 342 |
| Account Payable | PayableID | Wage, Purchase and so on | 125 |
| Customer Review | ReviewID | Customer Review | 26 |

# II. Inputs Design

**1. Input**

- **Customer information**

As long as the customer agrees to the collection of their personal information (of course, this is the default in the context of the campus card checkout), the customer information is an important part. We will be able to store each customer's personal information, allowing us to better understand our customers' needs. We can enter details including customer name, customer gender, phone number, email address, flavor, credit card number, and customer birthday with customer type. That’s because thinking of the policy from governments, such as discount for aging, we need to use these two fields to identify them.

- **Staff information**

Our restaurant staff is an important basis for ensuring the normal operation of the restaurant, including serving customers, confirming orders and cleaning facilities. The staff database provides restaurant managers and investors with easy access to basic staff information. We can enter detailed information, including the staff's name, phone number, address, job title, etc. The staff information sheet can be linked to the order sheet to determine the performance of each staff and facilitate the evaluation of staff performance and be linked to Staff Working table to determine and record their wages. Personal hygiene inspections by school departments and government agencies are also required

- **Supplier information**

Suppliers provide food raw materials for our restaurants, and information about suppliers is also an important part of the food safety and hygiene inspection. It is necessary for us to establish supplier data sheet, including supplier name, supplier property, city, province, contact name and contact phone number. When it is necessary to output management reports, call the information in the supplier data table, and contact the supplier according to the information in the table to supplement when there is a shortage of raw materials. At the same time, we add a item table to record what the supplier can provide concretely, so that we can identify quickly what we want if there is shortage.

- **Recipe information**

An ordinary hamburger consists of two slices of bread and a certain amount of beef, a hot dog consists of a bun and a sausage, how many kilograms of potatoes does a medium-sized French fries need? In order to prepare the product, we have to determine the exact ingredients needed, which is the responsibility of the kitchen. Also create a recipe table to store the types and quantities of items needed for each dish. This data is vital to the standardization and scale of our food production and is an important part of our internal records. We can enter the details, including the ID of the desired raw material, its corresponding menu ID, the quantity required for the raw material, etc. In fact, the table that stores this information is an association table that corresponds to information about multiple raw materials and becomes part of a menu.

- **Purchase information**

After we have determined the ingredient composition of the food through recipe, we also need to consider what to do if a certain ingredient is in short supply, or lack of packages. Naturally, there is a need for procurement. Through the purchase information sheet, we replenish stock to ensure a smooth supply. The input information will include InventoryID, InventooryDate, InventoryType, AccountID, OrderDate, DueDate, InventoryQuantity, InventoryMargin, ItemID, InventorySupplierID, etc., It's linked to our restaurant accounts, supplier information forms and inventory information.

- **Order information**

When all this is done, the sale begins. It is necessary to keep a record of each sale, including the name and quantity of the product ordered, the service personnel, the price, etc. We can enter the details, including OrderID, OrderDate, CustomerID, OrderPayment, MenuItemID, StaffID, AccountID, StaffID.

# III. Outputs Design

Our database system is designed to generate five different output interfaces that capture data through various input streams to provide key insights for restaurant management, customer understanding and operational efficiency.

- **Operational Report**

This report provides a comprehensive view of the restaurant's daily operations. It includes aggregated data from a variety of sources, such as sales, raw material use. The report provides managers with a quick snapshot of key performance indicators (KPIs), such as total sales, popular items, to facilitate immediate decisions regarding staffing levels and inventory management.

- **Customer Flow Intelligent Forecast Report**

This report analyzes historical sales data, seasonal trends and customer demographic information to predict future customer traffic patterns. The report helps managers optimize staffing, manage inventory levels, and plan marketing campaigns based on expected demand. The main data input is sales information linked to customer data.

- **Inventory Report**

This report draws from the Inventory table, this report provides a real-time overview of current stock levels. It includes the InventorylID, InventoryName, SupplierID, CurrentStock, DueDate, and Unit, directly from the Inventory table. It helps manage the restaurant stock, preventing shortages and reducing waste. The report is linked to the Supplier table for easy contact information when reordering is necessary and the Purchase table to track incoming deliveries.

- **Order Report**

This report provides an overview of the daily and weekly order performance. It includes information about the number of order, payments methods and total revenue. It is essential for financial management and enables analysis of consumption trends in real time.

- **Customer Report**

This report is designed to provide insights into customer demographics, preferences, and behavior patterns. The Customer Report includes Customer name, CustomerGender, CustomerPhone, CustomerEmail, and CustomerCard from the Customer table. This data can be used for targeted marketing campaigns, personalized menu recommendations, and loyalty programs. The Customer Review table can provide additional context, capturing customer feedback and sentiment.

- **Popular Dishes Analysis Report**

This report uses order and recipe information to assess the populay of each menu item. It links the MenuItemID in the Order table and the Recipe table. So it can inform menu optimization, pricing strategies, and promotions. For example, by analyzing which dishes are frequently ordered and the costs of ingredients associated with them, management can make decisions about menu offerings based on data.

# IV. Data Flow Diagram

### 1. Context Level DFD

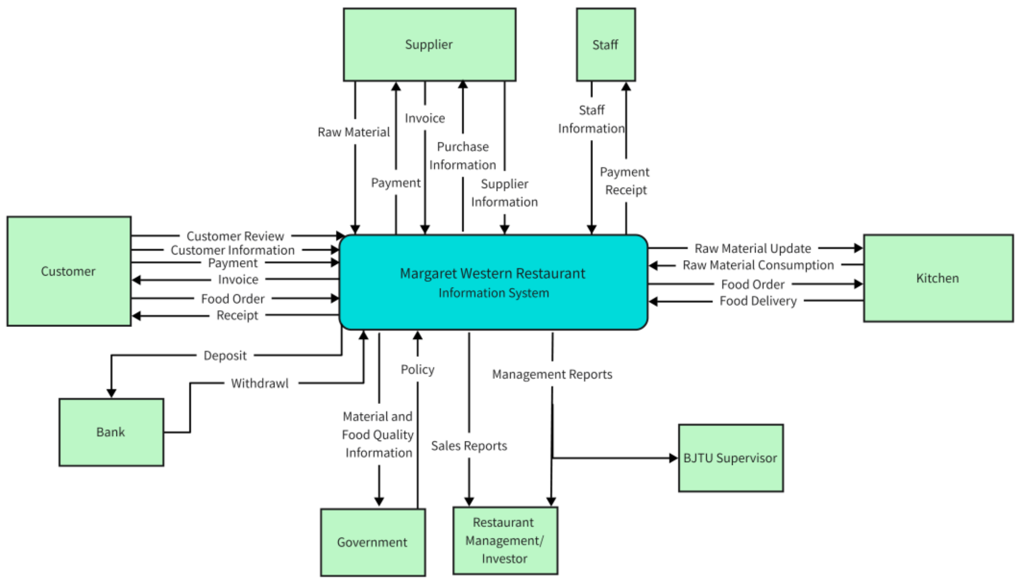



*Figure 1. Context Level DFD*

## 2. Level 0 DFD

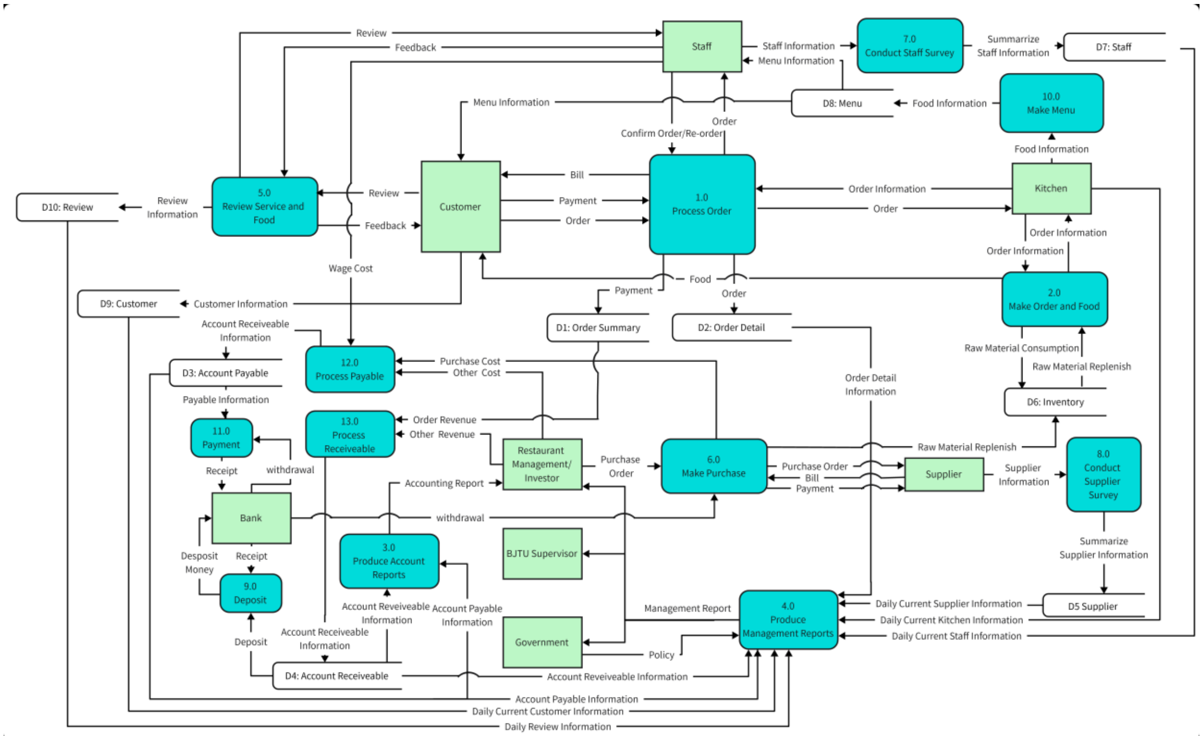



*Figure 2. Level 0 DFD*

# V. Entity Relationship Diagram

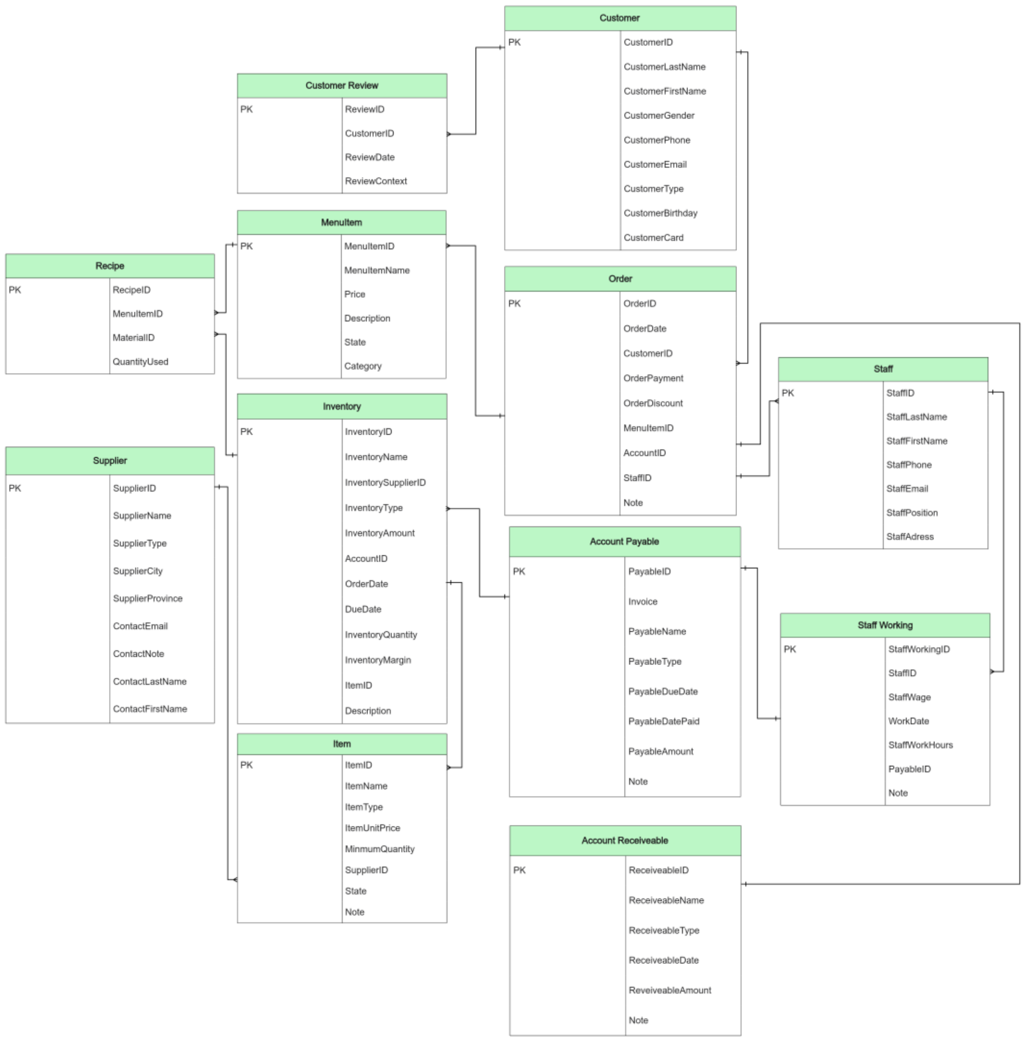



*Figure 3. ERD of the System*

# VI. Interfaces Design

## 1. System Interface

Our MIS system design combines Python GUI and Access database. The front-end uses Python's Tkinter library to build the user interface, which provides an intuitive interface to support data entry, querying, editing and report generation. The system is integrated with Access database through pyodbc library, and the data is stored in Access database to ensure efficient data management and query. Users input data through the form interface and Python script is responsible for data validation and error handling to ensure data integy. The query interface supports multi-ceria filtering, and the generated reports can be exported to Excel or PDF format. In addition, the system provides basic rights management functions to control user access rights through Python. The whole design focuses on the system's ease of use, data consistency and scalability.

### 1.1 Login interface

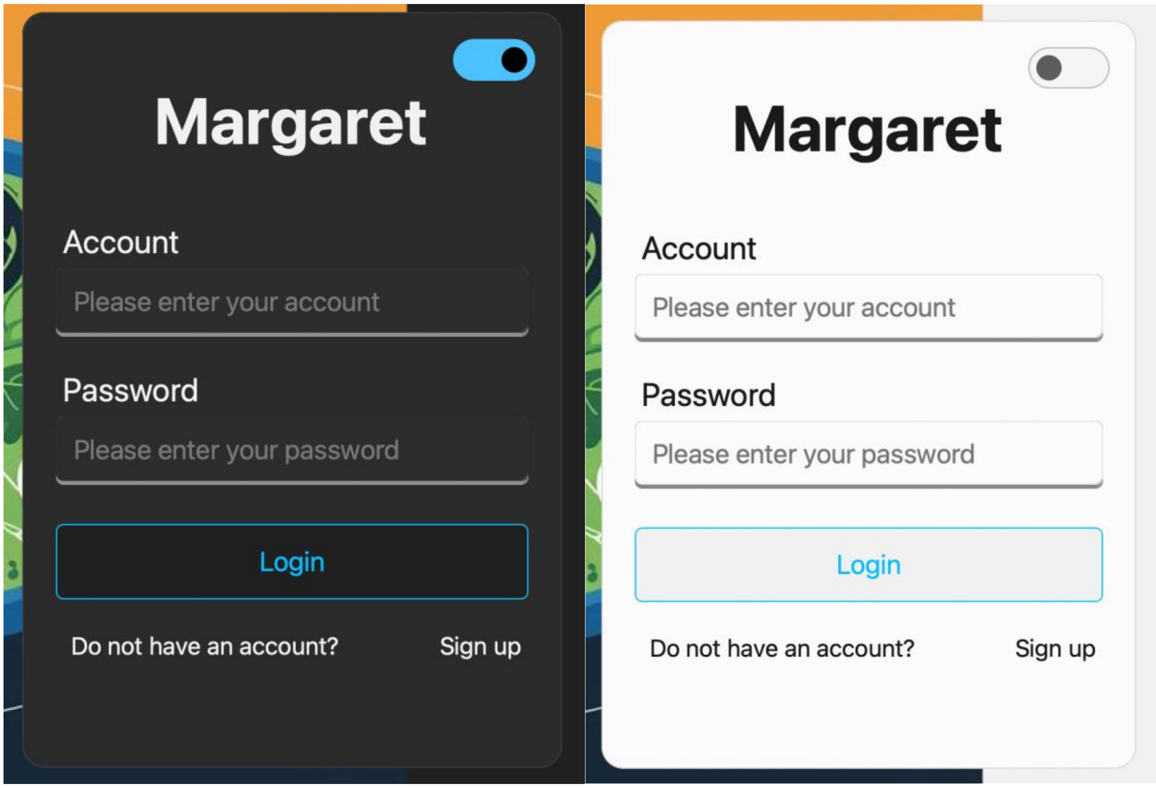



*Figure 4. Login interface*

Considering the permissions and secuy issues, we design user login interface, which has both light and dark modes, as well as password retrieval and registration functions. Through the login interface, the system can identify the user's permission and ensure that the corresponding user can only see the content corresponding to his permission. For example, staff can only make orders, managers can make purchases and view reports.

### 1.2 System main interface

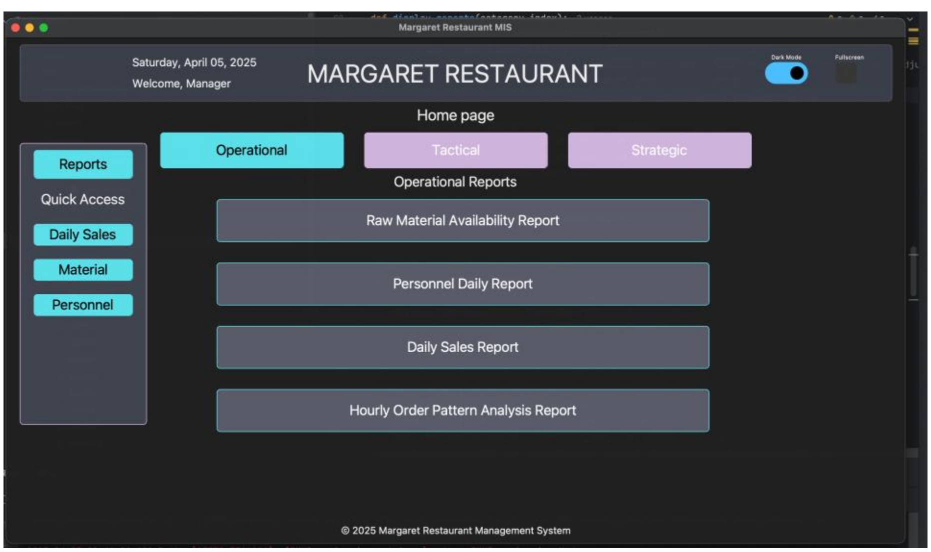



*Figure 5. System main interface(Dark mode）*

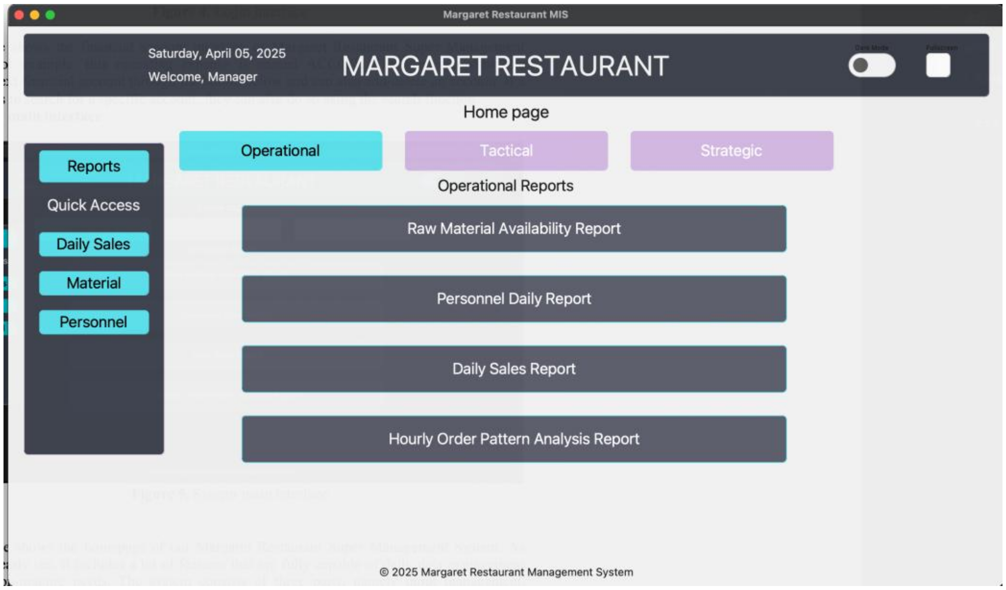



*Figure 5. System main interface(Light mode）*

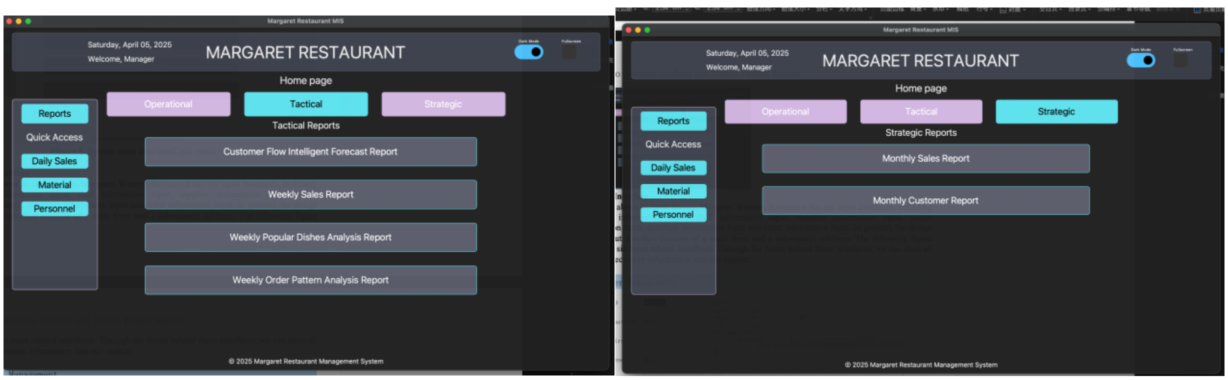



*Figure 6. Switch report type*

The above is the homepage of our restaurant management information system. The interface design is simple and the functions are clear. The top displays the restaurant name, current date and user information. The middle provides three report categories (operational, tactical, strategic), and users can switch and view reports of different categories through buttons. There is a side menu on the left, providing quick access to functions such as "Daily Sales", "Materials" and "Personnel". The interface also includes theme switching and full-screen mode buttons, allowing users to switch between light and dark modes according to their needs. The overall design adopts a modern style, with soft color combinations and clear layout, enabling users to operate the system intuitively and efficiently.

**2. Input Interface**

We have designed Input Interface for each form to ensure maintainability and modification of the form. We used two styles to create the forms, one is for forms that are used less frequently, we added the basic functionality of adding, deleting, changing and checking. The other group is forms that are frequently used for daily operations, and we added the display of other auxiliary information.We give the following case：

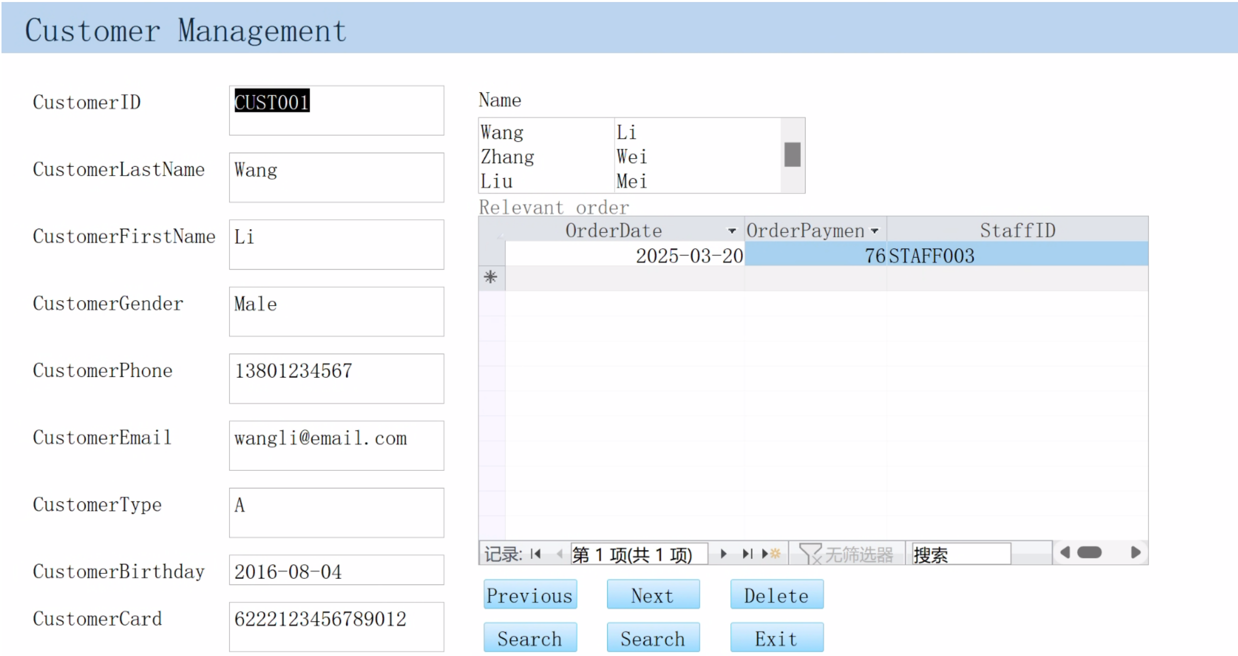



*Figure 6. Customer Input Interface*

The Customer Management UI is an example of a common everyday form. The customer management screen of the system will enter the basic information of the customer. As shown in the figure, each customer has a unique ID, which records personal information such as the customer's name, gender, phone number, e-mail address, customer's birthday, customer type, and card number. On the other hand, for each user's ID, we also display Relevant order information for the administrator's convenience and quick query.

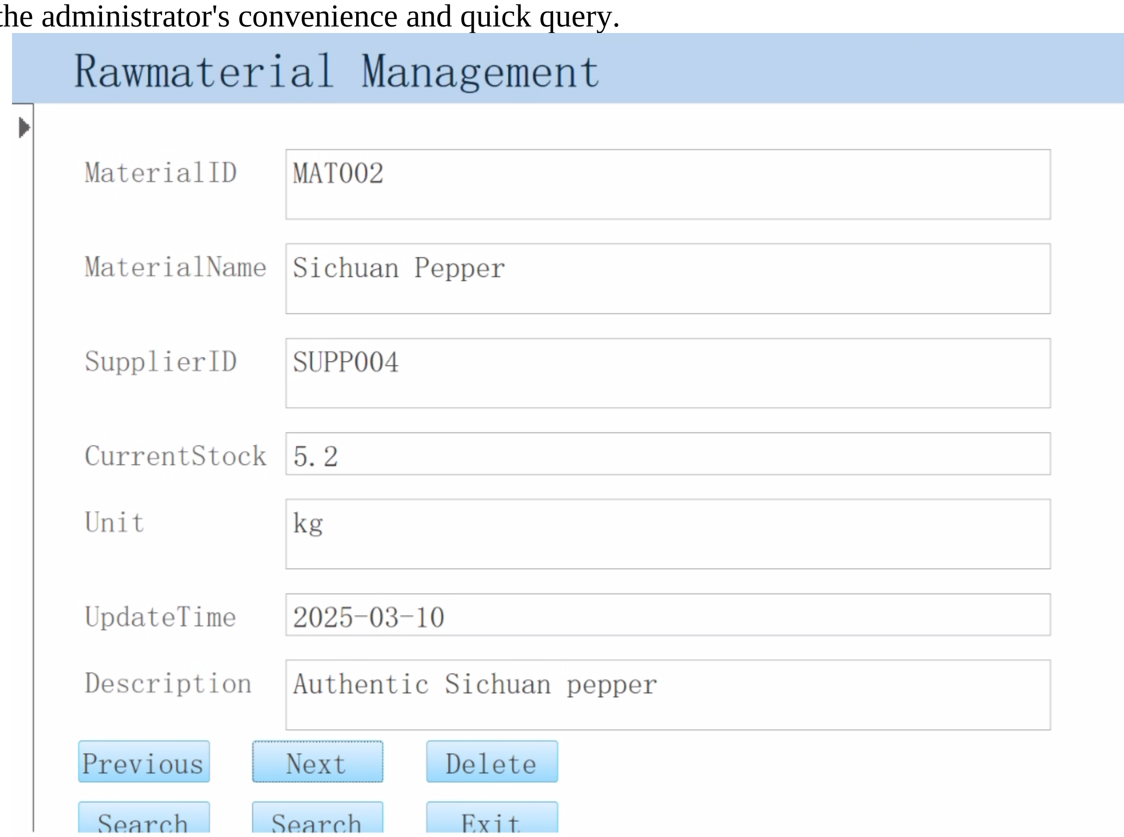



*Figure 7. Rawmaterial Input Interface*

The system outputs the Raw Material Management interface, as shown in Figure X. This interface displays detailed information about a raw material, including its Material ID, Material Name, Supplier ID, Current Stock, Unit, Update Time, and Description. The interface also provides navigation options, such as "Previous," "Next," "Delete," and "Search," allowing users to manage and update raw material information efficiently.

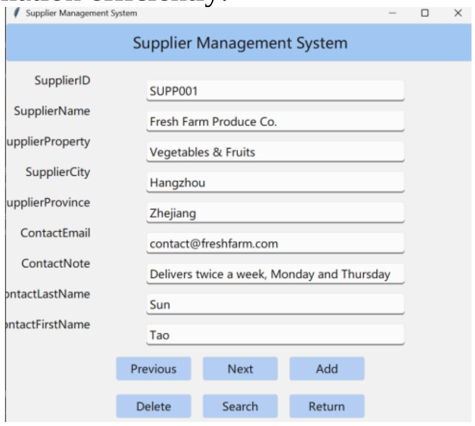



*Figure 8. Supplier Information Input Interface*

This diagram shows the Supplier management interface of the system. As shown in the figure, each supplier has a unique ID which records their supply information such as name, supply product, contact information, contact information, etc.

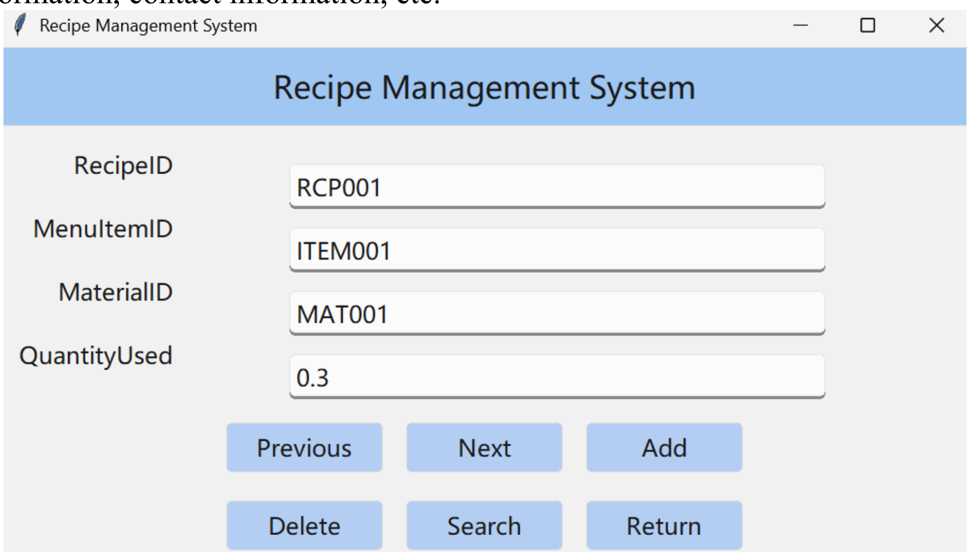



*Figure 9. Recipe Input Interface*

This diagram shows the system's video recipe management interface. As shown in the figure, when a food is identified, we will insert the information of the raw materials used according to its ID, corresponding to the ID and quantity of each raw material

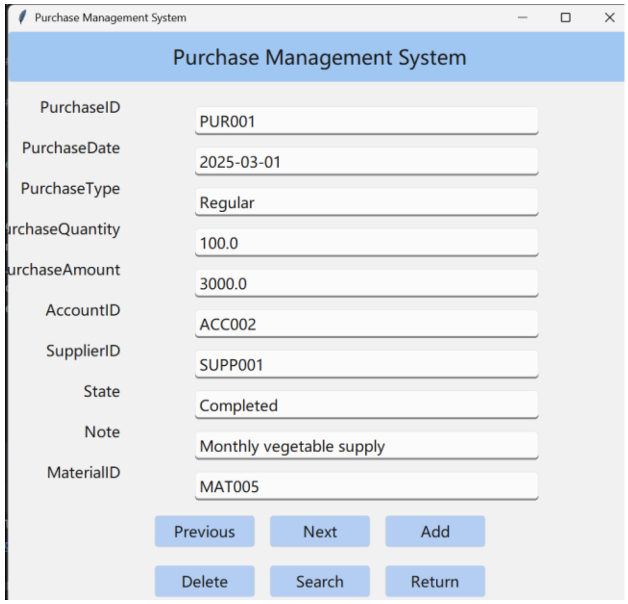


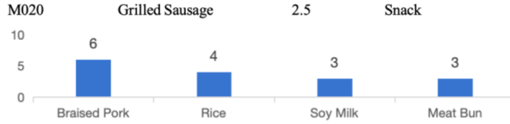

*Figure 10. Purchase Input Interface*

The system outputs the Purchase Management System interface, as shown in Figure X. This interface displays key details of a purchase order, such as Purchase ID, Date, Quantity, Amount, Supplier, and Material ID. Users can navigate through records, add, delete, or search for purchase entries.

## 3. Output Interface

Based on the system, we are able to generate a series of reports to assist operations. Some examples are given below

| MenuItemID | MenuItemName | Price | Category | Sales |
|---|---|---|---|---|
| M001 | Vegetable Bun | 1.5 | Main dish | 2 |
| M002 | Meat Bun | 2 | Main dish | 3 |
| M003 | Millet Porridge | 2 | Main dish | 2 |
| M004 | Soy Milk | 1.5 | Beverage | 3 |
| M005 | Egg Pancake | 3.5 | Main dish | 2 |
| M006 | Fried Rice | 8 | Main dish | 2 |
| M007 | Kung Pao Chicken | 10 | Meat dish | 1 |
| M008 | Tomato & Egg | 8 | Veg dish | 1 |
| M009 | Braised Pork | 12 | Meat dish | 6 |
| M010 | Stir-fried Broccoli | 6 | Veg dish | 2 |
| M011 | Boiled Fish | 15 | Meat dish | 1 |
| M012 | Rice | 1.5 | Main dish | 4 |
| M013 | Seaweed Soup | 3 | Soup | 2 |
| M014 | Hot & Sour Soup | 4 | Soup | 1 |
| M015 | Cola | 2.5 | Beverage | 2 |
| M016 | Mineral Water | 2 | Beverage | 1 |
| M017 | Mapo Tofu | 8 | Veg dish | 2 |
| M018 | Yu-Shiang Pork | 10 | Meat dish | 2 |
| M019 | Steamed Bun | 1 | Main dish | 1 |
| M020 | Grilled Sausage | 2.5 | Snack | 2 |



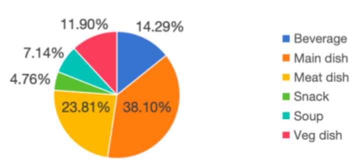




*Figure 12. Weekly Most Popular Dishes reportOutput Interface*

The system generates a weekly Most Popular Dishes report, as illustrated in Figure 13. This report offers comprehensive data on the sales of menu items, including the name, price, category, and total sales for each dish. The report also includes a visual representation of the sales distribution, which utilizes a bar graph to illustrate the sales of the dishes. In addition, pie charts categorize dishes based on their type (e.g., entrees, meat dishes, vegetarian dishes, beverages, soups, and snacks), showing their relative sales proportions. This report is a valuable tool for management, providing insights into customer preferences, optimizing menu offerings, and making informed decisions based on data. It facilitates improved service and inventory planning, contributing to the company's overall success.

**Personnel Daily Report** 2025/4/5 20:58:57

| StaffID | WorkDate | StaffWorkHours | Note |
|---|---|---|---|
| S001 | 2025-4-1 | 8 | Full attendance |
| S002 | 2025-4-1 | 8 | Full attendance |
| S003 | 2025-4-1 | 8 | Full attendance |
| S004 | 2025-4-1 | 8 | Full attendance |
| S005 | 2025-4-1 | 7 | 1 hour leave |
| S006 | 2025-4-1 | 8 | Full attendance |
| S007 | 2025-4-1 | 8 | Full attendance |
| S008 | 2025-4-1 | 8 | Full attendance |
| S009 | 2025-4-1 | 8 | Full attendance |
| S010 | 2025-4-1 | 6 | 2 hours leave |
| S011 | 2025-4-1 | 8 | Full attendance |
| S012 | 2025-4-1 | 8 | Full attendance |
| S013 | 2025-4-1 | 8 | Full attendance |
| S014 | 2025-4-1 | 8 | Full attendance |
| S015 | 2025-4-1 | 8 | Full attendance |
| S016 | 2025-4-1 | 8 | Full attendance |
| S017 | 2025-4-1 | 8 | Full attendance |
| S018 | 2025-4-1 | 8 | Full attendance |
| S019 | 2025-4-1 | 8 | Full attendance |
| S020 | 2025-4-1 | 8 | Full attendance |

*Figure 13. Personnel Daily Report*

The system outputs the Personnel Daily Report, as shown in Figure 13. The report provides detailed data on work attendance, including each employee's attendance status, working hours, and related notes. This report helps management better understand the employees' work status and provides data support for future decisions and management.

Customer Review Reports

| CustomerID | Date | ReviewContent |
|---|---|---|
| C001 | 2025/4/1 | Good fried rice, would order again. |
| C002 | 2025/4/1 | Kung Pao Chicken was spicy and delicious. |
| C003 | 2025/4/1 | The tomato and egg dish is always reliable. |
| C004 | 2025/4/1 | Braised pork was tender and flavorful. |
| C005 | 2025/4/1 | Broccoli was fresh and perfectly cooked. |
| C006 | 2025/4/1 | Yu-Shiang Pork has great flavor. |
| C007 | 2025/4/1 | Mapo Tofu is authentic Sichuan style. |
| C008 | 2025/4/1 | Hot & Sour Soup is perfect for cold days. |

*Figure 14. Customer Review Reports*

This report is a part of the customer management Report. Through browsing and summarizing the customer evaluation content, it can identify the customer satisfaction as well as the shortcomings and improvement directions of the restaurant

# VII. Design Details

Our restaurant management system is implemented using Microsoft Access. We chose it because it is easy to use, has a fast development capability, and integrates well with the Microsoft Office suite. To ensure data secuy and prevent accidental data loss, the Access database is regularly backed up to Microsoft Office's cloud service.

### 1. User Authentication and Access Control

Upon launching the application, users will be presented with a login screen that requires a valid username and password for authentication. This secuy layer is crucial for controlling access to sensitive data and functions within the system. To manage user access, a "user" table is maintained in the database. After a successful login, the system determines the user's permission group based on the "GroupID" in the "user" table. This permission group specifies which forms and reports are accessible, ensuring that each user can only interact with data and functions relevant to their role.

### 2. System Navigation and Data Management

Once authenticated, users will see a home page that provides convenient access to various modules of the system, including forms and reports.

Forms: Forms are used for data input, modification, and deletion. Users can add new records, update existing records, and delete records from the underlying database tables. These forms are designed to provide a user-friendly interface for interacting with data. Data validation is implemented in the forms to ensure data integy and prevent errors.

Reports: Reports provide structured and formatted representations of data extracted from the database. Users can generate reports by selecting specific conditions and filtering data to meet their specific needs. Reports are used to provide management insights and support decision-making.

### 3. Special Design

Due to python's rich extensibility and compatibility, we used the ui integration interface designed using Python and looked forward to packaging it as an exe. It should be pointed out that compared with the complete use of access, our python program realizes interface automation and uses a small amount of code to generate input in a uniform style, avoiding repeated design of the ui of each input. When tables and fields need to be added or modified, they can only be added in access. Without reconstructing python code, this is conducive to the later development and maintenance of the system, and convenient for iterative design. At the same time, we added light color mode and night mode to the ai to adapt to different user groups, and support full-screen display to adapt to different displays

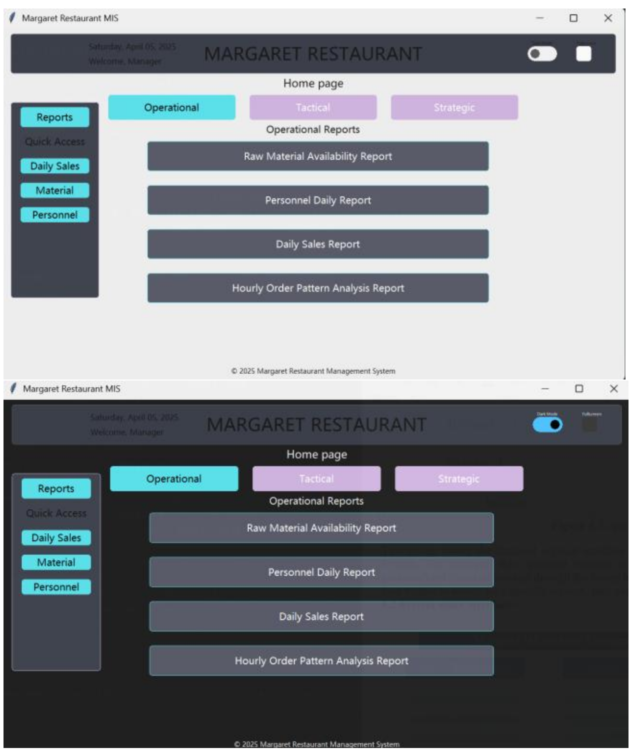



*Figure 16. System UI (Light and Darkness mode)*

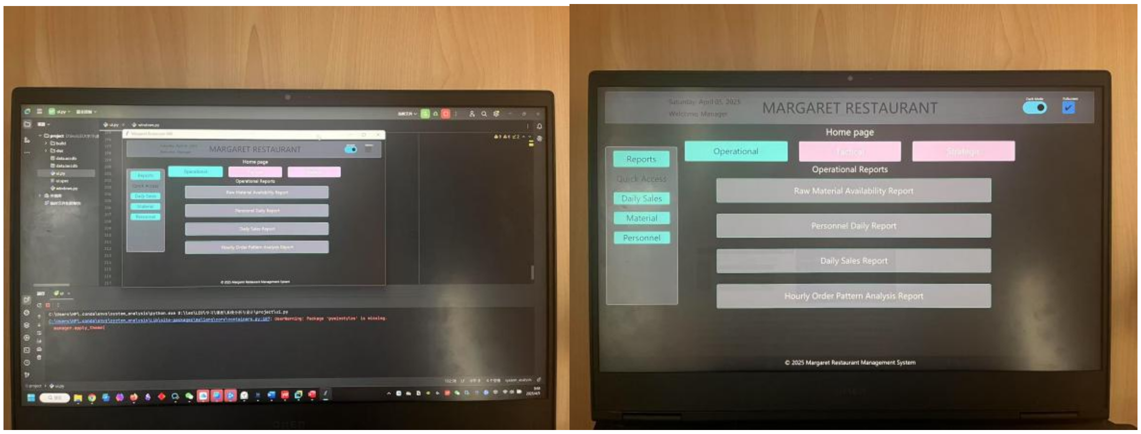



*Figure 17. System UI (Medium and full screen mode)*

Inventory management optimization: We use an inventory table to track the inventory level of each raw material. This inventory table is associated with the raw material table and the purchase table. The daily updated raw material inventory records based on formula consumption and new purchases are recorded in the inventory table. This approach helps to reduce the complexity of the MenuItem and Order tables, thereby reducing the overall burden of the system and improving operational efficiency.

Detailed customer and supplier records: The database has comprehensive records of both customer and supplier information. The Customer table stores detailed customer information for further customized services and customer retention. The Supplier table contains key supplier information, including contact details, to ensure reliable supply chain management.

Comprehensive Reporting Suite: The system provides a comprehensive reporting suite to meet daily operational needs and tactical/strategic decision-making. These reports cover key aspects of restaurant operations:

- Sales: Analyze sales trends, identify the most popular menu items, and track revenue.
- Customers: Understand customer demographics, preferences, and consumption habits.
- Staff: Monitor staff performance, track order processing times, and manage labor costs.
- Products (Menu Items): Evaluate the profitability and populay of menu items.

# VIII. Updated Gantt Chart

| | TASK NAME | START DATE | END DATE | DURATION (WORK DAYS) | TEAM MEMBER | PERCENT COMPLETE |
|---|---|---|---|---|---|---|
| Initial Project Proposal | | | | | | |
| | Identify Restaunrant Business | 2/26 | 2/28 | 3 | All | 100% |
| | State Restaunrant Background and Problems | 3/1 | 3/3 | 3 | Gao Tianrun | 100% |
| | Identify Organization Purpose and Objectives | 3/3 | 3/4 | 2 | Dong Renxi/Wang Kexin | 100% |
| | Determine System Functions | 3/4 | 3/5 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| | Identify Output Managerial Information | 3/5 | 3/6 | 2 | Zhu Yifan | 100% |
| | Define Automation Benefits | 3/7 | 3/7 | 1 | All | 100% |
| Feasibility Study / Planning Phase | | | | | | |
| | Identify Business Problem and scope | 3/10 | 3/11 | 2 | Gao Tianrun/Zhu Yifan | 100% |
| | Technical Feasibility Analysis | 3/11 | 3/13 | 3 | Dong Renxi | 100% |
| | Economic Feasibility Analysis | 3/11 | 3/14 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| | Schedule Feasibility Analysis | 3/12 | 3/15 | 4 | Wang Kexin | 100% |
| | Operational Feasibility Analysis | 3/12 | 3/15 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| | Initial Design – Reports / Outputs | 3/15 | 3/17 | 3 | All | 100% |
| Application Design | | | | | | |
| | Generate DFD and ERD | 3/19 | 3/20 | 2 | Zhu Yifan | 100% |
| | Design The User Interfaces | 3/19 | 3/21 | 3 | Gao Tianrun | 100% |
| | Design The System Interfaces | 3/20 | 3/21 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| | Design and Integrate The Database | 3/22 | 3/22 | 1 | Hu Shaohua | 100% |
| | Prototype For Design Details | 3/23 | 3/24 | 1 | Gao Tianrun/Zhu Yifan | 100% |
| | Design and Integrate The System Controls | 3/25 | 3/26 | 2 | Dong Renxi/Wang Kexin | 100% |
| Prototyping / Data Definitions – Design Phase | | | | | | |
| | Define Tables | 3/28 | 4/2 | 4 | Gao Tianrun | 0% |
| | Define Table Fields | 3/30 | 4/2 | 3 | Wang Kexin | 0% |
| | Define Table Indexes | 3/31 | 4/2 | 3 | Dong Renxi | 0% |
| | Define Queries | 4/2 | 4/4 | 3 | Liu Zhuodong | 0% |
| | Define Queries Fields | 4/2 | 4/5 | 4 | Hu Shaohua | 0% |
| | Print out Actual Reports | 4/5 | 4/6 | 2 | Zhu Yifan | 0% |
| Implementation Phase & Final Report | | | | | | |
| | Construct Software Components | 4/9 | 4/12 | 3 | Gao Tianrun/Zhu Yifan | 0% |
| | Verify and Test | 4/11 | 4/13 | 1 | Gao Tianrun | 0% |
| | Convert Data | 4/12 | 4/14 | 1 | Zhu Yifan | 0% |
| | Train users and Document System | 4/13 | 4/15 | 2 | Liu Zhuodong | 0% |
| | Install System | 4/14 | 4/16 | 3 | Hu Shaohua | 0% |
| Documentation | | | | | | |
| | Write Executive Overview | 4/16 | 4/16 | 1 | Wang Kexin | 0% |
| | Written components | 4/17 | 4/17 | 1 | Dong Renxi | 0% |
| | Database Application | 4/18 | 4/18 | 1 | All | 0% |

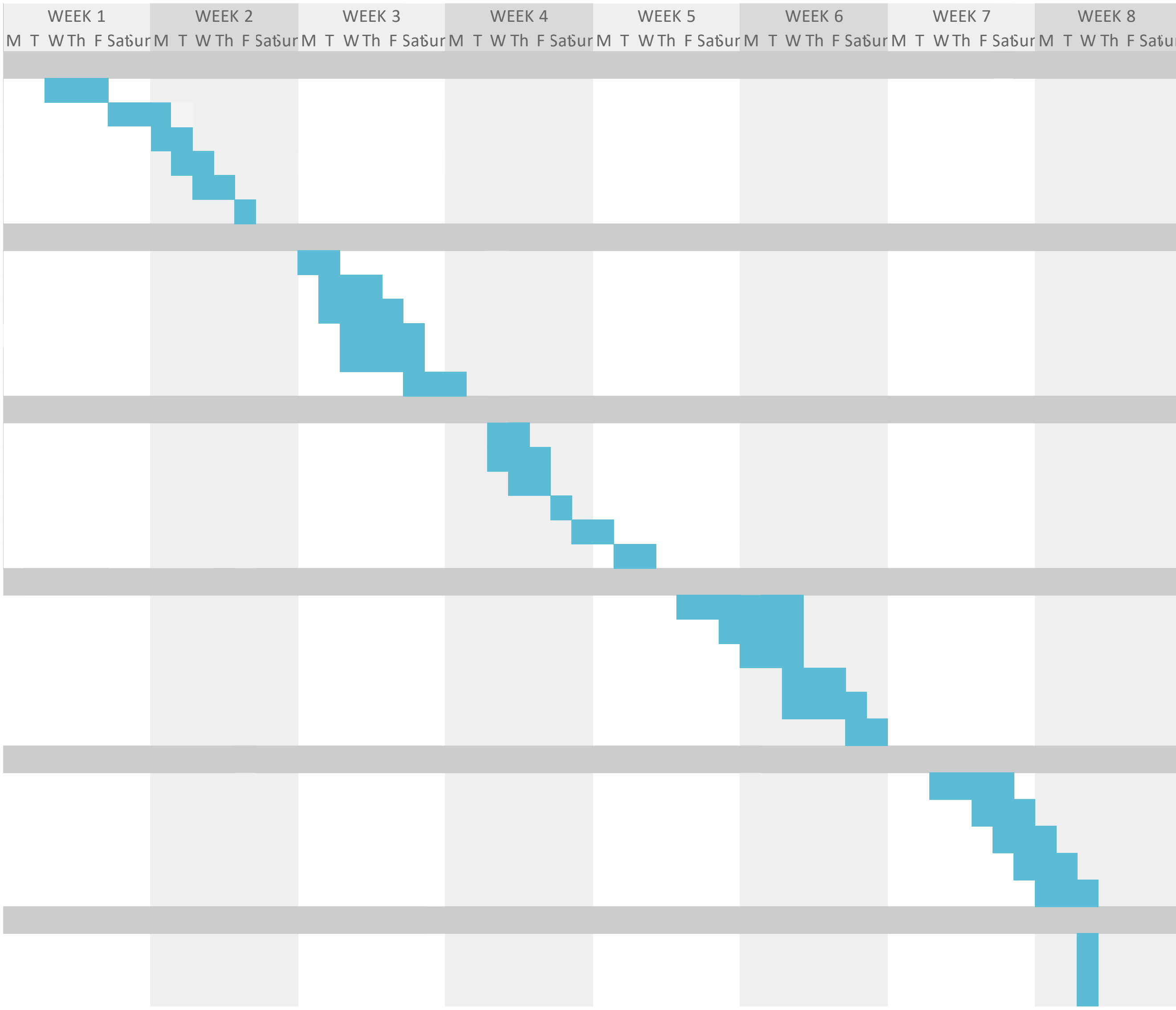



*Figure 18. Team Gantt Chart*

# PART 4: Prototyping & Data Definitions

# I. Forms and Related Tables

Our system has a total of 12 tables. Among them, the **Recipe table** serves as a bridge, establishing connections between MenuItems and Inventory. The rest 11 tables are shown in Figure 1.The ERD diagram comprehensively and clearly presents all tables and the relationships between them.

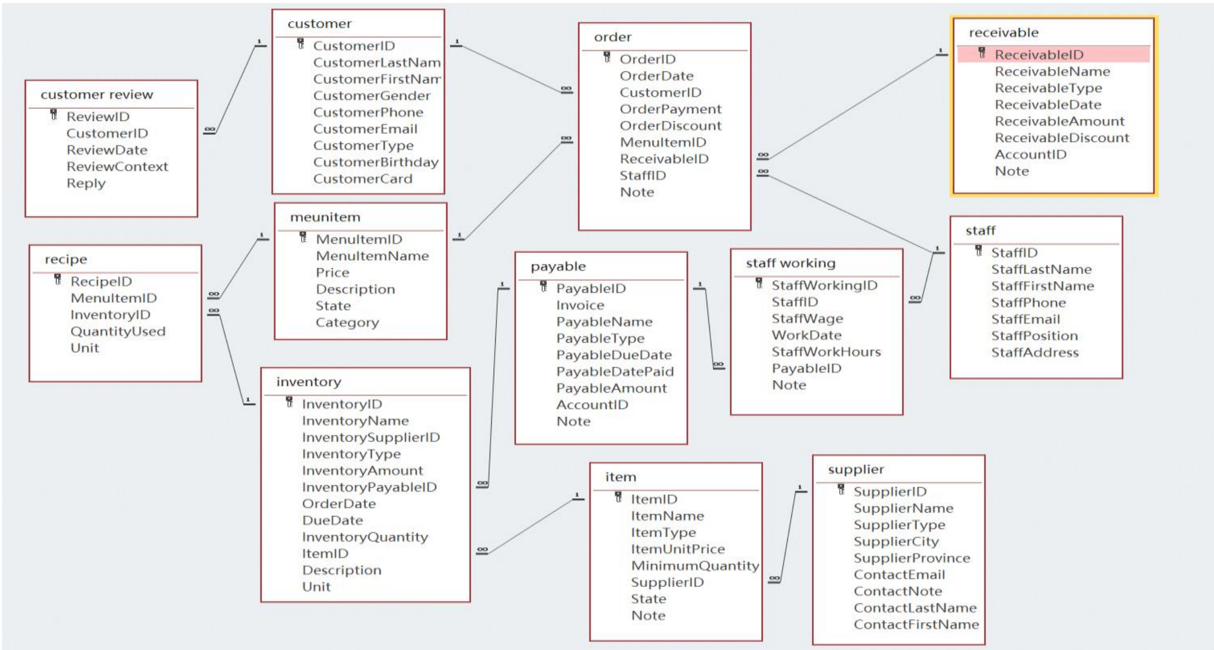



*Figure 1. ERD of the System*

**(1) Payables Table**

The **Payables Table** contains basic information about the amounts owed by the restaurant to staff and suppliers, such as salaries and inventory purchases. This table is for managers only, as only they have permission to modify the payables information. However, it can be read by all staff members, customers, and managers. The **Payables Table** is connected with the Staff Working Table and the Inventory Table. There are a total of 9 attributes in the **Payables Table**. The data types of PayableID, Invoice, PayableName, PayableType, AccountID, and Note are all "Short Text." The data type of PayableDueDate and PayableDatePaid is "Date/Time," while PayableAmount has the data type of "Number." Table 1 and 2 show the detailed information about the Payables table.

*Table 1. Payables Table*

| PayableID | Invoice | PayableNam | PayableTyp | PayableDue | PayableDat | PayableAmo | AccountID | Note |
|---|---|---|---|---|---|---|---|---|
| PAY1001 | SUPINV-1001 | Global Foods | Supplier Pay | 2025/3/30 | 2025/3/29 | 30 | AP-GOODS | Payment for |
| PAY1002 | SUPINV-1002 | Meat Packers | Supplier Pay | 2025/3/31 | 2025/3/30 | 250 | AP-GOODS | Payment for |
| PAY1003 | SUPINV-1003 | Fresh Dairy ( | Supplier Pay | 2025/4/1 | | 90 | AP-GOODS | Payment for |
| PAY1004 | SUPINV-1004 | Global Foods | Supplier Pay | 2025/3/30 | 2025/3/29 | 45 | AP-GOODS | Payment for |
| PAY1005 | SUPINV-1005 | Fresh Dairy ( | Supplier Pay | 2025/4/2 | | 25 | AP-GOODS | Payment for |
| PAY1006 | SUPINV-1006 | Global Foods | Supplier Pay | 2025/3/31 | 2025/3/30 | 60 | AP-GOODS | Payment for |
| PAY1007 | SUPINV-1007 | Global Foods | Supplier Pay | 2025/3/31 | 2025/3/30 | 35 | AP-GOODS | Payment for |
| PAY1008 | SUPINV-1008 | Meat Packers | Supplier Pay | 2025/4/1 | | 320 | AP-GOODS | Payment for |
| PAY1009 | SUPINV-1009 | Fresh Dairy ( | Supplier Pay | 2025/4/2 | | 40 | AP-GOODS | Payment for |
| PAY1010 | SUPINV-1010 | Global Foods | Supplier Pay | 2025/3/31 | 2025/3/30 | 55 | AP-GOODS | Payment for |
| PAY1011 | SUPINV-1011 | Global Foods | Supplier Pay | 2025/3/30 | 2025/3/29 | 35 | AP-GOODS | Payment for |
| PAY1012 | SUPINV-1012 | Meat Packers | Supplier Pay | 2025/3/31 | 2025/3/30 | 150 | AP-GOODS | Payment for |
| PAY1013 | SUPINV-1013 | Global Foods | Supplier Pay | 2025/3/27 | 2025/3/26 | 15 | AP-GOODS | Payment for |
| PAY1014 | SUPINV-1014 | Meat Packers | Supplier Pay | 2025/3/31 | 2025/3/30 | 90 | AP-GOODS | Payment for |
| PAY1015 | SUPINV-1015 | Meat Packers | Supplier Pay | 2025/4/2 | | 450 | AP-GOODS | Payment for |
| PAY1016 | SUPINV-1016 | Global Foods | Supplier Pay | 2025/3/30 | 2025/3/29 | 70 | AP-GOODS | Payment for |
| PAY1017 | SUPINV-1017 | Meat Packers | Supplier Pay | 2025/3/31 | 2025/3/30 | 200 | AP-GOODS | Payment for |
| PAY1018 | SUPINV-1018 | Global Foods | Supplier Pay | 2025/3/30 | 2025/3/29 | 80 | AP-GOODS | Payment for |
| PAY1019 | SUPINV-1019 | Meat Packers | Supplier Pay | 2025/4/2 | | 300 | AP-GOODS | Payment for |
| PAY1020 | SUPINV-1020 | Global Foods | Supplier Pay | 2025/3/31 | 2025/3/30 | 75 | AP-GOODS | Payment for |
| PAY1021 | SUPINV-1021 | Global Foods | Supplier Pay | 2025/4/1 | | 120 | AP-GOODS | Payment for |
| PAY1022 | SUPINV-1022 | Global Foods | Supplier Pay | 2025/3/31 | 2025/3/30 | 50 | AP-GOODS | Payment for |
| PAY1023 | SUPINV-1023 | Fresh Dairy ( | Supplier Pay | 2025/4/1 | | 180 | AP-GOODS | Payment for |
| PAY1024 | SUPINV-1024 | Meat Packers | Supplier Pay | 2025/3/31 | 2025/3/30 | 110 | AP-GOODS | Payment for |
| PAY1025 | SUPINV-1025 | Fresh Dairy ( | Supplier Pay | 2025/4/3 | | 20 | AP-GOODS | Payment for |

*Table 2. Data Type of Payables Table*

| Field Name | Data Type |
|---|---|
| PayableID | Short Text |
| Invoice | Short Text |
| PayableName | Short Text |
| PayableType | Short Text |
| PayableDueDate | Date/Time |
| PayableDatePaid | Date/Time |
| PayableAmount | Number |
| AccountID | Short Text |
| Note | Short Text |

(2) Receivables Table

The **Receivables Table** has account receivables from customers at the restaurant, and this table can be read by managers. The **Receivables Table** is connected with the **Order Table**. There are a total of 8 attributes in the **Receivable Table**. The data types of ReceivableID, ReceivableName, ReceivableType, AccountID, and Note are all “Short Text.” The data type of ReceivableDate is “Date/Time,” and that of ReceivableAmount and ReceivableDiscount is “Number.” Table 3 and Table 4 show the detailed information about the **Receivables Table**.

*Table 1. Receivables Table*

| ReceivableID | ReceivableType | AccountID | ReceivableName | ReceivableDat | Receivable | ReceivableD | Note |
|---|---|---|---|---|---|---|---|
| RCV1001 | Sales Revenue | ACC-SALES | C035 | 2025/3/2 | 15.75 | 0 | 1001 |
| RCV1002 | Sales Revenue | ACC-SALES | C091 | 2025/3/3 | 28.5 | 0 | 1002 |
| RCV1003 | Sales Revenue | ACC-SALES | C005 | 2025/3/3 | 8.99 | 0 | 1003 |
| RCV1004 | Sales Revenue | ACC-SALES | C077 | 2025/3/5 | 22 | 0 | 1004 |
| RCV1005 | Sales Revenue | ACC-SALES | C035 | 2025/3/7 | 11.25 | 0 | 1005 |
| RCV1006 | Sales Revenue | ACC-SALES | C101 | 2025/3/10 | 19.8 | 0 | 1006 |
| RCV1007 | Sales Revenue | ACC-SALES | C018 | 2025/3/11 | 6.5 | 0 | 1007 |
| RCV1008 | Sales Revenue | ACC-SALES | C052 | 2025/3/14 | 29.95 | 0 | 1008 |
| RCV1009 | Sales Revenue | ACC-SALES | C088 | 2025/3/16 | 14 | 0 | 1009 |
| RCV1010 | Sales Revenue | ACC-SALES | C023 | 2025/3/18 | 21.1 | 0 | 1010 |
| RCV1011 | Sales Revenue | ACC-SALES | C099 | 2025/3/20 | 9.75 | 0 | 1011 |
| RCV1012 | Sales Revenue | ACC-SALES | C041 | 2025/3/22 | 17.3 | 0 | 1012 |
| RCV1013 | Sales Revenue | ACC-SALES | C064 | 2025/3/25 | 25 | 0 | 1013 |
| RCV1014 | Sales Revenue | ACC-SALES | C009 | 2025/3/28 | 7.2 | 0 | 1014 |
| RCV1015 | Sales Revenue | ACC-SALES | C035 | 2025/3/30 | 18.6 | 0 | 1015 |
| RCV1016 | Sales Revenue | ACC-SALES | C072 | 2025/4/1 | 26.4 | 0 | 1016 |
| RCV1017 | Sales Revenue | ACC-SALES | C050 | 2025/4/3 | 5.5 | 0 | 1017 |
| RCV1018 | Sales Revenue | ACC-SALES | C081 | 2025/4/5 | 12.9 | 0 | 1018 |
| RCV1019 | Sales Revenue | ACC-SALES | C015 | 2025/4/8 | 23.75 | 0 | 1019 |
| RCV1020 | Sales Revenue | ACC-SALES | C095 | 2025/4/10 | 10 | 0 | 1020 |
| RCV1021 | Sales Revenue | ACC-SALES | Customer C045 | 2025/3/1 | 22.15 | 0 | From Order 1021 |
| RCV1022 | Sales Revenue | ACC-SALES | Customer C088 | 2025/3/28 | 14.5 | 1 | From Order 1022 |
| RCV1023 | Sales Revenue | ACC-SALES | Customer C012 | 2025/4/5 | 7.99 | 0 | From Order 1023 |
| RCV1024 | Sales Revenue | ACC-SALES | Customer C077 | 2025/3/15 | 28.3 | 2.5 | From Order 1024 |
| RCV1025 | Sales Revenue | ACC-SALES | Customer C031 | 2025/4/9 | 11.8 | 0 | From Order 1025 |
| RCV1026 | Sales Revenue | ACC-SALES | Customer C095 | 2025/3/3 | 19.25 | 0 | From Order 1026 |
| RCV1027 | Sales Revenue | ACC-SALES | Customer C004 | 2025/3/19 | 6.75 | 0 | From Order 1027 |
| RCV1028 | Sales Revenue | ACC-SALES | Customer C059 | 2025/4/1 | 25.5 | 0 | From Order 1028 |

*Table 4. Data Type of Receivables Table*

| Field Name | Data Type |
|---|---|
| ReceivableID | Short Text |
| ReceivableName | Short Text |
| ReceivableType | Short Text |
| ReceivableDate | Date/Time |
| ReceivableAmount | Number |
| ReceivableDiscount | Number |
| AccountID | Short Text |
| Note | Short Text |

(3) Customer Table

The **Customer** Table contains basic information about each customer who has purchased from the restaurant. This table can be read by all staff members and managers. The **Customer** Table is connected with the **Order** Table and **Customer Review** Table. There are a total of 9 attributes in the Customer Table. The data types of CustomerID, CustomerLastName, CustomerFirstName, CustomerGender, CustomerEmail, and CustomerType are all “Short Text”. The data types of CustomerPhone and CustomerCard are “Number”. The data type of CustomerBirthday is “Date/Time”. Table 5 and Table 6 show the detailed information about the **Customer** Table.

*Table 5. Customer Table*

| | CustomerID | CustomerLast | CustomerFirstN | CustomerGender | CustomerPhone | CustomerEmail | CustomerType | CustomerBirthday | CustomerCard |
|---|---|---|---|---|---|---|---|---|---|
| + | C001 | Li | Ming | Male | 15366781234 | ming@example.cc | Student | 2006/5/15 | 1001 |
| + | C002 | Wang | Fang | Female | 15566782345 | gfang@example.c | Student | 2005/8/23 | 1002 |
| + | C003 | Zhang | Wei | Male | 13866783456 | ngwei@example.c | Student | 2007/3/10 | 1003 |
| + | C004 | Liu | Na | Female | 13766784567 | iuna@example.co | Student | 2006/11/28 | 1004 |
| + | C005 | Chen | Qiang | Male | 15866785678 | nqiang@example. | Student | 2005/7/14 | 1005 |
| + | C006 | Yang | Li | Female | 13966786789 | ngli@example.cc | Student | 2007/9/2 | 1006 |
| + | C007 | Zhao | Xin | Male | 15266787890 | aoxin@example.c | Student | 2006/2/19 | 1007 |
| + | C008 | Huang | Lin | Female | 13866788901 | nglin@example.c | Student | 2005/12/7 | 1008 |
| + | C009 | Zhou | Hao | Male | 13766789012 | ouhao@example.c | Student | 2007/4/25 | 1009 |
| + | C010 | Wu | Fang | Female | 15866780123 | fang@example.cc | Student | 2006/10/13 | 1010 |
| + | C011 | Guo | Lei | Male | 139667812 | ouhao@example.c | Student | 2007/4/25 | 1009 |
| + | C012 | Ma | Li | Female | 15266782345 | ali@example.cor | Teacher | 1985/9/14 | 2002 |
| + | C013 | Lin | Qiang | Male | 13866783456 | qiang@example.c | Teacher | 1982/2/28 | 2003 |
| + | C014 | Zhu | Ting | Female | 13766784567 | uting@example.c | Teacher | 1990/11/7 | 2004 |
| + | C015 | Xu | Peng | Male | 15866785678 | peng@example.cc | Teacher | 1979/7/16 | 2005 |
| + | C016 | Sun | Ling | Female | 13966786789 | nling@example.c | Teacher | 1987/4/3 | 2006 |
| + | C017 | Hu | Ming | Male | 15266787890 | ming@example.cc | Staff | 1975/5/21 | 3001 |
| + | C018 | Gao | Yan | Female | 13866788901 | oyan@example.cc | Staff | 1980/8/12 | 3002 |
| + | C019 | Xie | Gang | Male | 13766789012 | egang@example.c | Staff | 1978/3/25 | 3003 |
| + | C020 | Tang | Ying | Female | 15866780123 | gying@example.c | Staff | 1982/10/9 | 3004 |
| + | C021 | Song | Jie | Male | 13611234501 | songjie@mail.cor | Staff | 1985/4/18 | 1021 |
| + | C022 | Han | Fei | Female | 11098765432 | olei@example.cc | Teacher | 1988/6/22 | 2001 |
| + | C023 | Feng | Tao | Male | 15933456703 | engtao@mail.cor | Staff | 1992/12/3 | 1023 |
| + | C024 | Deng | Juan | Female | 13444567804 | engjuan@mail.co | Teacher | 1989/7/29 | 1024 |
| + | C025 | Cao | Yang | Male | 15055678905 | aoyang@mail.cor | Student | 2007/8/11 | 1025 |
| + | C026 | Cheng | Min | Female | 18766789006 | hengmin@mail.co | Staff | 1995/3/5 | 1026 |
| + | C027 | Yuan | Bo | Male | 13577890107 | yuanbo@mail.com | Teacher | 1981/6/19 | 1027 |
| + | C028 | Jiang | Li | Female | 15188901208 | jiangli@mail.cor | Student | 2005/9/30 | 1028 |

*Table 6. Data Type of Customer Table*

| Field Name | Data Type |
|---|---|
| CustomerID | Short Text |
| CustomerLastName | Short Text |
| CustomerFirstName | Short Text |
| CustomerGender | Short Text |
| CustomerPhone | Number |
| CustomerEmail | Short Text |
| CustomerType | Short Text |
| CustomerBirthday | Date/Time |
| CustomerCard | Number |

(4) Customer Review Table

The **Customer Review** Table provides specific content about each customer service review. It can be accessed and read by managers and relevant staff for quality improvement analysis. This table is connected with the **Customer** Table. There are 5 attributes in the Customer Review Table. The data types of ReviewID, CustomerID, ReviewContext, and Reply are all “Short Text”. The data type of ReviewDate is “Date/Time”. Table 7 and Table 8 show the detailed information about the **Customer Review** Table.

*Table 7. Customer Review Table*

| ReviewID | CustomerID | ReviewDate | ReviewContext | Reply |
|---|---|---|---|---|
| CR001 | C045 | 2025/3/5 | Burger was juicy and delicious! Fries were hot too. | |
| CR002 | C012 | 2025/3/8 | Service was a bit slow during the lunch rush. | |
| CR003 | C088 | 2025/3/10 | Loved the spicy chicken wings! Perfect heat. | |
| CR004 | C025 | 2025/3/11 | It was okay, standard fast food. | |
| CR005 | C071 | 2025/3/14 | Tables needed cleaning, but the food was fine. | |
| CR006 | C009 | 2025/3/15 | Fries were cold and soggy. Disappointing. | |
| CR007 | C055 | 2025/3/16 | Friendly staff at the counter. | |
| CR008 | C093 | 2025/3/18 | The veggie burger was surprisingly tasty! | |
| CR009 | C018 | 2025/3/20 | Quick service, got my order in 5 minutes. | |
| CR010 | C064 | 2025/3/21 | Good value combo meal. | |
| CR011 | C045 | 2025/3/22 | Tried the pizza this time, better than expected. | |
| CR012 | C030 | 2025/3/23 | Chicken nuggets were a bit dry. | |
| CR013 | C077 | 2025/3/24 | Clean restaurant and restrooms. | |
| CR014 | C101 | 2025/3/25 | Double cheeseburger hit the spot! | |
| CR015 | C011 | 2025/3/27 | Hot dog was pretty standard. | |
| CR016 | C082 | 2025/3/28 | Milkshake was thick and creamy, perfect. | |
| CR017 | C059 | 2025/3/29 | They forgot the sauce for my nuggets. | |
| CR018 | C022 | 2025/3/30 | Reasonable prices for the portion size. | |
| CR019 | C096 | 2025/3/31 | The fried chicken was crispy and flavorful. | |
| CR020 | C041 | 2025/4/1 | Long wait time, seemed understaffed. | |
| CR021 | C005 | 2025/4/2 | Apple pie was nice and hot. | |
| CR022 | C070 | 2025/4/3 | Just average, nothing special. | |
| CR023 | C038 | 2025/4/4 | The coleslaw was fresh and creamy. | |
| CR024 | C088 | 2025/4/5 | Staff wasn't very friendly this time. | |
| CR025 | C015 | 2025/4/6 | Great place for a quick bite. | |
| CR026 | C051 | 2025/4/7 | Onion rings were greasy. | |
| CR027 | C099 | 2025/4/8 | My order was accurate and quick. | |

*Table 8. Data Type of Customer Review Table*

| Field Name | Data Type |
|---|---|
| ReviewID | Short Text |
| CustomerID | Short Text |
| ReviewDate | Date/Time |
| ReviewContext | Short Text |
| Reply | Short Text |

(5) Customer Management Form

There is a **Customer Management** form for editing the Customer table. In this form, managers and waiters can add, delete, and save edit customer records as well as search and navigate through customer information. The Customer Management form shown in Figure 2 displays customer details including ID, name, contact information and customer type, with functional buttons for record operations.

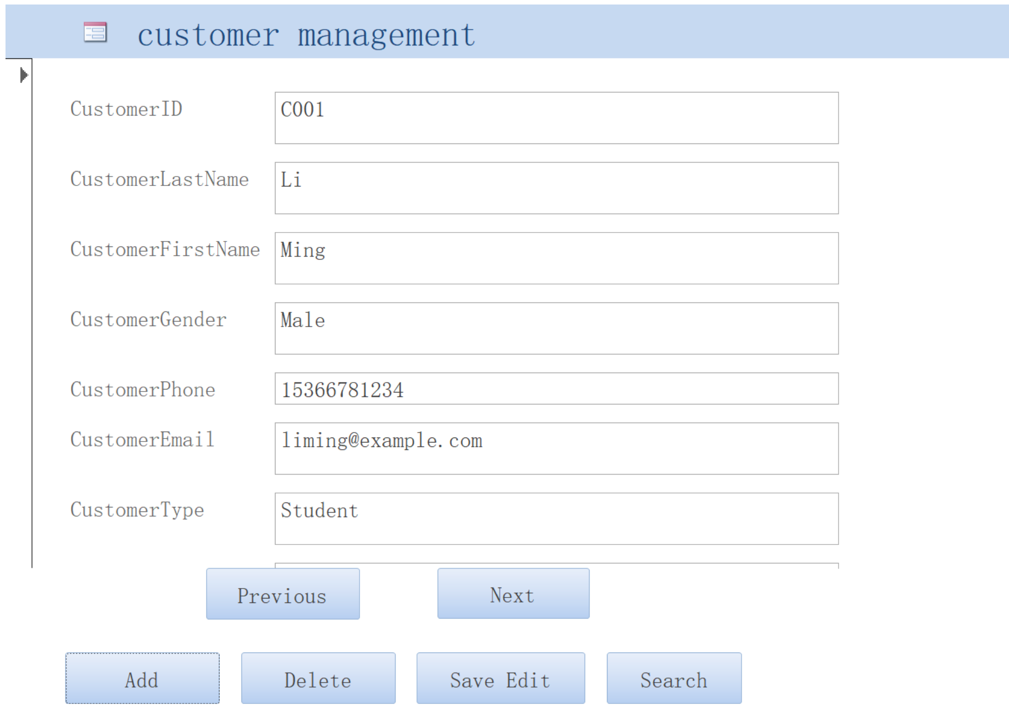



*Figure 2. Customer Management Form*

(6) Customer Review Form

There is a **Customer Review** form for editing the Customer Review table. In this form, managers can add, delete, and save edit review records as well as search and navigate through customer feedback. The Customer Review form shown in Figure 3 displays review details including ReviewID, CustomerID, Reviewdate, and review content, with functional buttons for record operations and a Reply option for staff responses.

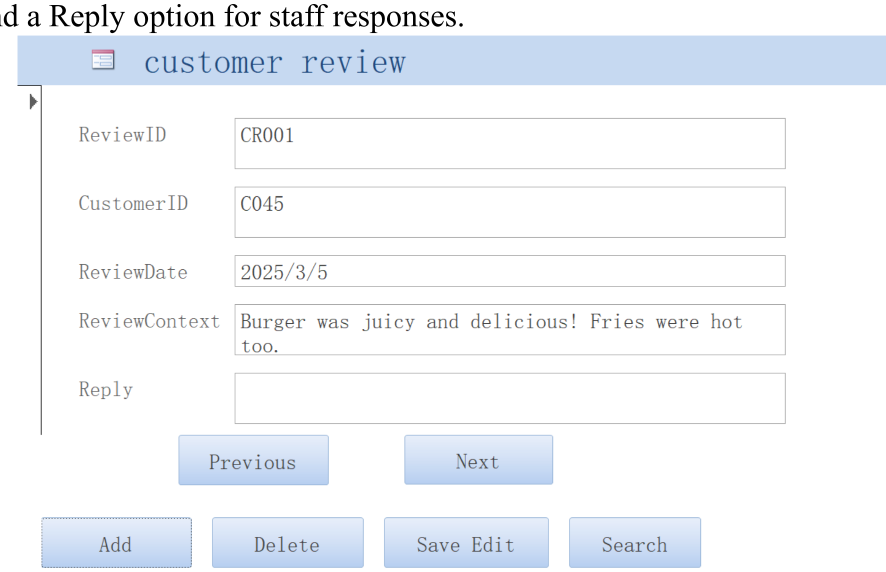



*Figure 3. Customer Review Form*

(7) Inventory Table

The **Inventory** Table stores comprehensive details about the inventory items in the restaurant. It is used to record updates to each item of inventory. This table can be accessed by chefs and managers. The Inventory table helps to construct the inventory report, which can provide the quantity and time of raw materials and packages update. It is connected with the **Item** Table, **Account Payable** Table, and **Recipe** Table.

Particularly, the **Inventory** table is connected with the **MenuItem** table through a bridge table (**Recipe** table). There are a total of 12 attributes in the Inventory Table. The data types of InventoryID, InventoryName, InventorySupplierID, InventoryType, InventoryPayableID, ItemID, Description and Unit are “Short Text”. InventoryAmount and InventoryQuantity are of the “Number” data type. OrderDate and DueDate have the “Date/Time” data type. Table 9 and Table 10 show the detailed information about the **Inventory Table.**

*Table 9. Inventory Table*

| InventoryID | InventoryName | InventorySupplierID | InventoryType | InventoryAmount | InventoryPayableID |
|---|---|---|---|---|---|
| INV001 | Burger Bun (Pack of 12) | SUP01 | Ingredient-Dry | 30 | PAY1001 |
| INV002 | Beef Patty (Frozen, 4oz) | SUP02 | Ingredient-Meat | 250 | PAY1002 |
| INV003 | heddar Cheese Slices (Stack | SUP03 | Ingredient-Dairy | 90 | PAY1003 |
| INV004 | Pickle Slices (Jar) | SUP01 | Ingredient-Produce | 45 | PAY1004 |
| INV005 | Onion (Bag, Yellow) | SUP03 | Ingredient-Produce | 25 | PAY1005 |
| INV006 | Ketchup (Bulk Dispenser Bag) | SUP01 | Ingredient-Sauce | 60 | PAY1006 |
| INV007 | Mustard (Bulk Bottle) | SUP01 | Ingredient-Sauce | 35 | PAY1007 |
| INV008 | rispy Chicken Fillet (Frozer | SUP02 | Ingredient-Meat | 320 | PAY1008 |
| INV009 | Lettuce (Iceberg, Bag) | SUP03 | Ingredient-Produce | 40 | PAY1009 |
| INV010 | Mayonnaise (Bulk Tub) | SUP01 | Ingredient-Sauce | 55 | PAY1010 |
| INV011 | Sesame Seed Bun (Pack of 12) | SUP01 | Ingredient-Dry | 35 | PAY1011 |
| INV012 | Frozen Fries (Bulk Bag) | SUP02 | Ingredient-Frozen | 150 | PAY1012 |
| INV013 | Salt (Fine Grain, Bag) | SUP01 | Ingredient-Dry | 15 | PAY1013 |
| INV014 | Frozen Onion Rings (Bag) | SUP02 | Ingredient-Frozen | 90 | PAY1014 |
| INV015 | Raw Chicken Pieces (Case) | SUP02 | Ingredient-Meat | 450 | PAY1015 |
| INV016 | Chicken Breading Mix (Bag) | SUP01 | Ingredient-Dry | 70 | PAY1016 |
| INV017 | Frozen Chicken Nuggets (Bag) | SUP02 | Ingredient-Frozen | 200 | PAY1017 |
| INV018 | Dipping Sauce (BBQ, Box) | SUP01 | Ingredient-Sauce | 80 | PAY1018 |
| INV019 | Raw Chicken Wings (Case) | SUP02 | Ingredient-Meat | 300 | PAY1019 |
| INV020 | Spicy Wing Sauce (Bottle) | SUP01 | Ingredient-Sauce | 75 | PAY1020 |
| INV021 | Pizza Dough Ball (Frozen) | SUP01 | Ingredient-Frozen | 120 | PAY1021 |
| INV022 | Pizza Tomato Sauce (Can) | SUP01 | Ingredient-Sauce | 50 | PAY1022 |
| INV023 | zarella Cheese (Shredded, B | SUP03 | Ingredient-Dairy | 180 | PAY1023 |
| INV024 | Pepperoni Slices (Pack) | SUP02 | Ingredient-Meat | 110 | PAY1024 |
| INV025 | Fresh Basil (Container) | SUP03 | Ingredient-Produce | 20 | PAY1025 |
| INV026 | Hot Dog Sausage (Pack) | SUP02 | Ingredient-Meat | 150 | PAY1026 |
| INV027 | Hot Dog Bun (Pack of 8) | SUP01 | Ingredient-Dry | 25 | PAY1027 |
| INV028 | Cola Syrup (Bag-in-Box) | SUP05 | gredient-BeverageBa | 90 | PAY1028 |

| InventoryID | OrderDate | DueDate | InventoryQuantity | ItemID | Description | Unit |
|---|---|---|---|---|---|---|
| INV001 | 2025/2/28 | 2025/3/30 | 100 | ITM001 | Standard size burger buns | Pack |
| INV002 | 2025/3/1 | 2025/3/31 | 500 | ITM002 | Frozen beef patties, 4oz each | Piece |
| INV003 | 2025/3/2 | 2025/4/1 | 1000 | ITM003 | Individually wrapped slices | Piece |
| INV004 | 2025/2/28 | 2025/3/30 | 5 | ITM004 | Dill pickle slices | Kg |
| INV005 | 2025/3/3 | 2025/4/2 | 10 | ITM005 | Fresh yellow onions | Kg |
| INV006 | 2025/3/1 | 2025/3/31 | 10 | ITM006 | Standard tomato ketchup | Litre |
| INV007 | 2025/3/1 | 2025/3/31 | 5 | ITM007 | Yellow mustard | Litre |
| INV008 | 2025/3/2 | 2025/4/1 | 400 | ITM008 | Breaded chicken fillets | Piece |
| INV009 | 2025/3/3 | 2025/4/2 | 8 | ITM009 | Pre-shredded iceberg lettuce | Kg |
| INV010 | 2025/3/1 | 2025/3/31 | 8 | ITM010 | Standard mayonnaise | Litre |
| INV011 | 2025/2/28 | 2025/3/30 | 80 | ITM011 | Buns with sesame seeds | Pack |
| INV012 | 2025/3/1 | 2025/3/31 | 20 | ITM012 | Shoestring cut frozen potatoes | Kg |
| INV013 | 2025/2/25 | 2025/3/27 | 5 | ITM013 | Table salt | Kg |
| INV014 | 2025/3/1 | 2025/3/31 | 10 | ITM014 | Battered onion rings | Kg |
| INV015 | 2025/3/3 | 2025/4/2 | 300 | ITM015 | :ed raw chicken pieces for fry: | Piece |
| INV016 | 2025/2/28 | 2025/3/30 | 15 | ITM016 | Seasoned flour mix for chicken | Kg |
| INV017 | 2025/3/1 | 2025/3/31 | 2000 | ITM017 | Bulk bag of frozen nuggets | Piece |
| INV018 | 2025/2/28 | 2025/3/30 | 500 | ITM018 | Individual BBQ sauce packets | Packet |
| INV019 | 2025/3/3 | 2025/4/2 | 600 | ITM019 | Raw chicken wing sections | Piece |
| INV020 | 2025/3/1 | 2025/3/31 | 6 | ITM020 | Buffalo style hot sauce | Litre |
| INV021 | 2025/3/2 | 2025/4/1 | 150 | ITM021 | ersonal size frozen dough ball | Piece |
| INV022 | 2025/3/1 | 2025/3/31 | 5 | ITM022 | Bulk canned pizza sauce | Litre |
| INV023 | 2025/3/2 | 2025/4/1 | 10 | ITM023 | hredded low-moisture mozzarell | Kg |
| INV024 | 2025/3/1 | 2025/3/31 | 800 | ITM024 | Pre-sliced pepperoni | Piece |
| INV025 | 2025/3/4 | 2025/4/3 | 0.5 | ITM025 | Fresh basil leaves | Kg |
| INV026 | 2025/3/2 | 2025/4/1 | 200 | ITM026 | Standard beef/pork hot dogs | Piece |
| INV027 | 2025/2/28 | 2025/3/30 | 50 | ITM027 | Standard hot dog buns | Pack |
| INV028 | 2025/3/1 | 2025/3/31 | 5 | ITM028 | Concentrate for soda fountain | Litre |

*Table 10. Data Type of Inventory Table*

| Field Name | Data Type |
|---|---|
| InventoryID | Short Text |
| InventoryName | Short Text |
| InventorySupplierID | Short Text |
| InventoryType | Short Text |
| InventoryAmount | Number |
| InventoryPayableID | Short Text |
| OrderDate | Date/Time |
| DueDate | Date/Time |
| InventoryQuantity | Number |
| ItemID | Short Text |
| Description | Short Text |
| Unit | Short Text |

(8) Inventory Management Form

There is an **Inventory Management** form for editing the Inventory table. In this form, managers can add, delete, save edit and search inventory records, as well as navigate through them using Previous/Next buttons. The Inventory Management form shown in Figure 3 displays inventory details including InventoryID, InventoryName, InventorySupplierID, InventoryType, InventoryAmount and InventoryPayableID.

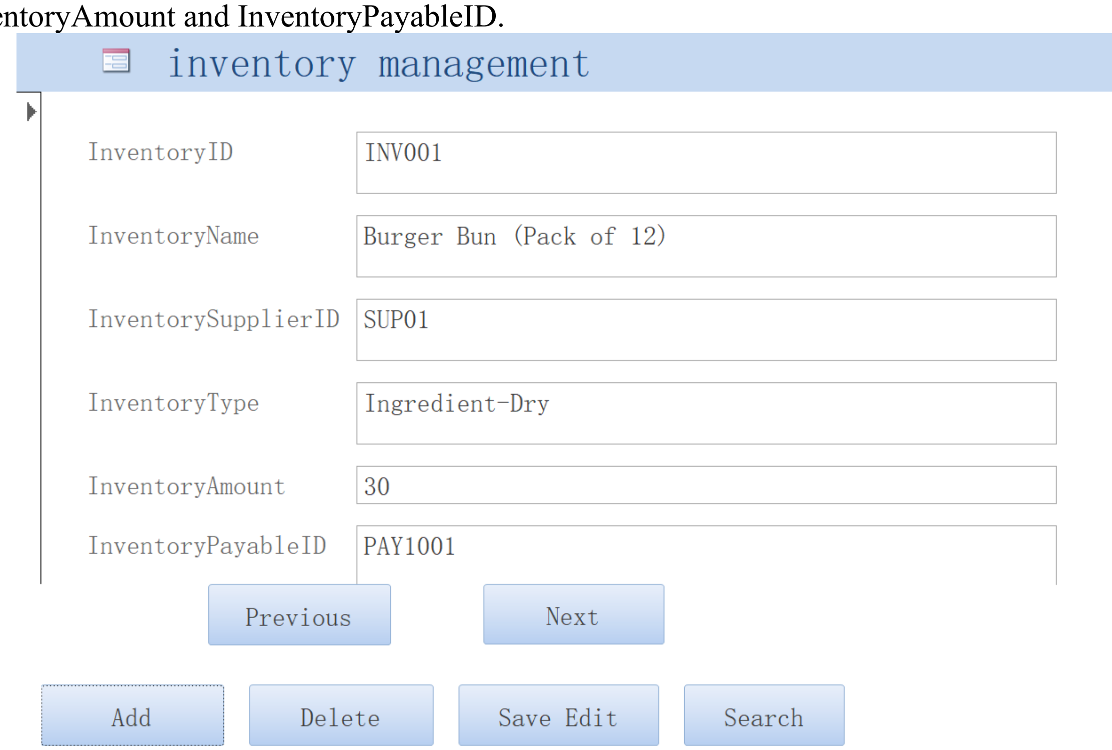



*Figure 4. Inventory Management Form*

(9) Item table

The **Item** table is used to record the whole items the restaurant need, from raw materials to packages. The table can be read by all customers, waiters, chefs and managers. The **Item** table is connected with the Inventory Table and Supplier Table. There are 8 attributes in the Item Table.

The data types of ItemID, ItemName, ItemType, SupplierID, State, and Note are “Short Text”. ItemUnitPrice and MinimumQuantity have the data type “Number”. Table 11 and Table 12 show the detailed information about the **Item Table.**

*Table 11. Item Table*

| ItemID | ItemName | ItemType | ItemUnitPrice | MinimumQuantity | SupplierID | State | Note |
|---|---|---|---|---|---|---|---|
| ITM001 | Burger Bun (Standard) | Ingredient-Dry | 0.3 | 20 | SUP01 | Active | Standard 4-inch buns, purchased in pac |
| ITM002 | Beef Patty (Frozen, 4oz) | Ingredient-Meat | 0.5 | 100 | SUP02 | Active | 80/20 blend frozen patties |
| ITM003 | Cheddar Cheese Slice (Ind | Ingredient-Dairy | 0.09 | 200 | SUP03 | Active | Standard American cheddar slices |
| ITM004 | Pickle Slices (Dill) | Ingredient-Produce | 9 | 2 | SUP01 | Active | Purchased in bulk jars/tubs (Price per |
| ITM005 | Onion (Yellow) | Ingredient-Produce | 2.5 | 5 | SUP03 | Active | Purchased in bulk bags (Price per Kg) |
| ITM006 | Ketchup | Ingredient-Sauce | 6 | 2 | SUP01 | Active | Standard tomato ketchup, bulk dispense |
| ITM007 | Mustard (Yellow) | Ingredient-Sauce | 7 | 1 | SUP01 | Active | Standard yellow mustard, bulk bottle ( |
| ITM008 | Crispy Chicken Fillet (Br | Ingredient-Meat | 0.8 | 80 | SUP02 | Active | Pre-breaded chicken fillets for frying |
| ITM009 | Lettuce (Iceberg, Shredde | Ingredient-Produce | 5 | 3 | SUP03 | Active | Purchased pre-shredded in bags (Price |
| ITM010 | Mayonnaise | Ingredient-Sauce | 6.88 | 1 | SUP01 | Active | Standard mayonnaise, bulk tub (Price p |
| ITM011 | Sesame Seed Bun | Ingredient-Dry | 0.44 | 15 | SUP01 | Active | Buns with sesame seeds, purchased in p |
| ITM012 | French Fries (Frozen, Sho | Ingredient-Frozen | 7.5 | 5 | SUP02 | Active | Standard shoestring cut frozen fries ( |
| ITM013 | Salt (Fine Grain) | Ingredient-Dry | 3 | 1 | SUP01 | Active | Standard table salt (Price per Kg) |
| ITM014 | Onion Rings (Battered, Fr | Ingredient-Frozen | 9 | 3 | SUP02 | Active | Pre-battered frozen onion rings (Price |
| ITM015 | Raw Chicken Piece (Mixed, | Ingredient-Meat | 1.5 | 50 | SUP02 | Active | Assorted raw chicken pieces (Price per |
| ITM016 | Chicken Breading Mix | Ingredient-Dry | 4.67 | 3 | SUP01 | Active | Seasoned flour mix for breading chicke |
| ITM017 | Chicken Nugget (Frozen) | Ingredient-Frozen | 0.1 | 300 | SUP02 | Active | Standard frozen chicken nuggets (Price |
| ITM018 | Dipping Sauce (BBQ Packet | Ingredient-Sauce | 0.16 | 100 | SUP01 | Active | Individual BBQ sauce packets |
| ITM018R | Dipping Sauce (Ranch Pack | Ingredient-Sauce | 0.17 | 100 | SUP01 | Active | Individual Ranch sauce packets (Matche |
| ITM019 | Raw Chicken Wing (Section | Ingredient-Meat | 0.5 | 100 | SUP02 | Active | Raw chicken wing sections (Price per F |
| ITM020 | Spicy Wing Sauce (Buffalo | Ingredient-Sauce | 12.5 | 1 | SUP01 | Active | Bulk buffalo style hot sauce (Price pe |
| ITM021 | Pizza Dough Ball (Persona | Ingredient-Frozen | 0.8 | 30 | SUP01 | Active | Frozen dough balls for personal pizzas |
| ITM022 | Pizza Tomato Sauce | Ingredient-Sauce | 10 | 1 | SUP01 | Active | Bulk canned pizza sauce (Price per Lit |
| ITM023 | Mozzarella Cheese (Shredd | Ingredient-Dairy | 18 | 3 | SUP03 | Active | Low-moisture shredded mozzarella (Pric |
| ITM024 | Pepperoni Slices | Ingredient-Meat | 0.14 | 150 | SUP02 | Active | Standard pre-sliced pepperoni (Price p |
| ITM025 | Fresh Basil | Ingredient-Produce | 40 | 0.1 | SUP03 | Active | Fresh basil leaves (Price per Kg) |
| ITM026 | Hot Dog Sausage | Ingredient-Meat | 0.75 | 40 | SUP02 | Active | Standard beef/pork hot dogs |
| ITM027 | Hot Dog Bun | Ingredient-Dry | 0.5 | 10 | SUP01 | Active | Standard hot dog buns, purchased in pa |

*Table 12. Data Type of Item Table*

| Field Name | Data Type |
|---|---|
| ItemID | Short Text |
| ItemName | Short Text |
| ItemType | Short Text |
| ItemUnitPrice | Number |
| MinimumQuantity | Number |
| SupplierID | Short Text |
| State | Short Text |
| Note | Short Text |

(10) Item Management Form

There is an **Item Management** form for editing the Item table. In this form, managers can add, delete, save edit and search item records, as well as navigate through them using Previous/Next buttons. The Item Management form shown in Figure 5 displays item details including ItemID, ItemName, ItemType, ItemUnitPrice, MinimumQuantity, SupplierID and State.

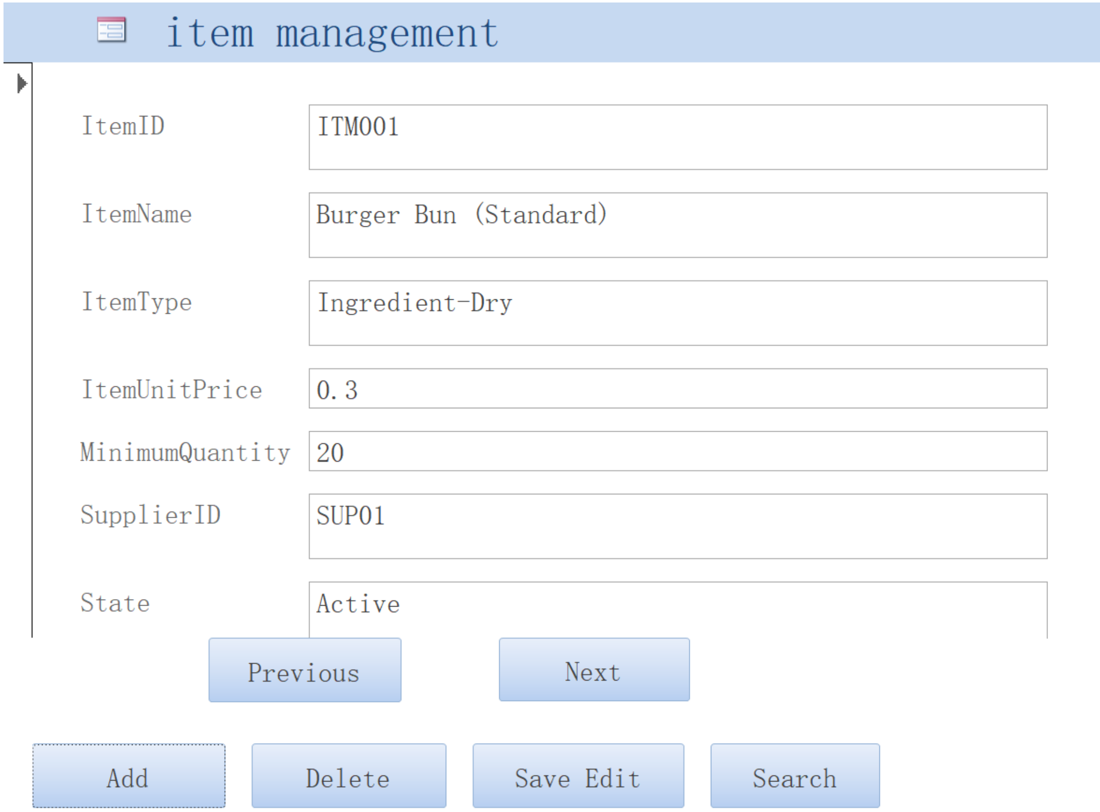



*Figure 5. Item Management Form*

(11) MenuItem Table

The **MenuItem** Table contains all dishes and drinks served by the restaurant, including their descriptions and pricing. This table is visible to customers for browsing but can only be modified by managers and kitchen staff. The MenuItem Table connects to both the **Recipe** Table and **Order** Table. There are a total of 6 attributes in the MenuItem Table. The data types of MenuItemID, MenuItemName, State, Description and Category are all "Short Text", while data type of Price is "Nubmer" . Tables 29 and 30 show the detailed information about the **MenuItem** table.

*Table 29 MenuItem Table*

| MenuItemID | MenuItemNa | Price | Descriptio | State | Category | Click to Add |
|---|---|---|---|---|---|---|
| M01 | Classic Chee: | 5.5 | Single beef | Available | Burger | |
| M02 | Double Chees | 7.5 | Two beef pat | Available | Burger | |
| M03 | Crispy Chick | 6.8 | Fried crispy | Available | Burger | |
| M04 | French Fries | 2.5 | Classic gold | Available | Side | |
| M05 | French Fries | 3.5 | Larger porti | Available | Side | |
| M06 | Onion Rings | 3.8 | Battered and | Available | Side | |
| M07 | Fried Chicke | 6 | Two pieces o | Available | Fried Chicke | |
| M08 | Chicken Nugg | 4.5 | Six tender a | Available | Fried Chicke | |
| M09 | Spicy Chicke | 7 | Five crispy | Available | Fried Chicke | |
| M10 | Pepperoni Pi: | 8.5 | Personal-siz | Available | Pizza | |
| M11 | Margherita P | 8 | Personal-siz | Available | Pizza | |
| M12 | Classic Hot | 4 | Grilled saus | Available | Hot Dog | |
| M13 | Cola (Regula | 2.2 | Standard col | Available | Drink | |
| M14 | Diet Cola (R | 2.2 | Diet version | Available | Drink | |
| M15 | Lemonade | 2.5 | Refreshing s | Available | Drink | |
| M16 | Chocolate Mi | 3.9 | Creamy choco | Available | Drink | |
| M17 | Fried Apple | 1.8 | Crispy fried | Available | Dessert | |
| M18 | Soft Serve C | 1.5 | Classic vani | Available | Dessert | |
| M19 | Coleslaw | 2 | Creamy shred | Available | Side | |
| M20 | Veggie Burge | 6.5 | Plant-based | Available | Burger | |

*Table 30 Data Type of MenuItem Table*

| Field Name | Data Type |
|---|---|
| MenuItemID | Short Text |
| MenuItemName | Short Text |
| Price | Number |
| Description | Short Text |
| State | Short Text |
| Category | Short Text |

(12) MenuItem Management Form

There is a **MenuItem Management form** for editing the MenuItem table. In this form, managers can add, delete, save edit and search menu item records, as well as navigate through them using Previous/Next buttons. The MenuItem Management form displays details including MenuItemID, MenuItemName, Price, Description, State and Category.

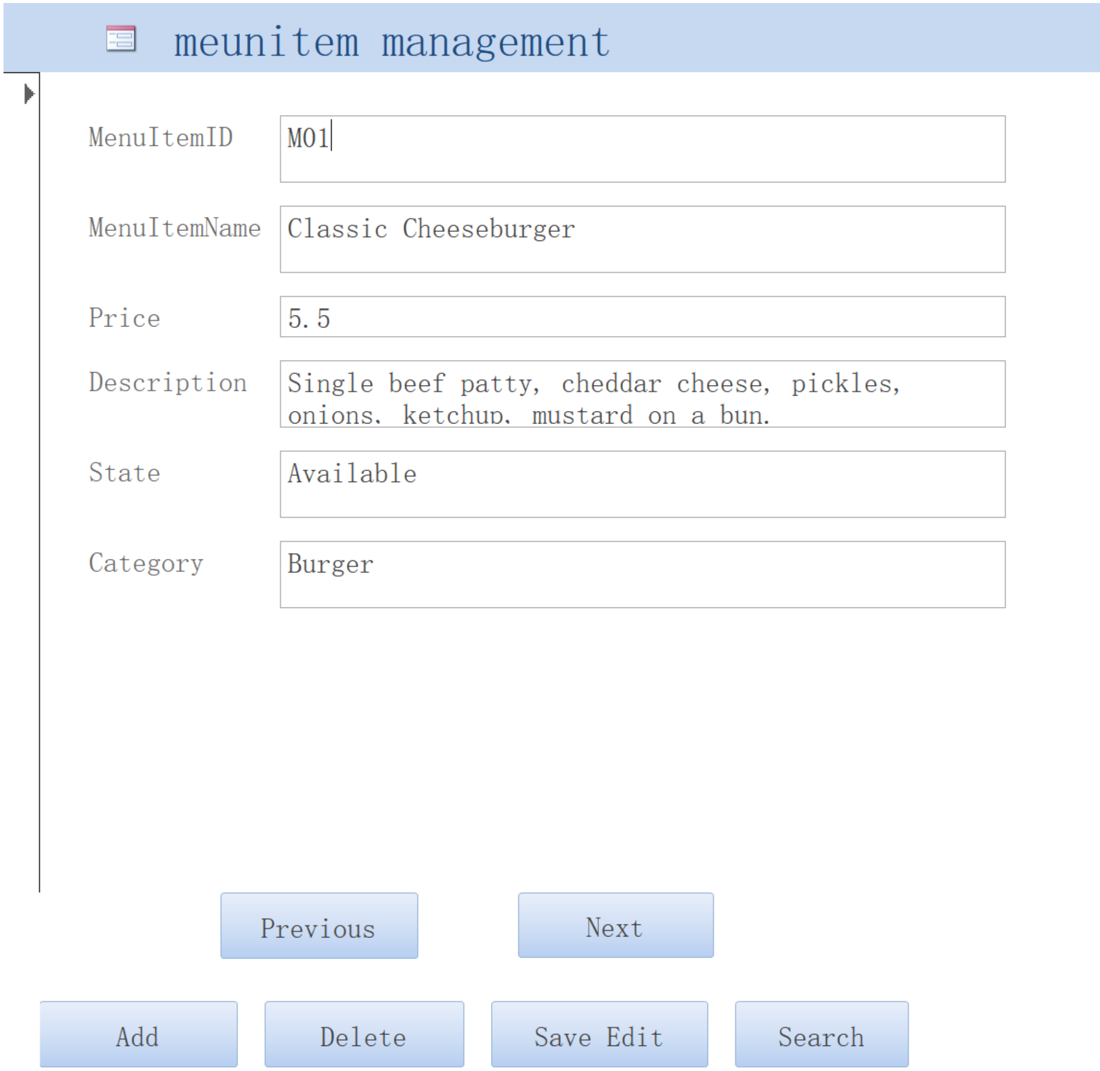



*Figure 6. MenuItem Management Form*

(13) Order Table

The **Order** Table records all customer transactions and payment information in the restaurant. This table is created by staff members but can be viewed by managers and customers (for their own orders only). The Order Table is connected with the Customer Table, MenuItem Table, Staff Table and StaffID Table. There are a total of 9 attributes in the Order Table. The data types of CustomerID, MenuItemID, ReceivableID, StaffID and Note are all "Short Text." The data type of OrderDate is "Date/Time," while OrderID, OrderPayment and OrderDiscount use "Number." Tables 31 and 32 show the detailed information about the Order table.

*Table 31 Order Table*

| OrderID | OrderDate | CustomerID | OrderPayme | OrderDisco | MenuItemID | Receivable | StaffID | Note | Click to Add |
|---|---|---|---|---|---|---|---|---|---|
| 1001 | 2025/3/2 | C035 | 15.75 | 0 | M05 | RCV1001 | S02 | | |
| 1002 | 2025/3/3 | C091 | 28.5 | 2.5 | M12 | RCV1002 | S05 | Takeaway | |
| 1003 | 2025/3/3 | C005 | 8.99 | 0 | M01 | RCV1003 | S02 | | |
| 1004 | 2025/3/5 | C077 | 22 | 0 | M15 | RCV1004 | S01 | | |
| 1005 | 2025/3/7 | C035 | 11.25 | 1 | M08 | RCV1005 | S03 | Allergic to | |
| 1006 | 2025/3/10 | C101 | 19.8 | 0 | M03 | RCV1006 | S04 | | |
| 1007 | 2025/3/11 | C018 | 6.5 | 0 | M19 | RCV1007 | S01 | | |
| 1008 | 2025/3/14 | C052 | 29.95 | 5 | M11 | RCV1008 | S05 | Birthday dis | |
| 1009 | 2025/3/16 | C088 | 14 | 0 | M07 | RCV1009 | S03 | | |
| 1010 | 2025/3/18 | C023 | 21.1 | 0 | M02 | RCV1010 | S02 | | |
| 1011 | 2025/3/20 | C099 | 9.75 | 0 | M05 | RCV1011 | S04 | | |
| 1012 | 2025/3/22 | C041 | 17.3 | 1.5 | M10 | RCV1012 | S01 | | |
| 1013 | 2025/3/25 | C064 | 25 | 0 | M16 | RCV1013 | S03 | Extra spicy | |
| 1014 | 2025/3/28 | C009 | 7.2 | 0 | M20 | RCV1014 | S05 | | |
| 1015 | 2025/3/30 | C035 | 18.6 | 0 | M12 | RCV1015 | S02 | | |
| 1016 | 2025/4/1 | C072 | 26.4 | 2 | M14 | RCV1016 | S01 | | |
| 1017 | 2025/4/3 | C050 | 5.5 | 0 | M01 | RCV1017 | S04 | | |
| 1018 | 2025/4/5 | C081 | 12.9 | 0 | M06 | RCV1018 | S03 | | |
| 1019 | 2025/4/8 | C015 | 23.75 | 0 | M13 | RCV1019 | S05 | No onions | |
| 1020 | 2025/4/10 | C095 | 10 | 0 | M09 | RCV1020 | S02 | | |
| 1021 | 2025/3/1 | C045 | 22.15 | 0 | M11 | RCV1021 | S03 | | |
| 1022 | 2025/3/28 | C088 | 14.5 | 1 | M05 | RCV1022 | S01 | | |
| 1023 | 2025/4/5 | C012 | 7.99 | 0 | M01 | RCV1023 | S04 | Takeaway | |
| 1024 | 2025/3/15 | C077 | 28.3 | 2.5 | M18 | RCV1024 | S02 | | |
| 1025 | 2025/4/9 | C031 | 11.8 | 0 | M07 | RCV1025 | S05 | | |

*Table 32 Data Type of Order Table*

| Field Name | Data Type |
|---|---|
| OrderID | Number |
| OrderDate | Date/Time |
| CustomerID | Short Text |
| OrderPayment | Number |
| OrderDiscount | Number |
| MenuItemID | Short Text |
| ReceivableID | Short Text |
| StaffID | Short Text |
| Note | Short Text |

(14) Order Management Form

There is an **Order Management** form for editing the Order table. In this form, staff can add, delete, save edit and search order records, as well as navigate through them using Previous/Next buttons. The Order Management form displays details including OrderID, OrderDate, CustomerID, OrderPayment, OrderDiscount, MenuItemID, ReceivableID, StaffID and Note.

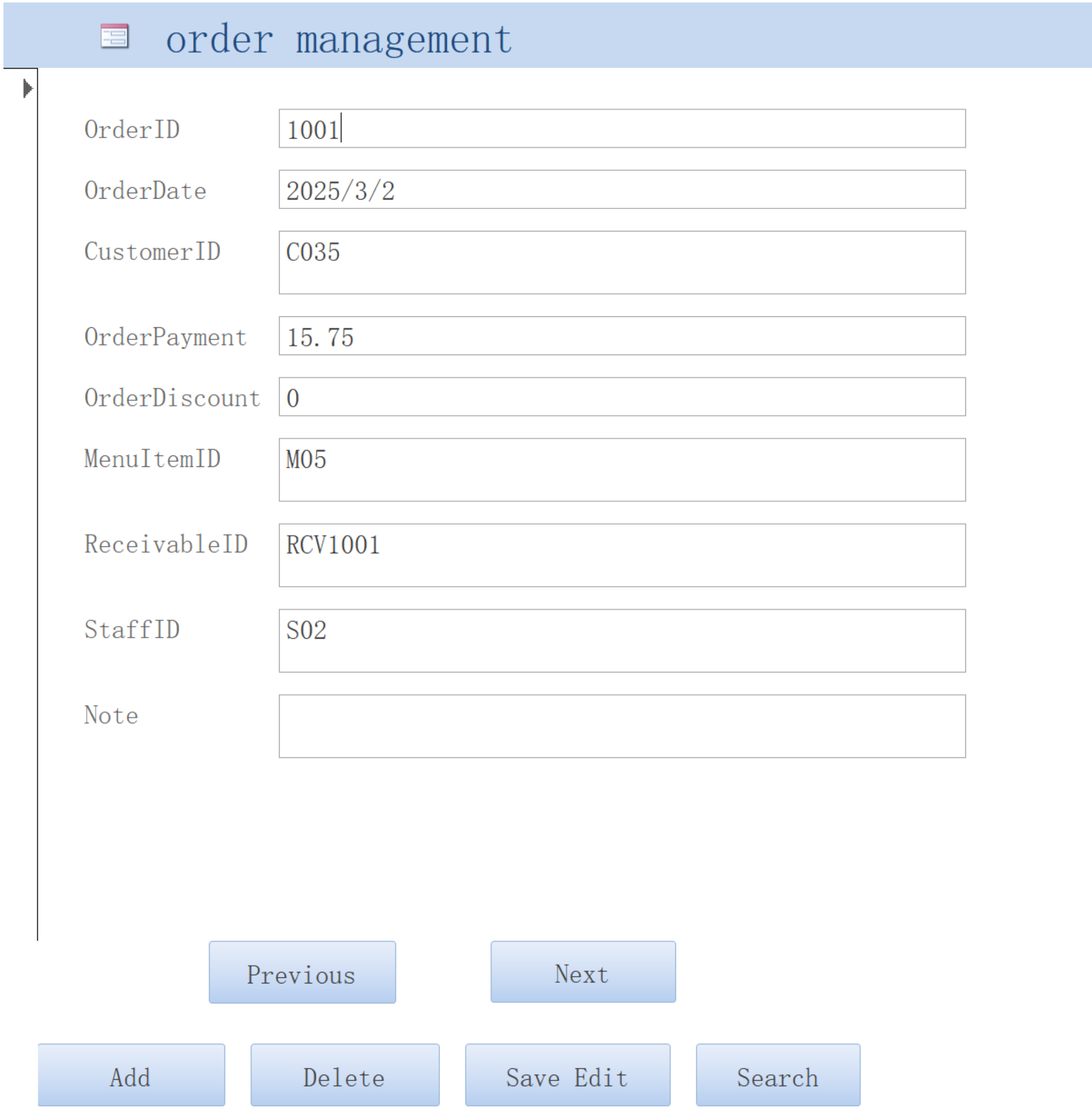



*Figure 7. Order Management Form*

(15) Recipe Table

The **Recipe Table** specifies the ingredients and quantities required to prepare each menu item. This table is primarily used by kitchen staff and can be modified only by managers. The Recipe Table serves as a bridge between the MenuItem Table and Inventory Table. There are a total of 5 attributes in the Recipe Table. The data types of RecipeID, MenuItemID, InventoryID and Unit are all "Short Text," while QuantityUsed is "Number". Tables 33 and 34 shows the detailed information about the Recipe table.

*Table 33 Recipe Table*

| RecipeID | MenuItemID | InventoryI | QuantityUs | Unit | Click to Add |
|---|---|---|---|---|---|
| R001 | M01 | INV001 | 1 | Piece | |
| R002 | M01 | INV002 | 1 | Piece | |
| R003 | M01 | INV003 | 1 | Piece | |
| R004 | M01 | INV004 | 5 | Grams | |
| R005 | M01 | INV005 | 5 | Grams | |
| R006 | M01 | INV006 | 10 | ml | |
| R007 | M01 | INV007 | 5 | ml | |
| R008 | M02 | INV001 | 1 | Piece | |
| R009 | M02 | INV002 | 2 | Piece | |
| R010 | M02 | INV003 | 2 | Piece | |
| R011 | M02 | INV004 | 5 | Grams | |
| R012 | M02 | INV005 | 5 | Grams | |
| R013 | M02 | INV006 | 10 | ml | |
| R014 | M02 | INV007 | 5 | ml | |
| R015 | M03 | INV011 | 1 | Piece | |
| R016 | M03 | INV008 | 1 | Piece | |
| R017 | M03 | INV009 | 15 | Grams | |
| R018 | M03 | INV010 | 15 | ml | |
| R019 | M04 | INV012 | 100 | Grams | |
| R020 | M04 | INV013 | 1 | Grams | |
| R021 | M05 | INV012 | 150 | Grams | |
| R022 | M05 | INV013 | 1.5 | Grams | |
| R023 | M06 | INV014 | 80 | Grams | |
| R024 | M06 | INV013 | 1 | Grams | |
| R025 | M07 | INV015 | 2 | Piece | |

*Table 34 Data Type of Recipe Table*

| Field Name | Data Type |
|---|---|
| RecipeID | Short Text |
| MenuItemID | Short Text |
| InventoryID | Short Text |
| QuantityUsed | Number |
| Unit | Short Text |

(16) Recipe Management Form

There is a **Recipe Management form** for editing the Recipe table. In this form, kitchen managers can add, delete, save edit and search recipe records, as well as navigate through them using Previous/Next buttons. The Recipe Management form displays details including RecipeID, MenuItemID, InventoryID, QuantityUsed and Unit.

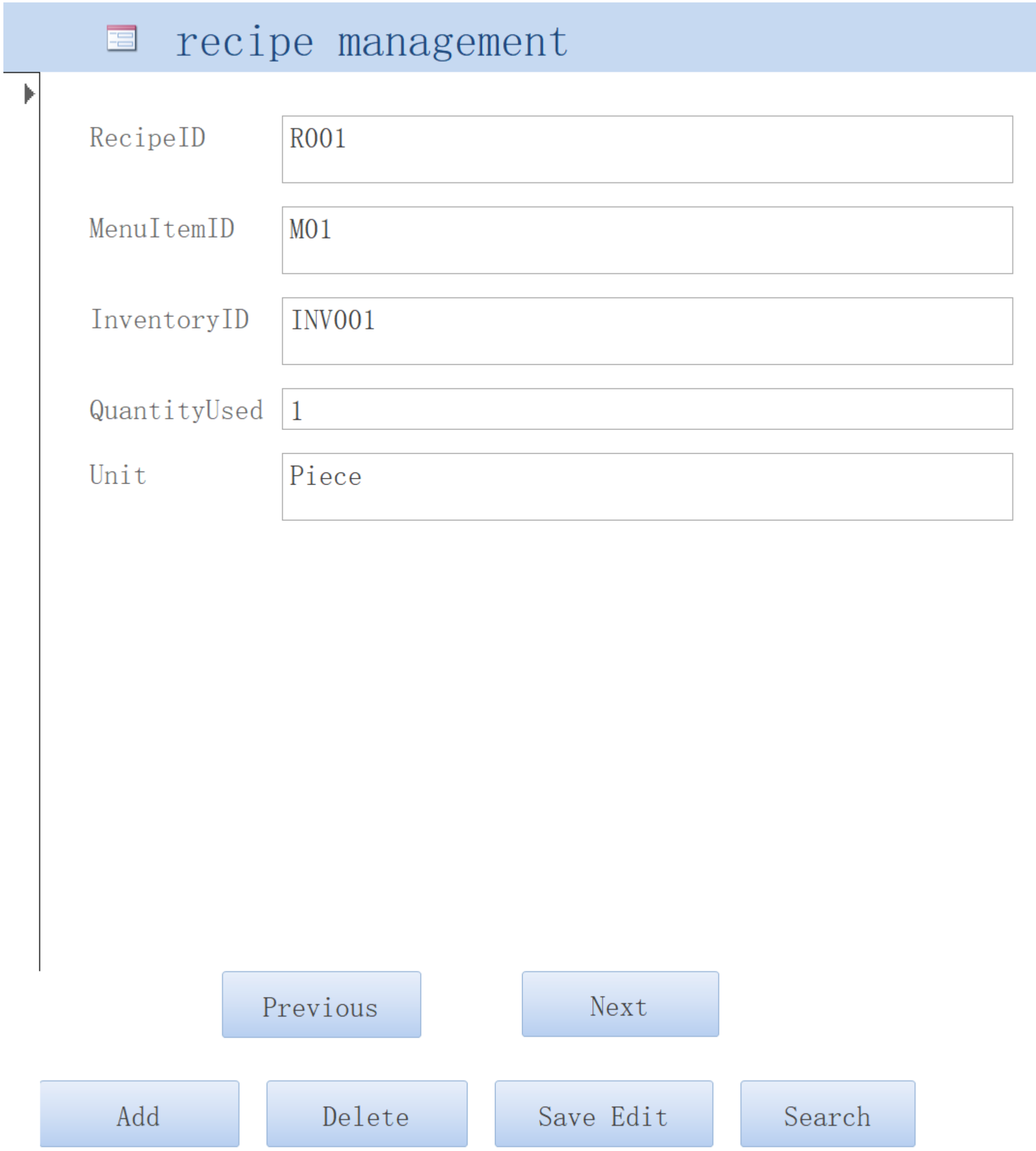



*Figure 8. Recipe Management Form*

(17) Staff Table

The **Staff Table** maintains all employee records including their contact information and positions. This table is managed by HR and senior managers, with limited view access for other staff. The Staff Table connects to both the Order Table and Staff Working Table. There are a total of 7 attributes in the Staff Table. The data types of StaffID, StaffLastName, StaffFirstName, StaffEmail, StaffPosition and StaffAddress are all "Short Text" , while StaffPhone uses "Number". Tables 35 and 36 show the detailed information about the Staff table.

*Table 35 Staff Table*

| StaffID | StaffLastN | StaffFirst | StaffPhone | StaffEmail | StaffPosit | StaffAddre | Click to Add |
|---|---|---|---|---|---|---|---|
| S01 | Chen | Wei | 13812345678 | wei.chen@res | Waiter | 123 Guangmin; | |
| S02 | Li | Mei | 13987654321 | mei.li@resta | Cashier | 45 Renmin Av | |
| S03 | Zhang | Hao | 13711223344 | hao.zhang@re | Chef | 78 Jiefang S | |
| S04 | Wang | Fang | 13655447788 | fang.wang@re | Waiter | 90 Zhongshan | |
| S05 | Liu | Jian | 13599881122 | jian.liu@res | Manager | 56 Heping Ln. | |

*Table 36 Data Type of Staff Table*

| Field Name | Data Type |
|---|---|
| StaffID | Short Text |
| StaffLastName | Short Text |
| StaffFirstName | Short Text |
| StaffPhone | Number |
| StaffEmail | Short Text |
| StaffPosition | Short Text |
| StaffAddress | Short Text |

(18) Staff Management Form

There is a **Staff Management form** for editing the Staff table. In this form, HR managers can add, delete, save edit and search staff records, as well as navigate through them using Previous/Next buttons. The Staff Management form displays details including StaffID, StaffLastName, StaffFirstName, StaffPhone, StaffEmail, StaffPosition and StaffAddress.

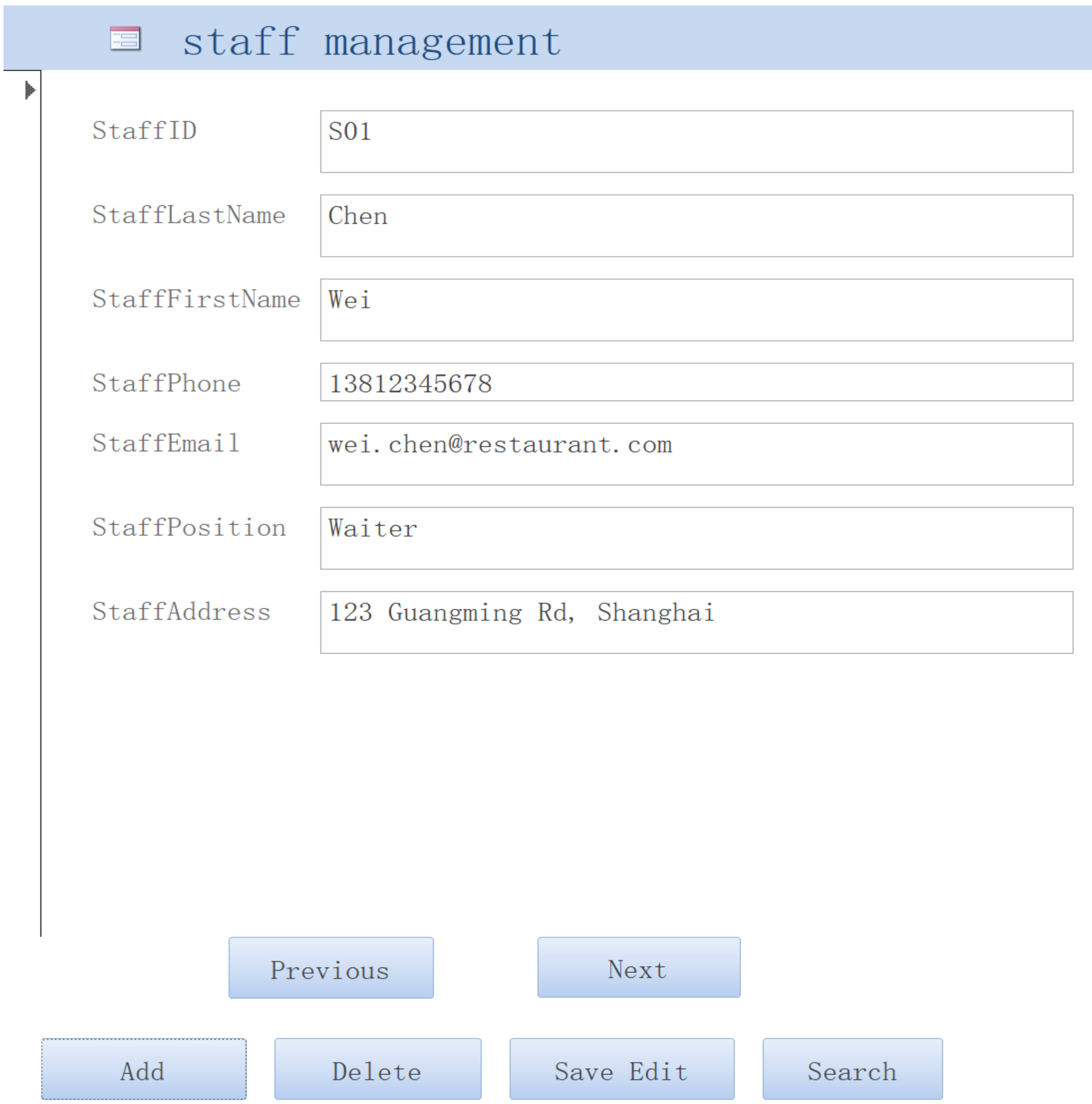



*Figure 9. Staff Management Form*

(19) Staff Working Table

The **Staff Working Table** tracks employee working hours and wages for payroll calculation. This table is exclusively managed by HR and accounting staff. The Staff Working Table connects to both the Staff Table and Payables Table. There are a total of 7 attributes in the Staff Working Table. The data types of StaffWorkingID, StaffID, PayableID and Note are all "Short Text." The data type of WorkDate is "Date/Time," while StaffWage and StaffWorkHours use "Number" . Tables 37 and 38 show the detailed information about the Staff Working table.

*Table 37 Staff Working Table*

| StaffWorki | StaffID | StaffWage | WorkDate | StaffWorkH | PayableID | Note | Click to Add |
|---|---|---|---|---|---|---|---|
| SW001 | S01 | 120 | 2025/3/1 | 8 | PAY3001 | Weekend Shif | |
| SW002 | S02 | 128 | 2025/3/1 | 8 | PAY3002 | | |
| SW003 | S03 | 200 | 2025/3/1 | 8 | PAY3003 | | |
| SW004 | S05 | 240 | 2025/3/1 | 8 | PAY3004 | Weekend Duty | |
| SW005 | S01 | 105 | 2025/3/2 | 7 | PAY3005 | | |
| SW006 | S04 | 120 | 2025/3/2 | 8 | PAY3006 | Weekend Shif | |
| SW007 | S03 | 175 | 2025/3/2 | 7 | PAY3007 | | |
| SW008 | S01 | 127.5 | 2025/3/3 | 8.5 | PAY3008 | | |
| SW009 | S02 | 120 | 2025/3/3 | 7.5 | PAY3009 | | |
| SW010 | S04 | 112.5 | 2025/3/4 | 7.5 | PAY3010 | | |
| SW011 | S03 | 212.5 | 2025/3/4 | 8.5 | PAY3011 | | |
| SW012 | S05 | 255 | 2025/3/5 | 8.5 | PAY3012 | Meeting Prep | |
| SW013 | S01 | 120 | 2025/3/5 | 8 | PAY3013 | | |
| SW014 | S02 | 128 | 2025/3/6 | 8 | PAY3014 | | |
| SW015 | S04 | 120 | 2025/3/7 | 8 | PAY3015 | | |
| SW016 | S03 | 187.5 | 2025/3/8 | 7.5 | PAY3016 | Weekend Shif | |
| SW017 | S01 | 135 | 2025/3/8 | 9 | PAY3017 | Weekend Shif | |
| SW018 | S05 | 225 | 2025/3/9 | 7.5 | PAY3018 | Weekend Duty | |
| SW019 | S02 | 136 | 2025/3/10 | 8.5 | PAY3019 | | |
| SW020 | S04 | 127.5 | 2025/3/10 | 8.5 | PAY3020 | | |
| SW021 | S01 | 112.5 | 2025/3/11 | 7.5 | PAY3021 | | |
| SW022 | S03 | 200 | 2025/3/12 | 8 | PAY3022 | | |
| SW023 | S05 | 270 | 2025/3/12 | 9 | PAY3023 | Inventory Ch | |
| SW024 | S02 | 128 | 2025/3/13 | 8 | PAY3024 | | |
| SW025 | S04 | 112.5 | 2025/3/14 | 7.5 | PAY3025 | | |

*Table 38 Data Type of Staff Working Table*

| Field Name | Data Type |
|---|---|
| StaffWorkingID | Short Text |
| StaffID | Short Text |
| StaffWage | Number |
| WorkDate | Date/Time |
| StaffWorkHours | Number |
| PayableID | Short Text |
| Note | Short Text |

(20) Staff Working Management Form

There is a **Staff Working Management Form** for editing the Staff Working table. In this form, accounting staff can add, delete, save edit and search working records, as well as navigate through them using Previous/Next buttons. The Staff Working Management form displays details including StaffWorkingID, StaffID, StaffWage, WorkDate, StaffWorkHours, PayableID and Note.

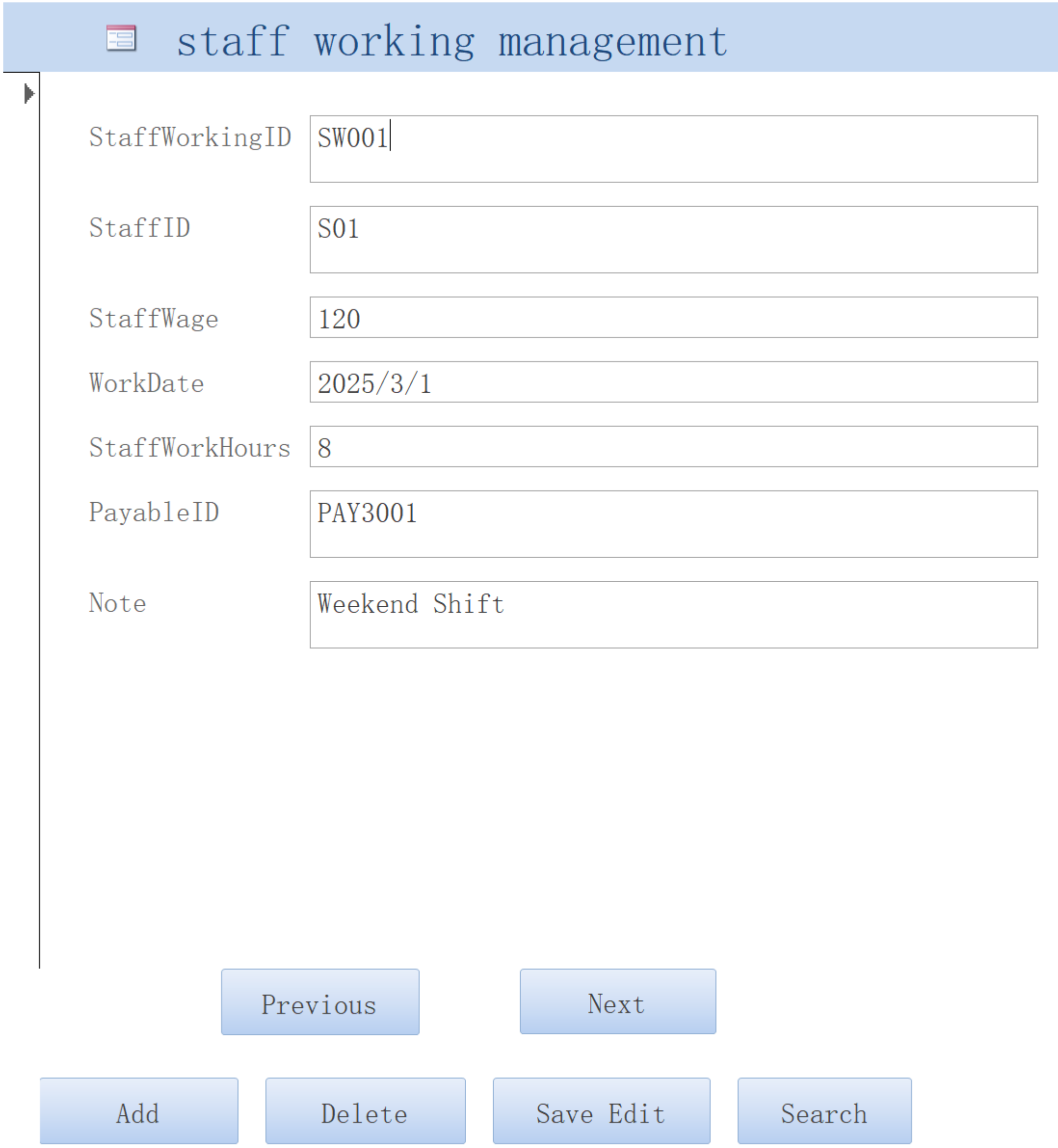



*Figure 10. Staff Working Management Form*

(21) Supplier Table

The **Supplier Table** contains information about all vendors providing ingredients and materials to the restaurant. This table is maintained by the purchasing department and accessible to managers. The Supplier Table connects to the Item Table. There are a total of 9 attributes in the Supplier Table. The data types of SupplierID, SupplierName, SupplierType, SupplierCity, SupplierProvince, ContactEmail, ContactLastName, ContactNote and ContactFirstName are all "Short Text". Tables 39 and 40 show the detailed information about the Supplier table.

*Table 39 Supplier Table*

| SupplierID | SupplierNa | SupplierTy | SupplierCi | SupplierPr | ContactEma | ContactNot | ContactLas | ContactFir | Click to Add |
|---|---|---|---|---|---|---|---|---|---|
| SUP01 | Global Foods | Food Distrib | Shanghai | Shanghai | sales@global | Primary supp | Wang | Li | |
| SUP02 | Meat Packers | Meat & Froze | Nanjing | Jiangsu | orders@meatp | Main supplie | Zhao | David | |
| SUP03 | Fresh Dairy ( | Dairy & Prod | Suzhou | Jiangsu | fresh@dairyc | Supplier for | Liu | Mei | |
| SUP04 | Packaging So | Packaging Su | Hangzhou | Zhejiang | info@packagi | Supplier for | Chen | Sam | |
| SUP05 | Beverage Sys | Beverage Sup | Shanghai | Shanghai | support@bevs | Supplier for | Sun | Grace | |
| SUP06 | Service Pro | Equipment & ( | Shanghai | Shanghai | service@serv | Supplier for | Qian | Peter | |

*Table 40 Data Type of Supplier Table*

| Field Name | Data Type |
|---|---|
| SupplierID | Short Text |
| SupplierName | Short Text |
| SupplierType | Short Text |
| SupplierCity | Short Text |
| SupplierProvince | Short Text |
| ContactEmail | Short Text |
| ContactNote | Short Text |
| ContactLastName | Short Text |
| ContactFirstName | Short Text |

(22) Supplier Management Form

There is a **Supplier Management Form** for editing the Supplier table. In this form, purchasing managers can add, delete, save edit and search supplier records, as well as navigate through them using Previous/Next buttons. The Supplier Management form displays details including SupplierID, SupplierName, SupplierType, SupplierCity, SupplierProvince, ContactEmail, ContactNote and ContactLastName.

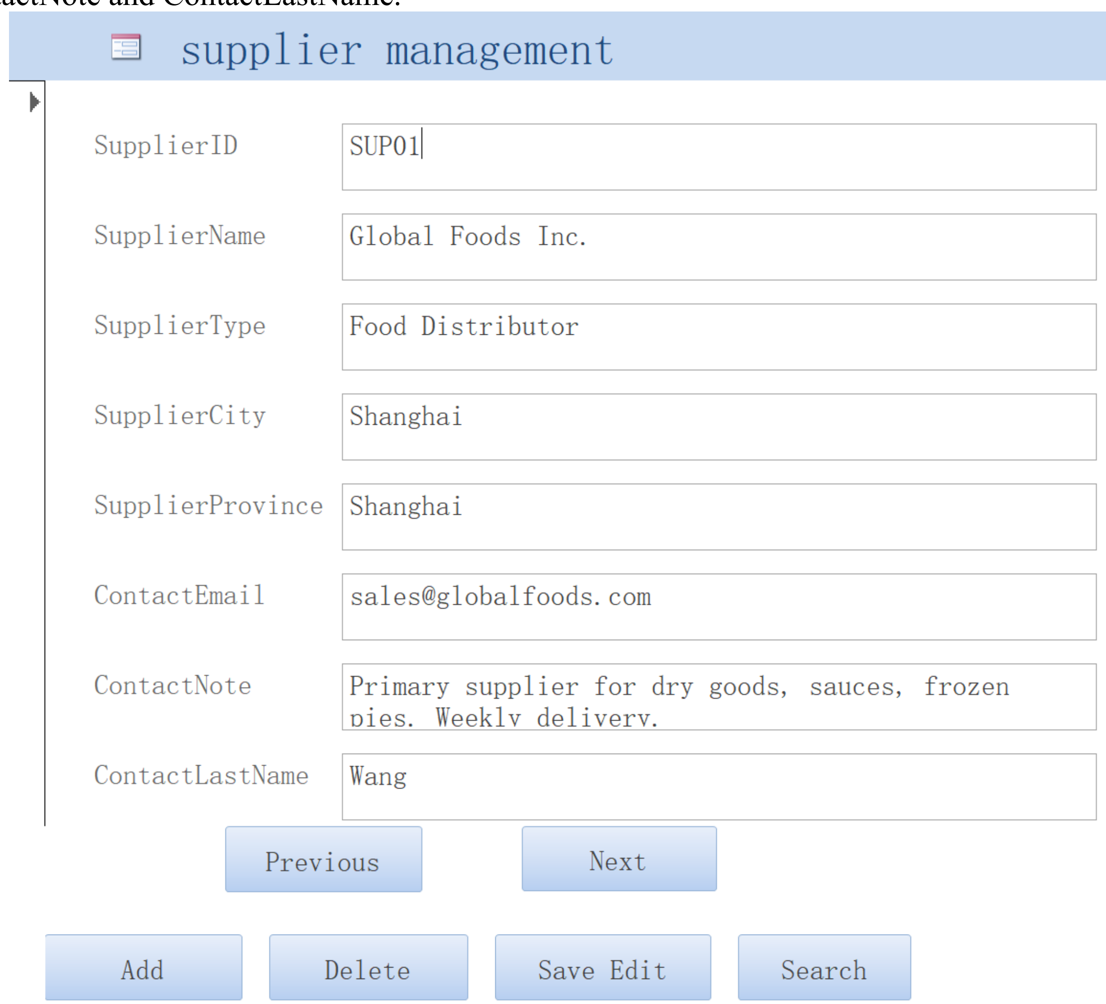



*Figure 11. Supplier Management Form*

# II. Reports

# 1. Operational Reports

Operational reports are reports used to sort out and analyze the daily operations of an enterprise. It presents multiple operational data to help management monitor and evaluate operational results, thus driving decision making and process optimization.

## Daily Sales Report

The Daily Sales Report provides data on daily sales. Its purpose is to help managers understand the number of daily sales orders, total sales, total discounts, and net sales. Attributes of the report include date, number of orders, total sales, total discounts, and net sales, all of which are quantitative indicators of daily sales business performance.

Daily Sales Report

2025年4月7日
22:03:47

| Date | Orders | Total Sales | Total Discounts | Net Sales |
|---|---|---|---|---|
| 2025-03-01 | 5 | 70.3 | 0 | 70.3 |
| 2025-03-02 | 5 | 55.7 | 0 | 55.7 |
| 2025-03-03 | 7 | 110.24 | 5.8 | 104.44 |
| 2025-03-04 | 4 | 89.5 | 0 | 89.5 |
| 2025-03-05 | 6 | 89.1 | 0 | 89.1 |
| 2025-03-06 | 5 | 79.4 | 0.5 | 78.9 |
| 2025-03-07 | 5 | 76.95 | 2.2 | 74.75 |
| 2025-03-08 | 5 | 71.09 | 0 | 71.09 |
| 2025-03-09 | 5 | 108.3 | 7.5 | 100.8 |
| 2025-03-10 | 6 | 99.55 | 1.7 | 97.85 |
| 2025-03-11 | 6 | 103.4 | 2 | 101.4 |
| 2025-03-12 | 3 | 46.7 | 0 | 46.7 |
| 2025-03-13 | 4 | 57.35 | 0.5 | 56.85 |
| 2025-03-14 | 5 | 91.35 | 7 | 84.35 |
| 2025-03-15 | 4 | 93.7 | 3.5 | 90.2 |
| 2025-03-16 | 4 | 52.4 | 1 | 51.4 |
| 2025-03-17 | 4 | 78.7 | 7.5 | 71.2 |
| 2025-03-18 | 6 | 133.3 | 2 | 131.3 |
| 18 | | | | |

共 1 页，第 1 页

*Figure 12. Daily Sales Report*

**SQL**

```
SELECT Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd")
AS [Week Start], Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-
```

```
mm-dd") AS [Week End], Count(o.OrderID) AS Orders, Sum(o.OrderPayment) AS [Total Sales], Sum(o.OrderDiscount) AS [Total Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS [Net Sales]

FROM [order] AS o

WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#

GROUP BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd"), Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-mm-dd")

ORDER BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd");
```

*Table 41. Daily Sales Query Result*

| Week Start | Week End | Orders | Total Sale | Total Disc | Net Sales |
|---|---|---|---|---|---|
| 2025-02-23 | 2025-03-01 | 5 | 70.3 | 0 | 70.3 |
| 2025-03-02 | 2025-03-08 | 37 | 571.98 | 8.5 | 563.48 |
| 2025-03-09 | 2025-03-15 | 33 | 600.35 | 22.2 | 578.15 |
| 2025-03-16 | 2025-03-22 | 14 | 264.4 | 10.5 | 253.9 |

## Personnel Management Report

This is a personnel management system report that records information about employees. The report includes employee number, employee name, position, date of entry, working hours, contact information, attendance status and other attributes. They are quantitative indicators of how well an employee is doing. The report can help managers grasp the basic situation of employees, working hours and attendance dynamics, which is convenient for human resource management and deployment.

Personnel management syste    2025年4月7日 22:06:43

| Employee ID | Employee Name | Position | Shift Date | Working Hours | Contact Info | Attendance Status |
|---|---|---|---|---|---|---|
| S01 | Wei Chen | Waiter | 2025/3/1 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/2 | 7 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/3 | 8.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/5 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/8 | 9 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/11 | 7.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/15 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/17 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/22 | 8.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/26 | 7.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/31 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/4/5 | 9 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/4/9 | 8 | 13812345678 | Present |
| S02 | Mei Li | Cashier | 2025/3/1 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/3 | 7.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/6 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/10 | 8.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/13 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/18 | 7.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/24 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/29 | 8.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/4/3 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/4/8 | 7.5 | 13987654321 | Present |
| S03 | Hao Zhang | Chef | 2025/3/1 | 8 | 13711223344 | Present |

*Figure 13. Personnel Management Report*

**SQL**

```
SELECT c.CustomerID AS [Customer ID], c.CustomerFirstName & " " & c.CustomerLastName AS [Customer Name], Sum(o.OrderPayment) AS [Total Spending], Count(o.OrderID) AS [Visit Count], Sum(IIf(o.Note <> "Takeaway" OR o.Note IS NULL, 1, 0)) AS [Dine-in Count], o.MenuItemID AS [Most Ordered Dish], Sum(o.OrderPayment)/Count(o.OrderID) AS [Customer Value]

FROM customer AS c INNER JOIN [order] AS o ON c.CustomerID = o.CustomerID

WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#

GROUP BY c.CustomerID, c.CustomerFirstName, c.CustomerLastName, o.MenuItemID

ORDER BY Count(o.OrderID) DESC , c.CustomerID;
```

*Table 42. Personnel Management Query Result*

| Customer ID | Customer Name | Total Spending | Visit Count | Dine-in Count | Most Ordered Dish | Customer Value |
|---|---|---|---|---|---|---|
| C002 | Fang Wang | 5.6 | 1 | 1 | M04 | 5.6 |
| C004 | Na Liu | 16.9 | 1 | 1 | M10 | 16.9 |
| C005 | Qiang Chen | 8.99 | 1 | 1 | M01 | 8.99 |
| C007 | Xin Zhao | 10.5 | 1 | 1 | M04 | 10.5 |
| C007 | Xin Zhao | 15.8 | 1 | 1 | M07 | 15.8 |
| C008 | Lin Huang | 29.1 | 1 | 1 | M16 | 29.1 |
| C009 | Hao Zhou | 21.2 | 1 | 1 | M15 | 21.2 |
| C011 | Lei Guo | 12.7 | 1 | 1 | M05 | 12.7 |
| C011 | Lei Guo | 26.3 | 1 | 1 | M14 | 26.3 |
| C012 | Li Ma | 28.4 | 1 | 1 | M14 | 28.4 |
| C013 | Qiang Lin | 13.4 | 1 | 1 | M08 | 13.4 |
| C015 | Peng Xu | 17.1 | 1 | 1 | M13 | 17.1 |
| C017 | Ming Hu | 26.2 | 1 | 1 | M16 | 26.2 |
| C018 | Yan Gao | 6.5 | 1 | 1 | M19 | 6.5 |
| C021 | Jie Song | 13.1 | 1 | 0 | M08 | 13.1 |
| C021 | Jie Song | 16.5 | 1 | 1 | M10 | 16.5 |
| C022 | Fei Han | 6 | 1 | 1 | M02 | 6 |
| C022 | Fei Han | 28 | 1 | 1 | M20 | 28 |
| C023 | Tao Feng | 21.1 | 1 | 1 | M02 | 21.1 |
| C024 | Juan Deng | 23.3 | 1 | 1 | M12 | 23.3 |
| C026 | Min Cheng | 6.3 | 1 | 1 | M02 | 6.3 |
| C027 | Bo Yuan | 12.1 | 1 | 1 | M05 | 12.1 |
| C028 | Li Jiang | 23.9 | 1 | 1 | M19 | 23.9 |
| C028 | Li Jiang | 18.7 | 1 | 1 | M15 | 18.7 |

## Weekly Sales Report

This is a Weekly Sales Report showing sales data for the week of February 23 - March 1, 2025. Managers can enter the weekly data they want to query and get a report for that week. The report contains attributes such as week start date, weekend date, order quantity, total sales, total discount, and net sales. These data can help managers understand the basic situation of the sales business in the week, evaluate the sales performance, and provide a reference for the development of subsequent sales strategies.

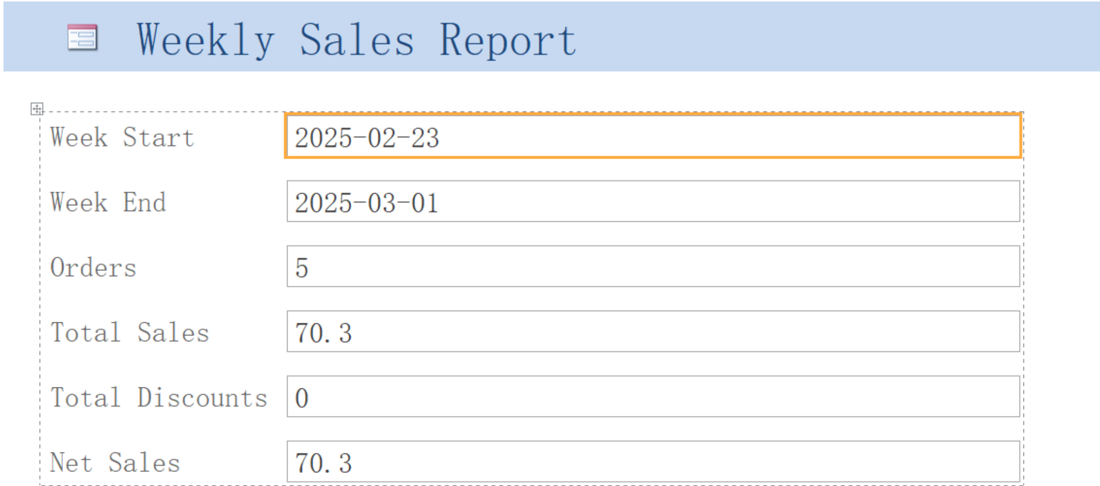



*Figure 14. Weekly Sales Report*

**SQL**

```
SELECT Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd") AS [Week Start], Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-mm-dd") AS [Week End], Count(o.OrderID) AS Orders, Sum(o.OrderPayment) AS [Total Sales], Sum(o.OrderDiscount) AS [Total Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS [Net Sales]

FROM [order] AS o

WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#

GROUP BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd"), Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-mm-dd")

ORDER BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd");
```

*Table 43. Weekly Sales Query Result*

| Week Start | Week End | Orders | Total Sale | Total Disc | Net Sales |
|---|---|---|---|---|---|
| 2025-02-23 | 2025-03-01 | 5 | 70.3 | 0 | 70.3 |
| 2025-03-02 | 2025-03-08 | 37 | 571.98 | 8.5 | 563.48 |
| 2025-03-09 | 2025-03-15 | 33 | 600.35 | 22.2 | 578.15 |
| 2025-03-16 | 2025-03-22 | 14 | 264.4 | 10.5 | 253.9 |

## Raw Material Availability Report

Before opening the raw material availability report, the manager can enter the name of a specific supplier and get information about that supplier.

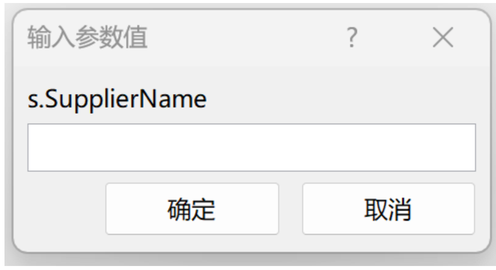



*Figure 15. Raw Material Availability Search Interface*

This is the raw material availability report. The report covers the order number, supplier, supplied raw materials, raw material specifications, quantity units, quantity, unit price, remarks and other attributes. The supplier shown in the report is Global Foods Inc. The remarks column gives a brief description of the status of raw materials, etc. The report helps the manager to grasp the raw material inventory and supply status and ensure the raw material demand of production and operation.

Raw material availability 2025年4月7日 22:15:09

| Order ID | Supplier | Supplied Ingredient | Ingredient Specification | Quantity | Unit | Unit Price (CNY) | Notes |
|---|---|---|---|---|---|---|---|
| 1021 | Global Foods Inc. | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1219 | Global Foods Inc. | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1163 | Global Foods Inc. | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1070 | Global Foods Inc. | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1133 | Global Foods Inc. | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1196 | Global Foods Inc. | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1100 | Global Foods Inc. | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1149 | Global Foods Inc. | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1058 | Global Foods Inc. | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1001 | Global Foods Inc. | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1171 | Global Foods Inc. | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1083 | Global Foods Inc. | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1026 | Global Foods Inc. | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1207 | Global Foods Inc. | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1003 | Global Foods Inc. | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |

*Figure 16. Raw Material Availability Report*

**SQL:**

```
SELECT o.OrderID AS [Order ID], s.SupplierName AS Supplier, i.InventoryName AS [Supplied Ingredient], i.Description AS [Ingredient Specification], i.InventoryQuantity AS Quantity, i.Unit AS Unit, (i.InventoryAmount / i.InventoryQuantity) AS [Unit Price (CNY)], i.Description AS Notes

FROM [order] AS o INNER JOIN inventory AS i ON i.OrderDate = o.OrderDate;
```

*Table 43. Raw Material Availability Report*

| Order ID | Supplier | Supplied I | Ingredient Specificatio | Quantity | Unit | Unit Price | Notes |
|---|---|---|---|---|---|---|---|
| 1021 | al Foods Inc | Beef Patty ( | Frozen beef patties, 4oz | 500 | Piece | 0. 5 | Frozen beef patties, 4oz each |
| 1219 | al Foods Inc | Beef Patty ( | Frozen beef patties, 4oz | 500 | Piece | 0. 5 | Frozen beef patties, 4oz each |
| 1163 | al Foods Inc | Beef Patty ( | Frozen beef patties, 4oz | 500 | Piece | 0. 5 | Frozen beef patties, 4oz each |
| 1070 | al Foods Inc | Beef Patty ( | Frozen beef patties, 4oz | 500 | Piece | 0. 5 | Frozen beef patties, 4oz each |
| 1133 | al Foods Inc | Beef Patty ( | Frozen beef patties, 4oz | 500 | Piece | 0. 5 | Frozen beef patties, 4oz each |
| 1196 | al Foods Inc | Cheddar Chee | Individually wrapped slic | 1000 | Piece | 0. 09 | Individually wrapped slices |
| 1100 | al Foods Inc | Cheddar Chee | Individually wrapped slic | 1000 | Piece | 0. 09 | Individually wrapped slices |
| 1149 | al Foods Inc | Cheddar Chee | Individually wrapped slic | 1000 | Piece | 0. 09 | Individually wrapped slices |
| 1058 | al Foods Inc | Cheddar Chee | Individually wrapped slic | 1000 | Piece | 0. 09 | Individually wrapped slices |
| 1001 | al Foods Inc | Cheddar Chee | Individually wrapped slic | 1000 | Piece | 0. 09 | Individually wrapped slices |
| 1171 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1083 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1026 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1207 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1003 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1002 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1114 | al Foods Inc | Onion (Bag, | Fresh yellow onions | 10 | Kg | 2. 5 | Fresh yellow onions |
| 1021 | al Foods Inc | Ketchup (Bul | Standard tomato ketchup | 10 | Litre | 6 | Standard tomato ketchup |
| 1219 | al Foods Inc | Ketchup (Bul | Standard tomato ketchup | 10 | Litre | 6 | Standard tomato ketchup |
| 1163 | al Foods Inc | Ketchup (Bul | Standard tomato ketchup | 10 | Litre | 6 | Standard tomato ketchup |
| 1070 | al Foods Inc | Ketchup (Bul | Standard tomato ketchup | 10 | Litre | 6 | Standard tomato ketchup |
| 1133 | al Foods Inc | Ketchup (Bul | Standard tomato ketchup | 10 | Litre | 6 | Standard tomato ketchup |

## 2.Tactic Report

Tactical reports are reports used to aid decision making at the executive level of an organization. It presents specific data, analysis, and insights about the organization's daily operations, business processes, and short-term goals to help middle - and bottom-level managers implement specific tasks, optimize business processes, and achieve short-term performance indicators to drive the organization's strategic goals.

### 1.Weekly Most Popular Dishes report

| MenuItemID | MenuItemName | Price | Category | Sales |
|---|---|---|---|---|
| M001 | Vegetable Bun | 1.5 | Main dish | 2 |
| M002 | Meat Bun | 2 | Main dish | 3 |
| M003 | Millet Porridge | 2 | Main dish | 2 |
| M004 | Soy Milk | 1.5 | Beverage | 3 |
| M005 | Egg Pancake | 3.5 | Main dish | 2 |
| M006 | Fried Rice | 8 | Main dish | 2 |
| M007 | Kung Pao Chicken | 10 | Meat dish | 1 |
| M008 | Tomato & Egg | 8 | Veg dish | 1 |
| M009 | Braised Pork | 12 | Meat dish | 6 |
| M010 | Stir-fried Broccoli | 6 | Veg dish | 2 |
| M011 | Boiled Fish | 15 | Meat dish | 1 |
| M012 | Rice | 1.5 | Main dish | 4 |
| M013 | Seaweed Soup | 3 | Soup | 2 |
| M014 | Hot & Sour Soup | 4 | Soup | 1 |
| M015 | Cola | 2.5 | Beverage | 2 |
| M016 | Mineral Water | 2 | Beverage | 1 |
| M017 | Mapo Tofu | 8 | Veg dish | 2 |
| M018 | Yu-Shiang Pork | 10 | Meat dish | 2 |
| M019 | Steamed Bun | 1 | Main dish | 1 |
| M020 | Grilled Sausage | 2.5 | Snack | 2 |

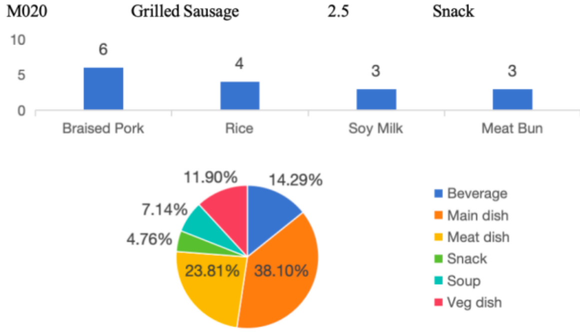



*Figure 17. Weekly Most Popular Dishes report*

The system generates a weekly Most Popular Dishes report, as illustrated in Figure 13. This report offers comprehensive data on the sales of menu items, including the name, price, category, and total sales for each dish. The report also includes a visual representation of the sales distribution, which utilizes a bar graph to illustrate the sales of the dishes. In addition, pie charts categorize dishes based on their type (e.g., entrees, meat dishes, vegetarian dishes, beverages, soups, and snacks), showing their relative sales proportions. This report is a valuable tool for management, providing insights into customer preferences, optimizing menu offerings, and making informed decisions based on data. It facilitates improved service and inventory planning, contributing to the company's overall success.

**SQL:**

**Count the frequency of each MenuItemID in the recent week:**

```
SELECT
   MenuItemID,
   COUNT(*) AS ItemFrequency
FROM [Order]
WHERE OrderDate >= DateAdd("d", -7, Date())
GROUP BY MenuItemID
ORDER BY COUNT(*) DESC;
```

**Obtain the 4 MenuItemIDs with the highest frequency:**

```
SELECT TOP 4
SELECT TOP 4
   MenuItemID,
   COUNT(*) AS ItemFrequency
FROM [Order]
WHERE OrderDate >= DateAdd("d", -7, Date())
GROUP BY MenuItemID
ORDER BY COUNT(*) DESC;
```

**Calculate the proportion of orders for each category within a week:**

```
SELECT
   m.Category,
   COUNT(o.OrderID) AS CategoryCount,
   ROUND(
    (COUNT(o.OrderID) * 1.0
     /
     (
      SELECT COUNT(*)
      FROM [Order]
      WHERE OrderDate >= DateAdd("d", -7, Date())
     )
    ) * 100
```

```
    , 2) & "%" AS PercentageOfTotal
FROM [Order] AS o
    INNER JOIN MenuItem AS m
        ON o.MenuItemID = m.MenuItemID
WHERE o.OrderDate >= DateAdd("d", -7, Date())
GROUP BY m.Category
ORDER BY COUNT(o.OrderID) DESC;
```

## 3. Strategic Reports

A strategy report is a report that underpins an organization's strategic decisions. It provides data, insights on the organization's macro direction to help senior management teams and decision makers plan long-term growth and determine resource allocation strategies to ensure that the organization stays on the right course and achieves sustainable growth in a complex market environment.

### Monthly Customer Report

This Monthly Customer Report presents customer data. Its role is to help managers gain insight into customer consumption, frequency of visits to the store, in-room dining, food preferences, and customer value. Report attributes include customer number, customer name, total amount spent, number of visits, number of dines, most frequently ordered dishes, and customer value. The customer number is used to identify the customer. The name of the customer shall be the address of the customer; The total consumption amount is the total consumption amount of the customer in the month; The number of visits is the number of visits in the month; The number of sit-down meals refers to the number of in-store meals in the month; The most frequently ordered dishes are the most ordered dishes in the month; Customer value reflects the comprehensive value of customers to merchants, which are indicators to quantify customer consumption behavior and value.

Monthly customer report 2025年4月7日 22:05:08

| Customer ID | Customer Name | Total Spending | Visit Count | Dine-in Count | Most Ordered Dish | Customer Value |
|---|---|---|---|---|---|---|
| C002 | Fang Wang | 5.6 | 1 | 1 | M04 | 5.6 |
| C004 | Na Liu | 16.9 | 1 | 1 | M10 | 16.9 |
| C005 | Qiang Chen | 8.99 | 1 | 1 | M01 | 8.99 |
| C007 | Xin Zhao | 10.5 | 1 | 1 | M04 | 10.5 |
| C007 | Xin Zhao | 15.8 | 1 | 1 | M07 | 15.8 |
| C008 | Lin Huang | 29.1 | 1 | 1 | M16 | 29.1 |
| C009 | Hao Zhou | 21.2 | 1 | 1 | M15 | 21.2 |
| C011 | Lei Guo | 12.7 | 1 | 1 | M05 | 12.7 |
| C011 | Lei Guo | 26.3 | 1 | 1 | M14 | 26.3 |
| C012 | Li Ma | 28.4 | 1 | 1 | M14 | 28.4 |
| C013 | Qiang Lin | 13.4 | 1 | 1 | M08 | 13.4 |
| C015 | Peng Xu | 17.1 | 1 | 1 | M13 | 17.1 |
| C017 | Ming Hu | 26.2 | 1 | 1 | M16 | 26.2 |
| C018 | Yan Gao | 6.5 | 1 | 1 | M19 | 6.5 |
| C021 | Jie Song | 13.1 | 1 | 0 | M08 | 13.1 |
| C021 | Jie Song | 16.5 | 1 | 1 | M10 | 16.5 |
| C022 | Fei Han | 6 | 1 | 1 | M02 | 6 |
| C022 | Fei Han | 28 | 1 | 1 | M20 | 28 |
| C023 | Tao Feng | 21.1 | 1 | 1 | M02 | 21.1 |
| C024 | Juan Deng | 23.3 | 1 | 1 | M12 | 23.3 |
| C026 | Min Cheng | 6.3 | 1 | 1 | M02 | 6.3 |
| C027 | Bo Yuan | 12.1 | 1 | 1 | M05 | 12.1 |
| C028 | Li Jiang | 18.7 | 1 | 1 | M15 | 18.7 |
| C028 | Li Jiang | 23.9 | 1 | 1 | M19 | 23.9 |

*Figure 18. Monthly Customer Report*

**SQL:**

```
SELECT c.CustomerID AS [Customer ID], c.CustomerFirstName & " " & c.CustomerLastName AS [Customer Name], Sum(o.OrderPayment) AS [Total Spending],
```

```
Count(o.OrderID) AS [Visit Count], Sum(IIf(o.Note <> "Takeaway" OR o.Note IS NULL, 1, 0)) AS [Dine-in Count], o.MenuItemID AS [Most Ordered Dish], Sum(o.OrderPayment)/Count(o.OrderID) AS [Customer Value]

FROM customer AS c INNER JOIN [order] AS o ON c.CustomerID = o.CustomerID

WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#

GROUP BY c.CustomerID, c.CustomerFirstName, c.CustomerLastName, o.MenuItemID

ORDER BY Count(o.OrderID) DESC , c.CustomerID;
```

*Table 44. Monthly Customer Query Result*

| Customer ID | Customer Name | Total Spending | Visit Count | Dine-in Count | Most Ordered Dish | Customer Value |
|---|---|---|---|---|---|---|
| C002 | Fang Wang | 5.6 | 1 | 1 | M04 | 5.6 |
| C004 | Na Liu | 16.9 | 1 | 1 | M10 | 16.9 |
| C005 | Qiang Chen | 8.99 | 1 | 1 | M01 | 8.99 |
| C007 | Xin Zhao | 10.5 | 1 | 1 | M04 | 10.5 |
| C007 | Xin Zhao | 15.8 | 1 | 1 | M07 | 15.8 |
| C008 | Lin Huang | 29.1 | 1 | 1 | M16 | 29.1 |
| C009 | Hao Zhou | 21.2 | 1 | 1 | M15 | 21.2 |
| C011 | Lei Guo | 12.7 | 1 | 1 | M05 | 12.7 |
| C011 | Lei Guo | 26.3 | 1 | 1 | M14 | 26.3 |
| C012 | Li Ma | 28.4 | 1 | 1 | M14 | 28.4 |
| C013 | Qiang Lin | 13.4 | 1 | 1 | M08 | 13.4 |
| C015 | Peng Xu | 17.1 | 1 | 1 | M13 | 17.1 |
| C017 | Ming Hu | 26.2 | 1 | 1 | M16 | 26.2 |
| C018 | Yan Gao | 6.5 | 1 | 1 | M19 | 6.5 |
| C021 | Jie Song | 13.1 | 1 | 0 | M08 | 13.1 |
| C021 | Jie Song | 16.5 | 1 | 1 | M10 | 16.5 |
| C022 | Fei Han | 6 | 1 | 1 | M02 | 6 |
| C022 | Fei Han | 28 | 1 | 1 | M20 | 28 |
| C023 | Tao Feng | 21.1 | 1 | 1 | M02 | 21.1 |
| C024 | Juan Deng | 23.3 | 1 | 1 | M12 | 23.3 |

### Monthly Sales Report

This Monthly Sales Report provides sales data. Its function and role is to help managers understand the number of monthly orders, total sales, total discounts, net sales and average order value. Attributes of the report include month, number of orders, total sales, total discounts, net sales, and average order value. They are both quantitative indicators of the performance of the monthly sales business.

Monthly Sales Report

2025年4月7日
22:06:05

| Month | Orders | Total Sales | Total Discounts | Net Sales | Average Order Value |
|---|---|---|---|---|---|
| 2025-03 | 89 | 1507.03 | 41.2 | 1465.83 | 16.9329213483146 |

1

共 1 页，第 1 页

**SQL:**

```
SELECT Format(o.OrderDate, "yyyy-mm") AS [Month], Count(o.OrderID) AS Orders,
```

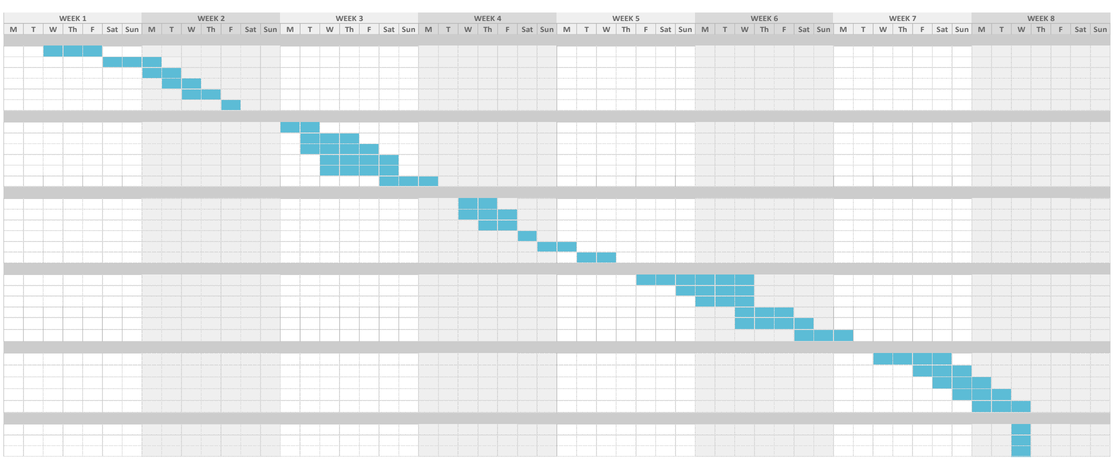
```
Sum(o.OrderPayment) AS [Total Sales], Sum(o.OrderDiscount) AS [Total Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS [Net Sales], Sum(o.OrderPayment)/Count(o.OrderID) AS [Average Order Value]

FROM [order] AS o

WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#

GROUP BY Format(o.OrderDate, "yyyy-mm")

ORDER BY Format(o.OrderDate, "yyyy-mm");
```

*Table 45. Monthly Sales Query Result*

| Month | Orders | Total Sale | Total Disc | Net Sales | Average Order Value |
|---|---|---|---|---|---|
| 2025-03 | 89 | 1507.03 | 41.2 | 1465.83 | 16.9329213483146 |

# III. Margaret Super Management SystemUpdated Gantt Chart

| TASK NAME | START DATE | END DATE | DURATION (WORK DAYS) | TEAM MEMBER | PERCENT COMPLETE |
|---|---|---|---|---|---|
| **tial Project Proposal** | | | | | |
| Identify Restaunrant Business | 2/26 | 2/28 | 3 | All | 100% |
| State Restaunrant Background and Problems | 3/1 | 3/3 | 3 | Gao Tianrun | 100% |
| Identify Organization Purpose and Objectives | 3/3 | 3/4 | 2 | Dong Renxi/Wang Kexin | 100% |
| Determine System Functions | 3/4 | 3/5 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| Identify Output Managerial Information | 3/5 | 3/6 | 2 | Zhu Yifan | 100% |
| Define Automation Benefits | 3/7 | 3/7 | 1 | All | 100% |
| **ısibility Study / Planning Phase** | | | | | |
| Identify Business Problem and scope | 3/10 | 3/11 | 2 | Gao Tianrun/Zhu Yifan | 100% |
| Technical Feasibility Analysis | 3/11 | 3/13 | 3 | Dong Renxi | 100% |
| Economic Feasibility Analysis | 3/11 | 3/14 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Schedule Feasibility Analysis | 3/12 | 3/15 | 4 | Wang Kexin | 100% |
| Operational Feasibility Analysis | 3/12 | 3/15 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Initial Design – Reports / Outputs | 3/15 | 3/17 | 3 | All | 100% |
| **plication Design** | | | | | |
| Generate DFD and ERD | 3/19 | 3/20 | 2 | Zhu Yifan | 100% |
| Design The User Interfaces | 3/19 | 3/21 | 3 | Gao Tianrun | 100% |
| Design The System Interfaces | 3/20 | 3/21 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| Design and Integrate The Database | 3/22 | 3/22 | 1 | Hu Shaohua | 100% |
| Prototype For Design Details | 3/23 | 3/24 | 1 | Gao Tianrun/Zhu Yifan | 100% |
| Design and Integrate The System Controls | 3/25 | 3/26 | 2 | Dong Renxi/Wang Kexin | 100% |
| **ɔtotyping / Data Definitions – Design Phase** | | | | | |
| Define Tables | 3/28 | 4/2 | 4 | Gao Tianrun | 100% |
| Define Table Fields | 3/30 | 4/2 | 3 | Wang Kexin | 100% |
| Define Table Indexes | 3/31 | 4/2 | 3 | Dong Renxi | 100% |
| Define Queries | 4/2 | 4/4 | 3 | Liu Zhuodong | 100% |
| Define Queries Fields | 4/2 | 4/5 | 4 | Hu Shaohua | 100% |
| Print out Actual Reports | 4/5 | 4/7 | 3 | Zhu Yifan | 100% |
| **plementation Phase & Final Report** | | | | | |
| Construct Software Components | 4/9 | 4/12 | 3 | Gao Tianrun/Zhu Yifan | 0% |
| Verify and Test | 4/11 | 4/13 | 1 | Gao Tianrun | 0% |
| Convert Data | 4/12 | 4/14 | 1 | Zhu Yifan | 0% |
| Train users and Document System | 4/13 | 4/15 | 2 | Liu Zhuodong | 0% |
| Install System | 4/14 | 4/16 | 3 | Hu Shaohua | 0% |
| **cumentation** | | | | | |
| Write Executive Overview | 4/16 | 4/16 | 1 | Wang Kexin | 0% |
| Written components | 4/17 | 4/17 | 1 | Dong Renxi | 0% |
| Database Application | 4/18 | 4/18 | 1 | All | 0% |



# PART 5: Final design and implementation

## I. Phase Overview

The fifth stage is mainly the system implementation stage. The implementation stage is one of the most important, time-consuming and costly stages in the SDLC. Therefore, our company has fully debugged all the functions of the system and conducted extensive technical training for the staff of Margaret Western Restaurant to ensure that the system can be put into practical use as soon as possible.

# II. Objectives of the application and how to realize them

The goal of our system is to help Margaret Western Restaurant reduce potential risks and increase operating profits through efficient management of inventory, customer information, financial information and data analysis. We have designed a simple, user-friendly and aesthetically pleasing interface, and developed a very easy-to-use exe program to read the access database. We have provided free user training and compiled detailed instructions to ensure that all employees can easily learn to use this system.

When users open this system, the first thing that comes to mind is a login interface. Employees enter the system based on their different identities and are granted corresponding permissions. Employees can enter raw material information, menu information, customer information and order information through this system. Some information, such as order information, can also be generated automatically. The system can generate various reports based on the existing data. Managers can view different types of reports to make the next business decision.

**Our system has enhanced the operational efficiency and profitability of restaurants through the following methods:**

**1. Through this system, we can clearly know the status of all raw materials.** The system will automatically identify raw materials that are about to be in short supply or expire and remind the staff to replenish the stock or take promotional measures to handle the raw materials. Meanwhile, the system can automatically calculate the estimated inventory level based on orders, and staff can combine on-site inventory to reduce the risk of raw material theft. These methods can minimize waste and customer churn caused by inventory issues to the greatest extent and enhance the operating profits of restaurants.

**2. Through the system's order management function, we can precisely calculate the sales data of dishes in different time periods.** Based on these data, we can more accurately estimate the usage of ingredients, adopt reasonable procurement plans, avoid the accumulation and waste of ingredients, reduce the cost of ingredient procurement, and thereby increase the profits of the restaurant.

**3. In terms of human resource management, the intelligent shift scheduling function of the system predicts the human resource demand at different times based on historical passenger flow data and automatically generates the optimal shift scheduling table.** This reduces staff redundancy, avoids unnecessary labor cost expenditures, and at the same time ensures that there are enough employees to provide high-quality services during peak hours, improving service efficiency and quality, and indirectly increasing the profits of the restaurant.

**4. With the help of the system's data analysis function, we conduct in-depth analysis of the sales situation of the dishes to clarify the populay and profit contribution of each dish.** For popular and highly profitable dishes, we increase the promotion efforts. For dishes with low profits, we optimize their production costs or adjust the menu prices. Meanwhile, the AI-based intelligent sales forecasting system can predict future sales situations, assist managers in formulating sales strategies, and enhance the overall profit level of the restaurant.

**5. The online reservation and takeout functions of the system have expanded the sales channels of restaurants.** The online reservation function facilitates customers to arrange meals in advance and attracts more customers to choose our restaurant. The takeout function has expanded the service scope of restaurants and increased the number of orders. These two functions work together, effectively increasing the sales volume and profits of the restaurant.

## Changes made to original design

**1. Technology route improvement:** from native access to the use of python GUI development, to the use of front-end and back-end development. Native access may encounter instability, as well as difficult to realize advanced functions, and access has been rarely updated, may not be conducive to the long-term use of enterprises. python GUI is more advanced, but on the one hand, it is difficult to become more beautiful, on the other hand, it is also difficult to embed the reports and forms, and for the user, the python gui operation is more difficult. Finally, we chose to take the python+html+js route. Because of the routine queries we found that due to the poor performance of access, the connection is often disconnected and unstable during the query process, so we use connection pooling and caching and other technologies to improve the stability.

**2. ERD chart modification:** We added Food Waste table to record food waste and form corresponding report, so as to better trace the source of waste.

**3. Report added:** The Menu Performance report evaluates the populay, profitability, and trends of menu items, supporting menu optimization. The Food Waste Report tracks and analyzes waste data, driving sustainability efforts. The Financial Summary presents a comprehensive overview of financial data and key metrics, driving sustainability efforts. The Financial Summary presents a comprehensive overview of financial data and key metrics, helping management quickly assess overall business health.

4. **Python UI:** Considering that the original system interface was implemented through access, which was rather professional and might not be friendly to employees with average computer skills, we designed a beautiful and easy-to-use operation interface exe program using python. Any employee can use this software easily. It is worth mentioning that the previous access database still plays an important role. It will be read by python as a database and serve as our backup solution.

## Verify and Test

We have developed the various functions required for Margaret's Western Restaurant respectively within these few weeks. When a single function is completed, we will conduct a functional test. Check whether this function can operate smoothly. If any problems are found, we will correct the errors and repeatedly optimize the functions until they achieve high-quality operation results.

After the individual function tests are perfected, we integrate all the functions for integration testing. This test mainly examines the interfaces between different modules, such as the interface between the customer relationship management module and the order management module. Ensure that all functions can work collaboratively as designed after integration.

Finally, we conducted a comprehensive system test on the entire Margaret Super Management System. We evaluate the overall performance of the system and verify whether all pre-set requirements are met, covering aspects such as system stability, data accuracy, and user-friendliness. We are constantly striving to optimize the system and provide a system that fully meets its expectations.

## Training

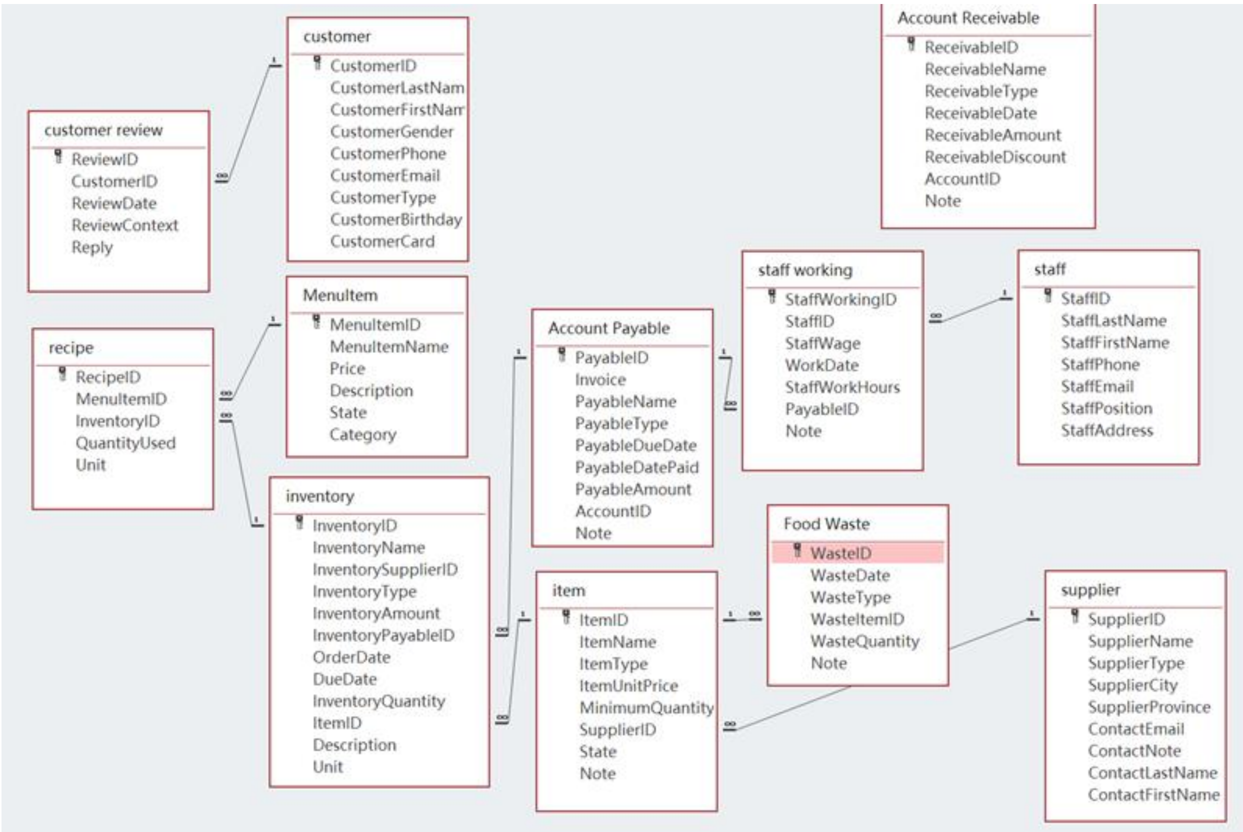
After confirming that the Margaret Super Management System was operating stably, our team members went to the Margaret Restaurant. We installed this system on the computers in the restaurant, including desktop computers and tablets.

Our team members input the historical data into the system through relevant forms such as the order management form, inventory management form and customer management form. We also demonstrated various reports generated by the system and illustrated how the system assists in daily operations, inventory control, and business decision-making.

We provided free training for the restaurant staff. The training content covers different aspects according to the roles of the employees. For the front desk staff, we train them on how to use the system on tablet computers and how to handle various operations related to orders. For kitchen staff, we trained them to identify the ingredient requirements and use the raw material warning system. Managers received training in using the system to generate reports, making decisions based on the system, and conducting data analysis with system management functions.

# III. Completed system

## DEFINITIONS




**Account Payable**

| Field Name | Data Type |
|---|---|
| PayableID | Short Text |
| Invoice | Short Text |
| PayableName | Short Text |
| PayableType | Short Text |
| PayableDueDate | Date/Time |
| PayableDatePaid | Date/Time |
| PayableAmount | Number |
| AccountID | Short Text |
| Note | Short Text |

**Account Receivable**

| Field Name | Data Type |
|---|---|
| ReceivableID | Short Text |
| ReceivableName | Short Text |
| ReceivableType | Short Text |
| ReceivableDate | Date/Time |
| ReceivableAmount | Number |
| ReceivableDiscount | Number |
| AccountID | Short Text |
| Note | Short Text |

**Customer**

| Field Name | Data Type |
|---|---|
| CustomerID | Short Text |
| CustomerLastName | Short Text |
| CustomerFirstName | Short Text |
| CustomerGender | Short Text |
| CustomerPhone | Number |
| CustomerEmail | Short Text |
| CustomerType | Short Text |
| CustomerBirthday | Date/Time |
| CustomerCard | Number |

**Customer Review**

| Field Name | Data Type |
|---|---|
| ReviewID | Short Text |
| CustomerID | Short Text |
| ReviewDate | Date/Time |
| ReviewContext | Short Text |
| Reply | Short Text |

**Food Waste**

| Field Name | Data Type |
|---|---|
| WasteID | AutoNumber |
| WasteDate | Short Text |
| WasteType | Short Text |
| WasteItemID | Short Text |
| WasteQuantity | Short Text |
| Note | Short Text |

**Inventory**

| Field Name | Data Type |
|---|---|
| InventoryID | Short Text |
| InventoryName | Short Text |
| InventorySupplierID | Short Text |
| InventoryType | Short Text |
| InventoryAmount | Number |
| InventoryPayableID | Short Text |
| OrderDate | Date/Time |
| DueDate | Date/Time |
| InventoryQuantity | Number |
| ItemID | Short Text |
| Description | Short Text |
| Unit | Short Text |

**Item**

| Field Name | Data Type |
|---|---|
| ItemID | Short Text |
| ItemName | Short Text |
| ItemType | Short Text |
| ItemUnitPrice | Number |
| MinimumQuantity | Number |
| SupplierID | Short Text |
| State | Short Text |
| Note | Short Text |

**MenuItem**

| Field Name | Data Type |
|---|---|
| MenuItemID | Short Text |
| MenuItemName | Short Text |
| Price | Number |
| Description | Short Text |
| State | Short Text |
| Category | Short Text |

**Order**

| Field Name | Data Type |
|---|---|
| OrderID | Number |
| OrderDate | Date/Time |
| CustomerID | Short Text |
| OrderPayment | Number |
| OrderDiscount | Number |
| MenuItemID | Short Text |
| ReceivableID | Short Text |
| StaffID | Short Text |
| Note | Short Text |
| State Description | Short Text |

**Recipe**

| Field Name | Data Type |
|---|---|
| RecipeID | Short Text |
| MenuItemID | Short Text |
| InventoryID | Short Text |
| QuantityUsed | Number |
| Unit | Short Text |

**Staff**

| Field Name | Data Type |
|---|---|
| StaffID | Short Text |
| StaffLastName | Short Text |
| StaffFirstName | Short Text |
| StaffPhone | Number |
| StaffEmail | Short Text |
| StaffPosition | Short Text |
| StaffAddress | Short Text |

**Staff Working**

| Field Name | Data Type |
|---|---|
| StaffWorkingID | Short Text |
| StaffID | Short Text |
| StaffWage | Number |
| WorkDate | Date/Time |
| StaffWorkHours | Number |
| PayableID | Short Text |
| Note | Short Text |

**Supplier**

| Field Name | Data Type |
|---|---|
| SupplierID | Short Text |
| SupplierName | Short Text |
| SupplierType | Short Text |
| SupplierCity | Short Text |
| SupplierProvince | Short Text |
| ContactEmail | Short Text |
| ContactNote | Short Text |
| ContactLastName | Short Text |
| ContactFirstName | Short Text |

# Tables and forms

## Payables Table & Corresponding Form

Account Payable

| Payable | Invoice | PayableName | PayableType | PayableDue | PayableDat | PayableAmo | AccountID | Note |
|---|---|---|---|---|---|---|---|---|
| PAY1002 | SUPINV-1002 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 | 250 | AP-GOODS | Payment for |
| PAY1003 | SUPINV-1003 | Fresh Dairy Co. | Supplier Payment | 2025/4/1 | | 90 | AP-GOODS | Payment for |
| PAY1004 | SUPINV-1004 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 | 45 | AP-GOODS | Payment for |
| PAY1005 | SUPINV-1005 | Fresh Dairy Co. | Supplier Payment | 2025/4/2 | | 25 | AP-GOODS | Payment for |
| PAY1006 | SUPINV-1006 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 | 60 | AP-GOODS | Payment for |
| PAY1007 | SUPINV-1007 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 | 35 | AP-GOODS | Payment for |
| PAY1008 | SUPINV-1008 | Meat Packers Ltd. | Supplier Payment | 2025/4/1 | | 320 | AP-GOODS | Payment for |
| PAY1009 | SUPINV-1009 | Fresh Dairy Co. | Supplier Payment | 2025/4/2 | | 40 | AP-GOODS | Payment for |
| PAY1010 | SUPINV-1010 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 | 55 | AP-GOODS | Payment for |
| PAY1011 | SUPINV-1011 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 | 35 | AP-GOODS | Payment for |
| PAY1012 | SUPINV-1012 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 | 150 | AP-GOODS | Payment for |
| PAY1013 | SUPINV-1013 | Global Foods Inc. | Supplier Payment | 2025/3/27 | 2025/3/26 | 15 | AP-GOODS | Payment for |
| PAY1014 | SUPINV-1014 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 | 90 | AP-GOODS | Payment for |
| PAY1015 | SUPINV-1015 | Meat Packers Ltd. | Supplier Payment | 2025/4/2 | | 450 | AP-GOODS | Payment for |
| PAY1016 | SUPINV-1016 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 | 70 | AP-GOODS | Payment for |
| PAY1017 | SUPINV-1017 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 | 200 | AP-GOODS | Payment for |
| PAY1018 | SUPINV-1018 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 | 80 | AP-GOODS | Payment for |
| PAY1019 | SUPINV-1019 | Meat Packers Ltd. | Supplier Payment | 2025/4/2 | | 300 | AP-GOODS | Payment for |
| PAY1020 | SUPINV-1020 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 | 75 | AP-GOODS | Payment for |
| PAY1021 | SUPINV-1021 | Global Foods Inc. | Supplier Payment | 2025/4/1 | | 120 | AP-GOODS | Payment for |
| PAY1022 | SUPINV-1022 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 | 50 | AP-GOODS | Payment for |
| PAY1023 | SUPINV-1023 | Fresh Dairy Co. | Supplier Payment | 2025/4/1 | | 180 | AP-GOODS | Payment for |
| PAY1024 | SUPINV-1024 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 | 110 | AP-GOODS | Payment for |
| PAY1025 | SUPINV-1025 | Fresh Dairy Co. | Supplier Payment | 2025/4/3 | | 20 | AP-GOODS | Payment for |
| PAY1026 | SUPINV-1026 | Meat Packers Ltd. | Supplier Payment | 2025/4/1 | | 150 | AP-GOODS | Payment for |
| PAY1027 | SUPINV-1027 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 | 25 | AP-GOODS | Payment for |
| PAY1028 | SUPINV-1028 | Beverage Systems | Supplier Payment | 2025/3/31 | 2025/3/30 | 90 | AP-GOODS | Payment for |
| PAY1029 | SUPINV-1029 | Beverage Systems | Supplier Payment | 2025/3/31 | 2025/3/30 | 95 | AP-GOODS | Payment for |

payable management

PayableID: PAY1001

Invoice: SUPINV-1001

PayableName: Global Foods Inc.

PayableType: Supplier Payment

PayableDueDate: 2025/3/30

PayableDatePaid: 2025/3/29

PayableAmount: 30

AccountID: AP-GOODS

Note: Payment for Inventory INV001

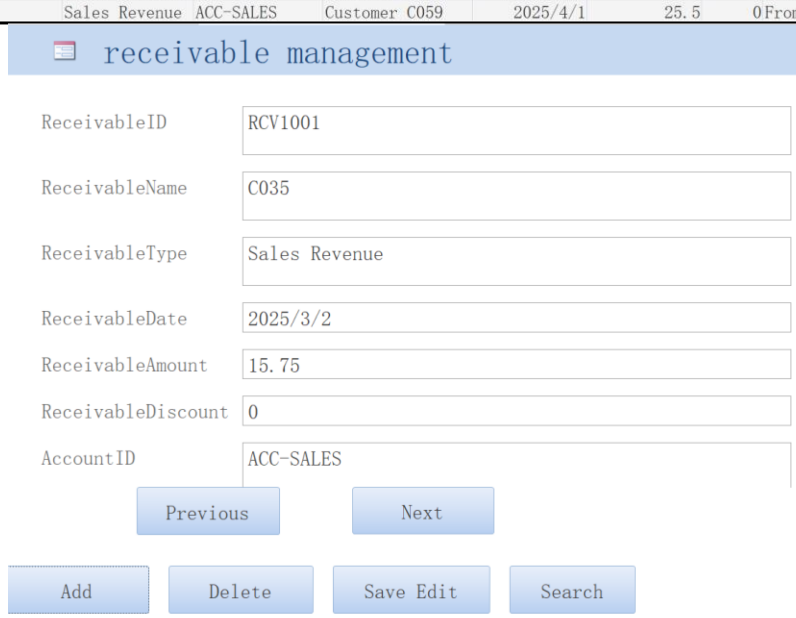
Previous | Next

Add | Delete | Save Edit | Search

## Receivables Table & Corresponding Form

Account Receivable

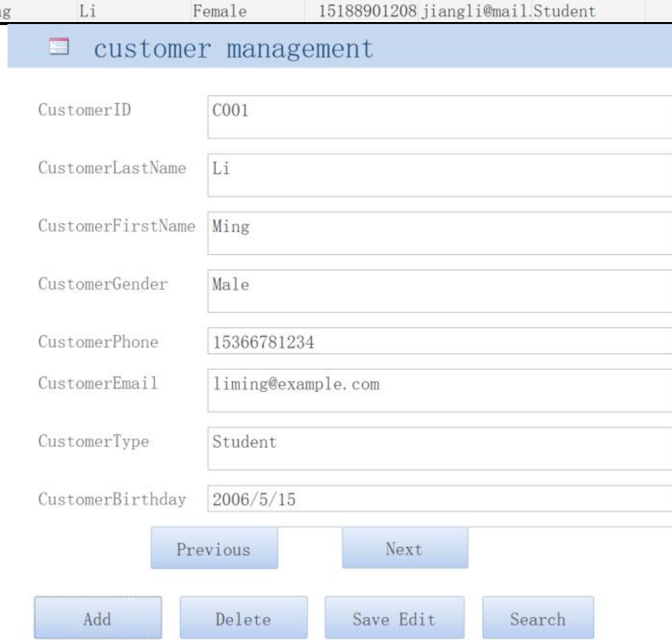
| ReceivableID | ReceivableTy | AccountID | ReceivableName | Receivable | Receivable | Rece | Note |
|---|---|---|---|---|---|---|---|
| RCV1001 | Sales Revenue | ACC-SALES | C035 | 2025/3/2 | 15.75 | 0 | 1001 |
| RCV1002 | Sales Revenue | ACC-SALES | C091 | 2025/3/3 | 28.5 | 0 | 1002 |
| RCV1003 | Sales Revenue | ACC-SALES | C005 | 2025/3/3 | 8.99 | 0 | 1003 |
| RCV1004 | Sales Revenue | ACC-SALES | C077 | 2025/3/5 | 22 | 0 | 1004 |
| RCV1005 | Sales Revenue | ACC-SALES | C035 | 2025/3/7 | 11.25 | 0 | 1005 |
| RCV1006 | Sales Revenue | ACC-SALES | C101 | 2025/3/10 | 19.8 | 0 | 1006 |
| RCV1007 | Sales Revenue | ACC-SALES | C018 | 2025/3/11 | 6.5 | 0 | 1007 |
| RCV1008 | Sales Revenue | ACC-SALES | C052 | 2025/3/14 | 29.95 | 0 | 1008 |
| RCV1009 | Sales Revenue | ACC-SALES | C088 | 2025/3/16 | 14 | 0 | 1009 |
| RCV1010 | Sales Revenue | ACC-SALES | C023 | 2025/3/18 | 21.1 | 0 | 1010 |
| RCV1011 | Sales Revenue | ACC-SALES | C099 | 2025/3/20 | 9.75 | 0 | 1011 |
| RCV1012 | Sales Revenue | ACC-SALES | C041 | 2025/3/22 | 17.3 | 0 | 1012 |
| RCV1013 | Sales Revenue | ACC-SALES | C064 | 2025/3/25 | 25 | 0 | 1013 |
| RCV1014 | Sales Revenue | ACC-SALES | C009 | 2025/3/28 | 7.2 | 0 | 1014 |
| RCV1015 | Sales Revenue | ACC-SALES | C035 | 2025/3/30 | 18.6 | 0 | 1015 |
| RCV1016 | Sales Revenue | ACC-SALES | C072 | 2025/4/1 | 26.4 | 0 | 1016 |
| RCV1017 | Sales Revenue | ACC-SALES | C050 | 2025/4/3 | 5.5 | 0 | 1017 |
| RCV1018 | Sales Revenue | ACC-SALES | C081 | 2025/4/5 | 12.9 | 0 | 1018 |
| RCV1019 | Sales Revenue | ACC-SALES | C015 | 2025/4/8 | 23.75 | 0 | 1019 |
| RCV1020 | Sales Revenue | ACC-SALES | C095 | 2025/4/10 | 10 | 0 | 1020 |
| RCV1021 | Sales Revenue | ACC-SALES | Customer C045 | 2025/3/1 | 22.15 | 0 | From Order 1021 |
| RCV1022 | Sales Revenue | ACC-SALES | Customer C088 | 2025/3/28 | 14.5 | 1 | From Order 1022 |
| RCV1023 | Sales Revenue | ACC-SALES | Customer C012 | 2025/4/5 | 7.99 | 0 | From Order 1023 |
| RCV1024 | Sales Revenue | ACC-SALES | Customer C077 | 2025/3/15 | 28.3 | 2.5 | From Order 1024 |
| RCV1025 | Sales Revenue | ACC-SALES | Customer C031 | 2025/4/9 | 11.8 | 0 | From Order 1025 |
| RCV1026 | Sales Revenue | ACC-SALES | Customer C095 | 2025/3/3 | 19.25 | 0 | From Order 1026 |
| RCV1027 | Sales Revenue | ACC-SALES | Customer C004 | 2025/3/19 | 6.75 | 0 | From Order 1027 |
| RCV1028 | Sales Revenue | ACC-SALES | Customer C059 | 2025/4/1 | 25.5 | 0 | From Order 1028 |




## Customer Table & Corresponding Form

Customer

| CustomerID | CustomerLa | CustomerFi | CustomerGe | CustomerPh | CustomerEm | CustomerTy | CustomerBi | CustomerCa |
|---|---|---|---|---|---|---|---|---|
| C001 | Li | Ming | Male | 15366781234 | liming@examp | Student | 2006/5/15 | 1001 |
| C002 | Wang | Fang | Female | 15566782345 | wangfang@exa | Student | 2005/8/23 | 1002 |
| C003 | Zhang | Wei | Male | 13866783456 | zhangwei@exa | Student | 2007/3/10 | 1003 |
| C004 | Liu | Na | Female | 13766784567 | liuna@exampl | Student | 2006/11/28 | 1004 |
| C005 | Chen | Qiang | Male | 15866785678 | chenqiang@ex | Student | 2005/7/14 | 1005 |
| C006 | Yang | Li | Female | 13966786789 | yangli@examp | Student | 2007/9/2 | 1006 |
| C007 | Zhao | Xin | Male | 15266787890 | zhaoxin@exam | Student | 2006/2/19 | 1007 |
| C008 | Huang | Lin | Female | 13866788901 | huanglin@exa | Student | 2005/12/7 | 1008 |
| C009 | Zhou | Hao | Male | 13766789012 | zhouhao@exam | Student | 2007/4/25 | 1009 |
| C010 | Wu | Fang | Female | 15866780123 | wufang@examp | Student | 2006/10/13 | 1010 |
| C011 | Guo | Lei | Male | 139667812 | zhouhao@exam | Student | 2007/4/25 | 1009 |
| C012 | Ma | Li | Female | 15266782345 | mali@example. | Teacher | 1985/9/14 | 2002 |
| C013 | Lin | Qiang | Male | 13866783456 | linqiang@exa | Teacher | 1982/2/28 | 2003 |
| C014 | Zhu | Ting | Female | 13766784567 | zhuting@exam | Teacher | 1990/11/7 | 2004 |
| C015 | Xu | Peng | Male | 15866785678 | xupeng@examp | Teacher | 1979/7/16 | 2005 |
| C016 | Sun | Ling | Female | 13966786789 | sunling@exam | Teacher | 1987/4/3 | 2006 |
| C017 | Hu | Ming | Male | 15266787890 | huming@examp | Staff | 1975/5/21 | 3001 |
| C018 | Gao | Yan | Female | 13866788901 | gaoyan@examp | Staff | 1980/8/12 | 3002 |
| C019 | Xie | Gang | Male | 13766789012 | xiegang@exam | Staff | 1978/3/25 | 3003 |
| C020 | Tang | Ying | Female | 15866780123 | tangying@exa | Staff | 1982/10/9 | 3004 |
| C021 | Song | Jie | Male | 13611234501 | songjie@mail. | Staff | 1985/4/18 | 1021 |
| C022 | Han | Fei | Female | 11098765432 | guolei@examp | Teacher | 1988/6/22 | 2001 |
| C023 | Feng | Tao | Male | 15933456703 | fengtao@mail. | Staff | 1992/12/3 | 1023 |
| C024 | Deng | Juan | Female | 13444567804 | dengjuan@mai | Teacher | 1989/7/29 | 1024 |
| C025 | Cao | Yang | Male | 15055678905 | caoyang@mail. | Student | 2007/8/11 | 1025 |
| C026 | Cheng | Min | Female | 18766789006 | chengmin@mai | Staff | 1995/3/5 | 1026 |
| C027 | Yuan | Bo | Male | 13577890107 | yuanbo@mail. | Teacher | 1981/6/19 | 1027 |
| C028 | Jiang | Li | Female | 15188901208 | jiangli@mail. | Student | 2005/9/30 | 1028 |




**Customer Review Table & Corresponding Form**

Customer Review

| ReviewID | CustomerID | ReviewDate | ReviewContext | Reply |
|---|---|---|---|---|
| CR001 | C045 | 2025/3/5 | Burger was juicy and delicious! Fries were hot too. | |
| CR002 | C012 | 2025/3/8 | Service was a bit slow during the lunch rush. | |
| CR003 | C088 | 2025/3/10 | Loved the spicy chicken wings! Perfect heat. | |
| CR004 | C025 | 2025/3/11 | It was okay, standard fast food. | |
| CR005 | C071 | 2025/3/14 | Tables needed cleaning, but the food was fine. | |
| CR006 | C009 | 2025/3/15 | Fries were cold and soggy. Disappointing. | |
| CR007 | C055 | 2025/3/16 | Friendly staff at the counter. | |
| CR008 | C093 | 2025/3/18 | The veggie burger was surprisingly tasty! | |
| CR009 | C018 | 2025/3/20 | Quick service, got my order in 5 minutes. | |
| CR010 | C064 | 2025/3/21 | Good value combo meal. | |
| CR011 | C045 | 2025/3/22 | Tried the pizza this time, better than expected. | |
| CR012 | C030 | 2025/3/23 | Chicken nuggets were a bit dry. | |
| CR013 | C077 | 2025/3/24 | Clean restaurant and restrooms. | |
| CR014 | C101 | 2025/3/25 | Double cheeseburger hit the spot! | |
| CR015 | C011 | 2025/3/27 | Hot dog was pretty standard. | |
| CR016 | C082 | 2025/3/28 | Milkshake was thick and creamy, perfect. | |
| CR017 | C059 | 2025/3/29 | They forgot the sauce for my nuggets. | |
| CR018 | C022 | 2025/3/30 | Reasonable prices for the portion size. | |
| CR019 | C096 | 2025/3/31 | The fried chicken was crispy and flavorful. | |
| CR020 | C041 | 2025/4/1 | Long wait time, seemed understaffed. | |
| CR021 | C005 | 2025/4/2 | Apple pie was nice and hot. | |
| CR022 | C070 | 2025/4/3 | Just average, nothing special. | |
| CR023 | C038 | 2025/4/4 | The coleslaw was fresh and creamy. | |
| CR024 | C088 | 2025/4/5 | Staff wasn't very friendly this time. | |
| CR025 | C015 | 2025/4/6 | Great place for a quick bite. | |
| CR026 | C051 | 2025/4/7 | Onion rings were greasy. | |
| CR027 | C099 | 2025/4/8 | My order was accurate and quick. | |
| CR028 | C029 | 2025/4/8 | Decent burger, but have had better. | |


**Food Waste Table & Corresponding Form**

Food Waste

| WasteID | WasteDate | WasteType | WasteItemI | WasteQuantit | Note |
|---|---|---|---|---|---|
| 1 | 2025/3/1 | Customer Waste | ITM002 | 5 | |
| 2 | 2025/3/2 | Kitchen Waste | ITM005 | 10 | expiration |
| 3 | 2025/3/2 | Kitchen Waste | ITM002 | 10 | expiration |
| 4 | 2025/3/3 | Customer Waste | ITM001 | 5 | |
| 5 | 2025/3/4 | Kitchen Waste | ITM005 | 10 | power cut |
| 6 | 2025/3/5 | Kitchen Waste | ITM002 | 10 | power cut |
| 7 | 2025/3/6 | Customer Waste | ITM009 | 5 | |
| 8 | 2025/3/6 | Kitchen Waste | ITM010 | 3 | expiration |
| 9 | 2025/3/6 | Kitchen Waste | ITM001 | 32 | expiration |
| 10 | 2025/3/6 | Customer Waste | ITM003 | 5 | forget throw |
| 11 | 2025/3/7 | Kitchen Waste | ITM004 | 9 | expiration |
| 12 | 2025/3/7 | Kitchen Waste | ITM004 | 2 | expiration |
| 13 | 2025/3/8 | Customer Waste | ITM013 | 1 | |
| 14 | 2025/3/11 | Kitchen Waste | ITM005 | 10 | expiration |
| 15 | 2025/3/18 | Kitchen Waste | ITM002 | 10 | expiration |
| 16 | 2025/3/18 | Customer Waste | ITM001 | 7 | |
| 17 | 2025/3/19 | Kitchen Waste | ITM005 | 10 | power cut |
| 18 | 2025/3/25 | Kitchen Waste | ITM002 | 10 | power cut |
| 19 | 2025/3/26 | Customer Waste | ITM009 | 15 | |
| 20 | 2025/3/26 | Kitchen Waste | ITM010 | 3 | expiration |
| 21 | 2025/3/27 | Kitchen Waste | ITM001 | 32 | expiration |
| 22 | 2025/4/1 | Customer Waste | ITM003 | 5 | |
| 23 | 2025/4/1 | Kitchen Waste | ITM004 | 9 | expiration |
| 24 | 2025/4/2 | Kitchen Waste | ITM004 | 4 | expiration |
| 25 | 2025/4/2 | Kitchen Waste | ITM002 | 10 | power cut |
| 26 | 2025/4/2 | Customer Waste | ITM020 | 15 | |
| 27 | 2025/4/6 | Customer Waste | ITM019 | 3 | |
| 28 | 2025/4/7 | Customer Waste | ITM019 | 5 | |

food waste

WasteID: 1

WasteDate: 2025/3/1

WasteType: Customer Waste

WasteItemID: ITM002

WasteQuantity: 5

Note:

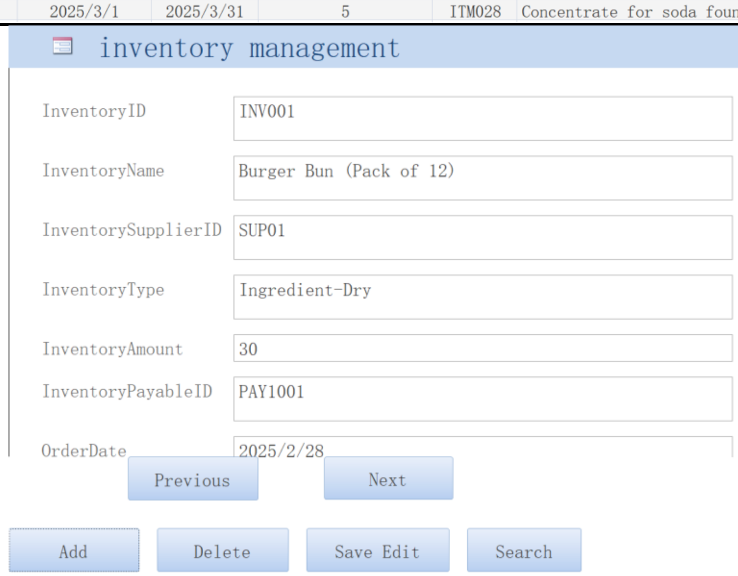
Previous | Next

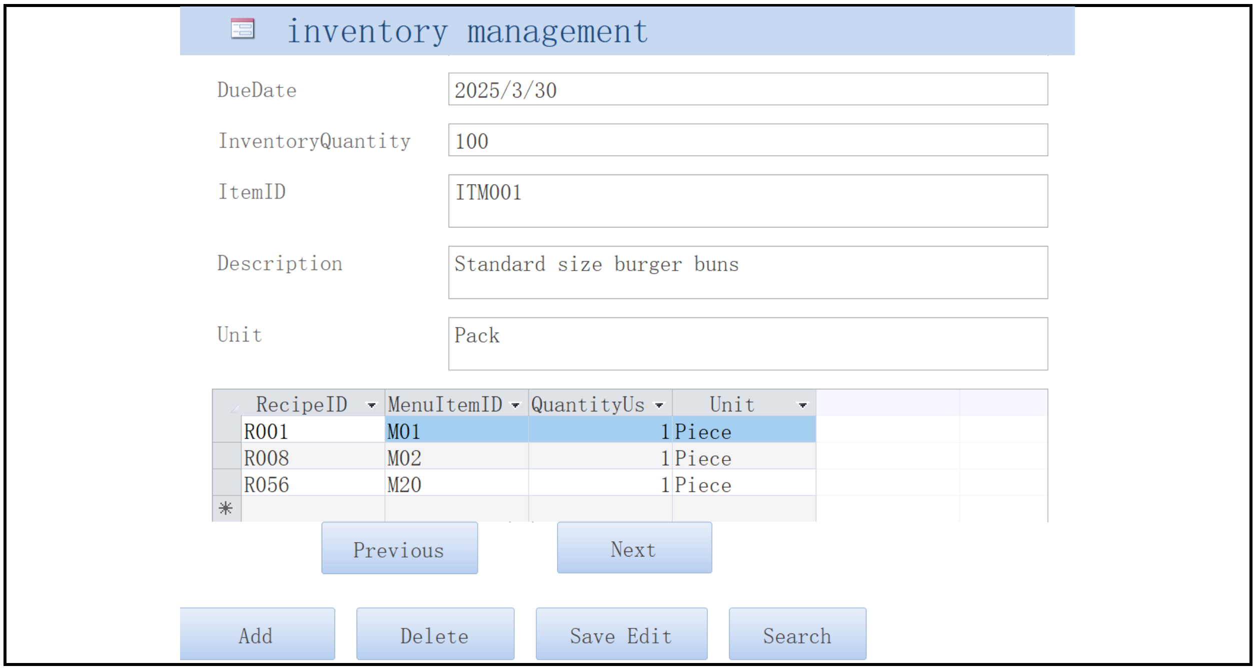
Add | Delete | Save Edit | Search

**Inventory Table & Corresponding Form**

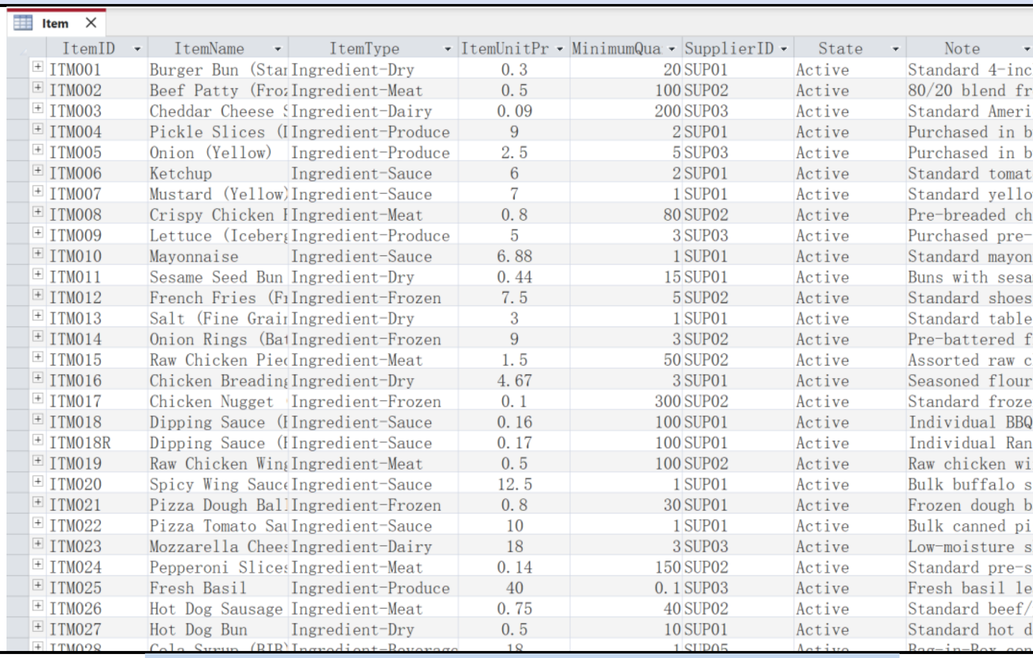
| | InventoryID | InventoryName | InventorySupplierID | InventoryType | InventoryAmount | InventoryPayableID |
|---|---|---|---|---|---|---|
| + | INV001 | Burger Bun (Pack of 12) | SUP01 | Ingredient-Dry | 30 | PAY1001 |
| + | INV002 | Beef Patty (Frozen, 4oz) | SUP02 | Ingredient-Meat | 250 | PAY1002 |
| + | INV003 | heddar Cheese Slices (Stack | SUP03 | Ingredient-Dairy | 90 | PAY1003 |
| + | INV004 | Pickle Slices (Jar) | SUP01 | Ingredient-Produce | 45 | PAY1004 |
| + | INV005 | Onion (Bag, Yellow) | SUP03 | Ingredient-Produce | 25 | PAY1005 |
| + | INV006 | Ketchup (Bulk Dispenser Bag) | SUP01 | Ingredient-Sauce | 60 | PAY1006 |
| + | INV007 | Mustard (Bulk Bottle) | SUP01 | Ingredient-Sauce | 35 | PAY1007 |
| + | INV008 | rispy Chicken Fillet (Frozer | SUP02 | Ingredient-Meat | 320 | PAY1008 |
| + | INV009 | Lettuce (Iceberg, Bag) | SUP03 | Ingredient-Produce | 40 | PAY1009 |
| + | INV010 | Mayonnaise (Bulk Tub) | SUP01 | Ingredient-Sauce | 55 | PAY1010 |
| + | INV011 | Sesame Seed Bun (Pack of 12) | SUP01 | Ingredient-Dry | 35 | PAY1011 |
| + | INV012 | Frozen Fries (Bulk Bag) | SUP02 | Ingredient-Frozen | 150 | PAY1012 |
| + | INV013 | Salt (Fine Grain, Bag) | SUP01 | Ingredient-Dry | 15 | PAY1013 |
| + | INV014 | Frozen Onion Rings (Bag) | SUP02 | Ingredient-Frozen | 90 | PAY1014 |
| + | INV015 | Raw Chicken Pieces (Case) | SUP02 | Ingredient-Meat | 450 | PAY1015 |
| + | INV016 | Chicken Breading Mix (Bag) | SUP01 | Ingredient-Dry | 70 | PAY1016 |
| + | INV017 | Frozen Chicken Nuggets (Bag) | SUP02 | Ingredient-Frozen | 200 | PAY1017 |
| + | INV018 | Dipping Sauce (BBQ, Box) | SUP01 | Ingredient-Sauce | 80 | PAY1018 |
| + | INV019 | Raw Chicken Wings (Case) | SUP02 | Ingredient-Meat | 300 | PAY1019 |
| + | INV020 | Spicy Wing Sauce (Bottle) | SUP01 | Ingredient-Sauce | 75 | PAY1020 |
| + | INV021 | Pizza Dough Ball (Frozen) | SUP01 | Ingredient-Frozen | 120 | PAY1021 |
| + | INV022 | Pizza Tomato Sauce (Can) | SUP01 | Ingredient-Sauce | 50 | PAY1022 |
| + | INV023 | :zarella Cheese (Shredded, B | SUP03 | Ingredient-Dairy | 180 | PAY1023 |
| + | INV024 | Pepperoni Slices (Pack) | SUP02 | Ingredient-Meat | 110 | PAY1024 |
| + | INV025 | Fresh Basil (Container) | SUP03 | Ingredient-Produce | 20 | PAY1025 |
| + | INV026 | Hot Dog Sausage (Pack) | SUP02 | Ingredient-Meat | 150 | PAY1026 |
| + | INV027 | Hot Dog Bun (Pack of 8) | SUP01 | Ingredient-Dry | 25 | PAY1027 |
| + | INV028 | Cola Syrup (Bag-in-Box) | SUP05 | gredient-BeverageBa | 90 | PAY1028 |

| | InventoryID | OrderDate | DueDate | InventoryQuantity | ItemID | Description | Unit |
|---|---|---|---|---|---|---|---|
| + | INV001 | 2025/2/28 | 2025/3/30 | 100 | ITM001 | Standard size burger buns | Pack |
| + | INV002 | 2025/3/1 | 2025/3/31 | 500 | ITM002 | Frozen beef patties, 4oz each | Piece |
| + | INV003 | 2025/3/2 | 2025/4/1 | 1000 | ITM003 | Individually wrapped slices | Piece |
| + | INV004 | 2025/2/28 | 2025/3/30 | 5 | ITM004 | Dill pickle slices | Kg |
| + | INV005 | 2025/3/3 | 2025/4/2 | 10 | ITM005 | Fresh yellow onions | Kg |
| + | INV006 | 2025/3/1 | 2025/3/31 | 10 | ITM006 | Standard tomato ketchup | Litre |
| + | INV007 | 2025/3/1 | 2025/3/31 | 5 | ITM007 | Yellow mustard | Litre |
| + | INV008 | 2025/3/2 | 2025/4/1 | 400 | ITM008 | Breaded chicken fillets | Piece |
| + | INV009 | 2025/3/3 | 2025/4/2 | 8 | ITM009 | Pre-shredded iceberg lettuce | Kg |
| + | INV010 | 2025/3/1 | 2025/3/31 | 8 | ITM010 | Standard mayonnaise | Litre |
| + | INV011 | 2025/2/28 | 2025/3/30 | 80 | ITM011 | Buns with sesame seeds | Pack |
| + | INV012 | 2025/3/1 | 2025/3/31 | 20 | ITM012 | Shoestring cut frozen potatoes | Kg |
| + | INV013 | 2025/2/25 | 2025/3/27 | 5 | ITM013 | Table salt | Kg |
| + | INV014 | 2025/3/1 | 2025/3/31 | 10 | ITM014 | Battered onion rings | Kg |
| + | INV015 | 2025/3/3 | 2025/4/2 | 300 | ITM015 | :ed raw chicken pieces for fry | Piece |
| + | INV016 | 2025/2/28 | 2025/3/30 | 15 | ITM016 | Seasoned flour mix for chicken | Kg |
| + | INV017 | 2025/3/1 | 2025/3/31 | 2000 | ITM017 | Bulk bag of frozen nuggets | Piece |
| + | INV018 | 2025/2/28 | 2025/3/30 | 500 | ITM018 | Individual BBQ sauce packets | Packet |
| + | INV019 | 2025/3/3 | 2025/4/2 | 600 | ITM019 | Raw chicken wing sections | Piece |
| + | INV020 | 2025/3/1 | 2025/3/31 | 6 | ITM020 | Buffalo style hot sauce | Litre |
| + | INV021 | 2025/3/2 | 2025/4/1 | 150 | ITM021 | ersonal size frozen dough ball | Piece |
| + | INV022 | 2025/3/1 | 2025/3/31 | 5 | ITM022 | Bulk canned pizza sauce | Litre |
| + | INV023 | 2025/3/2 | 2025/4/1 | 10 | ITM023 | hredded low-moisture mozzarell | Kg |
| + | INV024 | 2025/3/1 | 2025/3/31 | 800 | ITM024 | Pre-sliced pepperoni | Piece |
| + | INV025 | 2025/3/4 | 2025/4/3 | 0.5 | ITM025 | Fresh basil leaves | Kg |
| + | INV026 | 2025/3/2 | 2025/4/1 | 200 | ITM026 | Standard beef/pork hot dogs | Piece |
| + | INV027 | 2025/2/28 | 2025/3/30 | 50 | ITM027 | Standard hot dog buns | Pack |
| + | INV028 | 2025/3/1 | 2025/3/31 | 5 | ITM028 | Concentrate for soda fountain | Litre |



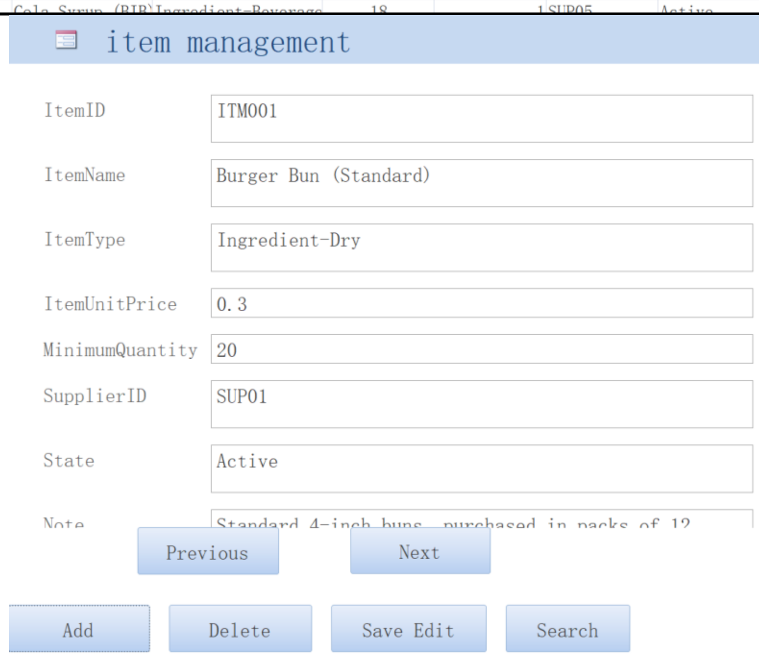

inventory management
DueDate
2025/3/30
InventoryQuantity
100
ItemID
ITM001
Description
Standard size burger buns
Unit
Pack
RecipeID
MenuItemID
QuantityUs
Unit
R001
M01
1
Piece
R008
M02
1
Piece
R056
M20
1
Piece
Previous
Next
Add
Delete
Save Edit
Search

## Item Table & Corresponding Form



Item

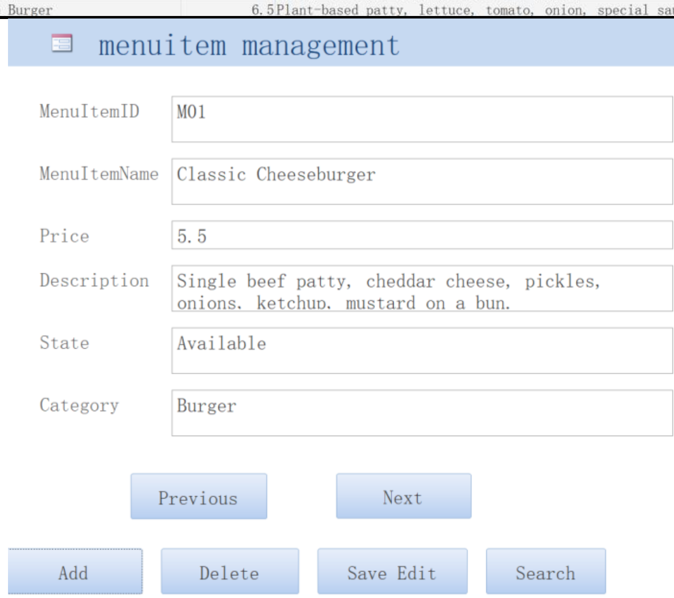
| ItemID | ItemName | ItemType | ItemUnitPr | MinimumQua | SupplierID | State | Note |
|---|---|---|---|---|---|---|---|
| ITM001 | Burger Bun (Star | Ingredient-Dry | 0.3 | 20 | SUP01 | Active | Standard 4-inch |
| ITM002 | Beef Patty (Froz | Ingredient-Meat | 0.5 | 100 | SUP02 | Active | 80/20 blend fro |
| ITM003 | Cheddar Cheese S | Ingredient-Dairy | 0.09 | 200 | SUP03 | Active | Standard Americ |
| ITM004 | Pickle Slices (I | Ingredient-Produce | 9 | 2 | SUP01 | Active | Purchased in bu |
| ITM005 | Onion (Yellow) | Ingredient-Produce | 2.5 | 5 | SUP03 | Active | Purchased in bu |
| ITM006 | Ketchup | Ingredient-Sauce | 6 | 2 | SUP01 | Active | Standard tomato |
| ITM007 | Mustard (Yellow) | Ingredient-Sauce | 7 | 1 | SUP01 | Active | Standard yellow |
| ITM008 | Crispy Chicken F | Ingredient-Meat | 0.8 | 80 | SUP02 | Active | Pre-breaded chi |
| ITM009 | Lettuce (Iceberg | Ingredient-Produce | 5 | 3 | SUP03 | Active | Purchased pre-s |
| ITM010 | Mayonnaise | Ingredient-Sauce | 6.88 | 1 | SUP01 | Active | Standard mayonn |
| ITM011 | Sesame Seed Bun | Ingredient-Dry | 0.44 | 15 | SUP01 | Active | Buns with sesar |
| ITM012 | French Fries (Fr | Ingredient-Frozen | 7.5 | 5 | SUP02 | Active | Standard shoest |
| ITM013 | Salt (Fine Grain | Ingredient-Dry | 3 | 1 | SUP01 | Active | Standard table |
| ITM014 | Onion Rings (Bat | Ingredient-Frozen | 9 | 3 | SUP02 | Active | Pre-battered fr |
| ITM015 | Raw Chicken Piec | Ingredient-Meat | 1.5 | 50 | SUP02 | Active | Assorted raw ch |
| ITM016 | Chicken Breading | Ingredient-Dry | 4.67 | 3 | SUP01 | Active | Seasoned flour |
| ITM017 | Chicken Nugget | Ingredient-Frozen | 0.1 | 300 | SUP02 | Active | Standard frozen |
| ITM018 | Dipping Sauce (I | Ingredient-Sauce | 0.16 | 100 | SUP01 | Active | Individual BBQ |
| ITM018R | Dipping Sauce (I | Ingredient-Sauce | 0.17 | 100 | SUP01 | Active | Individual Ranc |
| ITM019 | Raw Chicken Wing | Ingredient-Meat | 0.5 | 100 | SUP02 | Active | Raw chicken wir |
| ITM020 | Spicy Wing Sauce | Ingredient-Sauce | 12.5 | 1 | SUP01 | Active | Bulk buffalo st |
| ITM021 | Pizza Dough Ball | Ingredient-Frozen | 0.8 | 30 | SUP01 | Active | Frozen dough ba |
| ITM022 | Pizza Tomato Sau | Ingredient-Sauce | 10 | 1 | SUP01 | Active | Bulk canned piz |
| ITM023 | Mozzarella Chees | Ingredient-Dairy | 18 | 3 | SUP03 | Active | Low-moisture sh |
| ITM024 | Pepperoni Slices | Ingredient-Meat | 0.14 | 150 | SUP02 | Active | Standard pre-sl |
| ITM025 | Fresh Basil | Ingredient-Produce | 40 | 0.1 | SUP03 | Active | Fresh basil lea |
| ITM026 | Hot Dog Sausage | Ingredient-Meat | 0.75 | 40 | SUP02 | Active | Standard beef/p |
| ITM027 | Hot Dog Bun | Ingredient-Dry | 0.5 | 10 | SUP01 | Active | Standard hot do |
| ITM028 | Cola Syrup (BIB) | Ingredient-Beverage | 18 | 1 | SUP05 | Active | Bag-in-Box conc |




## MenuItem Table & Corresponding Form

MenuItem

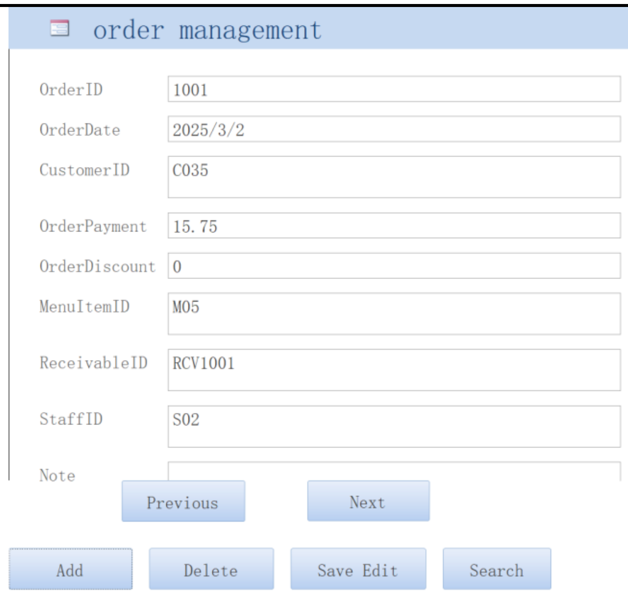
| MenuItemID | MenuItemName | Price | Description |
|---|---|---|---|
| M01 | Classic Cheeseburger | 5.5 | Single beef patty, cheddar cheese, pickles, onions, ketchup, mustard on a bun. |
| M02 | Double Cheeseburger | 7.5 | Two beef patties, double cheddar cheese, pickles, onions, ketchup, mustard on a |
| M03 | Crispy Chicken Burger | 6.8 | Fried crispy chicken fillet, lettuce, mayonnaise on a sesame seed bun. |
| M04 | French Fries (Regular) | 2.5 | Classic golden crispy french fries, lightly salted. |
| M05 | French Fries (Large) | 3.5 | Larger portion of our classic golden crispy french fries. |
| M06 | Onion Rings | 3.8 | Battered and deep-fried crispy onion rings. |
| M07 | Fried Chicken (2 pieces) | 6 | Two pieces of our signature crispy fried chicken (choice of original or spicy). |
| M08 | Chicken Nuggets (6 pieces) | 4.5 | Six tender and juicy chicken nuggets served with dipping sauce. |
| M09 | Spicy Chicken Wings (5 pieces) | 7 | Five crispy chicken wings coated in spicy buffalo sauce. |
| M10 | Pepperoni Pizza (Personal) | 8.5 | Personal-sized pizza with tomato sauce, mozzarella cheese, and pepperoni slice |
| M11 | Margherita Pizza (Personal) | 8 | Personal-sized pizza with tomato sauce, mozzarella cheese, and fresh basil. |
| M12 | Classic Hot Dog | 4 | Grilled sausage in a soft bun, served with optional ketchup and mustard. |
| M13 | Cola (Regular) | 2.2 | Standard cola soft drink (e.g., Coke/Pepsi). |
| M14 | Diet Cola (Regular) | 2.2 | Diet version of standard cola soft drink. |
| M15 | Lemonade | 2.5 | Refreshing still lemonade. |
| M16 | Chocolate Milkshake | 3.9 | Creamy chocolate milkshake topped with whipped cream. |
| M17 | Fried Apple Pie | 1.8 | Crispy fried pastry filled with warm apple filling. |
| M18 | Soft Serve Cone | 1.5 | Classic vanilla soft serve ice cream in a crispy cone. |
| M19 | Coleslaw | 2 | Creamy shredded cabbage and carrot salad. |
| M20 | Veggie Burger | 6.5 | Plant-based patty, lettuce, tomato, onion, special sauce on a bun. |




## Order Table & Corresponding Form

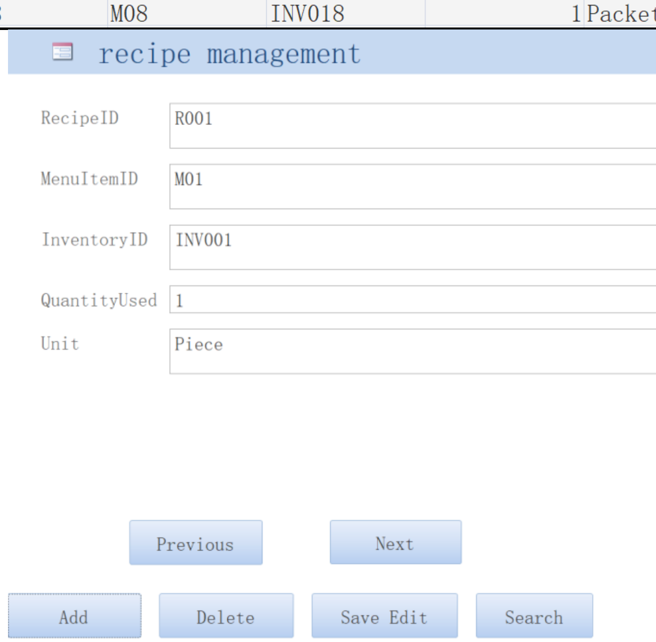
Order

| OrderID | OrderDate | CustomerID | OrderPayment | OrderDiscount | MenuItemID | Receivable | StaffID | Note | State Desc |
|---|---|---|---|---|---|---|---|---|---|
| 1001 | 2025/3/2 | C035 | 15.75 | 0 | M05 | RCV1001 | S02 | | Complete |
| 1002 | 2025/3/3 | C091 | 28.5 | 2.5 | M12 | RCV1002 | S05 | Takeaway | Complete |
| 1003 | 2025/3/3 | C005 | 8.99 | 0 | M01 | RCV1003 | S02 | | Complete |
| 1004 | 2025/3/5 | C077 | 22 | 0 | M15 | RCV1004 | S01 | | Complete |
| 1005 | 2025/3/7 | C035 | 11.25 | 1 | M08 | RCV1005 | S03 | Allergic to | Complete |
| 1006 | 2025/3/10 | C101 | 19.8 | 0 | M03 | RCV1006 | S04 | | Complete |
| 1007 | 2025/3/11 | C018 | 6.5 | 0 | M19 | RCV1007 | S01 | | Complete |
| 1008 | 2025/3/14 | C052 | 29.95 | 5 | M11 | RCV1008 | S05 | Birthday dis | Complete |
| 1009 | 2025/3/16 | C088 | 14 | 0 | M07 | RCV1009 | S03 | | Complete |
| 1010 | 2025/3/18 | C023 | 21.1 | 0 | M02 | RCV1010 | S02 | | Complete |
| 1011 | 2025/3/20 | C099 | 9.75 | 0 | M05 | RCV1011 | S04 | | Complete |
| 1012 | 2025/3/22 | C041 | 17.3 | 1.5 | M10 | RCV1012 | S01 | | Complete |
| 1013 | 2025/3/25 | C064 | 25 | 0 | M16 | RCV1013 | S03 | Extra spicy | Complete |
| 1014 | 2025/3/28 | C009 | 7.2 | 0 | M20 | RCV1014 | S05 | | Complete |
| 1015 | 2025/3/30 | C035 | 18.6 | 0 | M12 | RCV1015 | S02 | | Complete |
| 1016 | 2025/4/1 | C072 | 26.4 | 2 | M14 | RCV1016 | S01 | | Complete |
| 1017 | 2025/4/3 | C050 | 5.5 | 0 | M01 | RCV1017 | S04 | | Complete |
| 1018 | 2025/4/5 | C081 | 12.9 | 0 | M06 | RCV1018 | S03 | | Complete |
| 1019 | 2025/4/8 | C015 | 23.75 | 0 | M13 | RCV1019 | S05 | No onions | Complete |
| 1020 | 2025/4/10 | C095 | 10 | 0 | M09 | RCV1020 | S02 | | Complete |
| 1021 | 2025/3/1 | C045 | 22.15 | 0 | M11 | RCV1021 | S03 | | Complete |
| 1022 | 2025/3/28 | C088 | 14.5 | 1 | M05 | RCV1022 | S01 | | Complete |
| 1023 | 2025/4/5 | C012 | 7.99 | 0 | M01 | RCV1023 | S04 | Takeaway | Complete |
| 1024 | 2025/3/15 | C077 | 28.3 | 2.5 | M18 | RCV1024 | S02 | | Complete |
| 1025 | 2025/4/9 | C031 | 11.8 | 0 | M07 | RCV1025 | S05 | | Complete |
| 1026 | 2025/3/3 | C095 | 19.25 | 0 | M14 | RCV1026 | S03 | | Complete |
| 1027 | 2025/3/19 | C004 | 6.75 | 0 | M02 | RCV1027 | S01 | | Complete |
| 1028 | 2025/4/1 | C059 | 25.5 | 0 | M16 | RCV1028 | S04 | No onions | Complete |




**Recipe Table & Corresponding Form**

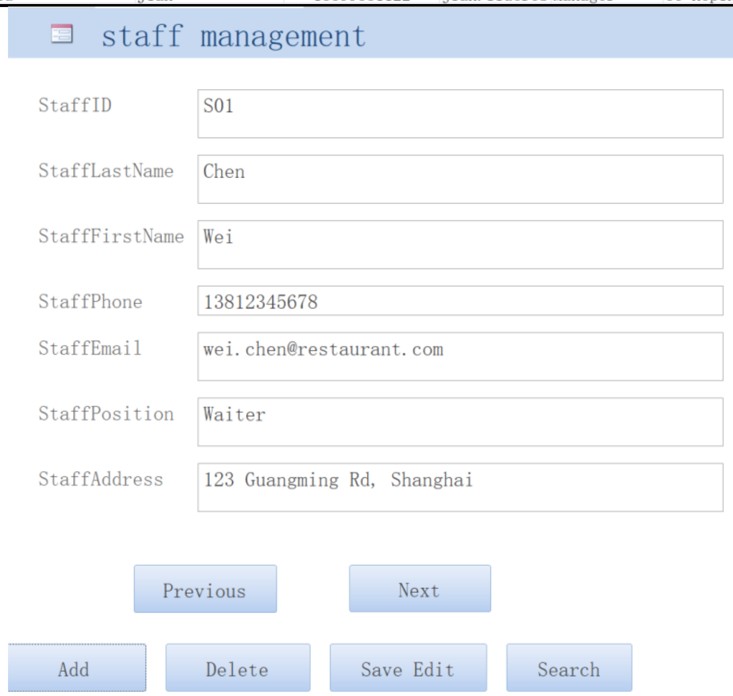
| RecipeID | MenuItemID | InventoryI | QuantityUs | Unit |
|---|---|---|---|---|
| R001 | M01 | INV001 | 1 | Piece |
| R002 | M01 | INV002 | 1 | Piece |
| R003 | M01 | INV003 | 1 | Piece |
| R004 | M01 | INV004 | 5 | Grams |
| R005 | M01 | INV005 | 5 | Grams |
| R006 | M01 | INV006 | 10 | ml |
| R007 | M01 | INV007 | 5 | ml |
| R008 | M02 | INV001 | 1 | Piece |
| R009 | M02 | INV002 | 2 | Piece |
| R010 | M02 | INV003 | 2 | Piece |
| R011 | M02 | INV004 | 5 | Grams |
| R012 | M02 | INV005 | 5 | Grams |
| R013 | M02 | INV006 | 10 | ml |
| R014 | M02 | INV007 | 5 | ml |
| R015 | M03 | INV011 | 1 | Piece |
| R016 | M03 | INV008 | 1 | Piece |
| R017 | M03 | INV009 | 15 | Grams |
| R018 | M03 | INV010 | 15 | ml |
| R019 | M04 | INV012 | 100 | Grams |
| R020 | M04 | INV013 | 1 | Grams |
| R021 | M05 | INV012 | 150 | Grams |
| R022 | M05 | INV013 | 1.5 | Grams |
| R023 | M06 | INV014 | 80 | Grams |
| R024 | M06 | INV013 | 1 | Grams |
| R025 | M07 | INV015 | 2 | Piece |
| R026 | M07 | INV016 | 50 | Grams |
| R027 | M08 | INV017 | 6 | Piece |
| R028 | M08 | INV018 | 1 | Packet |



## Staff Table & Corresponding Form

Staff

| StaffID | StaffLastName | StaffFirstName | StaffPhone | StaffEmail | StaffPosit | StaffAddre | Click to Add |
|---|---|---|---|---|---|---|---|
| S01 | Chen | Wei | 13812345678 | wei.chen@res | Waiter | 123 Guangmin | |
| S02 | Li | Mei | 13987654321 | mei.li@restau | Cashier | 45 Renmin Av | |
| S03 | Zhang | Hao | 13711223344 | hao.zhang@re | Chef | 78 Jiefang S | |
| S04 | Wang | Fang | 13655447788 | fang.wang@re | Waiter | 90 Zhongshan | |
| S05 | Liu | Jian | 13599881122 | jian.liu@res | Manager | 56 Heping Ln. | |



## Staff Working Table & Corresponding Form

Staff Working

| StaffWorkingID | StaffID | StaffWage | WorkDate | StaffWorkH | PayableID | Note |
|---|---|---|---|---|---|---|
| SW001 | S01 | 120 | 2025/3/1 | 8 | PAY3001 | Weekend Shif |
| SW002 | S02 | 128 | 2025/3/1 | 8 | PAY3002 | |
| SW003 | S03 | 200 | 2025/3/1 | 8 | PAY3003 | |
| SW004 | S05 | 240 | 2025/3/1 | 8 | PAY3004 | Weekend Duty |
| SW005 | S01 | 105 | 2025/3/2 | 7 | PAY3005 | |
| SW006 | S04 | 120 | 2025/3/2 | 8 | PAY3006 | Weekend Shif |
| SW007 | S03 | 175 | 2025/3/2 | 7 | PAY3007 | |
| SW008 | S01 | 127.5 | 2025/3/3 | 8.5 | PAY3008 | |
| SW009 | S02 | 120 | 2025/3/3 | 7.5 | PAY3009 | |
| SW010 | S04 | 112.5 | 2025/3/4 | 7.5 | PAY3010 | |
| SW011 | S03 | 212.5 | 2025/3/4 | 8.5 | PAY3011 | |
| SW012 | S05 | 255 | 2025/3/5 | 8.5 | PAY3012 | Meeting Prep |
| SW013 | S01 | 120 | 2025/3/5 | 8 | PAY3013 | |
| SW014 | S02 | 128 | 2025/3/6 | 8 | PAY3014 | |
| SW015 | S04 | 120 | 2025/3/7 | 8 | PAY3015 | |
| SW016 | S03 | 187.5 | 2025/3/8 | 7.5 | PAY3016 | Weekend Shif |
| SW017 | S01 | 135 | 2025/3/8 | 9 | PAY3017 | Weekend Shif |
| SW018 | S05 | 225 | 2025/3/9 | 7.5 | PAY3018 | Weekend Duty |
| SW019 | S02 | 136 | 2025/3/10 | 8.5 | PAY3019 | |
| SW020 | S04 | 127.5 | 2025/3/10 | 8.5 | PAY3020 | |
| SW021 | S01 | 112.5 | 2025/3/11 | 7.5 | PAY3021 | |
| SW022 | S03 | 200 | 2025/3/12 | 8 | PAY3022 | |
| SW023 | S05 | 270 | 2025/3/12 | 9 | PAY3023 | Inventory Ch |
| SW024 | S02 | 128 | 2025/3/13 | 8 | PAY3024 | |
| SW025 | S04 | 112.5 | 2025/3/14 | 7.5 | PAY3025 | |
| SW026 | S01 | 120 | 2025/3/15 | 8 | PAY3026 | Weekend Shif |
| SW027 | S03 | 225 | 2025/3/15 | 9 | PAY3027 | Weekend Shif |
| SW028 | S04 | 120 | 2025/3/16 | 8 | PAY3028 | Weekend Shif |

staff working management

StaffWorkingID: SW001

StaffID: S01

StaffWage: 120

WorkDate: 2025/3/1

StaffWorkHours: 8

PayableID: PAY3001

Note: Weekend Shift

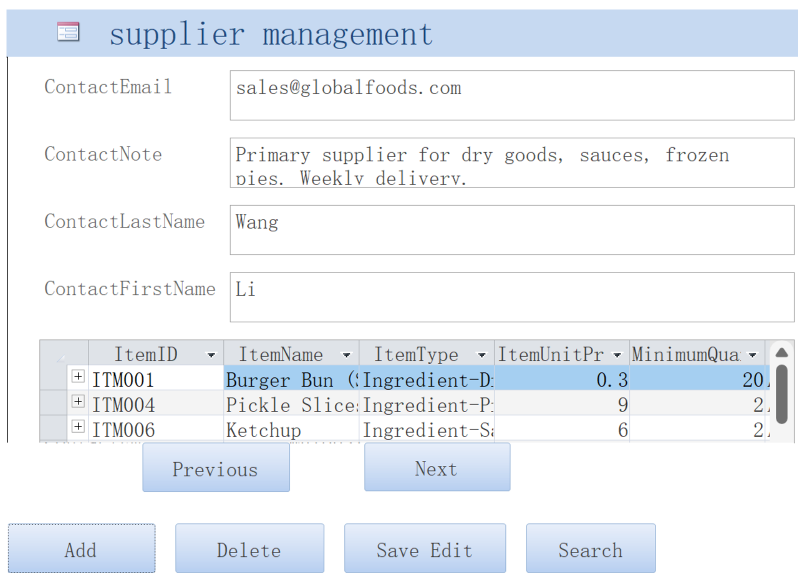
Previous | Next

Add | Delete | Save Edit | Search

## Supplier Table & Corresponding Form

Supplier

| SupplierID | SupplierName | SupplierType | SupplierCi | SupplierPr | ContactEma | ContactNot | ContactLas | ContactFir |
|---|---|---|---|---|---|---|---|---|
| SUP01 | Global Foods Inc. | Food Distributor | Shanghai | Shanghai | sales@global | Primary supp | Wang | Li |
| SUP02 | Meat Packers Ltd. | Meat & Frozen Goods Supp | Nanjing | Jiangsu | orders@meatp | Main supplie | Zhao | David |
| SUP03 | Fresh Dairy Co. | Dairy & Produce Supplier | Suzhou | Jiangsu | fresh@dairyc | Supplier for | Liu | Mei |
| SUP04 | Packaging Solutions | Packaging Supplier | Hangzhou | Zhejiang | info@packagi | Supplier for | Chen | Sam |
| SUP05 | Beverage Systems | Beverage Supplier | Shanghai | Shanghai | support@bevs | Supplier for | Sun | Grace |
| SUP06 | Service Pro | Equipment & Consumables | Shanghai | Shanghai | service@serv | Supplier for | Qian | Peter |

## Reports

**Aging Discount Report**

Aging Discount

2025年4月21日
19:11:20

| OrderID | OrderDate | CustomerID | CustomerFirstName | CustomerLastName | CustomerAge | CustomerGender | SalesAmount | DiscountAmount |
|---|---|---|---|---|---|---|---|---|
| 1019 | 2025/4/8 | C015 | Peng | Xu | 46 | Male | 23.75 | 0 |
| 1046 | 2025/3/20 | C015 | Peng | Xu | 46 | Male | 16 | 0 |
| 1185 | 2025/3/7 | C015 | Peng | Xu | 46 | Male | 17.1 | 1.2 |
| 1086 | 2025/3/11 | C017 | Ming | Hu | 50 | Male | 26.2 | 0 |
| 1154 | 2025/4/9 | C017 | Ming | Hu | 50 | Male | 20.6 | 1 |
| 1007 | 2025/3/11 | C018 | Yan | Gao | 45 | Female | 6.5 | 0 |
| 1121 | 2025/3/22 | C018 | Yan | Gao | 45 | Female | 23.1 | 1.75 |
| 1218 | 2025/4/10 | C018 | Yan | Gao | 45 | Female | 29.4 | 3.9 |
| 1034 | 2025/4/3 | C019 | Gang | Xie | 47 | Male | 9.5 | 0 |
| 1143 | 2025/4/5 | C019 | Gang | Xie | 47 | Male | 8.65 | 0 |
| 1063 | 2025/4/7 | C043 | Gang | Mo | 49 | Male | 25.2 | 2 |
| 1183 | 2025/3/29 | C043 | Gang | Mo | 49 | Male | 22.8 | 0 |
| 1053 | 2025/3/16 | C047 | Hao | Xia | 45 | Male | 14.8 | 1 |
| 1200 | 2025/4/1 | C047 | Hao | Xia | 45 | Male | 8.7 | 0 |
| 1072 | 2025/4/6 | C063 | Hao | Lai | 47 | Male | 22.6 | 0 |
| 1158 | 2025/3/16 | C063 | Hao | Lai | 47 | Male | 6.6 | 0 |
| 1094 | 2025/4/2 | C075 | Gang | Niu | 46 | Male | 7.75 | 0 |
| 1173 | 2025/4/5 | C075 | Gang | Niu | 46 | Male | 20.4 | 0 |
| 1079 | 2025/4/1 | C087 | Tao | Shen | 45 | Male | 27.8 | 2 |
| 1176 | 2025/4/10 | C087 | Tao | Shen | 45 | Male | 19.7 | 0 |
| 1085 | 2025/4/5 | C093 | Lei | Yao | 48 | Male | 6.4 | 0 |
| 1156 | 2025/3/21 | C093 | Lei | Yao | 48 | Male | 12.6 | 0 |

```
SELECT o.OrderID, o.OrderDate, c.CustomerID, c.CustomerFirstName, c.CustomerLastName, DateDiff("yyyy",
c.CustomerBirthday, Date()) AS CustomerAge, c.CustomerGender, o.OrderPayment AS SalesAmount,
o.OrderDiscount AS DiscountAmount
FROM Orders AS o INNER JOIN Customer AS c ON o.CustomerID = c.CustomerID
WHERE DateDiff("yyyy", c.CustomerBirthday, Date()) >= 45;
```

This is the "Aging Discount" records page, displaying all orders of qualified customers with associated discounts on a specified date (April 21, 2025, 19:11:20). The table contains various fields with in-depth details for every order, including Order ID, Order Date, Customer ID, First Name, Last Name, Customer Age, Gender, Sales Amount, and Discount Amount.

The "CustomerAge" field helps to identify the age of the customer, and based on pre-defined rules, the system offers discounts. The "DiscountAmount" field records the amount of discount offered. For instance, old customers (e.g., 50 years) can avail discounts, while young or ineligible customers receive no discount (discount amount = 0).

This report helps management to consider the application of age-discounting policies, their effectiveness, and as a basis for subsequent marketing planning. It is based on the customer and order tables and the logic behind creating the "DiscountAmount" field based on the age of the customer for dynamic discounting.

## Account Payable Information

Account Payable Information 2025年4月21日 23:33:36

| PayableID | Invoice | PayableName | PayableType | PayableDueDate | PayableDatePaid |
|---|---|---|---|---|---|
| PAY1001 | SUPINV-1001 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 |
| PAY1002 | SUPINV-1002 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 |
| PAY1003 | SUPINV-1003 | Fresh Dairy Co. | Supplier Payment | 2025/4/1 | |
| PAY1004 | SUPINV-1004 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 |
| PAY1005 | SUPINV-1005 | Fresh Dairy Co. | Supplier Payment | 2025/4/2 | |
| PAY1006 | SUPINV-1006 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 |
| PAY1007 | SUPINV-1007 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 |
| PAY1008 | SUPINV-1008 | Meat Packers Ltd. | Supplier Payment | 2025/4/1 | |
| PAY1009 | SUPINV-1009 | Fresh Dairy Co. | Supplier Payment | 2025/4/2 | |
| PAY1010 | SUPINV-1010 | Global Foods Inc. | Supplier Payment | 2025/3/31 | 2025/3/30 |
| PAY1011 | SUPINV-1011 | Global Foods Inc. | Supplier Payment | 2025/3/30 | 2025/3/29 |
| PAY1012 | SUPINV-1012 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 |
| PAY1013 | SUPINV-1013 | Global Foods Inc. | Supplier Payment | 2025/3/27 | 2025/3/26 |
| PAY1014 | SUPINV-1014 | Meat Packers Ltd. | Supplier Payment | 2025/3/31 | 2025/3/30 |

```
SELECT *
FROM [Account Receivable]
WHERE ([ReceivableName] = [Enter Receivable Name] OR [Enter Receivable Name] IS NULL)
  AND ([PayableDueDate] >= [Enter Start Due Date] OR [Enter Start Due Date] IS NULL)
  AND ([PayableDueDate] <= [Enter End Due Date] OR [Enter End Due Date] IS NULL)
  AND ([PayableDatePaid] >= [Enter Start Paid Date] OR [Enter Start Paid Date] IS NULL)
  AND ([PayableDatePaid] <= [Enter End Paid Date] OR [Enter End Paid Date] IS NULL)
  AND ([PayableAmount] >= [Enter Min Amount] OR [Enter Min Amount] IS NULL)
  AND ([PayableAmount] <= [Enter Max Amount] OR [Enter Max Amount] IS NULL);
```

This is the "Account Payable Information" records page, displaying all payable transactions as of a specified date (April 21, 2025, 23:33:36). The table includes multiple fields providing detailed information for each payable item, such as PayableID, Invoice, PayableName, PayableType, PayableDueDate, and PayableDatePaid.
The "PayableDueDate" field indicates when the payment is due, and the "PayableDatePaid" field records when the actual payment was made. This helps to track whether payments are made on time. For example, some payments are settled before the due date, while others may still be pending.
This report assists the finance team in managing cash flow, ensuring timely payments to suppliers, and maintaining good business relationships. It is generated based on accounts payable records and the logic of tracking payment timelines for each payable item.

## Daily Sales Report

### Daily Sales Report

2025年4月21日
19:12:50

| Date | Orders | Total Sales | Total Discounts | Net Sales |
|---|---|---|---|---|
| 2025-03-01 | 5 | 70.3 | 0 | 70.3 |
| 2025-03-02 | 5 | 55.7 | 0 | 55.7 |
| 2025-03-03 | 7 | 110.24 | 5.8 | 104.44 |
| 2025-03-04 | 4 | 89.5 | 0 | 89.5 |
| 2025-03-05 | 6 | 89.1 | 0 | 89.1 |
| 2025-03-06 | 5 | 79.4 | 0.5 | 78.9 |
| 2025-03-07 | 5 | 76.95 | 2.2 | 74.75 |
| 2025-03-08 | 5 | 71.09 | 0 | 71.09 |
| 2025-03-09 | 5 | 108.3 | 7.5 | 100.8 |
| 2025-03-10 | 6 | 99.55 | 1.7 | 97.85 |
| 2025-03-11 | 6 | 103.4 | 2 | 101.4 |
| 2025-03-12 | 3 | 46.7 | 0 | 46.7 |
| 2025-03-13 | 4 | 57.35 | 0.5 | 56.85 |
| 2025-03-14 | 5 | 91.35 | 7 | 84.35 |
| 2025-03-15 | 4 | 93.7 | 3.5 | 90.2 |
| 2025-03-16 | 4 | 52.4 | 1 | 51.4 |
| 2025-03-17 | 4 | 78.7 | 7.5 | 71.2 |
| 2025-03-18 | 6 | 133.3 | 2 | 131.3 |

共 1 页，第 1 页

```
SELECT Format(o.OrderDate, "yyyy-mm-dd") AS [Date], Count(o.OrderID) AS Orders, Sum(o.OrderPayment)
AS [Total Sales], Sum(o.OrderDiscount) AS [Total Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS
[Net Sales]
FROM Orders AS o
WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#
GROUP BY Format(o.OrderDate, "yyyy-mm-dd")
ORDER BY Format(o.OrderDate, "yyyy-mm-dd");
```

This page shows the daily sales report, aggregating and analyzing restaurant sales performance within a given time period. The timestamp of the current report is shown at the top (April 21, 2025, 19:12:50), and the table shows daily sales data by date with the following key fields: Date, Orders, Total Sales, Total Discounts, and Net Sales.

The "Orders" column takes the number of orders completed on the day; "Total Sales" shows revenue from the orders; "Total Discounts" is the total worth of discounts provided; and "Net Sales" shows actual revenue from sales (Total Sales - Discounts). This data allows the management to clearly understand sales activity by day along with the impact of discounts.

For example, on March 3, 2025, the restaurant completed 7 orders, a total sale of ¥110.24, discount of ¥5.8, and net sale of ¥104.44. Checking sales trends, estimating the effect of promotions, and being a valuable reference for financial settlement and operation analysis.

## Monthly Customer Report

Monthly customer report    2025年4月21日 19:13:12

| Customer ID | Customer Name | Total Spending | Visit Count | Dine-in Count | Most Ordered Dish | Customer Value |
|---|---|---|---|---|---|---|
| C002 | Fang Wang | 5.6 | 1 | 1 | M04 | 5.6 |
| C004 | Na Liu | 16.9 | 1 | 1 | M10 | 16.9 |
| C005 | Qiang Chen | 8.99 | 1 | 1 | M01 | 8.99 |
| C007 | Xin Zhao | 10.5 | 1 | 1 | M04 | 10.5 |
| C007 | Xin Zhao | 15.8 | 1 | 1 | M07 | 15.8 |
| C008 | Lin Huang | 29.1 | 1 | 1 | M16 | 29.1 |
| C009 | Hao Zhou | 21.2 | 1 | 1 | M15 | 21.2 |
| C011 | Lei Guo | 12.7 | 1 | 1 | M05 | 12.7 |
| C011 | Lei Guo | 26.3 | 1 | 1 | M14 | 26.3 |
| C012 | Li Ma | 28.4 | 1 | 1 | M14 | 28.4 |
| C013 | Qiang Lin | 13.4 | 1 | 1 | M08 | 13.4 |
| C015 | Peng Xu | 17.1 | 1 | 1 | M13 | 17.1 |
| C017 | Ming Hu | 26.2 | 1 | 1 | M16 | 26.2 |
| C018 | Yan Gao | 6.5 | 1 | 1 | M19 | 6.5 |
| C021 | Jie Song | 13.1 | 1 | 0 | M08 | 13.1 |
| C021 | Jie Song | 16.5 | 1 | 1 | M10 | 16.5 |
| C022 | Fei Han | 6 | 1 | 1 | M02 | 6 |
| C022 | Fei Han | 28 | 1 | 1 | M20 | 28 |
| C023 | Tao Feng | 21.1 | 1 | 1 | M02 | 21.1 |
| C024 | Juan Deng | 23.3 | 1 | 1 | M12 | 23.3 |
| C026 | Min Cheng | 6.3 | 1 | 1 | M02 | 6.3 |
| C027 | Bo Yuan | 12.1 | 1 | 1 | M05 | 12.1 |
| C028 | Li Jiang | 18.7 | 1 | 1 | M15 | 18.7 |
| C028 | Li Jiang | 23.9 | 1 | 1 | M19 | 23.9 |

SELECT c.CustomerID AS [Customer ID], c.CustomerFirstName & " " & c.CustomerLastName AS [Customer Name], Sum(o.OrderPayment) AS [Total Spending], Count(o.OrderID) AS [Visit Count], Sum(IIf(o.Note <> "Takeaway" OR o.Note IS NULL, 1, 0)) AS [Dine-in Count], o.MenuItemID AS [Most Ordered Dish], Sum(o.OrderPayment)/Count(o.OrderID) AS [Customer Value]
FROM customer AS c INNER JOIN Orders AS o ON c.CustomerID = o.CustomerID
WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#
GROUP BY c.CustomerID, c.CustomerFirstName, c.CustomerLastName, o.MenuItemID
ORDER BY Count(o.OrderID) DESC , c.CustomerID;

Monthly spend report of every customer's behavior and value is displayed on this page. At the top, there is the timestamp of the current report (April 21, 2025, 19:13:12). The table displays customer details such as Customer ID, Name, Total Spend, Visit Count, Dine-in Count, Most Ordered Dish, and Customer Value.
"Total Spending" reflects the customer's monthly total expenditure; "Visit Count" and "Dine-in Count" reflect the total number of visits and dine-in visits, respectively, to capture customer activity; "Most Ordered Dish" reflects the customer's most ordered dish; "Customer Value" is reflected in total spending, wherein the customer's economic value is determined.
The table helps the management identify loyal customers, tailor marketing activities, and menu optimization. Repeated customer IDs (e.g., C011, C021, C022, C028) ensure that the system accumulates multiple orders to ensure full behavior analysis. This page is crucial for customer relationship management and performance analysis.

## Monthly Sales Report

Monthly Sales Report

2025年4月21日
19:13:26

| Month | Orders | Total Sales | Total Discounts | Net Sales | Average Order Value |
|---|---|---|---|---|---|
| 2025-03 | 89 | 1507.03 | 41.2 | 1465.83 | 16.9329213483146 |

```
SELECT Format(o.OrderDate, "yyyy-mm") AS [Month], Count(o.OrderID) AS Orders, Sum(o.OrderPayment)
AS [Total Sales], Sum(o.OrderDiscount) AS [Total Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS
[Net Sales], Sum(o.OrderPayment)/Count(o.OrderID) AS [Average Order Value]
FROM Orders AS o
WHERE o.OrderDate Between #3/1/2025# AND #3/18/2025#
GROUP BY Format(o.OrderDate, "yyyy-mm")
ORDER BY Format(o.OrderDate, "yyyy-mm");
```

This page indicates the monthly summary of sales for the restaurant, with April 21, 2025, 19:13:26 as the data generated. The table indicates six significant fields: Month, Orders, Total Sales, Total Discounts, Net Sales, and Average Order Value.

For March 2025, the restaurant completed 89 orders with a total sale of ¥1507.03, total discount of ¥41.2, net sales of ¥1465.83, and an average order value of approximately ¥16.93. The report allows the management to compare monthly performance, analyze the impact of promotions on revenue, and provide insights for business strategy in the future. Data is aggregated from the order table on a monthly basis, which is suitable for tracking revenues and financial analysis.

## Personnel Management

Personnel management system

2025年4月21日
19:14:18

| Employee ID | Employee Name | Position | Shift Date | Working Hours | Contact Info | Attendance Status |
|---|---|---|---|---|---|---|
| S01 | Wei Chen | Waiter | 2025/3/1 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/2 | 7 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/3 | 8.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/5 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/8 | 9 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/11 | 7.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/15 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/17 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/22 | 8.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/26 | 7.5 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/3/31 | 8 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/4/5 | 9 | 13812345678 | Present |
| S01 | Wei Chen | Waiter | 2025/4/9 | 8 | 13812345678 | Present |
| S02 | Mei Li | Cashier | 2025/3/1 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/3 | 7.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/6 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/10 | 8.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/13 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/18 | 7.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/24 | 8 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/3/29 | 8.5 | 13987654321 | Present |
| S02 | Mei Li | Cashier | 2025/4/3 | 8 | 13987654321 | Present |

```
SELECT sw.StaffID AS [Employee ID], s.StaffFirstName & " " & s.StaffLastName AS [Employee Name],
s.StaffPosition AS [Position], sw.WorkDate AS [Shift Date], sw.StaffWorkHours AS [Working Hours],
s.StaffPhone AS [Contact Info], 'Present' AS [Attendance Status], '' AS [Shift Sales], '' AS Feedback, '' AS Rating
FROM staff AS s INNER JOIN [staff working] AS sw ON s.StaffID = sw.StaffID
ORDER BY sw.WorkDate DESC , s.StaffLastName;
```

This page monitors and manages employee attendance and shift data, showing each employee's attendance and working hours on a daily basis. The date and time of the present report (April 21, 2025, 19:14:18) are shown at the top. Employee attendance data are shown in the table by Employee ID, Name, Job Title, Shift Date, Working Hours, Contact Information, and Attendance Status.

For example, Wei Chen (Waiter) and Mei Li (Cashier) are marked with their respective working hours on various dates. Wei Chen worked 8.5 hours on March 3, 2025, with the attendance status "Present."

This page makes it easy to handle employee attendance data in an organized manner, helping to track work hours, calculate payroll, and schedule human resources. Data is derived from the personnel management system, referred to against the employee and shift tables, to ensure real-time accuracy.

## Raw Material Availability Report

Raw material availability report

2025年4月21日 19:18:00

| Order ID | Supplier | Supplied Ingredient | Ingredient Specification | Quantity | Unit | Unit Price (CNY) | Notes |
|---|---|---|---|---|---|---|---|
| 1021 | | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1219 | | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1163 | | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1070 | | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1133 | | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1196 | | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1100 | | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1149 | | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1058 | | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1001 | | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1171 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1083 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1026 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1207 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1003 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1002 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1114 | | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |

```
SELECT o.OrderID AS [Order ID], i.InventoryName AS [Supplied Ingredient], i.Description AS [Ingredient Specification], i.InventoryQuantity AS Quantity, i.Unit AS Unit, (i.InventoryAmount / i.InventoryQuantity) AS [Unit Price (CNY)], i.Description AS Notes
FROM Orders AS o INNER JOIN inventory AS i ON i.OrderDate = o.OrderDate;
```

Here is where raw material availability is displayed, which precedes each ingredient's supply history. The report timestamp as of today (April 21, 2025, 19:18:00) is displayed. The table displays raw material purchase data by order, Order ID, Supplier, Supplied Ingredient, Ingredient Specification, Quantity, Unit, Unit Price, and Notes. For example, "Beef Patty (Frozen, 4oz)" was purchased 5 times with 500 units for each purchase at ¥0.5 per unit; "Cheddar Cheese Slices (Stack)" was purchased 1000 units at ¥0.09 per unit; "Onion (Bag, Yellow)" was purchased in batches of 10 kg at ¥2.5 per kg.".

This report helps in monitoring the ingredient sourcing, specs, cost, and batches so that efficient supply chain management can be maintained. The data is used against the inventory table to update inventory, calculate the cost, and audit the purchase and hence is an integral part of procurement and warehousing management.

## Raw Material Used Report

Raw material used report

2025年4月21日
19:18:27

| Order ID | Supplied Ingredient | Ingredient Specification | Quantity | Unit | Unit Price (CNY) | Notes |
|---|---|---|---|---|---|---|
| 1021 | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1219 | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1163 | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1070 | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1133 | Beef Patty (Frozen, 4oz) | Frozen beef patties, 4oz each | 500 | Piece | 0.5 | Frozen beef patties, 4oz each |
| 1196 | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1100 | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1149 | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1058 | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1001 | Cheddar Cheese Slices (Stack) | Individually wrapped slices | 1000 | Piece | 0.09 | Individually wrapped slices |
| 1171 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1083 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1026 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1207 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1003 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1002 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |
| 1114 | Onion (Bag, Yellow) | Fresh yellow onions | 10 | Kg | 2.5 | Fresh yellow onions |

```
PARAMETERS [StartDate] DateTime, [EndDate] DateTime;
SELECT Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd") AS [Week Start],
Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-mm-dd") AS [Week End],
Count(o.OrderID) AS Orders, Sum(o.OrderPayment) AS [Total Sales], Sum(o.OrderDiscount) AS [Total
Discounts], Sum(o.OrderPayment - o.OrderDiscount) AS [Net Sales]
```

This page tracks and displays raw material usage in the restaurant, allowing management to monitor daily usage. The timestamp of the current report is displayed (April 21, 2025, 19:18:27). The table displays raw material usage records by order, including Order ID, Supplied Ingredient, Ingredient Specification, Quantity, Unit, Unit Price, and Notes.

Routine items like "Beef Patty (Frozen, 4oz)", "Cheddar Cheese Slices (Stack)", and "Onion (Bag, Yellow)" include quantities and unit prices. As an example, 500 frozen beef patties were eaten at ¥0.5 each and yellow onions were eaten in 10 kg lots at ¥2.5 per kg.

This page helps to control ingredient costs, analyze turnover of materials, and synchronize inventories. It is linked with the outbound management module to provide correct material use records,

## Weekly Sales Report

Weekly Sales Report

2025年4月21日
19:20:15

| Week End | Orders | Total Sales | Total Discounts | Net Sales |
|---|---|---|---|---|
| 2025-03-01 | 5 | 70.3 | 0 | 70.3 |
| 2025-03-08 | 37 | 571.98 | 8.5 | 563.48 |
| 2025-03-15 | 33 | 600.35 | 22.2 | 578.15 |
| 2025-03-22 | 35 | 650.35 | 24.45 | 625.9 |
| 2025-03-29 | 33 | 580.8 | 16.2 | 564.6 |
| 2025-04-05 | 16 | 341.1 | 10.3 | 330.8 |

```
FROM Orders AS o
WHERE o.OrderDate Between [StartDate] AND [EndDate]
GROUP BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd"),
Format(DateAdd("d", 7 - Weekday(o.OrderDate, 1), o.OrderDate), "yyyy-mm-dd")
ORDER BY Format(DateAdd("d", -Weekday(o.OrderDate, 1) + 1, o.OrderDate), "yyyy-mm-dd");
```

This page displays the weekly sales summary of the restaurant, allowing management to track weekly performance. The timestamp of the report is now shown (April 21, 2025, 19:20:15). Weekly sales information is tabulated by Weekend, showing Orders, Total Sales, Total Discounts, and Net Sales per week.

For instance, for the week of March 22, 2025, 35 orders were completed, totaling ¥650.35 sales, discounts of ¥24.45, and a net worth of ¥625.9 sales.

This report contributes to weekly performance measurement, tracking promotional campaign success, and providing valuable information for operational and financial decision-making. The information is rolled up weekly from the order table and is essential to routine performance review and financial operations.

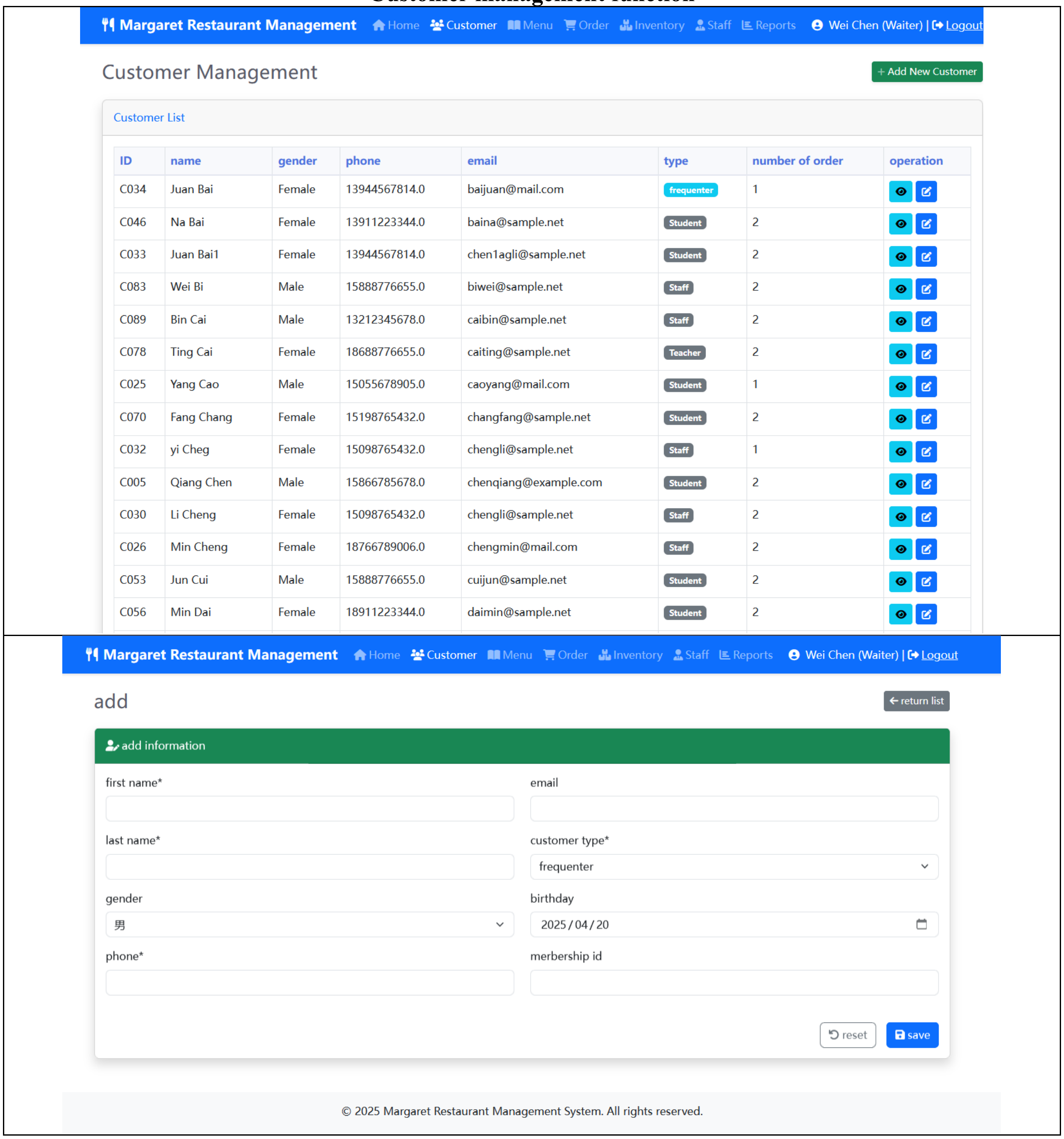
# IV. Introduction to Margaret's Graphical Operating System

In order to enable users to use our information system more easily, based on the access database, we have built a brand-new graphical operation program. This program can achieve more advanced data management functions by reading and analyzing the information of the access database.

## Basic management function

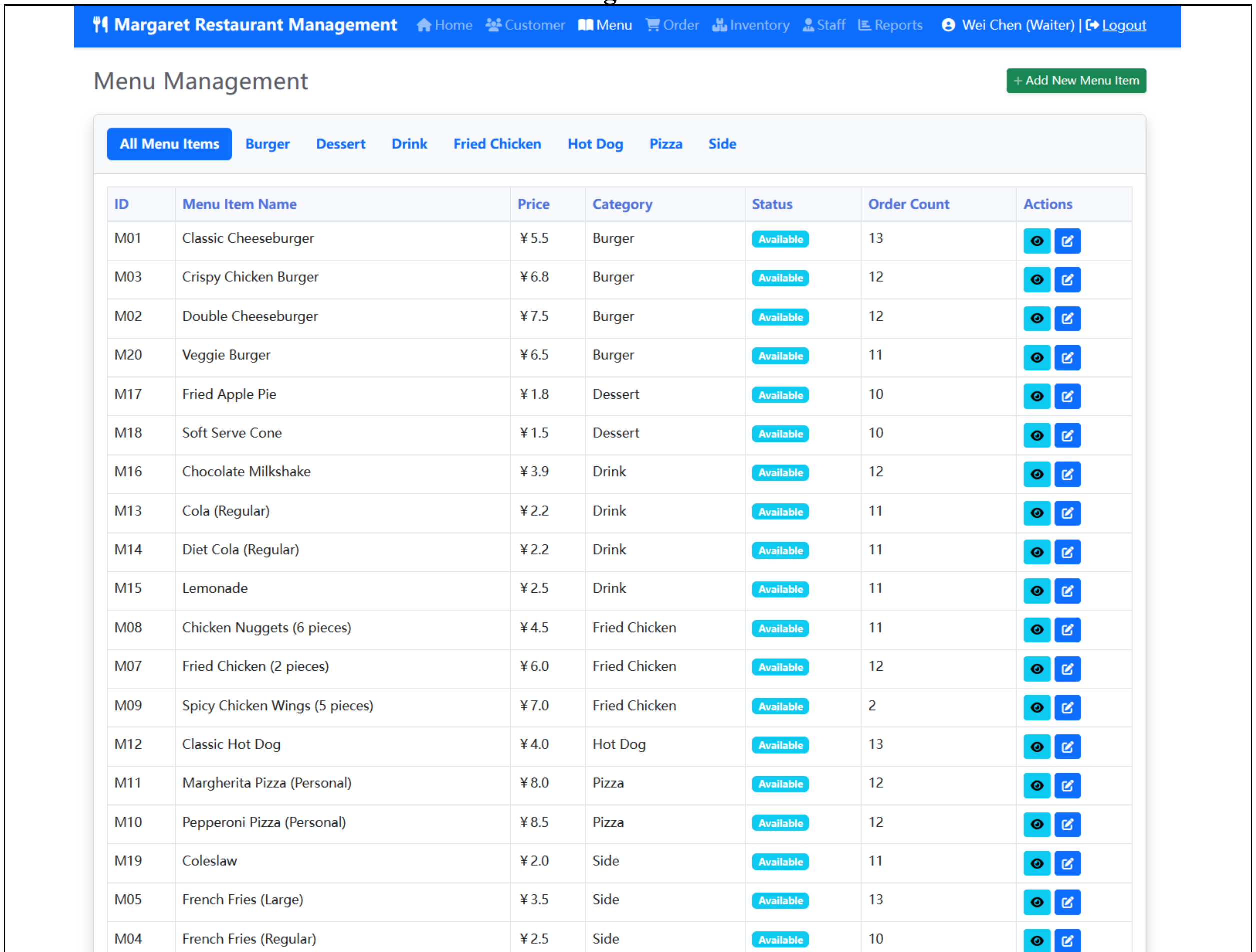
These functions are similar to the integration of table, form and report in access. Users can manage data in these interfaces, such as adding, modifying, querying and filtering. They conform to modern application design and can be easily used by users.

**Customer management function**

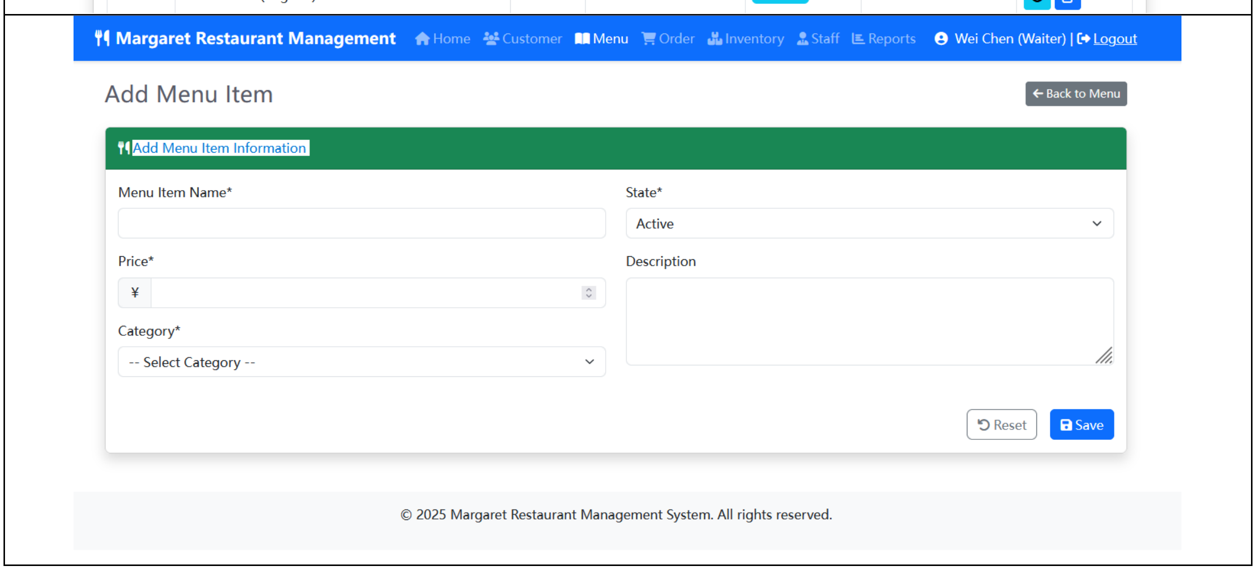



| ID | name | gender | phone | email | type | number of order | operation |
|---|---|---|---|---|---|---|---|
| C034 | Juan Bai | Female | 13944567814.0 | baijuan@mail.com | frequenter | 1 | |
| C046 | Na Bai | Female | 13911223344.0 | baina@sample.net | Student | 2 | |
| C033 | Juan Bai1 | Female | 13944567814.0 | chen1agli@sample.net | Student | 2 | |
| C083 | Wei Bi | Male | 15888776655.0 | biwei@sample.net | Staff | 2 | |
| C089 | Bin Cai | Male | 13212345678.0 | caibin@sample.net | Staff | 2 | |
| C078 | Ting Cai | Female | 18688776655.0 | caiting@sample.net | Teacher | 2 | |
| C025 | Yang Cao | Male | 15055678905.0 | caoyang@mail.com | Student | 1 | |
| C070 | Fang Chang | Female | 15198765432.0 | changfang@sample.net | Student | 2 | |
| C032 | yi Cheg | Female | 15098765432.0 | chengli@sample.net | Staff | 1 | |
| C005 | Qiang Chen | Male | 15866785678.0 | chenqiang@example.com | Student | 2 | |
| C030 | Li Cheng | Female | 15098765432.0 | chengli@sample.net | Staff | 2 | |
| C026 | Min Cheng | Female | 18766789006.0 | chengmin@mail.com | Staff | 2 | |
| C053 | Jun Cui | Male | 15888776655.0 | cuijun@sample.net | Student | 2 | |
| C056 | Min Dai | Female | 18911223344.0 | daimin@sample.net | Student | 2 | |

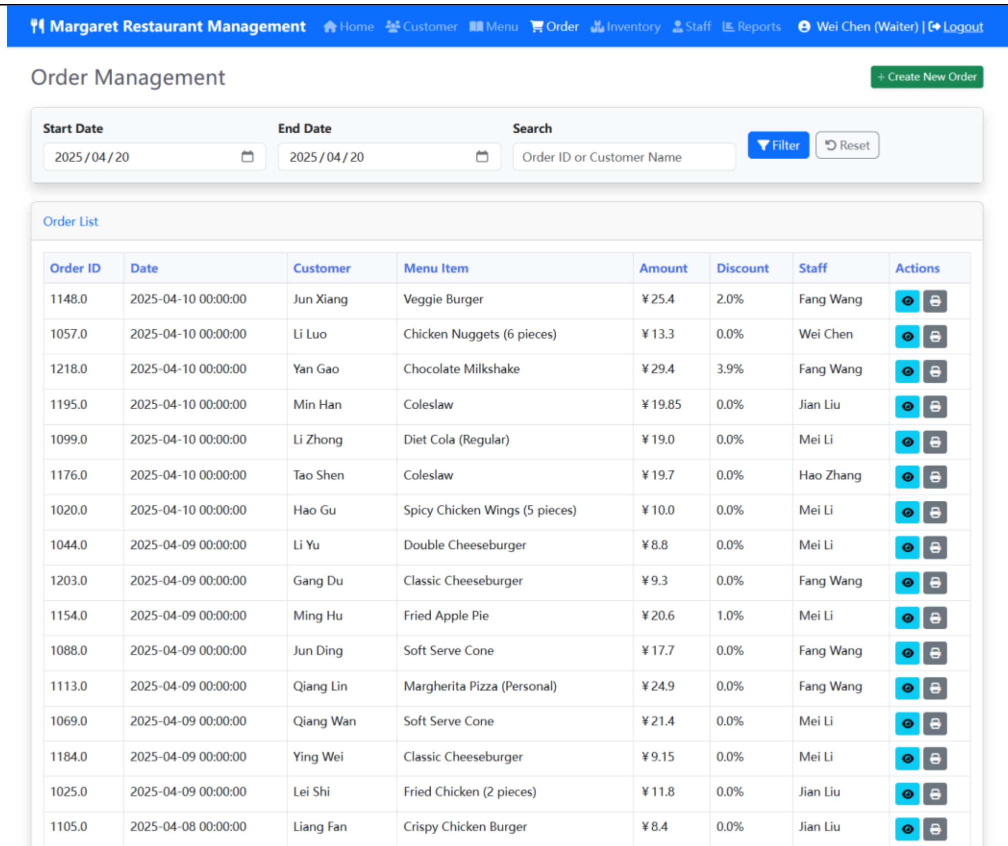
## Menu management function



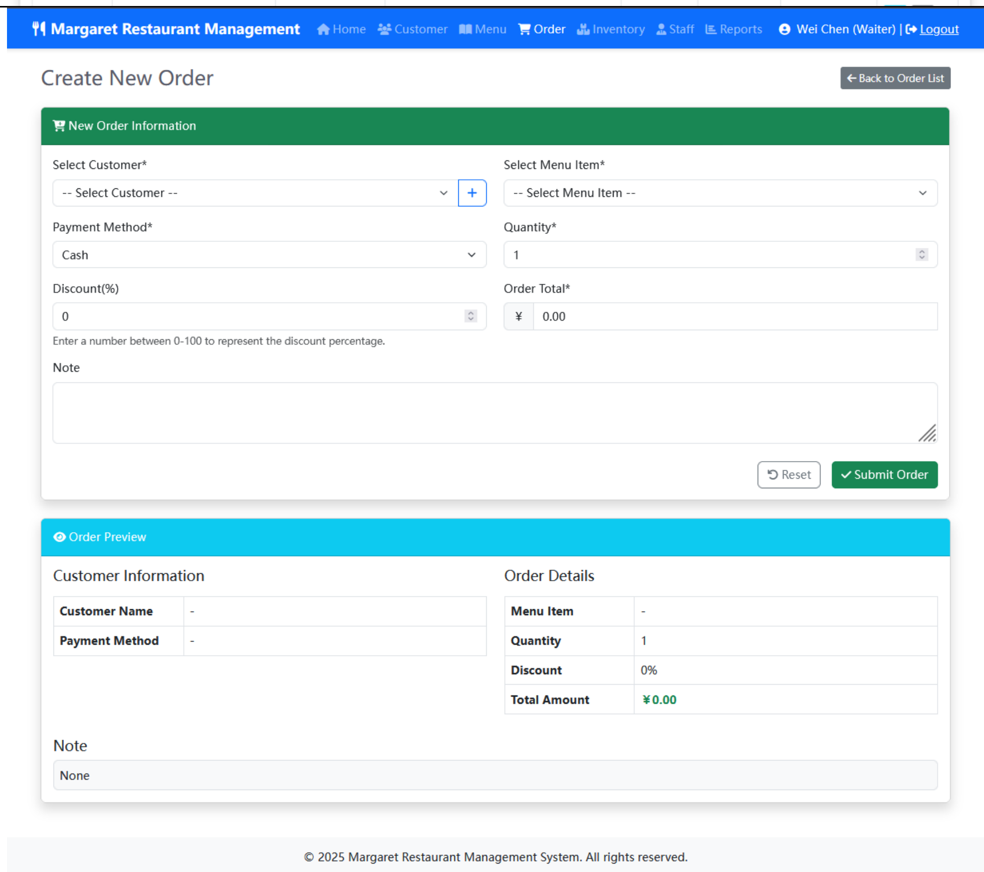

| ID | Menu Item Name | Price | Category | Status | Order Count | Actions |
|---|---|---|---|---|---|---|
| M01 | Classic Cheeseburger | ¥5.5 | Burger | Available | 13 | |
| M03 | Crispy Chicken Burger | ¥6.8 | Burger | Available | 12 | |
| M02 | Double Cheeseburger | ¥7.5 | Burger | Available | 12 | |
| M20 | Veggie Burger | ¥6.5 | Burger | Available | 11 | |
| M17 | Fried Apple Pie | ¥1.8 | Dessert | Available | 10 | |
| M18 | Soft Serve Cone | ¥1.5 | Dessert | Available | 10 | |
| M16 | Chocolate Milkshake | ¥3.9 | Drink | Available | 12 | |
| M13 | Cola (Regular) | ¥2.2 | Drink | Available | 11 | |
| M14 | Diet Cola (Regular) | ¥2.2 | Drink | Available | 11 | |
| M15 | Lemonade | ¥2.5 | Drink | Available | 11 | |
| M08 | Chicken Nuggets (6 pieces) | ¥4.5 | Fried Chicken | Available | 11 | |
| M07 | Fried Chicken (2 pieces) | ¥6.0 | Fried Chicken | Available | 12 | |
| M09 | Spicy Chicken Wings (5 pieces) | ¥7.0 | Fried Chicken | Available | 2 | |
| M12 | Classic Hot Dog | ¥4.0 | Hot Dog | Available | 13 | |
| M11 | Margherita Pizza (Personal) | ¥8.0 | Pizza | Available | 12 | |
| M10 | Pepperoni Pizza (Personal) | ¥8.5 | Pizza | Available | 12 | |
| M19 | Coleslaw | ¥2.0 | Side | Available | 11 | |
| M05 | French Fries (Large) | ¥3.5 | Side | Available | 13 | |
| M04 | French Fries (Regular) | ¥2.5 | Side | Available | 10 | |



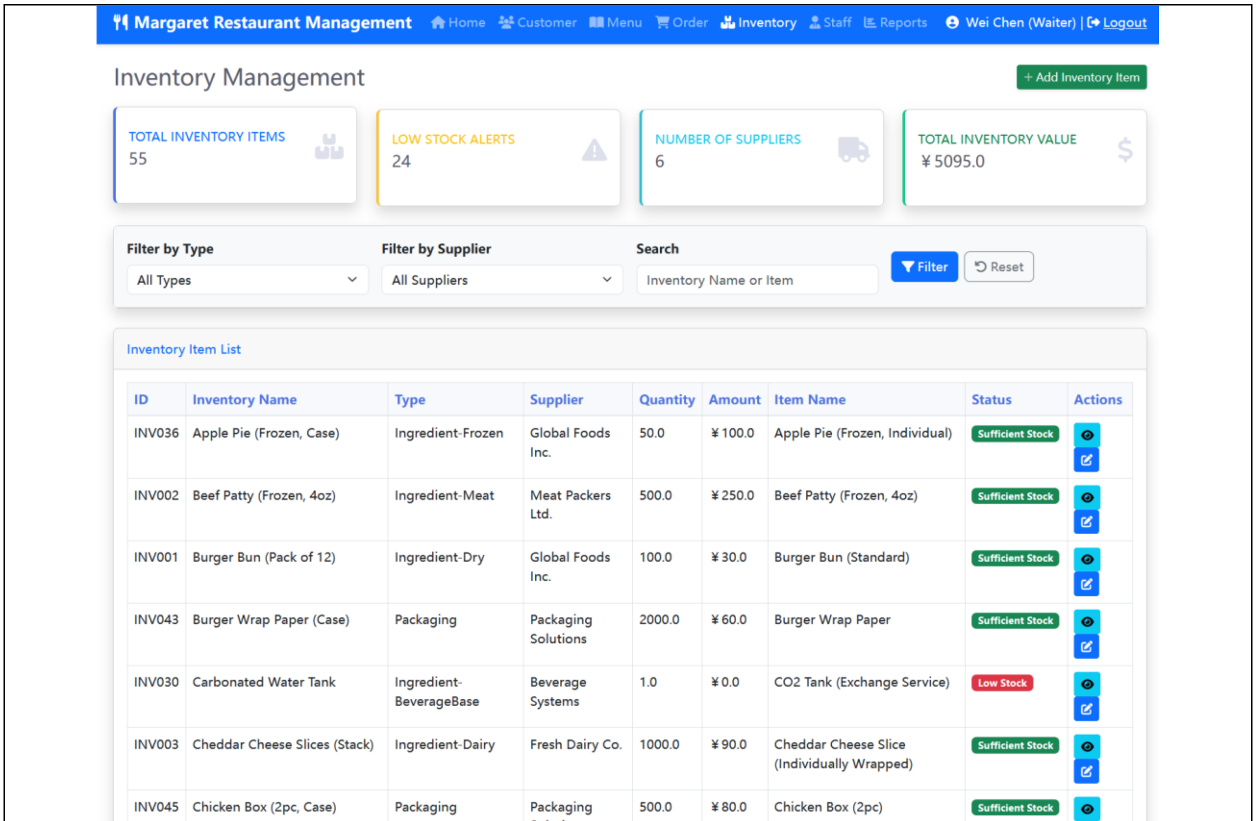

## Order management function

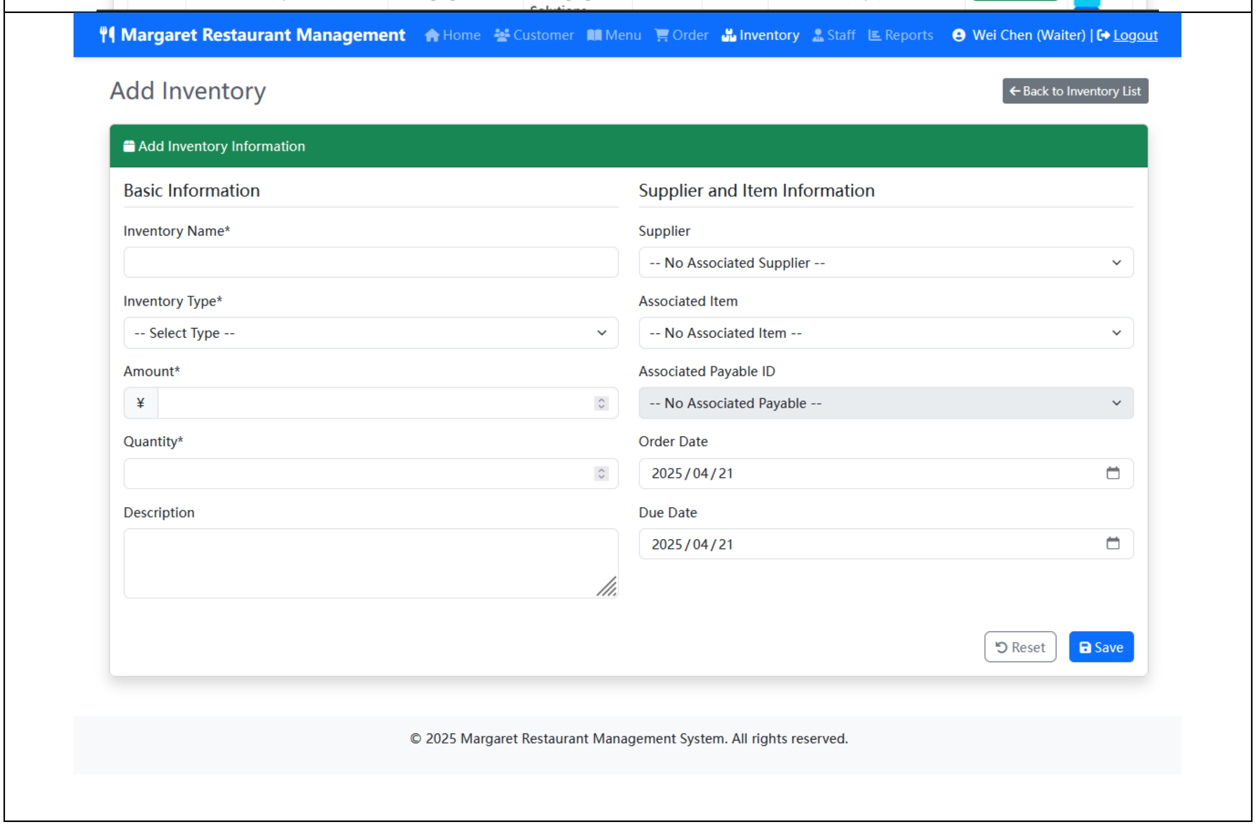



| Order ID | Date | Customer | Menu Item | Amount | Discount | Staff | Actions |
|---|---|---|---|---|---|---|---|
| 1148.0 | 2025-04-10 00:00:00 | Jun Xiang | Veggie Burger | ¥25.4 | 2.0% | Fang Wang | |
| 1057.0 | 2025-04-10 00:00:00 | Li Luo | Chicken Nuggets (6 pieces) | ¥13.3 | 0.0% | Wei Chen | |
| 1218.0 | 2025-04-10 00:00:00 | Yan Gao | Chocolate Milkshake | ¥29.4 | 3.9% | Fang Wang | |
| 1195.0 | 2025-04-10 00:00:00 | Min Han | Coleslaw | ¥19.85 | 0.0% | Jian Liu | |
| 1099.0 | 2025-04-10 00:00:00 | Li Zhong | Diet Cola (Regular) | ¥19.0 | 0.0% | Mei Li | |
| 1176.0 | 2025-04-10 00:00:00 | Tao Shen | Coleslaw | ¥19.7 | 0.0% | Hao Zhang | |
| 1020.0 | 2025-04-10 00:00:00 | Hao Gu | Spicy Chicken Wings (5 pieces) | ¥10.0 | 0.0% | Mei Li | |
| 1044.0 | 2025-04-09 00:00:00 | Li Yu | Double Cheeseburger | ¥8.8 | 0.0% | Mei Li | |
| 1203.0 | 2025-04-09 00:00:00 | Gang Du | Classic Cheeseburger | ¥9.3 | 0.0% | Fang Wang | |
| 1154.0 | 2025-04-09 00:00:00 | Ming Hu | Fried Apple Pie | ¥20.6 | 1.0% | Mei Li | |
| 1088.0 | 2025-04-09 00:00:00 | Jun Ding | Soft Serve Cone | ¥17.7 | 0.0% | Fang Wang | |
| 1113.0 | 2025-04-09 00:00:00 | Qiang Lin | Margherita Pizza (Personal) | ¥24.9 | 0.0% | Fang Wang | |
| 1069.0 | 2025-04-09 00:00:00 | Qiang Wan | Soft Serve Cone | ¥21.4 | 0.0% | Mei Li | |
| 1184.0 | 2025-04-09 00:00:00 | Ying Wei | Classic Cheeseburger | ¥9.15 | 0.0% | Mei Li | |
| 1025.0 | 2025-04-09 00:00:00 | Lei Shi | Fried Chicken (2 pieces) | ¥11.8 | 0.0% | Jian Liu | |
| 1105.0 | 2025-04-08 00:00:00 | Liang Fan | Crispy Chicken Burger | ¥8.4 | 0.0% | Jian Liu | |

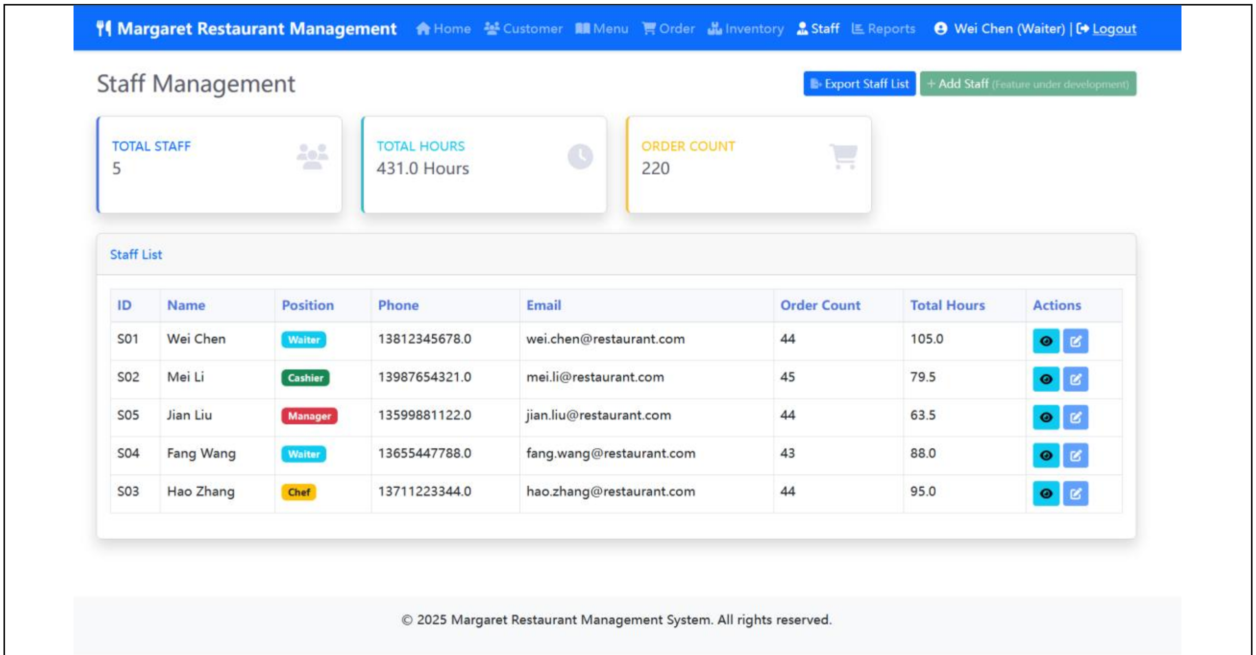


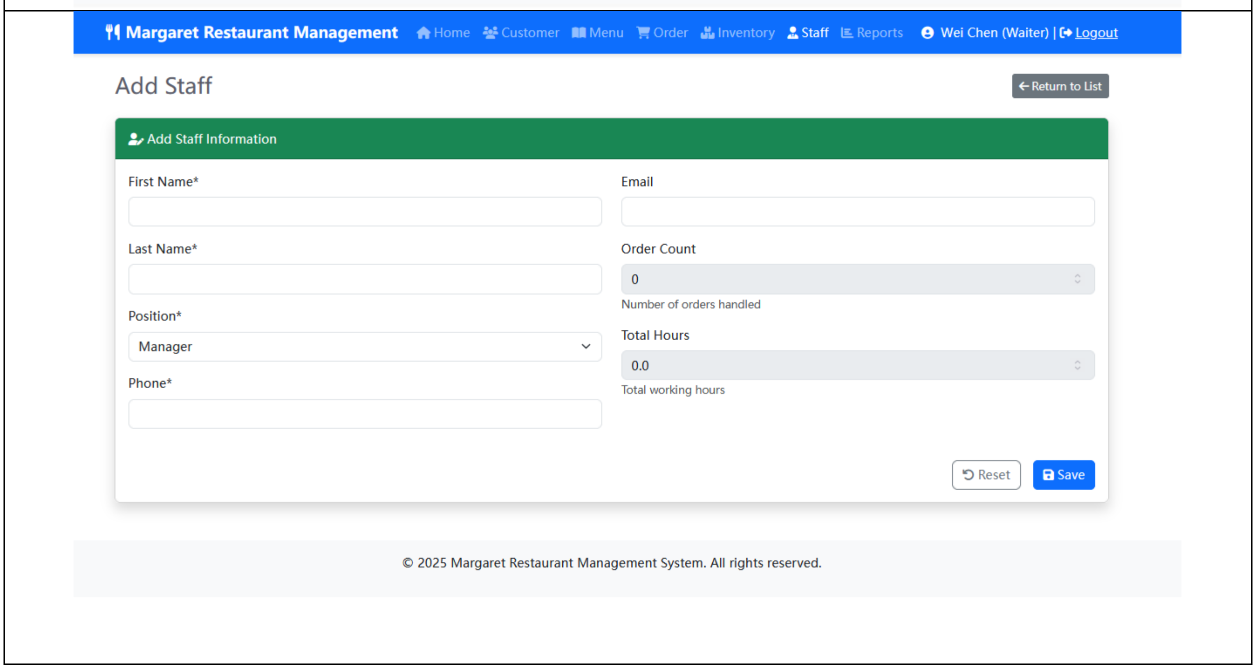

**Inventory management**






**Staff management**

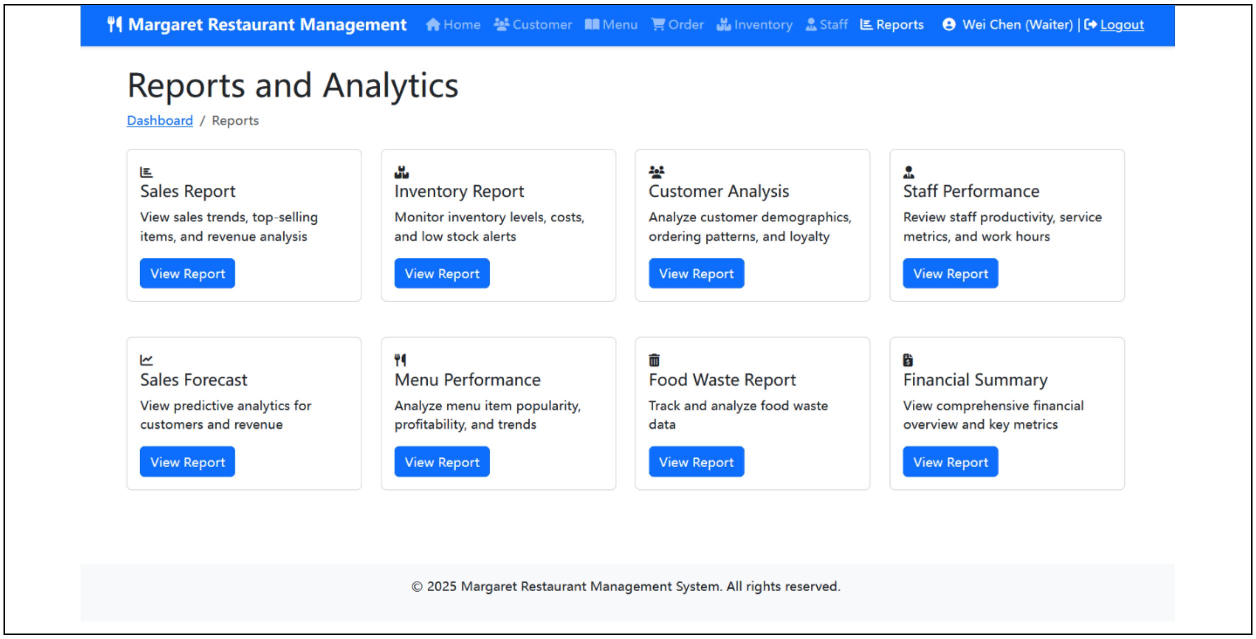



# Advanced data report

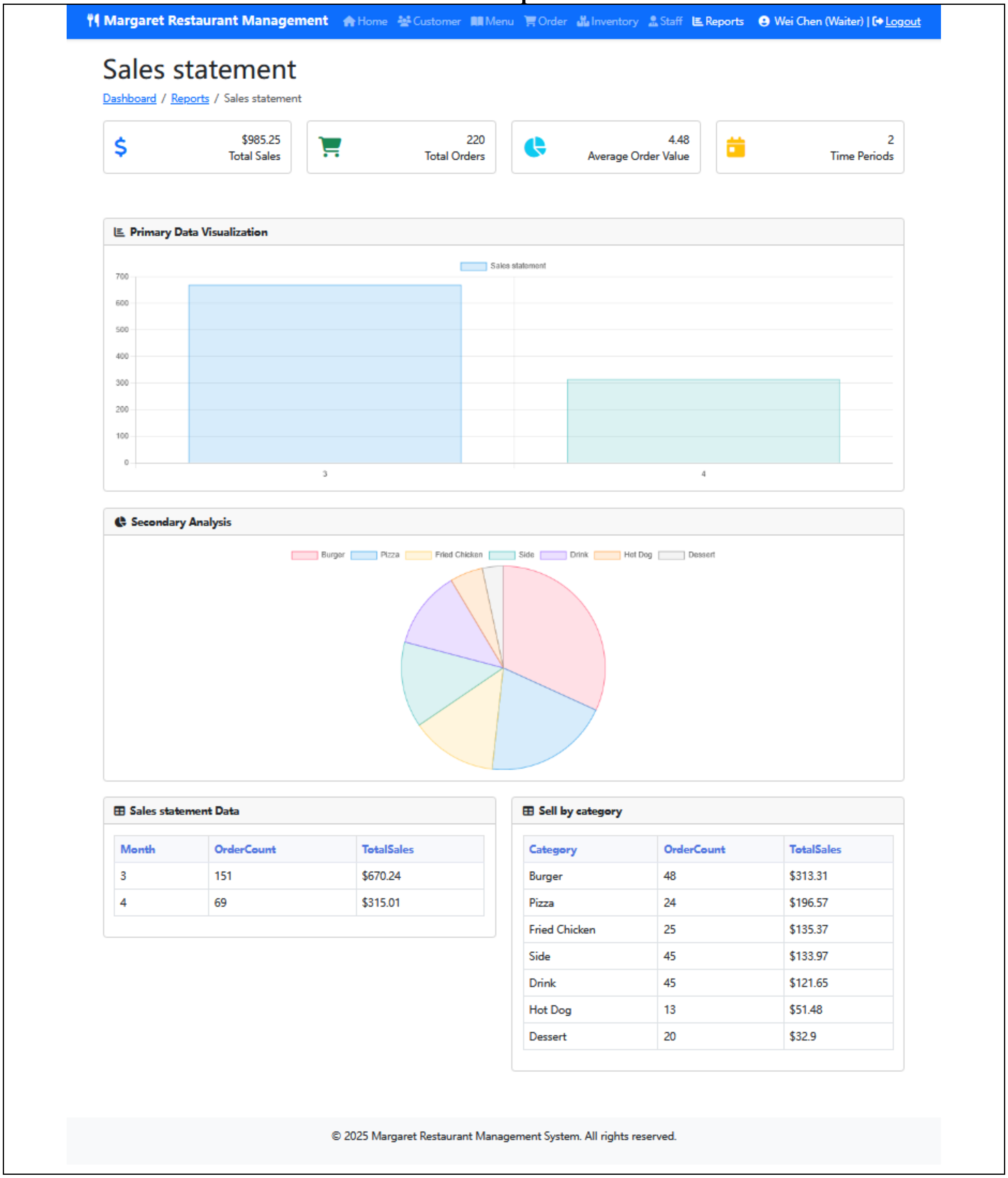
These are more advanced data analysis reports. They incorporate deeper insights into sales, customer, and inventory staff data. Meanwhile, they also combine graphical reports and AI functions such as sales forecasts. These reports can assist the management in making the next data decision.

**The Homepage of the Report**



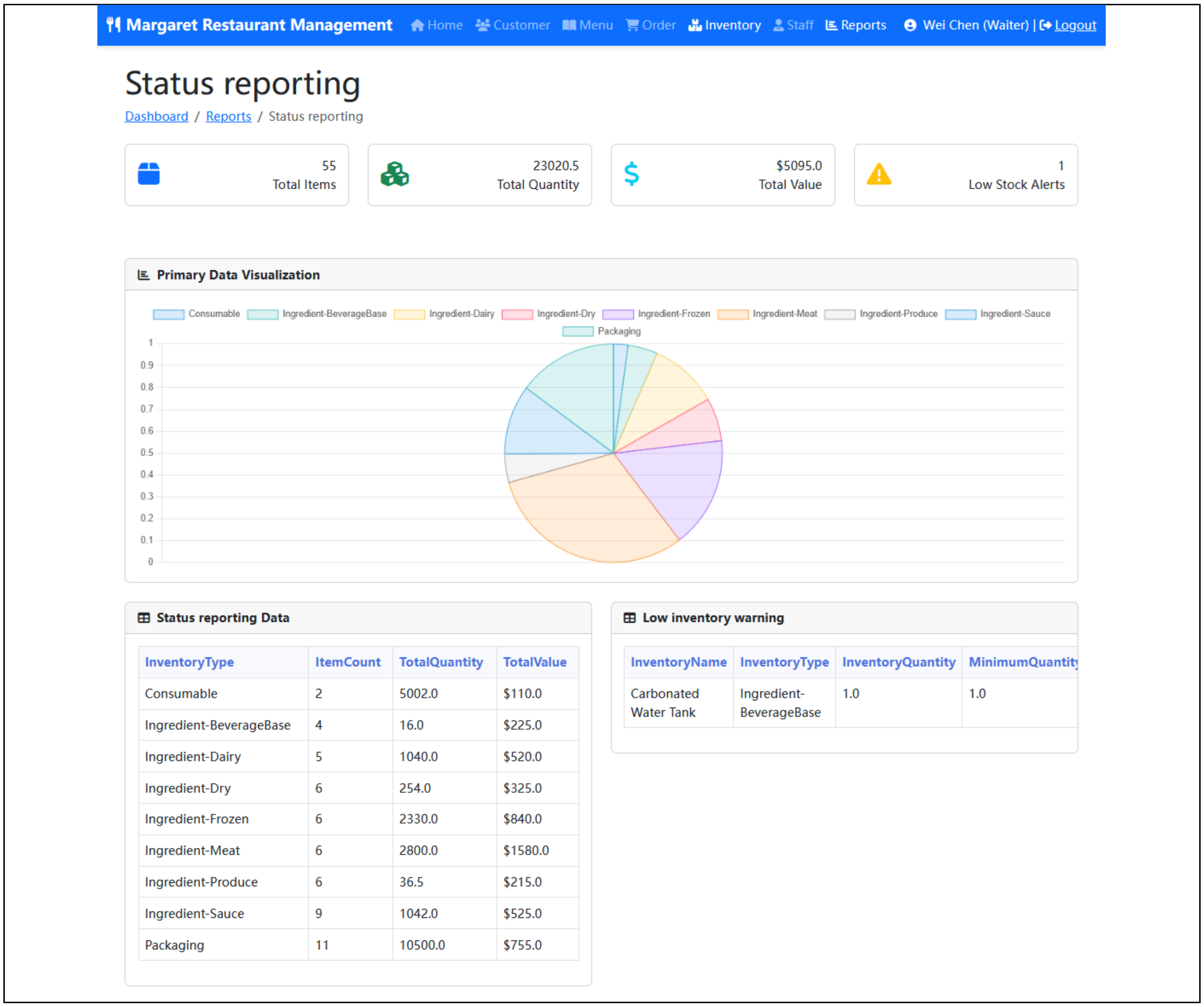
This page showcases various reports and analysis functions, helping managers conduct in-depth analysis and make decisions on the restaurant's operations. The page contains multiple visual report modules, each representing a different field of data analysis. The specific modules include Sales Report, Inventory Report, Customer Analysis, Staff Performance, Sales Forecast, Menu Performance, Food waste report and Financial Summary.

**Sales report**




The Sales Statement page presents the sales reports of the restaurant by combining bar charts, pie charts and detailed data tables, providing managers with a comprehensive sales analysis and trend view, helping them make more accurate decisions, optimize business strategies and enhance the overall profitability. Four key indicators are displayed at the top of the page: total sales, total number of orders, average order value and time period. These data provide managers with an overview of the overall operation of the restaurant, helping them understand the average revenue, total order volume and sales performance of each order, and conduct comparative analyses based on different time periods. The main part of the page contains the visualization of key data, presenting the sales trends for two time periods (March and April) through bar charts. The secondary analysis section uses a pie chart to show the sales proportion of each dish category. The sales data table at the bottom of the page lists the specific sales data for March and April, helping managers compare the performance of the two time periods. In addition, the sales table by category shows the specific sales data of each dish category.

**Inventory report**



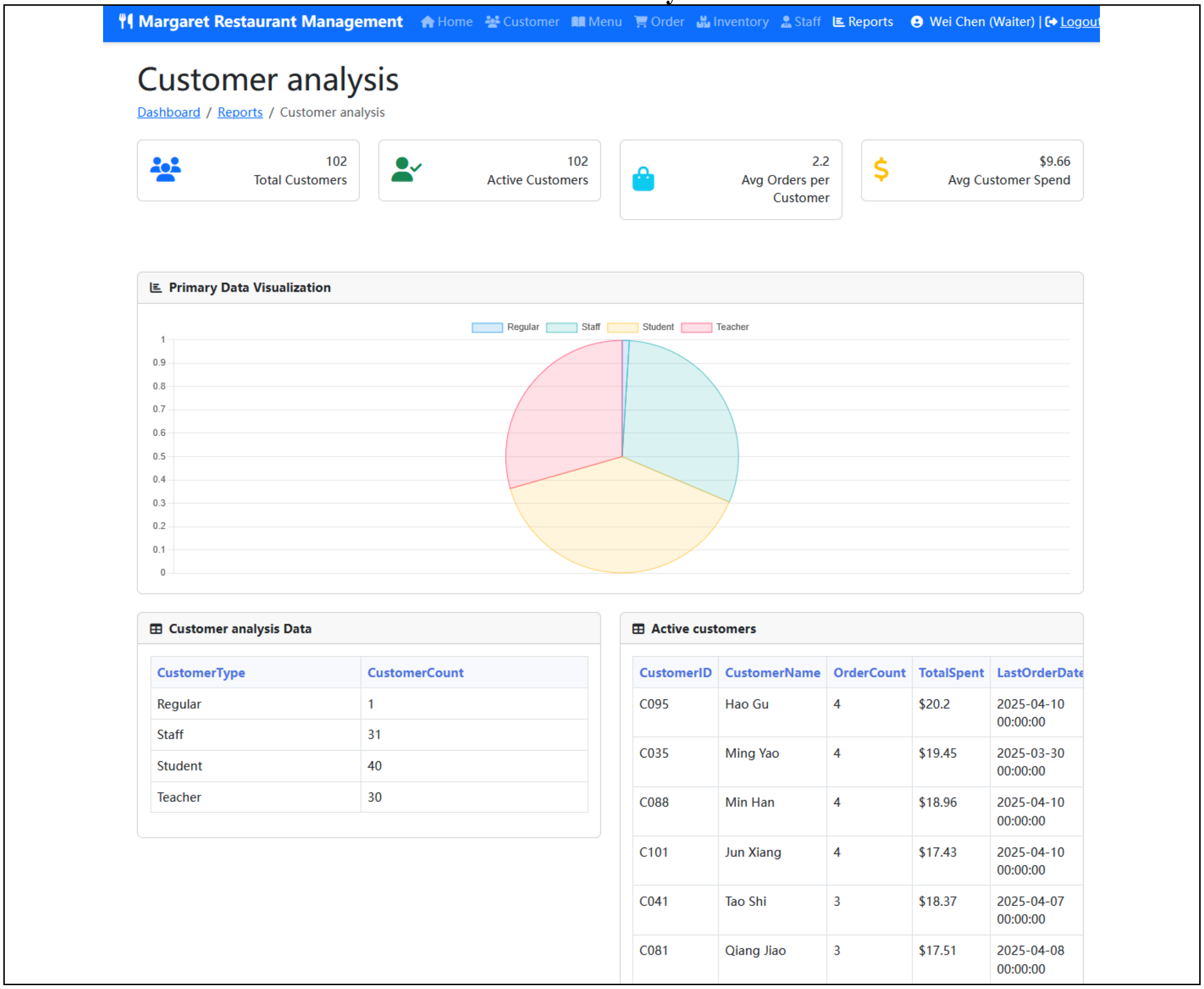

The Inventory report page helps managers effectively manage restaurant inventory and ensure smooth operations by providing detailed inventory information, visual charts and low inventory alerts.

Four key indicators are displayed at the top of the page: total inventory items, total inventory quantity, total inventory value, and low inventory alert. These data provide managers with an overview, helping them understand the overall situation of the inventory. The main data visualization part of the page uses pie charts to show the distribution of different types of inventories. Through these data, managers can intuitively see the proportion of different inventory types in the total inventory, thereby conducting effective inventory management and adjustment.

The inventory status data table below lists detailed information for each inventory type, including the inventory quantity, total value and inventory item quantity of each type. This information helps managers gain an in-depth understanding of the inventory status of each type to ensure adequate inventory.

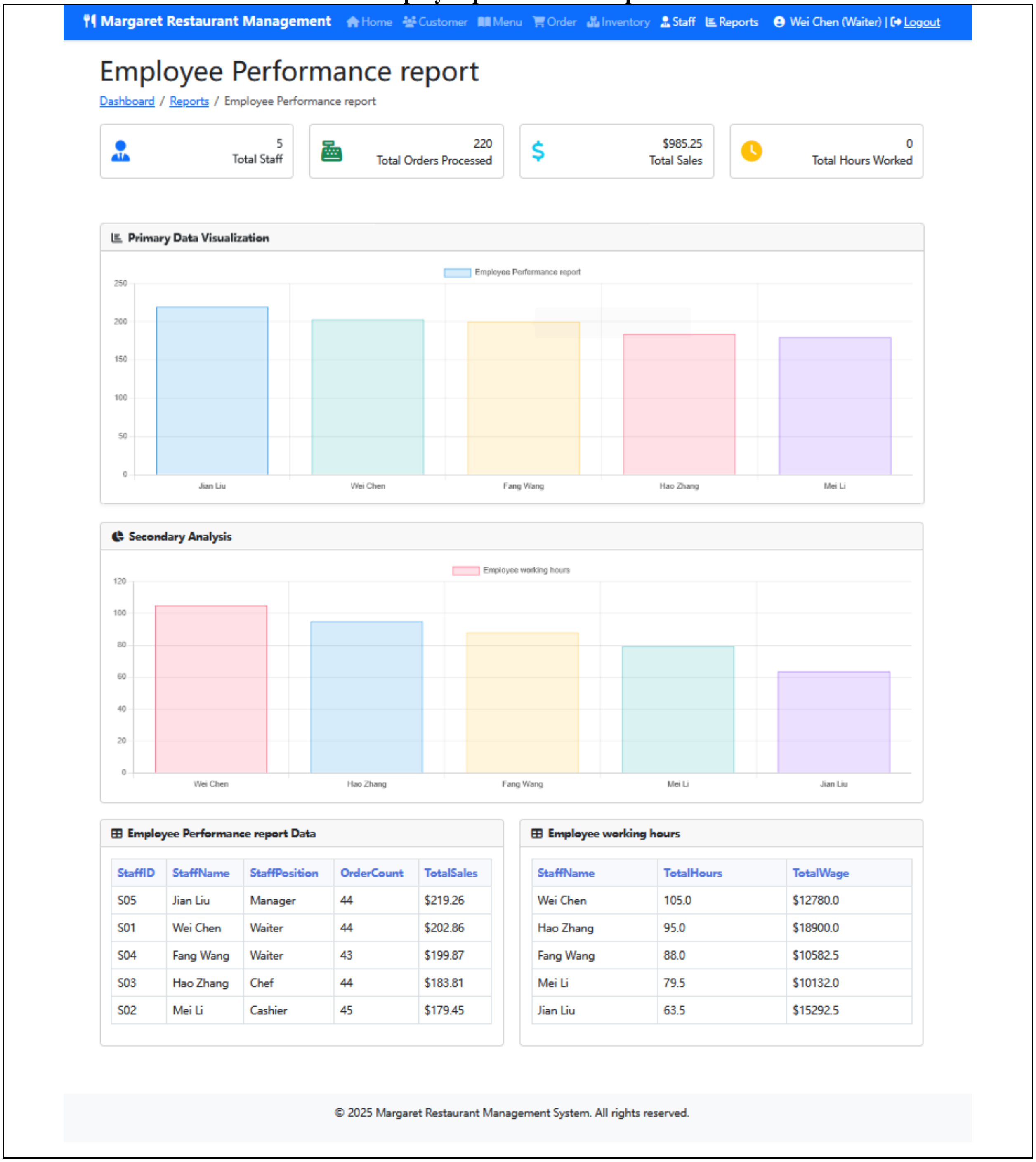
The low inventory warning section lists the items in the current inventory that are below the minimum inventory level, helping managers take replenishment measures in a timely manner. This section helps remind managers to replenish inventory in time and avoid stockouts.

**Customer analysis**




The Customer Analysis page helps restaurant managers fully understand customer behavior and optimize customer relationship management through data visualization, customer category statistics and active customer analysis. This page presents the customer analysis data of the restaurant, helping managers understand the basic information, consumption habits and activity levels of customers. Four key indicators are displayed at the top of the page: the total number of customers, the number of active customers, the average number of orders per customer, and the average consumption amount per customer. The main data visualization section of the page uses pie charts to show the distribution of different types of customers, including Regular customers, Staff, students, and teachers. Each category is displayed in a pie chart through blocks of different colors, helping managers intuitively understand the proportion of each customer category. The customer analysis data table below lists each type of customer and the corresponding number of customers. The Active Customers section shows the currently active customers and their detailed information, including customer ID, name, order quantity, total consumption amount, and the date of the last order. Managers can understand which customers are regulars of the restaurant and offer more personalized services or discounts to these customers.

**Employee performance report**




| StaffID | StaffName | StaffPosition | OrderCount | TotalSales |
|---|---|---|---|---|
| S05 | Jian Liu | Manager | 44 | $219.26 |
| S01 | Wei Chen | Waiter | 44 | $202.86 |
| S04 | Fang Wang | Waiter | 43 | $199.87 |
| S03 | Hao Zhang | Chef | 44 | $183.81 |
| S02 | Mei Li | Cashier | 45 | $179.45 |

| StaffName | TotalHours | TotalWage |
|---|---|---|
| Wei Chen | 105.0 | $12780.0 |
| Hao Zhang | 95.0 | $18900.0 |
| Fang Wang | 88.0 | $10582.5 |
| Mei Li | 79.5 | $10132.0 |
| Jian Liu | 63.5 | $15292.5 |

The Employee Performance Report page helps managers fully understand the job performance, contribution and efficiency of employees by providing data visualization, performance data tables and statistics of working hours, thereby optimizing employee management and incentive measures. This page presents the performance reports of the restaurant staff, helping managers evaluate the work performance and contributions of the employees. Four key indicators are provided at the top of the page: the total number of employees, the total number of orders processed, the total sales volume, and the total working hours. These data provide managers with an overall perspective, helping them understand the operation of the restaurant. The main data visualization section of the page presents the job performance of the employees through two bar charts. The first bar chart shows the order quantity and total sales volume of each employee. The employee performance report data table below lists detailed data for each employee, including employee ID, name, position, order quantity and total sales. The employee working hours table shows the total working hours and total salary of each employee.

**Orders and revenue forecasts**

Margaret Restaurant Management | Home | Customer | Menu | Order | Inventory | Staff | Reports | Wei Chen (Waiter) | Logout

# Orders and revenue forecasts

Dashboard / Reports / Orders and revenue forecasts

**Forecast Overview**

| Today's Orders | Today's Revenue | Tomorrow's Orders | Tomorrow's Revenue |
|---|---|---|---|
| 7 | $136.65 | 6.23 ↓ 11.0% | $112.00 ↓ 18.0% |

**Order Forecast**


**Revenue Forecast**


**Forecast Data**

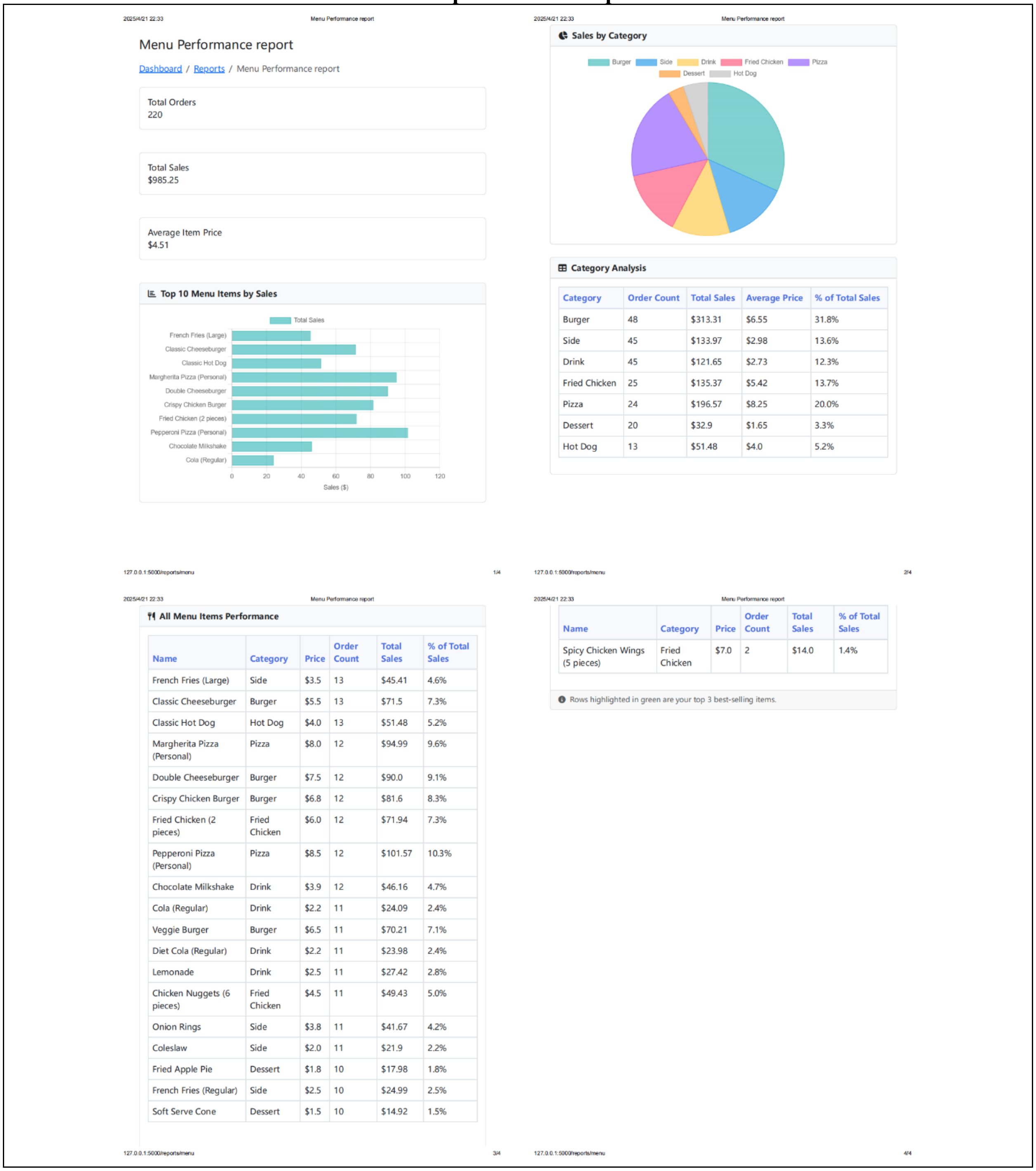
| Date | Order Count | Revenue | Data Type |
|---|---|---|---|
| 2025-04-11 | 6.23 | $112.00 | Forecast |
| 2025-04-12 | 5.24 | $92.10 | Forecast |
| 2025-04-13 | 5.6 | $104.40 | Forecast |
| 2025-04-14 | 5.58 | $105.70 | Forecast |
| 2025-04-15 | 6.04 | $105.10 | Forecast |
| 2025-04-16 | 6.11 | $98.60 | Forecast |
| 2025-04-17 | 6.76 | $116.40 | Forecast |
| 2025-04-04 | 6 | $80.20 | Historical |
| 2025-04-05 | 8 | $95.44 | Historical |
| 2025-04-06 | 7 | $128.80 | Historical |
| 2025-04-07 | 8 | $108.45 | Historical |
| 2025-04-08 | 7 | $110.95 | Historical |
| 2025-04-09 | 8 | $123.65 | Historical |
| 2025-04-10 | 7 | $136.65 | Historical |

Forecast data is calculated using advanced predictive algorithms based on historical trends. Results are estimates and may vary based on actual conditions.

The Orders and Revenue Forecasts page helps restaurant managers make effective predictions and plans for future operations by comparing predicted data with historical data. This page presents the restaurant's order and revenue forecasts, providing managers with expectations for orders and revenue in the coming days and helping them make more accurate operational decisions. The top of the page provides overview data, showing the current order quantity, revenue, and forecast data for tomorrow's orders and revenue.

The order prediction section of the page uses a line chart to show the changing trend between the historical order quantity and the predicted order quantity. In the chart, historical data is shown by solid lines, while predicted data is represented by dashed lines. The forecast data table below lists the order quantity, revenue and data types (forecast or historical data) for each date.

**Menu performance report**


2025/4/21 22:33 Menu Performance report

Menu Performance report

Dashboard / Reports / Menu Performance report

Total Orders
220

Total Sales
$985.25

Average Item Price
$4.51

Top 10 Menu Items by Sales



127.0.0.1:5000/reports/menu 1/4

2025/4/21 22:33 Menu Performance report

Sales by Category



Category Analysis

| Category | Order Count | Total Sales | Average Price | % of Total Sales |
|---|---|---|---|---|
| Burger | 48 | $313.31 | $6.55 | 31.8% |
| Side | 45 | $133.97 | $2.98 | 13.6% |
| Drink | 45 | $121.65 | $2.73 | 12.3% |
| Fried Chicken | 25 | $135.37 | $5.42 | 13.7% |
| Pizza | 24 | $196.57 | $8.25 | 20.0% |
| Dessert | 20 | $32.9 | $1.65 | 3.3% |
| Hot Dog | 13 | $51.48 | $4.0 | 5.2% |

127.0.0.1:5000/reports/menu 2/4

2025/4/21 22:33 Menu Performance report

All Menu Items Performance

| Name | Category | Price | Order Count | Total Sales | % of Total Sales |
|---|---|---|---|---|---|
| French Fries (Large) | Side | $3.5 | 13 | $45.41 | 4.6% |
| Classic Cheeseburger | Burger | $5.5 | 13 | $71.5 | 7.3% |
| Classic Hot Dog | Hot Dog | $4.0 | 13 | $51.48 | 5.2% |
| Margherita Pizza (Personal) | Pizza | $8.0 | 12 | $94.99 | 9.6% |
| Double Cheeseburger | Burger | $7.5 | 12 | $90.0 | 9.1% |
| Crispy Chicken Burger | Burger | $6.8 | 12 | $81.6 | 8.3% |
| Fried Chicken (2 pieces) | Fried Chicken | $6.0 | 12 | $71.94 | 7.3% |
| Pepperoni Pizza (Personal) | Pizza | $8.5 | 12 | $101.57 | 10.3% |
| Chocolate Milkshake | Drink | $3.9 | 12 | $46.16 | 4.7% |
| Cola (Regular) | Drink | $2.2 | 11 | $24.09 | 2.4% |
| Veggie Burger | Burger | $6.5 | 11 | $70.21 | 7.1% |
| Diet Cola (Regular) | Drink | $2.2 | 11 | $23.98 | 2.4% |
| Lemonade | Drink | $2.5 | 11 | $27.42 | 2.8% |
| Chicken Nuggets (6 pieces) | Fried Chicken | $4.5 | 11 | $49.43 | 5.0% |
| Onion Rings | Side | $3.8 | 11 | $41.67 | 4.2% |
| Coleslaw | Side | $2.0 | 11 | $21.9 | 2.2% |
| Fried Apple Pie | Dessert | $1.8 | 10 | $17.98 | 1.8% |
| French Fries (Regular) | Side | $2.5 | 10 | $24.99 | 2.5% |
| Soft Serve Cone | Dessert | $1.5 | 10 | $14.92 | 1.5% |

127.0.0.1:5000/reports/menu 3/4

2025/4/21 22:33 Menu Performance report

| Name | Category | Price | Order Count | Total Sales | % of Total Sales |
|---|---|---|---|---|---|
| Spicy Chicken Wings (5 pieces) | Fried Chicken | $7.0 | 2 | $14.0 | 1.4% |

Rows highlighted in green are your top 3 best-selling items.

127.0.0.1:5000/reports/menu 4/4

The Menu Presentation Report page provides managers with an in-depth analysis of the sales performance of menu items by combining bar charts, pie charts and detailed sales data tables, helping restaurants optimize the menu structure and pricing strategy and improve the overall performance of the restaurant.

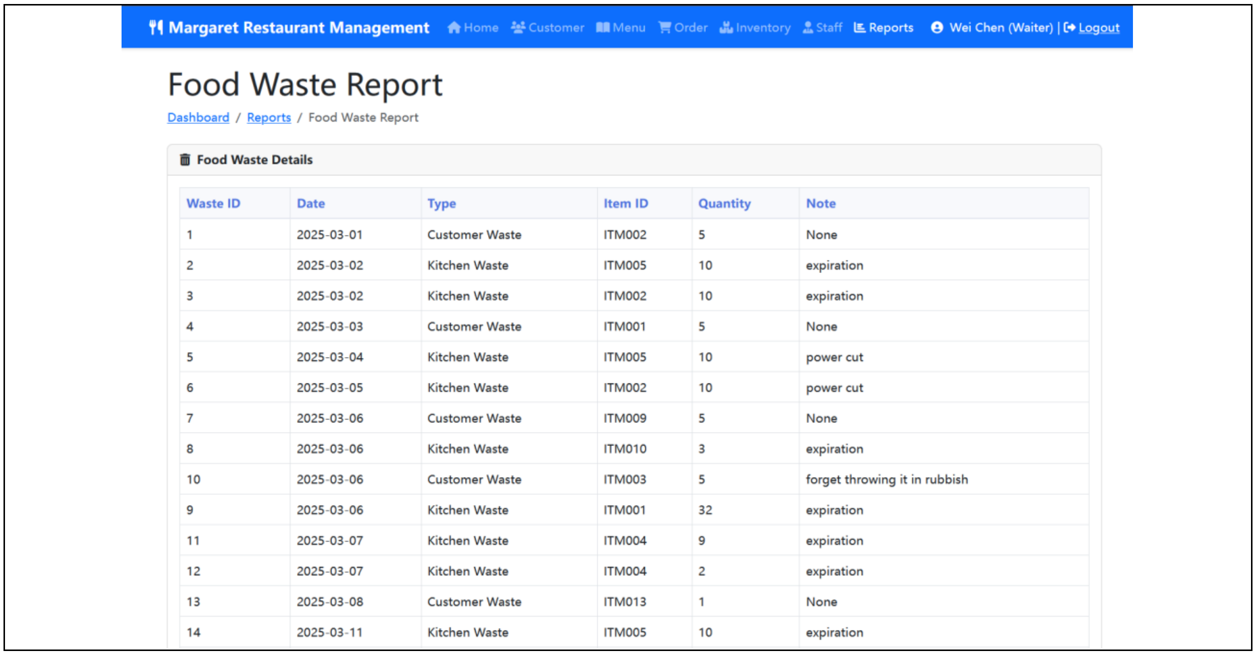
This page presents the sales report of the restaurant's menu items, providing managers with detailed data on the sales performance of the menu items and helping them optimize menu design and marketing strategies. The top of the page shows the total number of orders, the total sales volume, and the average price of each menu item, providing managers with an overview of the overall sales situation of the restaurant.

The "Top 10 Menu Items by Sales" section shows the top 10 best-selling dishes through a bar chart. The length of each bar indicates the sales volume of the corresponding menu item, helping managers understand which dishes are the most popular among customers.

The "Sales by Category" section uses pie charts to show the sales proportions of different categories of dishes, including hamburgers, beverages, fried chicken, desserts, hot dogs, etc., to help managers understand which

categories of dishes account for a large share of the total sales. For example, the sales of hamburgers and pizzas are relatively high.
The "Category Analysis" section provides detailed data for each category, including the number of orders, total sales, average price, and the percentage of total sales.
The "All Menu Items Performance" section lists the detailed performance of all the dishes in the restaurant, including the name, category, price, order quantity, total sales amount and sales proportion of each dish.

**Food waste report**


Margaret Restaurant Management Home Customer Menu Order Inventory Staff Reports Wei Chen (Waiter) | Logout

Food Waste Report

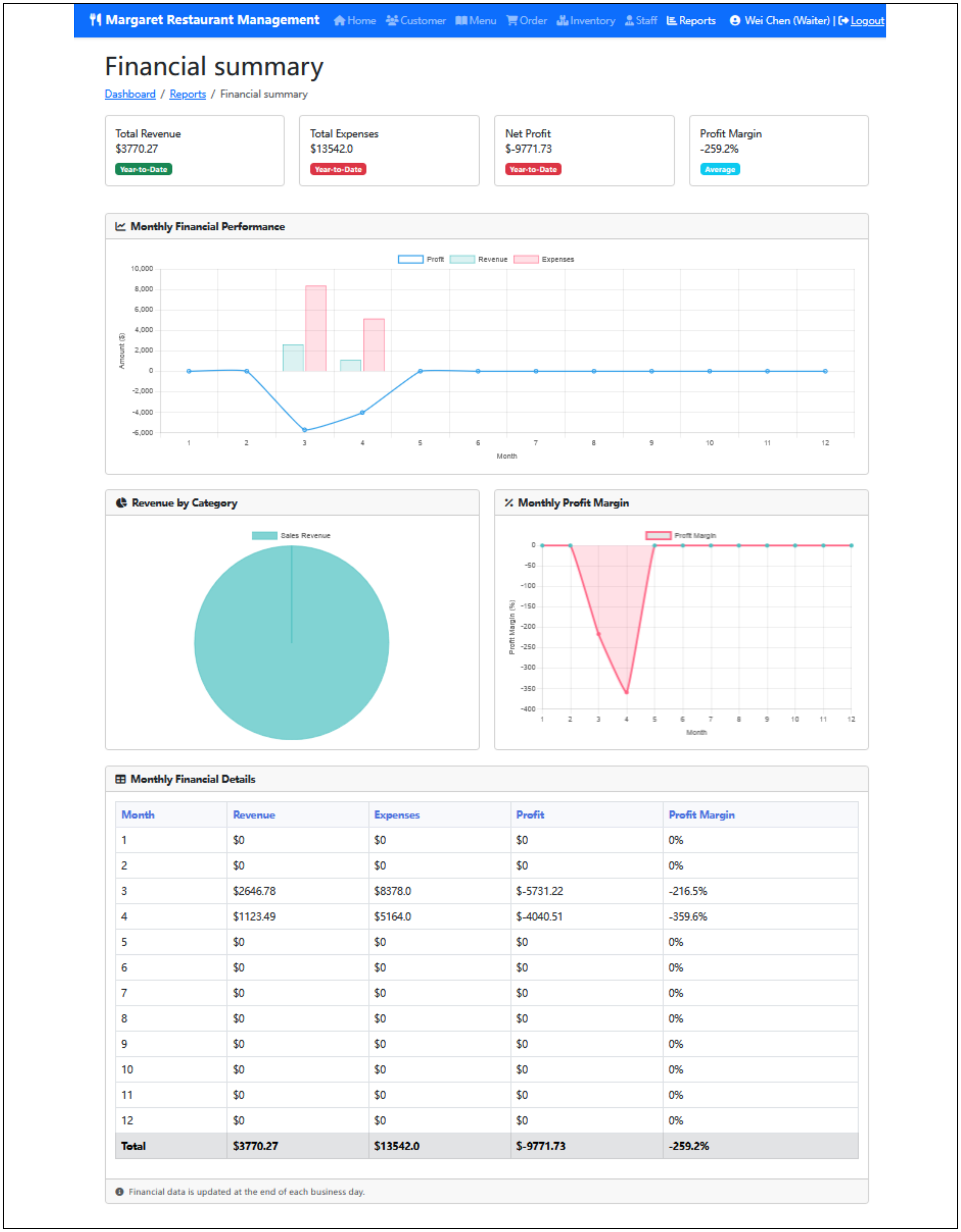
Dashboard / Reports / Food Waste Report

Food Waste Details

| Waste ID | Date | Type | Item ID | Quantity | Note |
|---|---|---|---|---|---|
| 1 | 2025-03-01 | Customer Waste | ITM002 | 5 | None |
| 2 | 2025-03-02 | Kitchen Waste | ITM005 | 10 | expiration |
| 3 | 2025-03-02 | Kitchen Waste | ITM002 | 10 | expiration |
| 4 | 2025-03-03 | Customer Waste | ITM001 | 5 | None |
| 5 | 2025-03-04 | Kitchen Waste | ITM005 | 10 | power cut |
| 6 | 2025-03-05 | Kitchen Waste | ITM002 | 10 | power cut |
| 7 | 2025-03-06 | Customer Waste | ITM009 | 5 | None |
| 8 | 2025-03-06 | Kitchen Waste | ITM010 | 3 | expiration |
| 10 | 2025-03-06 | Customer Waste | ITM003 | 5 | forget throwing it in rubbish |
| 9 | 2025-03-06 | Kitchen Waste | ITM001 | 32 | expiration |
| 11 | 2025-03-07 | Kitchen Waste | ITM004 | 9 | expiration |
| 12 | 2025-03-07 | Kitchen Waste | ITM004 | 2 | expiration |
| 13 | 2025-03-08 | Customer Waste | ITM013 | 1 | None |
| 14 | 2025-03-11 | Kitchen Waste | ITM005 | 10 | expiration |

This Food Waste Report comes from the Margaret Restaurant Management System and shows the details of food loss in the restaurant within a certain period of time. The report contains the loss number, date, type of loss (divided into loss caused by customers and loss caused by the kitchen), material number, quantity of loss and remarks information. The remarks record the reasons for loss such as expiration, power outage, forgetting and discarding. This report is helpful for restaurants to analyze food loss and take targeted measures to reduce waste.

## Financial summary




| Month | Revenue | Expenses | Profit | Profit Margin |
|---|---|---|---|---|
| 1 | $0 | $0 | $0 | 0% |
| 2 | $0 | $0 | $0 | 0% |
| 3 | $2646.78 | $8378.0 | $-5731.22 | -216.5% |
| 4 | $1123.49 | $5164.0 | $-4040.51 | -359.6% |
| 5 | $0 | $0 | $0 | 0% |
| 6 | $0 | $0 | $0 | 0% |
| 7 | $0 | $0 | $0 | 0% |
| 8 | $0 | $0 | $0 | 0% |
| 9 | $0 | $0 | $0 | 0% |
| 10 | $0 | $0 | $0 | 0% |
| 11 | $0 | $0 | $0 | 0% |
| 12 | $0 | $0 | $0 | 0% |
| **Total** | **$3770.27** | **$13542.0** | **$-9771.73** | **-259.2%** |

The Financial Summary page helps restaurant managers have a comprehensive understanding of the financial situation by providing detailed analyses of income, expenditure, profit and profit margin, and take timely measures to adjust business strategies and improve profitability. This page presents an overview of the restaurant's finances, helping managers understand financial conditions such as revenue, expenses, profits and profit margins. Four key indicators are displayed at the top of the page: total revenue, total expenditure, net profit and profit margin. These data provide managers with the overall financial health status of the restaurant.

The "Monthly Financial Performance" section presents the trends of monthly income, expenditure and profit changes through charts. The blue lines in the chart represent profits, the green bars represent revenues, and the red bars represent expenditures. Through this chart, managers can intuitively see the relationship among income, expenditure and profit each month. The "Revenue by Category" section presents the classification of restaurant revenues through pie charts. In this chart, the proportion of sales revenue in the total revenue is shown, helping managers understand the sources of revenue. The "Monthly Profit Margin" section shows the trend of the monthly profit margin. Through this chart, managers can view the changes in profit margins each month. The "Monthly Financial Details" section lists the specific income, expenditure, profit and profit margin data for each month. Through these specific data, managers can conduct an in-depth analysis of the restaurant's financial performance and identify which months 'financial situation requires special attention.

# V. Future recommendations

**Continuously optimize the system functions**

We will introduce more advanced data analysis algohms and tools to deeply mine the data accumulated by the system. In addition to the existing analysis, a comprehensive analysis of different seasons, special festivals and other factors can also be conducted to optimize the procurement plan and reduce food waste and inventory overstock. For instance, during the exam period, students' demand for energy-rich foods (such as hamburgers and coffee) may increase, and restaurants can adjust their purchase volume and inventory in advance.

We will further improve the inventory management function. When the inventory is below the safety threshold, the system automatically generates purchase orders and sends them to suppliers, achieving the automation of the procurement process, ensuring the timely supply of ingredients, and reducing inventory costs at the same time.

**Improve user experience**

We will further increase the membership service functions. We will add a membership level system and points redemption function. Members are classified into different levels based on indicators such as their consumption amount and frequency, and exclusive discounts are provided for members of different levels to enhance their loyalty.

**Expand the application of the system**

We will attempt to cooperate with nearby merchants. Based on the user data and operation data of the system, we can achieve resource sharing and get complementary advantages. For instance, cooperate with nearby fruit stores and coffee shops to launch joint packages or coupons, thereby expanding the service scope and customer base of the restaurant. Through the system, order collaboration and data sharing among cooperative merchants are realized to improve cooperation efficiency.

We will also support the subsequent chain development of the restaurant. If the restaurant has plans to open chain stores, the system should be scalable and capable of supporting the unified management of multiple stores. The headquarters can monitor and analyze the operation of each store and formulate unified marketing strategies and management standards.

Finally, we will also integrate into the campus life ecosystem. As a campus restaurant, we will enhance the integration with other service systems of the school (such as the campus card system) to achieve the integration of more functions. For instance, students can use the balance of their campus cards to order food.

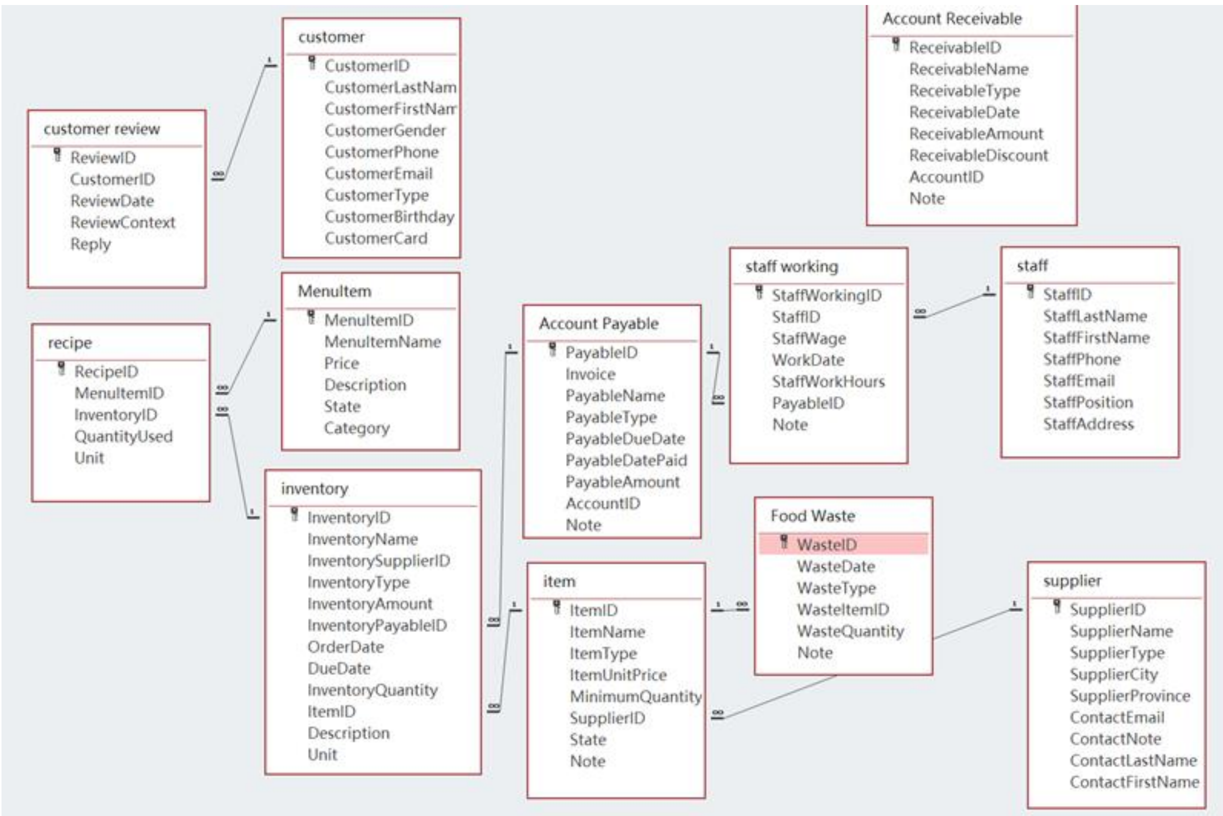
**Conclusion**

We successfully advanced the work in accordance with the project plan and completed the development of the Margaret super management system with high quality. After actual operation tests, the system performed outstandingly and effectively solved many management problems that the restaurant had previously faced. Through the application of the system, the restaurant will achieve a significant reduction in costs, a steady increase in sales, and an improvement in operational efficiency. During the project development process, we have accumulated rich practical experience, significantly improved our professional skills, and gained profound professional knowledge in the field of catering management system development, laying a solid foundation for undertaking more similar projects in the future.

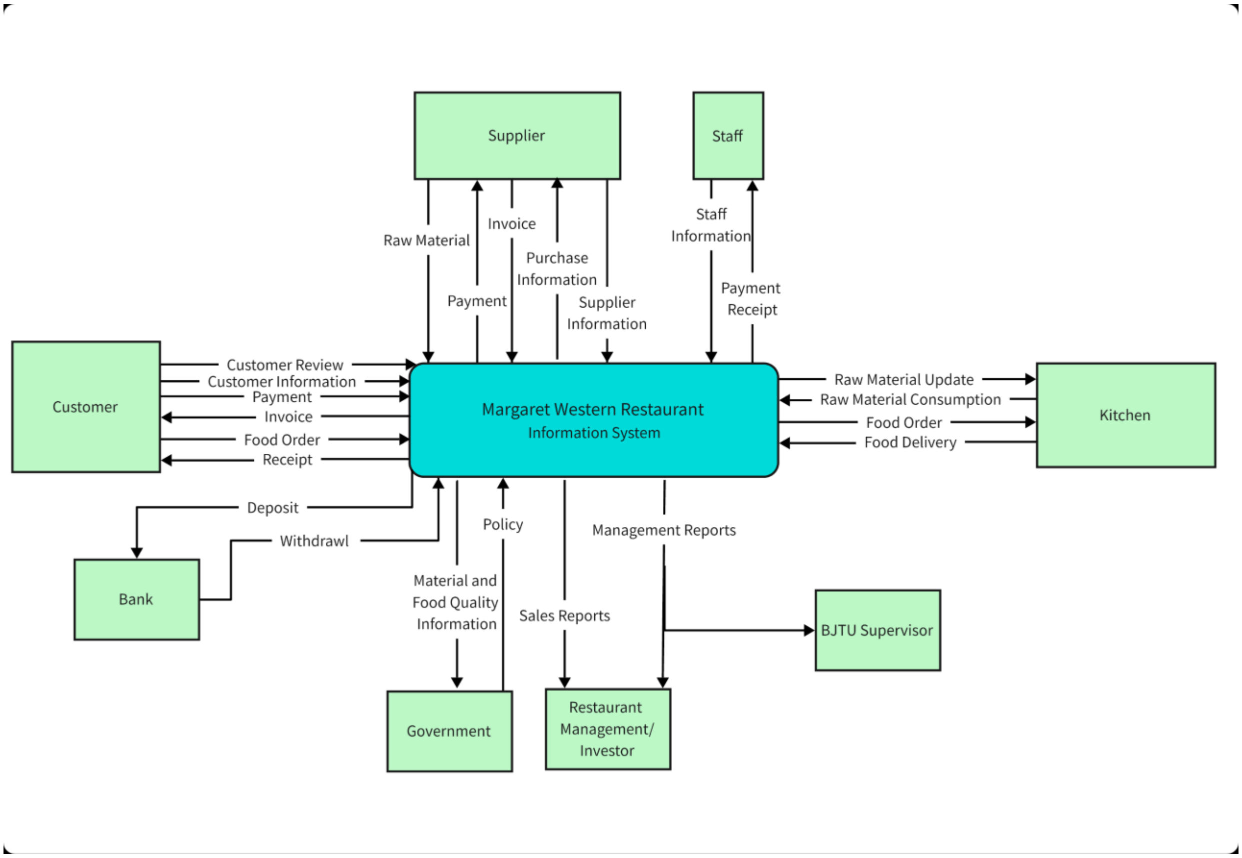
We will establish a communication platform with the management of Margaret Western Restaurant during the use of the system. After the system is deployed, we will arrange regular phone follow-ups to understand the usage experience of the system. Every quarter, technicians are dispatched to the restaurant for on-site investigation, to gain a thorough understanding of the system's operation status, solve technical problems, and collect new demands. If the restaurant encounters any problems during operation, the management can contact us at any time by phone (6666-8888), and we will immediately organize a technical team to provide solutions.

# VI Appendix

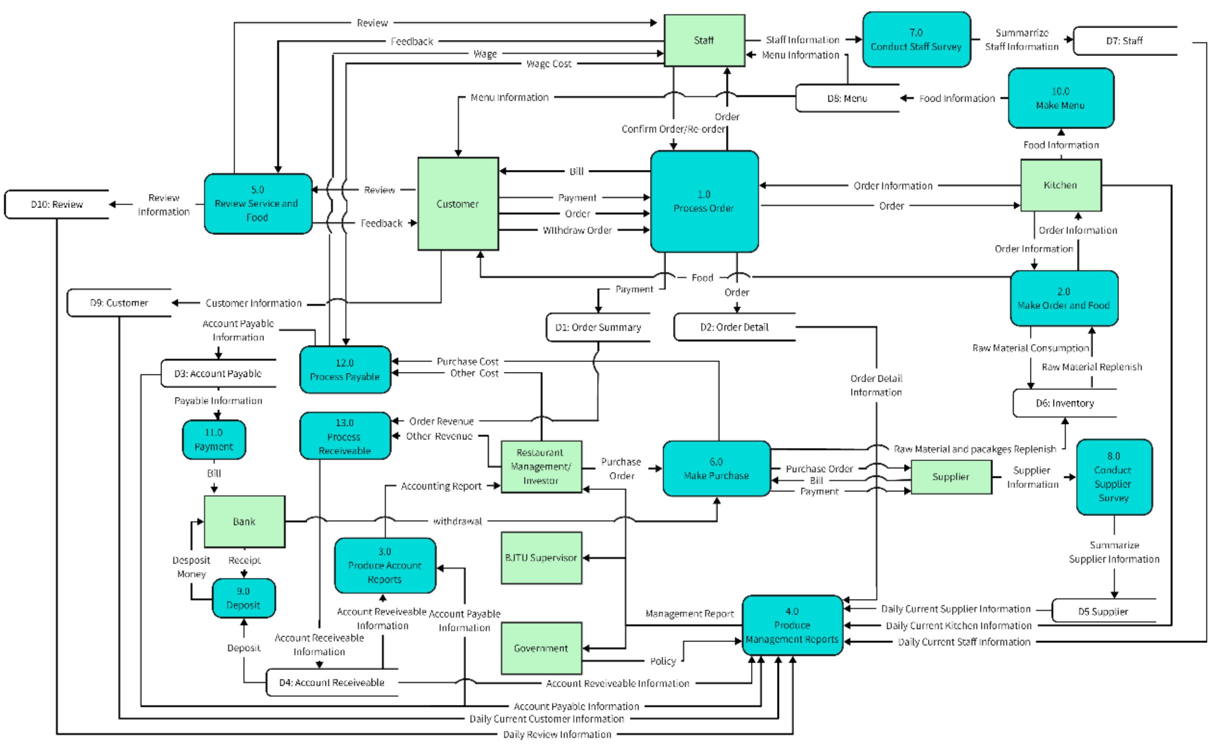
**Updated ERD**



**Updated DFD**




Context level DFD




Level 0 DFD

**Updated Gantt Chart of Margaret super management system**

| TASK NAME | START DATE | END DATE | DURATION (WORK DAYS) | TEAM MEMBER | PERCENT COMPLETE |
|---|---|---|---|---|---|
| **Initial Project Proposal** | | | | | |
| Identify Restaunrant Business | 2/26 | 2/28 | 3 | All | 100% |
| State Restaunrant Background and Problems | 3/1 | 3/3 | 3 | Gao Tianrun | 100% |
| Identify Organization Purpose and Objectives | 3/3 | 3/4 | 2 | Dong Renxi/Wang Kexin | 100% |
| Determine System Functions | 3/4 | 3/5 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| Identify Output Managerial Information | 3/5 | 3/6 | 2 | Zhu Yifan | 100% |
| Define Automation Benefits | 3/7 | 3/7 | 1 | All | 100% |
| **Feasibility Study / Planning Phase** | | | | | |
| Identify Business Problem and scope | 3/10 | 3/11 | 2 | Gao Tianrun/Zhu Yifan | 100% |
| Technical Feasibility Analysis | 3/11 | 3/13 | 3 | Dong Renxi | 100% |
| Economic Feasibility Analysis | 3/11 | 3/14 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Schedule Feasibility Analysis | 3/12 | 3/15 | 4 | Wang Kexin | 100% |
| Operational Feasibility Analysis | 3/12 | 3/15 | 4 | Hu Shaohua/Liu Zhuodong | 100% |
| Initial Design – Reports / Outputs | 3/15 | 3/17 | 3 | All | 100% |
| **Application Design** | | | | | |
| Generate DFD and ERD | 3/19 | 3/20 | 2 | Zhu Yifan | 100% |
| Design The User Interfaces | 3/19 | 3/21 | 3 | Gao Tianrun | 100% |
| Design The System Interfaces | 3/20 | 3/21 | 2 | Hu Shaohua/Liu Zhuodong | 100% |
| Design and Integrate The Database | 3/22 | 3/22 | 1 | Hu Shaohua | 100% |
| Prototype For Design Details | 3/23 | 3/24 | 1 | Gao Tianrun/Zhu Yifan | 100% |
| Design and Integrate The System Controls | 3/25 | 3/26 | 2 | Dong Renxi/Wang Kexin | 100% |
| **Prototyping / Data Definitions – Design Phase** | | | | | |
| Define Tables | 3/28 | 4/2 | 4 | Gao Tianrun | 100% |
| Define Table Fields | 3/30 | 4/2 | 3 | Wang Kexin | 100% |
| Define Table Indexes | 3/31 | 4/2 | 3 | Dong Renxi | 100% |
| Define Queries | 4/2 | 4/4 | 3 | Liu Zhuodong | 100% |
| Define Queries Fields | 4/2 | 4/5 | 4 | Hu Shaohua | 100% |
| Print out Actual Reports | 4/5 | 4/7 | 3 | Zhu Yifan | 100% |
| **Implementation Phase & Final Report** | | | | | |
| Construct Software Components | 4/9 | 4/12 | 3 | Gao Tianrun/Zhu Yifan | 100% |
| Verify and Test | 4/11 | 4/13 | 1 | Gao Tianrun | 100% |
| Convert Data | 4/12 | 4/14 | 1 | Zhu Yifan | 100% |
| Train users and Document System | 4/13 | 4/15 | 2 | Liu Zhuodong | 100% |
| Install System | 4/14 | 4/16 | 3 | Hu Shaohua | 100% |
| **Documentation** | | | | | |
| Write Executive Overview | 4/18 | 4/21 | 4 | Gao Tianrun | 100% |
| Written components | 4/19 | 4/21 | 3 | Gao Tianrun/Dong Renxi/Wang Kexin | 100% |
| Database Application | 4/20 | 4/21 | 2 | All | 100% |